



Static Analysis Results Interchange Format (SARIF) Version 2.2

Committee Specification Draft 01

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Additional artifacts:

This prose specification is one component of a Work Product that also includes:

- SARIF schema: <https://docs.oasis-open.org/sarif/sarif/v2.2/csd01/schema/sarif.json>.
Latest stage: <https://docs.oasis-open.org/sarif/sarif/v2.2/schema/sarif.json>.
- SARIF External Property File schema: <https://docs.oasis-open.org/sarif/sarif/v2.2/csd01/schema/sarif-external-property-file-schema-2.2.json>.
Latest stage: <https://docs.oasis-open.org/sarif/sarif/v2.2/schema/sarif-external-property-file-schema-2.2.json>.

Related work:

This specification replaces or supersedes:

- *Static Analysis Results Interchange Format (SARIF) Version 2.1.0 Plus Errata 01*. Edited by Michael C. Fanning and Laurence J. Golding. 12 July 2023. OASIS Standard incorporating Approved Errata. <https://docs.oasis-open.org/sarif/sarif/v2.1.0/errata01/os/sarif-v2.1.0-errata01-os-complete.html>. Latest stage: <https://docs.oasis-open.org/sarif/sarif/v2.1.0.html>.

Abstract:

This document defines a standard format for the output of static analysis tools. The format is referred to as the “Static Analysis Results Interchange Format” and is abbreviated as SARIF.

Status:

This document was last revised or approved by OASIS Static Analysis Results Interchange Format (SARIF) on the above date. The level of approval is also listed above. Check the “Latest stage” location noted above for possible later revisions of this document. Any other numbered Versions and other technical work produced by the Technical Committee (TC) are listed at https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=sarif#technical.

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Key words:

The key words “MUST”, “MUST NOT”, “REQUIRED”, “SHALL”, “SHALL NOT”, “SHOULD”, “SHOULD NOT”, “RECOMMENDED”, “NOT RECOMMENDED”, “MAY”, and “OPTIONAL” in this document are to be interpreted as described in BCP 14 [RFC2119] and [RFC8174] when, and only when, they appear in all capitals, as shown here.

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1 Introduction

Software developers use a variety of analysis tools to assess the quality of their programs. These tools report results which can indicate problems related to program qualities such as correctness, security, performance, compliance with contractual or legal requirements, compliance with stylistic standards, understandability, and maintainability. To form an overall picture of program quality, developers often need to aggregate the results produced by all of these tools. This aggregation is more difficult if each tool produces output in a different format.

This document defines a standard format for the output of static analysis tools, called the Static Analysis Results Interchange Format, or “SARIF”¹. The goals of the format are:

- Comprehensively capture the range of data produced by commonly used static analysis tools.
- Be a useful format for analysis tools to emit directly, and also an effective interchange format into which the output of any analysis tool can be converted.
- Be suitable for use in a variety of scenarios related to analysis result management and be extensible for use in new scenarios.
- Reduce the cost and complexity of aggregating the results of various analysis tools into common workflows.
- Capture information that is useful for assessing a project’s compliance with corporate policy or certification standards.
- Adopt a widely used serialization format that can be parsed by readily available tools.
- Represent analysis results for all kinds of artifacts, including source code and object code.

Although most static analysis tools analyze files on disk, SARIF can represent results detected in any URI-addressable artifact (for example, the text returned by an HTTP query). This specification uses the term “artifact” to refer to any item that a tool might analyze. It uses the more restrictive term “file” when referring specifically to a file on disk.

1.1 IPR Policy

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1.2 Terminology

The key words “MUST”, “MUST NOT”, “REQUIRED”, “SHALL”, “SHALL NOT”, “SHOULD”, “SHOULD NOT”, “RECOMMENDED”, “NOT RECOMMENDED”, “MAY”, and “OPTIONAL” in this document are to be interpreted as described in “Key words for use in RFCs to Indicate Requirement Levels” [BCP14] [RFC2119] and “Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words” [RFC8174] when, and only when, they appear in all capitals, as shown here.

For purposes of this document, the following terms and definitions apply:

analysis target artifact which an analysis tool is instructed to analyze

analysis tool tool that models and analyzes an artifact or the interaction between entities (cf. web analysis tool for an example)

artifact sequence of bytes addressable *via* a URI Examples: A physical file in a file system such as a source file, an object file, a configuration file or a data file; a specific version of a file in a version control system; a database table accessed *via* an HTTP request; an arbitrary stream of bytes returned from an HTTP request.

baseline set of results produced by a single run of a set of analysis tools on a set of artifacts NOTE: A result management system can compare the results of a subsequent run to a baseline produced by a baseline run to determine whether new results have been introduced.

¹Pronounced ‘sæ-rɪf’ (“a” as in “cat”, “i” as in “if”, emphasis on the first syllable).

baseline run run that produces a baseline to which subsequent runs can be compared

binary artifact artifact considered as a sequence of bytes

binary region region representing a contiguous range of zero or more bytes in a binary artifact

call stack sequence of nested function calls

camelCase name name that begins with a lowercase letter, in which each subsequent word begins with an uppercase letter
Example: `camelCase`, `version`, `fullName`.

code flow set of one or more thread flows which together specify a pattern of code execution relevant to detecting a result

column (number) 1-based index of a character within a line

configuration file file, typically textual, that configures the execution of an analysis tool or tool component

converter SARIF producer that transforms the output of an analysis tool from its native output format into the SARIF format

custom taxonomy taxonomy defined by and intended for use with a particular analysis tool

direct producer analysis tool which acts as a SARIF producer

driver tool component containing an analysis tool's or converter's primary executable, which controls the tool's or converter's execution, and which in the case of an analysis tool typically defines a set of analysis rules

embedded link syntactic construct which enables a message string to refer to a location within an artifact mentioned in a result

engineering system software development environment within which analysis tools execute NOTE: An engineering system might include a build system, a source control system, a result management system, a bug tracking system, a test execution system, and so on.

empty array array that contains no elements, and so has a length of 0

empty object object that contains no properties

empty string string that contains no characters, and so has a length of 0

(end) user person who uses the information in a log file to investigate, triage, or resolve results

extension tool component other than the driver (for example, a plugin, a configuration file, or a taxonomy)

external property file file containing the values of one or more externalized properties

externalizable property property that can be contained in an external property file

externalized property property stored outside of the SARIF log file to which it logically belongs

false positive result which an end user decides does not actually represent a problem

fingerprint stable value that can be used by a result management system to uniquely identify a result over time, even if a relevant artifact is modified

formatted message message string which contains formatting information such as Markdown formatting characters

fully qualified logical name string that fully identifies the programmatic construct specified by a logical location, typically by means of a hierarchical identifier. Example: The fully qualified logical name of the C# method `f(void)` in class C in namespace N is `"N.C.f(void)"`. Its logical name is `"f(void)"`.

hierarchical string string in the format `<component>{/<component>}*`

line contiguous sequence of characters, starting either at the beginning of an artifact or immediately after a newline sequence, and ending at and including the nearest subsequent newline sequence, if one is present, or else extending to the end of the artifact

line (number) 1-based index of a line within a file NOTE: Abbreviated to "line" when there is no danger of ambiguity with "line" in the sense of a sequence of characters.

localizable subject to being translated from one natural language to another

log file output file produced by an analysis tool, which enumerates the results produced by the tool

(log file) viewer SARIF consumer that reads a log file, displays a list of the results it contains, and allows an end user to view each result in the context of the artifact in which it occurs

logical location location specified by reference to a programmatic construct, without specifying the artifact within which that construct occurs Example: A class name, a method name, a namespace.

logical name string that partially identifies the programmatic construct specified by a logical location by specifying the most specific (often the rightmost) component of its fully qualified logical name. Example: The logical name of the C# method `f(void)` in class C in namespace N is `"f(void)"`. Its fully qualified logical name is `"N.C.f(void)"`.

message string human-readable string that conveys information relevant to an element in a SARIF file

nested artifact artifact that is contained within another artifact

nested logical location logical location that is contained within another logical location Example: A method within a class

in C++

newline sequence sequence of one or more characters representing the end of a line of text NOTE: Some systems represent a newline sequence with a single newline character; others represent it as a carriage return character followed by a newline character.

notification reporting item that describes a condition encountered by a tool during its execution

opaque neither human-readable nor machine-parseable into constituent parts

parent (artifact) artifact which contains one or more nested artifacts

physical location location specified by reference to an artifact, possibly together with a region within that artifact

plain text message message string which does not contain any formatting information

plugin tool component that defines additional rules

policy set of rule configurations that specify how results that violate the rules defined by a particular tool component are to be treated

problem result which indicates a condition that has the potential to detract from the quality of the program Example: A security vulnerability, a deviation from contractual or legal requirements, a deviation from stylistic standards.

property attribute of an object consisting of a name and a value associated with the name

property bag object consisting of an unordered set of non-standardized properties with arbitrary camelCase names

redactable property property that potentially contains sensitive information that a SARIF direct producer or a SARIF post-processor might wish to redact

region contiguous portion of an artifact

reporting item unit of output produced by a tool, either a result or a notification

reporting configuration the subset of reporting metadata that a tool can configure at runtime, before performing its scan Examples: severity level, rank

reporting descriptor container for reporting metadata

reporting metadata information that describes a class of related reporting items Examples: id, description

repository container for a related set of files in a version control system

response file file containing arguments for a tool, which are interpreted as if they had appeared directly on the command line

result reporting item that describes a condition present in an artifact

result file artifact in which an analysis tool detects a result

result management system software system that consumes the log files produced by analysis tools, produces reports that enable engineering teams to assess the quality of their software artifacts at a point in time and to observe trends in the quality over time, and performs functions such as filing bugs and displaying information about individual results NOTE: A result management system can interact with a log file viewer to display information about individual defects.

result matching process of determining whether two results are reporting the same condition in the code

root file SARIF log file to which one or more external property files logically belong

rule specific criterion for correctness verified by an analysis tool NOTE 1: Many analysis tools associate a rule id with each result they report, but some do not. NOTE 2: Some rules verify generally accepted criteria for correctness; others verify conventions in use in a particular team or organization. Examples: “Variables must be initialized before use.”, “Class names must begin with an uppercase letter.”.

rule configuration reporting configuration that applies to a rule

rule id stable value which an analysis tool associates with a rule NOTE: A rule id is more likely to remain stable if it is a symbolic or numeric value, as opposed to a descriptive string. Example: CA2001

rule metadata reporting metadata that describes a rule

run

1. invocation of a specified analysis tool on a specified version of a specified set of analysis targets, with a specified set of runtime parameters
2. set of results produced by such an invocation

SARIF consumer program that reads and interprets a SARIF log file

SARIF log file log file in the format defined by this document

SARIF post-processor SARIF producer that transforms an existing SARIF log file into a new SARIF log file, for example, by removing or redacting security-sensitive elements.

SARIF producer program that emits output in the SARIF format

stable value value which, once established, never changes over time

standard taxonomy taxonomy defined without reference to a particular analysis tool

(static analysis) tool program that examines artifacts to detect problems, without executing the program Example: Lint

taxon (pl. taxa) one of a set of categories which together comprise a taxonomy

taxonomy classification of analysis results into a set of categories

tag string that conveys additional information about the SARIF log file element to which it applies

text artifact artifact considered as a sequence of characters organized into lines and columns

text region region representing a contiguous range of zero or more characters in a text artifact

thread flow temporally ordered set of code locations specifying a possible execution path through the code, which occur within a single thread of execution, such as an operating system thread or a fiber

tool component component of an analysis tool or converter, either its driver or an extension, consisting of one or more files

top-level artifact artifact which is not contained within any other artifact

top-level logical location logical location that is not nested within another logical location Example: A global function in C++

translation rendering of a tool component's localizable strings into another language

triage decide whether a result indicates a problem that needs to be corrected

user see end user.

VCS version control system

viewer see log file viewer.

web analysis tool analysis tool that models and analyzes the interaction between a web client and a server.

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1.5 Trademarks

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2 Conventions

2.1 General

The following conventions are used within this document.

2.2 Format Examples

This document contains several partial examples of the JSON serialization of the SARIF format. The examples are formatted for clarity, as permitted by JSON [RFC8259], which allows “insignificant whitespace” before or after any token; implementations do not need to follow the whitespace convention used in these examples. The examples also employ typographical conventions that are not part of the JSON or SARIF formats:

- An ellipsis (...) is used to indicate that portions of the log file text required by this document have been omitted for brevity.
- A ‘#’ character introduces a comment that extends to the end of the line.
- When a JSON string is too long to fit on a line, it is broken into multiple lines.
- Some examples have italicized line numbers in the left margin.

2.3 Property Notation

A SARIF object consists of a set of properties. The value of a property can itself be an object, allowing arbitrary nesting. When necessary for clarity or to avoid ambiguity, we use the “dot” notation to refer to nested values. For example, the `physicalLocation` object defines a property `region` whose value is a `region` object, which in turn contains a `charLength` property. For clarity, we can refer to the `charLength` property as `physicalLocation.region.charLength`.

2.4 Syntax Notation

Where this document describes a syntactic construct, it uses the extended Backus-Naur form (EBNF) [ISO14977:1996].

In all EBNF definitions in this spec:

- The following syntax rules are assumed:

```
decimal digit = '0' | '1' | '2' | '3' | '4' | '5' | '6' | '7' | '8' | '9';
```

```
non negative integer =
```

```
"0"
```

```
| decimal digit - '0', { decimal digit };
```

- The following “special sequence” (see EBNF [ISO14977:1996], §4.19 and §5.11) refers to any character that can appear in a JSON string according to JSON [ECMA404]:

```
? JSON string character ?
```

2.5 Commonly Used Objects

This document uses the following notation for certain commonly used objects:

Notation	Commonly used object
the <code>sarifLog</code>	The <code>sarifLog</code> object of the SARIF log file.
the <code>run</code>	The run object (§3.14) containing the object under discussion.
the <code>tool</code>	The value of the <code>run.tool</code> (§3.14.6)
the <code>reportingDescriptor</code>	The <code>reportingDescriptor</code> object (§3.49) identified by the <code>reportingDescriptorReference</code> object (§3.52) under discussion.
the <code>toolComponent</code>	The <code>toolComponent</code> object (§3.19) identified by the <code>toolComponentReference</code> object (§3.54) under discussion.
the <code>result</code>	The <code>result</code> object (§3.27) containing the object under discussion.
this <code>object</code>	The object containing the property under discussion. NOTE: Usually when the description of a property refers to another property of the same object, the other property is referred to by its unqualified name. When necessary to avoid confusion, the name of the other property is qualified with " <code>thisObject.</code> " to emphasize that it is a property of the object under discussion. For an example, see §3.27.7.
the <code>locations</code>	The <code>locations</code> object (§3.27) or <code>notification</code> object (§3.58) with a <code>locations</code> array containing the location object (§3.28) under discussion.

3 File Format

3.1 General

SARIF defines an object model, the top level of which is the `sarifLog` object (§3.13), which contains the results of one or more analysis runs. The runs do not need to be produced by the same analysis tool.

A SARIF log file **SHALL** contain a serialization of the SARIF object model into the JSON format.

NOTE 1: In the future, other serializations might be defined.

The top-level value in the log file, representing the `sarifLog` object, **SHALL** conform to the JSON object grammar; that is, it **SHALL** consist of a comma-separated sequence of name/value pairs, enclosed in curly brackets, as specified by JSON [RFC8259].

The case of names (keys) of the name/value pairs is as defined in this specification. Using identical letters but different case leads to different names as per JSON [RFC8259], i.e. the identity of two keys is a case sensitive operation.

A SARIF log file **SHALL** be encoded in UTF-8 [RFC3629].

NOTE 2: JSON [RFC8259] requires this encoding for any JSON text “exchanged between systems that are not part of a closed ecosystem.”

Regardless of future added formats, SARIF files **SHOULD NOT** contain case only variations of required properties.

NOTE 3: SARIF files with properties name variations on e.g. runs like RUNS, Runs, or similar can confuse processors and human consumers alike.

3.2 SARIF File Naming Convention

The file name of a SARIF log file **SHOULD** end with the extension `".sarif"`.

EXAMPLE 1: `output.sarif`

The file name **MAY** end with the additional extension `".json"`.

EXAMPLE 2: `output.sarif.json`

3.3 artifactContent Object

3.3.1 General

Certain properties in this document represent the contents of portions of artifacts external to the log file, for example, artifacts that were scanned by an analysis tool. SARIF represents such content with an `artifactContent` object. Depending on the circumstances, the SARIF log file might need to represent this content as readable text, raw bytes, or both.

3.3.2 text Property

If the external artifact is a text artifact, an `artifactContent` object **SHOULD** contain a property named `text` whose value is a string containing the relevant text. Since SARIF log files are encoded in UTF-8 ([RFC3629]; see §3.1), this means that if the external artifact is a text artifact in any encoding other than UTF-8, the SARIF producer **SHALL** transcode the text to UTF-8 before assigning it to the `text` property. The SARIF producer **SHALL** escape any characters that JSON [RFC8259] requires to be escaped.

Notwithstanding any necessary transcoding and escaping, the SARIF producer **SHALL** preserve the text artifact's line breaking convention (for example, `"\n"` or `"\r\n"`).

If the external artifact is a binary artifact, the `text` property **SHALL** be absent.

3.3.3 binary Property

If the external artifact is a binary artifact, or if the SARIF producer cannot determine whether the external artifact is a text artifact or a binary artifact, an `artifactContent` object **SHALL** contain a property named `binary` whose value is a string containing the MIME Base64 encoding [RFC2045] of the bytes in the relevant portion of the artifact.

If the external artifact is a text artifact in an encoding other than UTF-8, the `binary` property **MAY** be present, in which case it **SHALL** contain the MIME Base64 encoding of the bytes representing the relevant text in its original encoding.

If the external artifact is a UTF-8 text artifact, the `binary` property **SHOULD** be absent. If it is present, it **SHALL** contain the MIME Base64 encoding of the UTF-8 bytes representing the relevant text.

3.3.4 rendered Property

An `artifactContent` object **MAY** contain a property named `rendered` whose value is a `multiFormatMessageString` object (§3.12) that provides a rendered view of the contents.

EXAMPLE 1: In this example, a `physicalLocation` object (§3.29) denotes a memory address. Its `region.snippet.rendered` property (§3.29.4, §3.30.13) offers a hex view of the relevant address range. The `markdown` property (§3.12.4) emphasizes a byte of particular interest.

```
{
  "address": {
    "baseAddress": 4202880,
    "offset": 64
  },
  "region": {
    "snippet": {
      "rendered": {
        "text": "00 00 01 00 00 00 00 00",
        "markdown": "00 00 **01** 00 00 00 00"
      }
    }
  }
}
```


3.4 artifactLocation Object

3.4.1 General

Certain properties in this document specify the location of an artifact. SARIF represents an artifact's location with an `artifactLocation` object. The most important member of an `artifactLocation` object is its `uri` property (§3.4.3). If the `uri` property contains a relative reference (the term used in the URI standard [RFC3986] for what is commonly called a “relative URI”), the `uriBaseId` property (§3.4.4) can sometimes be used to resolve the relative reference to an absolute URI.

3.4.2 Constraints

At least one of the `uri` property (§3.4.3) or the `index` property (§3.4.5) **SHALL** be present. In certain circumstances (see §3.4.4 and §3.4.5), they **MAY** both be present.

NOTE: Providing both `uri` and `index` makes the log file more readable at the expense of increased size. Providing only `index` reduces log file size but makes it less readable to an end user, who has to determine the URI by locating the `artifact` object (§3.24) at the index within `theRun.artifacts` (§3.14.15) specified by `index`.

If both `uri` and `index` are present, they **SHALL** both denote the same artifact. That is, let URI1 be the fully resolved URI of the artifact specified by an `artifactLocation` object as determined by the `uriBaseId` resolution procedure described in §3.4.4. Let URI2 be the fully resolved URI of the artifact specified by the `artifact` object indicated by `index`, determined in the same way. Then URI1 and URI2 **SHALL** be equivalent in the sense described in §3.10.1.

3.4.3 uri Property

Depending on the circumstances, an `artifactLocation` object either **SHALL**, **SHALL NOT**, or **MAY** contain a property named `uri` whose value is a string containing a URI [RFC3986] that specifies the location of the artifact.

If `thisObject` describes a nested artifact whose location within its parent container can be expressed by a path from the root of the container, then if `uri` is present, it **SHALL** specify a relative-path reference per section 4.2 of [RFC3986] expressing that path. A relative reference **SHALL NOT** begin with two slash characters (a ‘network-path’ reference per section 4.2 of [RFC3986]). A relative reference **SHALL NOT** begin with a single slash character (an ‘absolute-path’ reference per section 4.2 of [RFC3986]) unless doing so is required to distinguish between distinct items in archive formats, such as zip and tar.

NOTE 1: For example, `"/a.txt"` and `"a.txt"` can both exist as distinct files in the same archive.

NOTE 2: A relative path is useful to reference any artifact with a fixed location relative to a non-deterministic root, e.g., the relative version control path of a file as distinct from a local enlistment root. The `uriBaseId` (3.4.4) property can be used to express the non-deterministic absolute URI root. This approach assists in log file diffing and other scenarios where a clear distinction between data that is consistent or not between scan environments is helpful.

If the nested artifact is a member of an archive file (for example, zip [ZIP] or tar [TAR]), `uri` **SHOULD** specify the member name or path as specified by the archive.

If `thisObject` occurs as the value of a “top-level” property in `theRun.originalBaseIds` (§3.14.14), then `uri` **MAY** be absent. See §3.14.14 for an explanation and an example of this point. Otherwise:

If `index` (§3.4.5) is absent, `uri` **SHALL** be present.

NOTE 3: This ensures that there is a way to locate the artifact specified by the `artifactLocation` object.

If `thisObject` represents a nested artifact whose location within its parent container can be expressed only by means of a byte offset, then `uri` **SHALL NOT** be present.

NOTE 4: This implies that `index` will be present; see §3.4.5.

Otherwise, `uri` MAY be present.

3.4.4 `uriBaseId` Property

If this `artifactLocation` object describes a top-level artifact and the value of its `uri` property (§3.4.3) is a relative reference, the `artifactLocation` object **SHOULD** contain a property named `uriBaseId` whose value is a string which indirectly specifies the absolute URI with respect to which that relative reference is interpreted. If the `uri` property contains an absolute URI, the `uriBaseId` property **SHALL** be absent. If this `artifactLocation` object describes a nested artifact, `uriBaseId` **SHALL** be absent.

If a SARIF consumer requires an absolute URI (for example, to display the specified artifact to a user), then it needs to resolve `uriBaseId` to an absolute URI, which it can then combine with the relative reference stored in the `uri` property.

A SARIF consumer **SHALL** use the following procedure to resolve a `uriBaseId` to an absolute URI:

1. If the end user has configured the SARIF consumer with a value for the `uriBaseId` (for example, on the consumer's command line or through a user interface prompt), then the consumer **SHALL** use the configured value.

EXAMPLE 1: In this example the SARIF consumer's command line specifies that any `uriBaseId` property whose value is "SRCROOT" refers to the absolute URI "file:///C:/browser/src/":

```
C:> SarifAnalyzer --input log.sarif --uriBaseId SRCROOT="file:///C:/browser/src/"
```

2. If `uriBaseId` is not yet resolved and `theRun.originalUriBaseIds` (§3.14.14) is present, the consumer **SHALL** attempt to resolve the `uriBaseId` from the information in `originalUriBaseIds`, in the manner specified in §3.14.14.
3. If `uriBaseId` is not yet resolved, the consumer **MAY** use other information or heuristics to locate the artifact.

The `uriBaseId` property can be any string; it does not need to have any particular syntax or follow any particular naming convention. In particular, it does not need to designate a machine environment variable or similar value, although it might. The SARIF producer and any SARIF consumers need to agree on the meanings of any values for the `uriBaseId` property that appear in the log file.

EXAMPLE 2: In this example, the analysis tool has set the `uri` property of an `artifactLocation` object (§3.4) to a relative reference. The tool has also set the `uriBaseId` property to "%srcroot%". The analysis tool and the SARIF consumers have agreed upon a convention whereby this indicates that the relative reference is expressed relative to the root of the source tree in which the file appears.

```
"artifactLocation": {
  "uri": "drivers/video/hidef/driver.c",
  "uriBaseId": "%srcroot%"
}
```

NOTE: There are various reasons for providing the `uriBaseId` property:

- Portability: A log file that contains relative references together with `uriBaseId` properties can be interpreted on a machine where the files are located at a different absolute location.
- Determinism: A log file that uses `uriBaseId` properties has a better chance of being “deterministic”; that is, of being identical from run to run if none of its inputs have changed, even if those runs occur on machines where the files are located at different absolute locations. For more information on this point, see Appendix F.
- Security: The use of `uriBaseId` properties avoids the persistence of absolute path names in the log file. Absolute path names can reveal information that might be sensitive.
- Semantics: Assuming the reader of the log file (an end user or another tool) has the necessary context, they

can understand the meaning of the location specified by the `uri` property, for example, “this is a source file”.

For more guidance on the intended use of the `uriBaseId` property, see §3.4.7.

3.4.5 `index` Property

Depending on the circumstances, an `artifactLocation` object either **MAY**, **SHALL NOT**, **SHALL**, or **SHOULD** contain a property named `index` whose value is the array index (§3.7.4) within `theRun.artifacts` (§3.14.15) of the `artifact` object (§3.24), if any, that describes the artifact specified by this `artifactLocation` object.

If thisObject occurs as the `location` property (§3.24.2) of an `artifact` object in `theRun.artifacts`, then `index` **MAY** be present. If present, it **SHALL** equal the array index within `theRun.artifacts` of the containing `artifact` object.

Otherwise, if `theRun.artifacts` is absent or does not contain an element that describes the artifact specified by thisObject, then `index` **SHALL NOT** be present.

NOTE 1: `index` cannot be present in this case because there is no array element for it to point to. But this implies that `uri` is present, because otherwise there would be no way to locate the artifact specified by thisObject.

Otherwise, if the `uri` property (§3.4.3) is absent, then `index` **SHALL** be present.

NOTE 2: Again, this ensures that there is a way to locate the artifact specified by thisObject.

Otherwise (that is, if `uri` is present but there is a relevant `artifact` object in `theRun.artifacts`), `index` **SHOULD** be present.

NOTE 3: If `index` is absent, the SARIF consumer will not be able to locate the additional information contained in the `artifact` object about the artifact specified by thisObject.

EXAMPLE 1: In this example, `results[0].locations[0].physicalLocation.artifactLocation.index` specifies the `artifact` object located at `artifacts[0]`.

```
{
  # A run object (§3.14).
  "artifacts": [
    {
      "location": {
        "uri": "file:///C:/Code/main.c"
      },
      "sourceLanguage": "c"
    }
  ],
  "results": [
    {
      "ruleId": "CA2101",
      "level": "error",
      "locations": [
        {
          "physicalLocation": {
            "artifactLocation": {
              "uri": "file:///C:/Code/main.c",
              "index": 0
            },
            "region": {
              "startLine": 24,
              "startColumn": 9
            }
          }
        }
      ]
    }
  ]
}
```

3.4.6 `description` Property

An `artifactLocation` object **MAY** have a property named `description` whose value is a `message` object (§3.11) that describes this location.

EXAMPLE 1: In this example, the property values in `run.originalUriBaseIds` (§3.14.14), which are `artifactLocation` objects, have `description` properties. This allows a SARIF viewer to display helpful information when prompting a user to supply values for the base id symbols.

```
{
  "originalUriBaseIds": {
    "PROJROOT": {
      "uri": "file:///C:/browser/",
      "description": {
        "text": "The project root directory."
      }
    },
    "SRCROOT": {
      "uri": "file:///C:/browser/src/",
      "description": {
        "text": "The root of the source code tree."
      }
    },
    "BINROOT": {
      "uri": "file:///C:/browser/bin/",
      "description": {
        "text": "The build output directory."
      }
    }
  }
}
```

A run object (§3.14).
See §3.14.14.

3.4.7 Guidance on the Use of `artifactLocation` Objects

Some URIs are “deterministic” in the sense that they will be the same from one run to the next and are independent of machine-specific information such as volume names or drive letters. Internet addresses are typically deterministic.

In contrast, file system paths are typically non-deterministic. For example, a source code enlistment might exist at different paths on different machines.

`artifactLocation` objects **MAY** represent both deterministic and non-deterministic URIs. In either case, the `uri` property (§3.4.3) **SHOULD** be deterministic, either because it is a deterministic relative reference (for example, the relative path to a file from the root of the directory tree containing the analyzed source code) or because it is an absolute URI. If the URI is non-deterministic, the `uriBaseId` property (§3.4.4) **SHOULD** capture the non-deterministic portion of the URI, for example, the absolute path to the root of the directory tree containing the analyzed source code.

EXAMPLE 1: In this example, the location of a result detected by a tool is specified by a relative reference together with a `uriBaseId` that specifies the root of the source code enlistment.

```
{
  "originalUriBaseIds": {
    "SRCROOT": {
      "uri": "file:///C:/browser/src/"
    }
  },
  "results": [
    {
      "locations": [
        {
          "physicalLocation": {
            "artifactLocation": {
              "uri": "ui/window.cpp",
              "uriBaseId": "SRCROOT"
            }
          }
        }
      ]
    }
  ]
}
```

A run object (§3.14).
See §3.14.14.

See §3.14.23.
A result object (§3.27).
See §3.27.12.
A location object (§3.28).
See §3.28.3.
An artifactLocation object.

3.5 String Properties

3.5.1 Localizable Strings

Certain string-valued properties in this document, for example, `toolComponent.name` (§3.19.8), can be translated into other languages. We describe these properties as being “localizable.” The description of every localizable property will state that it is localizable.

3.5.2 Redactable Strings

Certain string-valued properties in this document (for example, `invocation.commandLine` (§3.20.2)) might contain sensitive information that a SARIF producer or a SARIF post-processor might choose to redact. We describe these properties as “redactable.” The description of every redactable property will state that it is redactable.

If a SARIF producer or a SARIF post-processor chooses to redact sensitive information in a redactable property, it **SHALL** replace the sensitive information with a string whose value is an element of `theRun.redactionTokens` (§3.14.28).

3.5.3 GUID-valued Strings

Certain string-valued properties in this document provide unique stable identifiers in the form of a GUID or UUID [RFC4122]. This document uses the term “GUID”.

EXAMPLE 1: "f81d4fae-7dec-11d0-a765-00a0c91e6bf6"

NOTE 1: The UUID standard [RFC4122] allows hex digits in either upper or lower case. It does not permit delimiters such as curly braces ("{" , "}") around the value.

The description of every GUID-valued property will state that it is GUID-valued.

NOTE 2: In the examples, the values shown for GUID-valued properties are valid GUIDs. In some cases, they are illustrative values such as "11111111-1111-1111-8888-111111111111" which are intended to make it easy to identify situations where two GUIDs in the same example are required to be the same. In these illustrative values, the third and fourth component are always "1111-8888", a sample value that conforms to the restrictions on the values of those components.

The same `guid` value on the root elements of two or more SARIF files indicates that the information content is the same.

NOTE 3: Hashing of text based formats is ambiguous for duplicate detection as the line ending conventions differ and impact the hash.

Examples of possible duplication sources are: File copies, stored byte streams.

Differing `guid` values on the root elements of two or more SARIF files indicate that the files are different.

Examples are reports from different nodes on the same system under test using identical tools or a retest run.

3.5.4 Hierarchical Strings

3.5.4.1 General

Certain string-valued properties and certain property names in this document (for example, the value of the `runAutomationDetails.id` property (§3.17.3), and the property names in a property bag (§3.8)) are said to be “hierarchical.” This means that the string consists of a sequence of forward-slash-separated components, with this syntax:

`hierarchical string = component, { "/", component };`

`component = { component character };`

`component character = ? JSON string character ? - "/";`

NOTE 1: The grammar prohibits a `component` from containing a forward slash. There is no escape mechanism to allow a `component` to include a forward slash.

For examples, see §3.8.2 and §3.17.3.

The description of every hierarchical string will state that it is hierarchical.

A SARIF consumer **SHALL** interpret the values of a hierarchical string as forming a logical hierarchy. The first component represents the top level of the hierarchy, the second component represents the second level, and so on.

NOTE 2: A hierarchical string does not need to include any forward slashes. The syntax permits a single string of non-forward-slash characters. The purpose of this section is to define the semantics of the forward slash character in those properties that respect it.

In string-valued properties and property names that are *not* described as hierarchical, the forward slash character has no special meaning, and a SARIF consumer **SHALL NOT** interpret it as dividing the value into hierarchical components.

3.5.4.2 Versioned Hierarchical Strings

Certain hierarchical strings in this document (for example, the property names in `result.fingerprints` (§3.27.16) and `result.partialFingerprints` (§3.27.17)) are said to be “versioned.” This means that if the last component of the string is of the form

`version component = "v", non negative integer;`

then a SARIF consumer **SHALL** consider that component to represent the version number of the entity specified by the string.

The description of every versioned hierarchical string will state that it is versioned.

In string-valued properties and property names that are described as hierarchical but *not* as versioned, a final component matching the syntax of `version component` has no special meaning, and a SARIF consumer **SHALL NOT** interpret it as a version number.

NOTE 1: A versioned hierarchical string does not need to include a version component. The syntax permits but does not require it.

A hierarchical string without a version component **SHALL** be considered older than any corresponding string with a version component.

EXAMPLE 1: In this example, the partial fingerprint whose property name is `"prohibitedWordHash"` is considered to have been computed with an older version of the “prohibited word hash” algorithm than the partial fingerprint whose property name is `"prohibitedWordHash/v1"`.

```
{
  "partialFingerprints": {      # A result object (§3.27).
    "prohibitedWordHash": "4efcc21977b55", # See §3.27.17.
    "prohibitedWordHash/v2": "097886bc876fe"
  }
}
```

NOTE 2: When a previously unversioned string is later versioned, as in the example above, it might be clearer to specify `"v2"` for the first explicitly versioned string.

3.6 Object Properties

Certain properties in this document are defined to be objects whose property names satisfy certain conditions. Examples are `run.originalUriBaseIds` (§3.14.14) and `reportingDescriptor.messageStrings` (§3.49.11). Unless otherwise specified in the description of a specific property, if any such object is empty, then either the property **SHALL** be represented as an empty object `{}`, or it **SHALL** be absent.

3.7 Array Properties

3.7.1 General

Certain properties in this document are defined to be arrays. Examples are the `invocation.toolExecutionNotification` property (§3.20.21) and the property bag `tags` property (§3.8.2).

3.7.2 Default Value

If an array-valued property is absent, it **SHALL** default to an empty array unless the property's description specifies otherwise.

3.7.3 Array Properties with Unique Values

Certain array-valued properties in this document are described as having “unique” elements. When a property is so described, it means that no two elements of the array **SHALL** have equal values. For purposes of this document, two array elements **SHALL** be considered equal when they satisfy the condition for equality described in the JSON Schema standard [JSONSCHEMA01], §4.3, “Instance equality”. In particular, two strings are considered equal when they consist of the same sequence of Unicode [UNICODE12] code points.

3.7.4 Array Indices

If any property in this document is described as an “array index,” it **SHALL** contain an integer that is a zero-based index into the specified array. If any such property is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set), unless the property's description specifies otherwise.

3.8 Property Bags

3.8.1 General

Certain properties in this document are defined to be “property bags”. A property bag is an object (§3.6) containing an unordered set of properties with arbitrary names.

The property names are hierarchical strings (§3.5.4). The components of the property names **SHOULD** be camelCase strings, but see §Appendix D. for exceptions.

The property values **MAY** be of any JSON type, including strings, numbers, arrays, objects, Booleans, and null. If a property value is a string, it **MAY** be an empty string.

In addition to those properties that are explicitly documented, every object defined in this document **MAY** contain a property named `properties` whose value is a property bag. This allows SARIF producers to include information about each object that is not explicitly specified in the SARIF format.

3.8.2 Tags

3.8.2.1 General

If a property bag contains a property named `tags`, the property value **SHALL** be an array of zero or more unique (§3.7.3), hierarchical (§3.5.4) strings. Two strings **SHALL** be considered the same if they consist of the same sequence of Unicode [UNICODE12] code points.

Tags **SHOULD NOT** be used to label a result or a rule as belonging to a category in a classification system such as the Common Weakness Enumeration [CWE] (for example, by adding a tag "CWE/622"). Instead, taxonomies (§3.19.3) **SHOULD** be used for this purpose.

Even when defining a custom classification system used within an engineering team, taxonomies **SHOULD** be used rather than tags when labeling a result or a rule.

EXAMPLE 1: Rather than adding the tag "shipBlocking" to a result, consider defining a taxonomy such as "Shipping Impact". This enables metadata such as a description and a help URI to be associated with each taxonomic category.

EXAMPLE 2: In this example, the SARIF producer tags an artifact with the string "openSource".

```
{
  "artifacts": [
    {
      "location": {
        "uri": "http://www.example.com/libraries/jsonParser.js"
      },
      "properties": {
        "tags": [
          "openSource"
        ]
      }
    }
  ],
  ...
}
```

NOTE: Anything a tag expresses can also be expressed with a named property bag entry, for example "openSource": true, but a tag is more concise.

3.8.2.2 Tag Metadata

A SARIF log file **MAY** provide additional information about any tag value by including a property whose name is the same as that tag value and whose value is any JSON value. If present, this property **SHALL** be located by searching first in the property bag that contains the tag, and then in the property bag of the containing run object (§3.14) theRun, if any.

EXAMPLE 1: Continuing the example from §3.8.2.1, suppose the tool wishes to provide additional information about using open source code. It might provide that information within the property bag containing the tag (the property bag belonging to the artifact object):

```
{
  "location": {
    "uri": "http://www.example.com/libraries/jsonParser.js"
  },
  "properties": {
    "tags": [
      "openSource"
    ],
    "openSource": {
      "informationUri":
        "http://www.example.com/procedures/usingOpenSource.html"
    }
  }
}
```

EXAMPLE 2: There might be several open source files. To avoid duplicating information, the tool might choose to place the tag metadata in the property bag belonging to theRun:

```
{
  "artifacts": [
    {
      "location": {
        "uri": "http://www.example.com/libraries/jsonParser.js"
      },
      "properties": {
        "tags": [
          "openSource"
        ]
      }
    },
    ...
  ],
  ...
  "properties": {
    "openSource": {
      "informationUri":
        "http://www.example.com/procedures/usingOpenSource.html"
    }
  }
}
```


3.9 Date/time Properties

Certain properties in this document specify a date and time. The value of every such property, if present, **SHALL** be a string in the following format, which is compatible with the ISO standard for date and time formats [ISO8601:2004]:

```
date time = date, [ "T", time, "Z" ] (* UTC time *);

date = year, "-", month, "-", day;

year = 4 * decimal digit;

month = 2 * decimal digit (* from 01 to 12 *);

day = 2 * decimal digit (* from 01 to 31 *);

time = hour, ":", minute, [ ":", second, [ ".", fraction ] ];

hour = 2 * decimal digit (* from 00 to 24, to represent midnight at the
                           end of a calendar day *);

minute = 2 * decimal digit (* from 00 to 59 *);

second = 2 * decimal digit (* from 00 to 60, to accommodate leap second *);

fraction = decimal digit, { decimal digit };
```

EXAMPLES:

```
2016-02-08
2016-02-08T16:08Z
2016-02-08T16:08:25Z
2016-02-08T16:08:25.943Z
2016-02-08T00:00:00Z
2016-02-08T16:08:00Z
2016-02-08T16:08:25Z
2016-02-08T16:08:25.943Z
```

The time component of every date/time-valued property **SHALL** be expressed in Coordinated Universal Time (UTC).

NOTE 1: The name of every date/time-valued property ends in “Utc” to emphasize that requirement.

The time components of date/time-valued properties in property bags (§3.8) **SHOULD** also be expressed in UTC.

NOTE 2: This might not always be possible if the property comes from a source that does not provide time zone information.

A SARIF producer **SHOULD NOT** provide more digits in `fraction` than warranted by the precision of the clock on the computer on which it runs.

A SARIF producer **SHOULD** express date/time properties, except for those that express product release dates, to a precision of at least whole seconds.

3.10 URI-valued Properties

3.10.1 General

Certain properties in this document specify either an absolute URI or a URI reference (the term used in the URI standard [RFC3986] to describe either an absolute URI or a relative reference). The value of every such property, if present, **SHALL** be a string in the format specified by the standard [RFC3986].

If a URI reference refers to a file stored in a version control system (VCS), its value **SHALL** include sufficient information (for example, a commit id) to enable the correct version of the target file to be retrieved from the VCS. If a URI reference refers to a file stored on a physical file system, it **MAY** be specified as a relative reference that omits root information details (such as hard drive letter and an arbitrarily named root directory associated with a source code enlistment).

NOTE 1: A URI reference (even a relative reference) might contain information that represents unwanted information disclosure, particularly in cases where a tool is analyzing files stored on a physical file system. For example, a file path might contain the account name of a developer.

The URI **SHALL** specify the location of the artifact at the time the analysis was performed.

Two URI references **SHALL** be considered equivalent if their normalized forms are the same, as described in the standard [RFC3986].

NOTE 2: Features of this normalized form include using upper-case hexadecimal digits for percent-encoded characters and expressing the scheme component in lower-case. For the full specification of the normalized URI form, see the standard [RFC3986].

For additional normalization requirements for URIs that use the "file" scheme, see §3.10.2.

When two URI references are not equivalent in this sense (that is, when their normalized forms are not the same), we will say that they are "distinct."

Aside from normalization, SARIF producers **SHALL NOT** make any other changes to the text of a URI reference; for example, they **SHALL NOT** convert the path to upper case or to lower case.

NOTE 3: This is especially important when the same SARIF file might be consumed on multiple platforms, for example, a platform such as Microsoft Windows®, whose NTFS file system is case-insensitive but case-preserving, and a platform such as Linux®, whose file system is case-sensitive. Consider a scenario where a tool runs on a Windows® system using NTFS, and the tool decides to lower-case the file names in the log. If the source files and the SARIF log were transferred to a Linux® system, the URI references in the log file would not match the path names on the destination system.

3.10.2 Normalizing File Scheme URIs

If a URI uses the "file" scheme [RFC8089] and the specified path is network-accessible, the SARIF producer **SHALL** include the host name.

EXAMPLE 1: A file-based URI that references a network share.

```
file://build.example.com/drops/Build-2018-04-19.01/src
```

If a URI uses the "file" scheme and the specified path is *not* network-accessible, the SARIF producer **SHOULD NOT** include the host name.

EXAMPLE 2: A file-based URI that references the local file system.

```
file:///C:/src
```

A SARIF producer **MAY** choose to omit the hostname (authority) from a file URI, for example, for security reasons. If it does so, then to maximize interoperability with previous versions of the URI specification, the URI **SHOULD** start with `file:///`, as in EXAMPLE 2. See the standard [RFC8089] for more information on this point.

SARIF producers **SHALL** create `file` scheme URIs by means of the following procedure or any procedure with the same result:

1. In the case of a direct producer, preserve the file system's casing, even if the file system is case-insensitive. In the case of a converter (which might not know the file system's casing), preserve the casing specified in the analysis tool's native output file.
2. Remove `.` path segments.
3. Remove empty path segments.
4. If the path contains `..` path segments, then in the case of a direct producer, resolve the path to a canonical absolute path, using an appropriate algorithm for the operating system on which the tool ran.

NOTE 1: This is necessary because, for example, the path `/d1/..f` naively converted to a URI is `file:///d1/..f`, which resolves to `file:///f` according to the URI standard [RFC3986]. But if `/d1` is a symbolic link to the directory `d2/d3`, then the correct URI is `file:///d2/f`.

NOTE 2: `..` path segments are dangerous because the semantics of the file system on which the SARIF log file was produced might not match the semantics of the file system on which it is consumed. For example, the presence of a symbolic link in the path might redirect the consumer to an unpredictable location.

5. Create a URI from the resulting path.
6. Optionally, divide the resulting URI into a base URI and a relative URI (preserving case in both parts), and create an entry for the base URI in `theRun.originalUriBaseIds` (§3.14.14).

NOTE 3: URI and path manipulation are complex topics. Many operating systems, languages, and frameworks provide methods to perform these operations, which is preferable to having every SARIF producer reimplement them. For example, in C#, the operation can be performed as follows:

```
using System;
using System.IO;
...
string path = ...;
string fullPath = Path.GetFullPath(path);
var uri = new Uri(fullPath, UriKind.Absolute);
string uriString = uri.AbsoluteUri;
```

SARIF consumers **SHALL NOT** normalize `..` segments out of a path. A consumer **SHOULD** reject paths that contain `..` segments, otherwise a consumer **SHALL** treat distinct portions of paths up to and including the rightmost `..` segment as unique directories on the file system, even if [RFC3986] normalization would produce identical paths.

EXAMPLE 3: Consider the following three URIs:

- `file:///d1/..f1`
- `file:///d1/..f2`
- `file:///d1/d2/..f3`

A consumer would treat `f1` and `f2` as residing in the same directory. So, for example, if a viewer prompted the user to supply the directory where `f1` resides, it could search for `f2` in the same directory, without prompting again. On the other hand, even though `f3` appears to reside in the same directory as `f1` and `f2`, the viewer would not assume that, and would prompt the user to supply the directory where `f3` resides.

3.10.3 URIs That use the SARIF Scheme

In certain circumstances, a URI can refer to an element of the current SARIF log file (for example, see §3.16.3). Such a URI uses the `sarif` scheme. The `sarif` URI scheme consists of only a scheme (with the value `sarif`) and a path component. The path component is interpreted as a JSON pointer [RFC6901] into the SARIF document containing the URI. The authority, query and fragment URI components **SHALL NOT** be present.

EXAMPLE 1: The URI "`sarif:/inlineExternalProperties/0`" refers to the 0th element of the array contained in the `inlineExternalProperties` property (§3.13.5) at the root of the log file.

3.10.4 Internationalized Resource Identifiers (IRIs)

If a URI-valued property refers to a resource identified by an Internationalized Resource Identifier (IRI) [RFC3987], the SARIF producer **SHALL** first transform the IRI into a URI, using the mapping mechanism specified in §3.1 of the standard [RFC3987], and then assign the transformed value to the property. The string value of a URI-valued property **SHALL NOT** include Unicode characters such as "é"; such characters are permitted in IRIs but are not permitted in URIs. §3.1 of the standard [RFC3987] describes how to replace such characters with "percent-encoded" equivalents to produce a valid URI.

EXAMPLE 1: Suppose a URI-valued property needs to refer to a resource identified by the string "`http://www.example.com/hu/sör.txt`". This string contains the character "ö", so it is a valid IRI but not a valid URI. Following the procedure in §3.1 of the standard [RFC3987], a SARIF producer would transform this string to the valid URI "`http://www.example.com/hu/s%C3%B6r.txt`" before assigning it to the property.

3.11 message Object

3.11.1 General

Certain objects in this document define messages intended to be viewed by a user. SARIF represents such a message with a message object, which offers the following features:

- Message strings in plain text ("plain text messages") (§3.11.3).
- Message strings that incorporate formatting information ("formatted messages") in GitHub Flavored Markdown [GFM] (§3.11.4).
- Message strings with placeholders for variable information (§3.11.5).
- Message strings with embedded links (§3.11.6).

3.11.2 Constraints

At least one of the `text` (§3.11.8) or `id` (§3.11.10) properties **SHALL** be present.

NOTE: This ensures that a SARIF consumer can locate the text of the message.

3.11.3 Plain Text Messages

A plain text message **SHALL NOT** contain formatting information, for example, HTML tags or white space whose purpose is to provide indentation or suggest some structure to the message.

If a plain text message consists of multiple paragraphs, it **MAY** contain line breaks (for example, "`\r\n`" or "`\n`", if the SARIF log file is serialized as JSON) to separate the paragraphs. Line breaks **MAY** follow any convention (for example, "`\n`" or "`\r\n`"). A SARIF post-processor **MAY** normalize line breaks to any desired convention, including escaping or removing the line breaks so that the entire message renders on a single line.

The message string **MAY** contain placeholders (§3.11.5) and embedded links (§3.11.6).

If the message consists of more than one sentence, its first sentence **SHOULD** provide a useful summary of the message, suitable for display in cases where UI space is limited.

NOTE 1: If a tool does not construct the message in this way, the initial portion of the message that a viewer displays where UI space is limited might not be understandable.

NOTE 2: The rationale for these guidelines is that the SARIF format is intended to make it feasible to merge the outputs of multiple tools into a single user experience. A uniform approach to message authoring enhances the quality of that experience.

A SARIF post-processor **SHOULD NOT** modify line break sequences (except perhaps to adapt them to a particular viewing environment).

3.11.4 Formatted Messages

3.11.4.1 General

Formatted messages **MAY** be of arbitrary length and **MAY** contain formatting information. The message string **MAY** also contain placeholders (§3.11.5) and embedded links (§3.11.6).

Formatted messages **SHALL** be expressed in GitHub-Flavored Markdown [GFM]. Since GFM is a superset of CommonMark [CMARK], any CommonMark Markdown syntax is acceptable.

3.11.4.2 Security Implications

For security reasons, SARIF producers and consumers **SHALL** adhere to the following:

- SARIF producers **SHALL NOT** emit messages that contain HTML, even though all variants of Markdown permit it.
- Deeply nested markup can cause a stack overflow in the Markdown processor [GFMENG]. To reduce this risk, SARIF consumers **SHALL** use a Markdown processor that is hardened against such attacks.

NOTE: One example is the GitHub fork of the cmark Markdown processor [GFMCMARK].

- To reduce the risk posed by possibly malicious SARIF files that do contain arbitrary HTML (including, for example, `javascript:` links), SARIF consumers **SHALL** either disable HTML processing (for example, by using an option such as the `--safe` option in the cmark Markdown processor) or run the resulting HTML through an HTML sanitizer.

SARIF consumers that are not prepared to deal with the security implications of formatted messages **SHALL NOT** attempt to render them and **SHALL** instead fall back to the corresponding plain text messages.

3.11.5 Messages with Placeholders

A message string **MAY** include one or more “placeholders.” The syntax of a placeholder is:

```
placeholder = "{", index, "}";
```

```
index = non negative integer;
```

`index` represents a zero-based index into the array of strings contained in the `arguments` property (§3.11.11).

When a SARIF consumer displays the message, it **SHALL** replace every occurrence of the placeholder `{n}` with the string value at index `n` in the `arguments` array. Within both plain text and formatted message strings, the characters “{” and “}” **SHALL** be represented by the character sequences “{” and “}” respectively.

Within a given message object:

- The plain text and formatted message strings **MAY** contain different numbers of placeholders.
- A given placeholder index **SHALL** have the same meaning in the plain text and formatted message strings (so they can be replaced with the same element of the `arguments` array).

EXAMPLE 1: Suppose a message object's `text` property (§3.11.8) contains this string:

"The variable \"{0}\" defined on line {1} is never used. Consider removing \"{0}\"."

There are two distinct placeholders, {0} and {1} (although {0} occurs twice). Therefore, the `arguments` array will have at least two elements, the first corresponding to {0} and the second corresponding to {1}.

EXAMPLE 2: In this example, the SARIF consumer will replace the placeholder {0} in `message.text` with the value "pBuffer" from the 0 element of `message.arguments`.

```
{
  "results": [
    {
      "ruleId": "CA2101",
      "message": {
        "text": "Variable '{0}' is uninitialized.",
        "arguments": [ "pBuffer" ]
      }
    }
  ]
}
```

A run object (§3.14).
 # See §3.14.23.
 # A result object (§3.27).
 # See §3.27.5.
 # See §3.27.11.
 # See §3.11.8.
 # See §3.11.11.

3.11.6 Messages with Embedded Links

A message string **MAY** include one or more links to locations within artifacts mentioned in the enclosing `result` object (§3.27). We refer to these links as “embedded links”.

Within a formatted message (§3.11.4), an embedded link **SHALL** conform to the syntax of a GitHub Flavored Markdown link (see [GFM], §6.6, “Links”).

NOTE 1: The GFM link syntax is very flexible. Since a SARIF viewer that renders formatted messages will presumably rely on a full-featured GFM processor, there is no need to restrict the embedded link syntax in SARIF formatted messages.

Within a plain text message (§3.11.3), an embedded link **SHALL** conform to the following syntax (which is a greatly restricted subset of the GFM link syntax) before JSON encoding:

escaped link character = "\" | "[" | "]";

normal link character = ? JSON string character ? – escaped link character;

link character = normal link character | ("\", escaped link character);

link text = { link character };

link destination = ? Any valid URI ?;

embedded link = "[", link text, "]"(", link destination, ")";

`link text` is the message text visible to the user.

Literal square brackets (“[” and “]”) in the link text of a plain text message **SHALL** be escaped with a backslash (“\”).

NOTE 2: When a SARIF log file is serialized as JSON, JSON encoding doubles the backslash.

EXAMPLE 1: Consider this embedded link whose link text contains square brackets and backslashes:

```
"message": {
  "text": "Prohibited term used in [para\\[0\\]\\]\\span\\[2\\]](1)."
```

A SARIF viewer would render it as follows:

Prohibited term used in para[0]\spans[2].

Literal square brackets and (doubled) backslashes **MAY** appear anywhere else in a plain text message without being escaped.

In both plain text and formatted messages, if `link destination` is a non-negative integer, it **SHALL** refer to a `location` object (§3.28) whose `id` property (§3.28.2) equals the value of `link destination`. In this case, the `result` **SHALL** contain exactly one `location` object with that `id`.

NOTE 3: Negative values are forbidden because their use would suggest some non-obvious semantic difference between positive and negative values.

EXAMPLE 2: In this example, a plain text message contains an embedded link to a location with a file. The `result` object contains exactly one `location` object whose `id` property matches the `link destination`.

```
{
  "ruleId": "TNT0001",
  "message": {
    "text": "Tainted data was used. The data came from [here](3)."
```

The `link destination` in embedded links in both plain text messages and formatted messages **MAY** use the `sarif` URI scheme (§3.10.3). This allows a message to refer to any content elsewhere in the SARIF log file.

EXAMPLE 1: A `result.message` (§3.27.11) can refer to another result in the same run (or, for that matter, in another run within the same log file) as follows:

"There was [another result](sarif:/runs/0/results/42) found by this code flow."

A SARIF viewer executing in an IDE might respond to a click on such a link by selecting the target result in an error list window and navigating the editor to that result's location.

Because the "sarif" URI scheme uses JSON pointer [RFC6901], which locates array elements by their array index, these URIs are potentially fragile if the SARIF log file is transformed by a post-processor.

EXAMPLE 2: If a post-processor concatenates two runs into a single log file, the links within the run at index 1 will be incorrect, and will need to be updated from "sarif:/runs/0/..." to "sarif:/runs/1/..."

EXAMPLE 3: If a post-processor removes results from a run, any links that refer to results at indices following the removed results will need to be adjusted. For example, `sarif:/runs/0/results/54` might need to be adjusted to `sarif:/runs/0/results/42`.

When a tool displays on the console a result message containing an embedded link, it **MAY** reformat the link (for example, by removing the square brackets around the `link text`). If the `link destination` is an integer, and hence specifies a location object belonging to `theResult`, the tool **SHOULD** replace the integer with a string representation of the specified location.

EXAMPLE 4: Suppose a tool chooses to display the result message from Example 3, which contains an integer-valued `link destination`, on the console. The output might be:

Tainted data was used. The data came from here: `C:\code\input.c(25, 19)`.

Note that in addition to providing a string representation of the location, the tool removed the `[...] (...)` link syntax and separated the link text from the location with a colon. Finally, the tool recognized that the location's URI used the `file` scheme and chose to display it as a file system path rather than a URI.

URLs **MAY** contain unescaped closing parentheses `'`) and thus any parser applied to such content (`link destination`) is responsible for preserving the semantics of a link expression.

EXAMPLE 5: The following text if parsed should result in the following token sequence:

'Foo [unbalanced](https://example.org/aFgH)x_ ' (incoming text)

- | | |
|-----------------------------------|-----------------------|
| 1. 'Foo ' | (as text) |
| 3. 'unbalanced' | (as link-text) |
| 4. 'https://example.org/aFgH)x_ ' | (as link-destination) |
| 5. ' quux. ' | (as text) |

3.11.7 Message String Lookup

A message object can directly contain message strings in its `text` (§3.11.8) and `markdown` (§3.11.9) properties. It can also indirectly refer to message strings through its `id` (§3.11.10) property.

When a SARIF consumer needs to locate a message string from a message object, it **SHALL** follow the procedure specified in this section. The `run` object **SHALL** contain enough information for the procedure to succeed.

The lookup **SHALL** occur entirely within the context of a single `toolComponent` object (§3.19) which we refer to as `theComponent`. If the SARIF consumer is displaying messages in the language specified by `theRun.language` (§3.14.7), then `theComponent` is the tool component that defines the message. If the consumer is displaying messages in any other language – in which case a translation (§3.19.4) is in use – then `theComponent` is the tool component that contains the translation.

In this procedure, we refer to the message object whose string is being looked up as `theMessage`.

At various points in this procedure, we state that the consumer uses an object's "text property or markdown property, as appropriate." This means that if the consumer can render formatted messages, it **MAY** use the `markdown` property, if present; otherwise it **SHALL** use the `text` property, but if the consumer cannot render formatted messages, it **SHALL** use the `text` property.

The procedure is:

IF `theMessage.text` is present and the desired language is `theRun.language` THEN

 Use the `text` or `markdown` property of `theMessage` as appropriate.

IF the string has not yet been found THEN

 IF `theMessage` occurs as the value of `result.message` (§3.27.11) THEN

LET `theRule` be the `reportingDescriptor` object (§3.49), an element of `theComponent.rules` (§3.19.23), which defines the rule that was violated by this result.

IF `theRule` exists AND `theRule.messageStrings` (§3.49.11) is present AND contains a property whose name equals `theMessage.id` THEN

LET `theMFMS` be the `multiformatMessageString` object (§3.12) that is the value of that property.

Use the `text` or `markdown` property of `theMFMS` as appropriate.

ELSE IF `theMessage` occurs as the value of `notification.message` (§3.58.5) THEN

LET `theDescriptor` be the `reportingDescriptor` object (§3.49), an element of `theComponent.notifications` (§3.19.23), which describes this notification.

IF `theDescriptor` exists AND `theDescriptor.messageStrings` is present AND contains a property whose name equals `theMessage.id` THEN

LET `theMFMS` be the `multiformatMessageString` object that is the value of that property.

Use the `text` or `markdown` property of `theMFMS` as appropriate.

IF the string has not yet been found THEN

IF `theComponent.globalMessageStrings` (§3.19.22) is present AND contains a property whose name equals `theMessage.id` THEN

LET `theMFMS` be the `multiformatMessageString` object that is the value of that property.

Use the `text` or `markdown` property of `theMFMS` as appropriate.

IF the string has not yet been found THEN

The lookup procedure fails (which means the SARIF log file is invalid).

3.11.8 **text** Property

A message object **MAY** contain a property named `text` whose value is a non-empty string containing a plain text message (§3.11.3).

3.11.9 **markdown** Property

A message object **MAY** contain a property named `markdown` whose value is a non-empty string containing a formatted message (§3.11.4) expressed in GitHub-Flavored Markdown [GFM].

If the `markdown` property is present, the `text` property (§3.11.8) **SHALL** also be present.

NOTE: This ensures that the message is viewable even in contexts that do not support the rendering of formatted text.

SARIF consumers that cannot (or choose not to) render formatted text **SHALL** ignore the `markdown` property and use the `text` property instead.

3.11.10 **id** Property

A message object **MAY** contain a property named `id` whose value is a non-empty string containing the identifier for the desired message. See §3.11.7 for details of the message string lookup procedure.

3.11.11 arguments Property

If the message string specified by any of the properties `text` (§3.11.8), `markdown` (§3.11.9), or `id` (§3.11.10) contains any placeholders (§3.11.5), the message object **SHALL** contain a property named `arguments` whose value is an array of strings. §3.11.5 specifies how a SARIF consumer combines the contents of the `arguments` array with the message string to construct the message that it presents to the end user, and provides an example.

If none of the properties `text`, `markdown`, or `id` contains any placeholders, then `arguments` **MAY** be absent.

The `arguments` array **SHALL** contain as many elements as required by the maximum placeholder index among all the message strings specified by the `text`, `markdown`, and `id` properties.

EXAMPLE 1: If the highest numbered placeholder in the `text` message string is `{3}` and the highest numbered placeholder in the `markdown` message string is `{5}`, the `arguments` array must contain at least 6 elements.

3.12 multiformatMessageString Object

3.12.1 General

A `multiformatMessageString` object groups together all available textual formats for a message string.

3.12.2 Localizable multiformatMessageStrings

Certain `multiformatMessageString`-valued properties in this document, for example, `reportingDescriptor.short` (§3.49.9), can be translated into other languages. We describe these properties as being “localizable.” The description of every localizable property will state that it is localizable.

3.12.3 text Property

A `multiformatMessageString` object **SHALL** contain a property named `text` whose value is a non-empty string containing a plain text representation of the message including any links.

NOTE: This property is required to ensure that the message is viewable even in contexts that do not support the rendering of formatted text.

3.12.4 markdown Property

A `multiformatMessageString` object **MAY** contain a property named `markdown` whose value is a non-empty string containing a formatted message (§3.11.4) expressed in GitHub-Flavored Markdown [GFM].

SARIF consumers that cannot (or choose not to) render formatted text **SHALL** ignore the `markdown` property and use the `text` property (§3.12.3) instead.

3.13 sarifLog Object

3.13.1 General

A `sarifLog` object specifies the version of the file format and contains the output from one or more runs.

EXAMPLE 1:

```
{
  "version": "2.1.0", # See §3.13.2.
  "runs": [          # See §3.13.4.
    {
      ...           # A run object (§3.14).
    },
    ...
    {
      ...           # Another run object.
    }
  ]
}
```

```
}
]
}
```

3.13.2 **version** Property

A `sarifLog` object **SHALL** contain a property named `version` whose value is a string designating the version of the SARIF specification to which this log file conforms. This string **SHALL** have the value `"2.1.0"`.

Although the order in which properties appear in a JSON object value is not semantically significant, the `version` property **SHOULD** appear first.

NOTE: This will make it easier for parsers to handle multiple versions of the SARIF format if new versions are defined in the future.

3.13.3 **\$schema** Property

A `sarifLog` object **MAY** contain a property named `\$schema` whose value is a string containing an absolute URI from which a JSON schema document [JSONSCHEMA01] describing the version of the SARIF format to which this log file conforms can be obtained.

If the `\$schema` property is present, the JSON schema obtained from the specified URI **SHALL** describe the version of the SARIF format specified by the `version` property (§3.13.2).

NOTE 1: The purpose of the `\$schema` property is to allow JSON schema validation tools to locate an appropriate schema against which to validate the log file. This is useful, for example, for tool authors who wish to ensure that logs produced by their tools conform to the SARIF format.

NOTE 2: The SARIF schema is available at <https://docs.oasis-open.org/sarif/sarif/v2.1.0/errata01/csd01/schemas/sarif-schema-2.1.0.json>.

3.13.4 **runs** Property

A `sarifLog` object **SHALL** contain a property named `runs` whose value is either `null` or an array of zero or more `run` objects (§3.14).

The value of `runs` **SHALL** be an array with at least one element except in the following circumstances:

- If a SARIF producer finds no data with which to populate `runs`, then its value **SHALL** be an empty array.

NOTE 1: This would happen if, for example, the log file were the output of a query on a result management system, and the query did not match any runs stored in the result management system.

- If a SARIF producer tries to populate `runs` but fails, then its value **SHALL** be `null`.

NOTE 2: This would happen if, for example, the log file were the output of a query on a result management system, and the query was malformed.

3.13.5 **inlineExternalProperties** Property

A `sarifLog` object **MAY** contain a property named `inlineExternalProperties` whose value is an array of zero or more unique (§3.7.3) `externalProperties` objects (§4.3).

NOTE: This property allows multiple runs to share large data sets in a single, self-contained log file.

EXAMPLE 1: In this example, two tools analyze the same set of image files, stored in `sarifLog.inlineExternalProperties[0].artifacts`. The first tool locates the inline `externalProperties` object by means of a URI with the `sarif` scheme (see §3.10.3). The second tool locates the object by means of its `guid` property (§4.3.4).

```
{
  "version": "2.2",
  "$schema": "https://docs.oasis-open.org/sarif/sarif/v2.2/schema/sarif.json",
  "inlineExternalProperties": [
    {
      "guid": "00001111-2222-1111-8888-555566667777", # See §4.3.4.
      "artifacts": [ # See §4.3.6.
        {
          "location": {
            "uri": "apple.png"
          },
          "mimeType": "image/png"
        },
        {
          "location": {
            "uri": "banana.png"
          },
          "mimeType": "image/png"
        }
      ]
    }
  ],
  "runs": [ # See §3.13.4.
    { # A run object (§3.14). # See §3.14.6.
      "tool": {
        "driver": {
          "name": "ImageAccessibilityScanner"
        }
      },
      "externalPropertyFileReferences": { # See §3.14.2.
        "artifacts": [
          {
            "location": {
              "uri": "sarif:/inlineExternalPropertyFiles/0"
            }
          }
        ]
      },
      "results": [
        ...
      ]
    },
    {
      "tool": {
        "driver": {
          "name": "ImageSuitabilityScanner"
        }
      },
      "externalPropertyFileReferences": {
        "artifacts": [
          {
            "guid": "00001111-2222-1111-8888-555566667777"
          }
        ]
      },
      "results": [
        ...
      ]
    }
  ]
}
```

3.13.6 `guid` Property

A `sarifLog` object **SHOULD** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) that provides a unique, stable identifier for the `sarifLog` designating that the log itself has a conceptual identity as a bundle of tool runs for tracking a SARIF log through a distributed results processing pipeline.

3.14 `run` Object

3.14.1 General

A run object describes a single run of an analysis tool and contains the output of that run.

EXAMPLE 1:

```

{
  "tool": {           # See §3.14.6.
    ...             # A tool object (§3.18).
  },
  "results": [       # See §3.14.23.
    {
      ...           # A result object (§3.27).
    },
    ...
    {
      ...           # Another result object.
    }
  ]
}

```

3.14.2 externalPropertyFileReferences Property

A run object **MAY** contain a property named `externalPropertyFileReferences` whose value is an `externalPropertyFileReferences` object (§3.15) that specifies the locations of the external property files (see §3.15.2) associated with this log file.

3.14.3 automationDetails Property

A run object **MAY** contain a property named `automationDetails` whose value is a `runAutomationDetails` object (§3.17) that describes this run.

For an example, see §3.17.1.

3.14.4 runAggregates Property

A run object **MAY** contain a property named `runAggregates` whose value is an array of zero or more unique (§3.7.3) `runAutomationDetails` objects (§3.17) each of which describes an aggregate of runs to which this run belongs.

For an example, see §3.17.1.

3.14.5 baselineGuid Property

A run object **MAY** contain a property named `baselineGuid` whose value is a GUID-valued string (§3.5.3) which **SHALL** equal the `automationDetails.guid` property (§3.14.3, §3.17.4) of some previous run.

NOTE: This ensures that only “similar” runs are compared.

If `baselineGuid` is present, the `result.baselineState` property (§3.27.24) of every `result` object (§3.27) in `theRun` **SHALL** be computed with respect to the run specified by `baselineGuid`.

3.14.6 tool Property

A run object **SHALL** contain a property named `tool` whose value is a `tool` object (§3.18) that describes the analysis tool that was run.

3.14.7 **language** Property

A run object **MAY** contain a property named `language` whose value is a string specifying the language of the localizable strings (§3.5.1) in `theRun` (except for localizable strings that occur within `theRun.translations` (§3.14.9)), in the format specified by the language tags standard [RFC5646]. If this property is absent, it **SHALL** default to `"en-US"`.

EXAMPLE 1: The language is region-neutral English:

```
"language": "en"
```

EXAMPLE 2: The language is French as spoken in France:

```
"language": "fr-FR"
```

3.14.8 **taxonomies** Property

A run object **MAY** contain a property named `taxonomies` whose value is an array of zero or more unique (§3.7.3) `toolComponent` objects (§3.19) each of which represents a standard taxonomy (§3.19.3).

NOTE: Analysis tools can define their own custom taxonomies; see §3.19.3 and §3.19.25.

3.14.9 **translations** Property

A run object **MAY** contain a property named `translations` whose value is an array of zero or more unique (§3.7.3) `toolComponent` objects (§3.19) each of which represents a translation (§3.19.4).

3.14.10 **policies** Property

A run object **MAY** contain a property named `policies` whose value is an array of zero or more unique (§3.7.3) `toolComponent` objects (§3.19) each of which represents a policy (§3.19.5).

3.14.11 **invocations** Property

A run object **MAY** contain a property named `invocations` whose value is an array of zero or more `invocation` objects (§3.20) that together describe a single run of a single analysis tool.

NOTE: Normally, an analysis tool runs as a single process, and the `invocations` array requires only one element. The `invocations` property is defined as an array, rather than as a single `invocation` object, to accommodate tools which execute a sequence of programs to produce results. For example, a tool might run one program to determine the set of artifacts to analyze and another program to analyze those artifacts.

The elements of the `invocations` array **SHOULD**, as far as possible, be arranged in chronological order according to the start time of each process. If some of the processes run in parallel, this might not be possible.

3.14.12 **conversion** Property

If a run object was produced by a converter, it **MAY** contain a property named `conversion` whose value is a `conversion` object (§3.22) that describes how the converter transformed the analysis tool's native output format into the SARIF format.

A direct producer **SHALL NOT** emit the `conversion` property.

3.14.13 versionControlProvenance Property

A run object **MAY** contain a property named `versionControlProvenance` whose value is an array of zero or more unique (§3.7.3) `versionControlDetails` objects (§3.23). Each array entry specifies a revision in a repository containing files that were scanned during the run.

NOTE 1: This property allows an engineering system to reproduce a scan by retrieving the specified revision of the required files from each repository before repeating the analysis run.

NOTE 2: This property is an array, rather than a single `versionControlDetails` object, to support scenarios where a tool scans files from multiple repositories in a single run.

NOTE 3: This document refers to a container for a related set of files in a VCS as a “repository.” Different VCSs might use different terms.

NOTE 4: This document refers to a fixed revision of a set of files as a “revision”. Different VCSs use different terms; for example, Git calls it a “commit”.

EXAMPLE 1: In this example, an analysis tool has scanned files from one repository: the GitHub repository `example/browser`.

```
{
  "versionControlProvenance": [
    {
      "repositoryUri": "https://github.com/example/browser",
      "revisionId": "1a0c6554caa37144459cb97cb15429b27831476e",
      "branch": "master"
    }
  ]
}
```

3.14.14 originalUriBaseIds Property

A run object **MAY** contain a property named `originalUriBaseIds` whose value is an object (§3.6) each of whose property names designates a URI base id (§3.4.4) and each of whose property values is an `artifactLocation` object (§3.4) that specifies (in the manner described below) the absolute URI [RFC3986] of that URI base id on the machine where the SARIF producer ran.

If the `artifactLocation` object’s `uri` property (§3.4.3) is a relative reference, its `uriBaseId` property (§3.4.4) **SHALL** be present. Otherwise (that is, if `uri` is an absolute URI, or if it is absent), `uriBaseId` **SHALL** be absent.

If the actual value of `uri` would have been an absolute URI, `uri` **MAY** be omitted.

NOTE 1: A SARIF producer might omit such an absolute URI, or a SARIF postprocessor might remove it, for various reasons:

- To avoid revealing sensitive information such as a user name in a URI, for example, `file:///C:/Users/Mary/code/TheProject/`.
- To produce deterministic output (see §Appendix F.) by avoiding path names that differ depending on the machine where the analysis tool runs.

EXAMPLE 1: In this example, the “top-level” property `PROJECTROOT` specifies a URI containing a username:

```
"originalUriBaseIds": {
  "PROJECTROOT": {
    "uri": "file:///C:/Users/Mary/code/TheProject/",
    "description": {
      "text": "The root directory for all project files."
    }
  },
  "SRCROOT": {
    "uri": "src/",
    "uriBaseId": "PROJECTROOT",
    "description": {
```

```

    "text": "The root of the source tree."
  }
}

```

A post-processor might remove `uri` to avoid revealing a username. The advantage of this approach over removing the entire `PROJECTROOT` property is that it retains the `description` property:

```

"originalUriBaseIds": {
  "PROJECTROOT": {
    "description": {
      "text": "The root directory for all project files."
    }
  },
  "SRCROOT": {
    "uri": "src/",
    "uriBaseId": "PROJECTROOT",
    "description": {
      "text": "The root of the source tree."
    }
  }
}

```

The values of the `uriBaseId` properties in the `artifactLocation` objects in `originalUriBaseIds` SHALL NOT form a loop, in the sense described in the URI base id resolution procedure below.

The values of the `uri` properties in the `artifactLocation` objects in `originalUriBaseIds`:

- SHALL end with a single forward slash .
- SHALL NOT include a query or fragment component as defined in URI Generic Syntax [RFC3986].
- SHALL NOT include `".."` path segments.

NOTE 2: The rationale for these restrictions is to allow the `uriBaseId` resolution procedure described below to work by simple concatenation of the `uri` properties in `originalUriBaseIds`. The prohibition of `".."` path segments ensures that the resolution procedure works with file scheme URIs, without concern for the presence of symbolic links. See §3.10.2 for more information on this point.

This property allows SARIF consumers to resolve any relative references which appear in any `artifactLocation` objects elsewhere in the run, as long as the consumer runs either on the same machine as the producer, or on a machine with an identical file system layout. This is useful for individual developers who wish to run analysis tools and examine the results in a viewer. It is also useful for teams which share a convention for their file system layout.

A SARIF consumer SHALL use the following procedure to resolve a URI base id from the information in `originalUriBaseIds`:

NOTE 3: This procedure is part of an overall URI base id resolution procedure described in §3.4.4.

NOTE 4: In this procedure, we refer to the resolved URI value by the variable name `resolvedUri`.

1. Set `resolvedUri` to an empty string.
2. Fetch the `artifactLocation` object whose property name within `originalUriBaseIds` is the value of `uriBaseId`. If there is no such property, the resolution procedure fails.
3. Prepend `artifactLocation.uri` to `resolvedUri`.
4. If `artifactLocation.uri` is an absolute URI, `resolvedUri` is the final resolved URI, and the procedure succeeds.

Otherwise:

5. If `uriBaseId` is absent, the resolution procedure fails.

NOTE 3: This would not occur in a valid SARIF file, but the file might not be valid.

6. If the value of `uriBaseId` has already been encountered during this resolution procedure (that is, if there is a loop in the sequence of URI base ids), the resolution procedure fails.

NOTE 4: This would not occur in a valid SARIF file, but the file might not be valid.

7. Otherwise (that is, if `uriBaseId` is present and its value has not previously been encountered during this resolution), return to Step 2.

EXAMPLE 2: In this example, the URI base id "SRCROOT" on the machine where the SARIF producer ran was "file:///C:/code/MyProject/src/". The producer detected a result in a file whose location relative to that URI base id was "lib/memory.c". A viewer which wished to display that file would first attempt to locate it on the local file system at "C:\code\MyProject\src\lib\memory.c". If the file did not exist at that location, the viewer might prompt the user for the location.

```
{
  "originalUriBaseIds": {
    "PROJECTROOT": {
      "uri": "file:///C:/code/TheProject/"
    },
    "SRCROOT": {
      "uri": "src/",
      "uriBaseId": "PROJECTROOT"
    }
  },
  "results": [
    {
      "ruleId": "CA1001",
      "locations": [
        {
          "physicalLocation": {
            "artifactLocation": {
              "uri": "lib/memory.c",
              "uriBaseId": "SRCROOT"
            }
          }
        }
      ]
    }
  ]
}
```

3.14.15 **artifacts** Property

A run object **MAY** contain a property named **artifacts** whose value is an array of zero or more unique (§3.7.3) **artifact** objects (§3.24) each of which represents an artifact relevant to the run.

The array **SHOULD** contain elements representing at least those artifacts in which results were detected, but it **MAY** contain elements representing all artifacts examined by the tool (whether or not results were detected in those artifacts), or any subset of those artifacts. It **MAY** also include other artifacts relevant to the run, such as attachments (§3.27.26).

NOTE: **artifact** objects contain information that is useful for viewers. Viewers will be able to provide the most information to users if the **artifacts** property is present and contains information for every artifact in which results were detected.

EXAMPLE 1:

```
"artifacts": [
  {
    "location": {
      "uri": "file:///C:/Code/main.c"
    },
    "sourceLanguage": "c",
    "hashes": {
      "sha-256": "b13ce2678a8807ba0765ab94a0ecd394f869bc81"
    }
  }
]
```

In some cases, an artifact might be nested within another artifact (for example, a compressed container), referred to as its “parent.” An artifact that is not nested within another artifact is referred to as a “top-level artifact”. An artifact that is nested within another artifact is referred to as a “nested artifact”. Within the **artifacts** array, an **artifact** object representing a nested artifact is linked to its parent *via* its **parentIndex** property (§3.24.3). For an example, see §3.24.3.

If a nested artifact appears in the `artifacts` array, then the `artifacts` array **SHALL** also contain elements describing each of its parents, up to and including the top-level artifact.

3.14.16 `specialLocations` Property

A run object **MAY** contain a property named `specialLocations` whose value is a `specialLocations` object (§3.25) that defines locations of special significance to SARIF consumers.

3.14.17 `logicalLocations` Property

A run object **MAY** contain a property named `logicalLocations` whose value is an array of zero or more unique (§3.7.3) `logicalLocation` objects (§3.33) each of which represents a logical location relevant to one or more results detected during the run.

In some cases, a logical location might be nested within another logical location (for example, a class nested within a namespace), referred to as its “parent.” A logical location that is not nested within another logical location is referred to as a “top-level logical location”. A logical location that is nested within another logical location is referred to as a “nested logical location”. Within the `logicalLocations` array, a `logicalLocation` object representing a nested logical location is linked to its parent *via* its `parentIndex` property (§3.33.8).

If a nested logical location appears in the `logicalLocations` array, then the `logicalLocations` array **SHALL** also contain elements describing each of its parents, up to and including the top-level logical location.

EXAMPLE 1: In this example, a result was detected in the C++ class `namespaceA::namespaceB::classC`. The `logicalLocations` array contains not only an element describing the class, but also elements describing its containing namespaces.

```
"logicalLocations": [
  {
    "name": "classC",
    "fullyQualifiedName": "namespaceA::namespaceB::classC",
    "kind": "type",
    "parentIndex": 1
  },
  {
    "name": "namespaceB",
    "fullyQualifiedName": "namespaceA::namespaceB",
    "kind": "namespace",
    "parentIndex": 2
  },
  {
    "fullyQualifiedName": "namespaceA",
    "kind": "namespace"
  }
]
```

NOTE: The detailed information in `logicalLocations` is useful, even though much of it is captured in `logicalLocation.fullyQualifiedName` (§3.33.5), because it allows results management systems and other SARIF consumers to organize analysis results, for example, by asking questions such as “How many results were found in the namespace `namespaceA::namespaceB`?”. Programs can ask these questions without having to know how to parse the `fullyQualifiedName` string.

3.14.18 `addresses` Property

A run object **MAY** contain a property named `addresses` whose value is an array of zero or more unique (§3.7.3) `address` objects (§3.32) representing addresses that appear in `physicalLocation` objects (§3.29) within the run.

In some cases, an address might be nested within another address (for example, an offset within a table within a section). An address that is nested within another address is referred to as a “nested address”. Within the `addresses` array, an `address` object representing a nested address is linked to its parent *via* its `parentIndex` property (§3.32.13).

If a nested address appears in the `addresses` array, then `addresses` **SHALL** also contain elements describing each of its parents, up to and including the top-level address.

3.14.19 **threadFlowLocations** Property

A run object **MAY** contain a property named `threadFlowLocations` whose value is an array of zero or more unique (§3.7.3) `threadFlowLocation` objects (§3.38) representing locations that appear in `threadFlow` objects (§3.37) within `theRun`.

The `threadFlowLocations` array may contain all or any subset of the `threadFlowLocation` objects in the run.

NOTE: Defining `threadFlowLocation` objects within `run.threadFlowLocations` can reduce the size of the log file if certain locations occur frequently, either within a single thread flow (for example, if the thread flow represents a loop) or across thread flows (for example, if all thread flows start at the program entry point and share their first few locations).

3.14.20 **graphs** Property

A run object **MAY** contain a property named `graphs` whose value is an array of zero or more unique (§3.7.3) `graph` objects (§3.39). A `graph` object represents a directed graph: a network of nodes and directed edges that describes some aspect of the structure of the code (for example, a call graph).

A `graph` object defined at the run level **MAY** be referenced by a `graphTraversal` object (§3.42) defined in the `graphTraversals` property (§3.27.20) of any `result` object (§3.27) in `theRun`.

3.14.21 **webRequests** Property

A run object **MAY** contain a property named `webRequests` whose value is an array of zero or more unique (§3.7.3) `webRequest` objects (§3.46) representing HTTP requests that appear in `result` objects (§3.27) within `theRun`.

NOTE: This property is primarily useful to web analysis tools.

3.14.22 **webResponses** Property

A run object **MAY** contain a property named `webResponses` whose value is an array of zero or more unique (§3.7.3) `webResponse` objects (§3.47) representing HTTP responses that appear in `result` objects (§3.27) within `theRun`.

NOTE: This property is primarily useful to web analysis tools.

3.14.23 **results** Property

Depending on the circumstances, a run object either **SHALL** or **MAY** contain a property named `results` whose value, again depending on circumstances, is either `null` or an array of zero or more `result` objects (§3.27) each of which represents a single result detected in the course of the run.

NOTE: The `results` array is not defined to contain unique (§3.7.3) elements because some tools report a line number but not a column number for a result's location. Such a tool might report the same result twice on the same line, in some cases producing multiple identical `result` objects.

If the tool failed to start, and if the engineering system responsible for running the tool synthesized a SARIF file to record the failure, then `results` **MAY** be present. If it is present, its value **SHALL** be `null`. See §3.20.13, `invocation.processStartFailureMessage`, for more about this scenario.

If the tool started but failed to begin its analysis (for example, because its command line was invalid), then again `results` **MAY** be present, and if present **SHALL** be `null`.

In all other circumstances, `results` **SHALL** be present and **SHALL** contain all results detected by the tool. If the tool did not detect any results, `results` **SHALL** be an empty array.

If `results` is absent, it **SHALL** default to `null`.

3.14.24 `defaultEncoding` Property

A run object **MAY** contain a property named `defaultEncoding` whose value is a case-sensitive string that provides a default for the `encoding` property (§3.24.9) of any `artifact` object (§3.24) in the `run.artifacts` (§3.14.15) that refers to a text artifact. The string **SHALL** be one of the character set names defined by IANA [IANA-ENC].

If this property is absent, it **SHALL** be interpreted as meaning that there is no default file encoding. In that case, the encoding of any `artifact` object that does not contain an `encoding` property **SHALL** be taken to be unknown.

For an example, see §3.24.9.

3.14.25 `defaultSourceLanguage` Property

A run object **MAY** contain a property named `defaultSourceLanguage` whose value is a hierarchical string (§3.5.4) that provides a default value for the `sourceLanguage` property (§3.24.10) of any `artifact` object (§3.24) in the `run.artifacts` (§3.14.15) which refers to a text artifact that contains source code.

If `defaultSourceLanguage` is present, its value **SHOULD** conform to the conventions defined in §3.24.10.2.

If `defaultSourceLanguage` is absent, it **SHALL** be taken to mean that there is no default source language. In that case, the source language of any `artifact` object that does not contain a `sourceLanguage` property **SHALL** be taken to be unknown. In that case, a SARIF viewer **MAY** use any method or heuristic to determine the source language of each file, for example by examining the file's file name extension or MIME type, or by prompting the user.

3.14.26 `newlineSequences` Property

A run object **MAY** contain a property named `newlineSequences` whose value is an array of one or more unique (§3.7.3) strings each of which specifies a character sequence that the tool treated as a line break during this run.

If this property is absent, it **SHALL** default to the array [`"\r\n"`, `"\n"`].

The order of the elements in the array is significant. It **SHALL** mean that at potential line breaks, the tool “greedily” attempted to match each element of the array in order.

EXAMPLE 1: If `newlineSequences` has the value [`"\r\n"`, `"\r"`, `"\n"`], the character sequence `"\r\n"` counts as one line break, not two.

NOTE: This property is useful for SARIF consumers that are sensitive to the value of the line number properties `startLine` (§3.30.5) and `endLine` (§3.30.7) in `region` objects (§3.30). It ensures that the consumer counts lines in the same way as the producer. A SARIF viewer might use this property when highlighting a region to ensure that it highlights the correct lines. More critically, a tool that applies fixes (see §3.55), especially one that applies them automatically, can use this property to ensure that it inserts and removes content on the correct lines.

EXAMPLE 2: In this example, the SARIF producer accepts the Unicode characters NEXT LINE (U+0085) and LINE SEPARATOR (U+2028) as line separators in addition to the usual values.

```
{
    # A run object (§3.14).
    ...
    "newlineSequences": [ "\r\n", "\n", "\u0085", "\u2028" ],
    ...
}
```

3.14.27 `columnKind` Property

If a SARIF producer processes text artifacts and `theRun.results` (§3.14.23) is non-empty, the `run` object **SHALL** contain a property named `columnKind` whose value is a string that specifies the unit in which the analysis tool measures columns. If a SARIF producer processes text artifacts and `theRun.results` is empty, `columnKind` **MAY** be present.

`columnKind` **SHALL** have one of the following values, with the specified meanings:

- `"utf16CodeUnits"`: Each UTF-16 code unit is considered to occupy one column. This means that a surrogate pair is considered to occupy two columns.
- `"unicodeCodePoints"`: Each Unicode code point (abstract character) is considered to occupy one column. This means that even a character that is represented in UTF-16 by a surrogate pair is considered to occupy one column.
- `"bytes"`: column numbers refer to byte offsets from the start of the line, where the first byte in a line has value 1. Note: this is for consistency with column numbering in text regions (§3.30.2), and is different from binary regions (§3.30.3), where the first byte offset in an **artifact** is 0.

If the SARIF producer does not process text artifacts, `columnKind` **SHALL** be absent.

If a SARIF consumer uses a column measurement unit other than that specified by `columnKind`, and if the consumer is required to interact with the artifact's contents (for example, by displaying the artifact in an editor and highlighting a region), the consumer **SHALL** recompute column numbers in its (the consumer's) native measurement unit.

3.14.28 `redactionTokens` Property

If the value of any redactable property (§3.5.2) in `theRun` has been redacted, `theRun` **SHALL** contain a property named `redactionTokens` whose value is an array of zero or more unique (§3.7.3) strings any of which can be used to replace redacted text. If no text in `theRun` has been redacted, `redactionTokens` **SHALL** be absent.

If `redactionTokens` contains a single element, that element **SHOULD** be the string `"[REDACTED]"`; if it contains more than one, each additional element **SHOULD** be of the form `"[REDACTED-n]"` where *n* is a positive integer.

NOTE 1: The rationale for recommending the alternate form only for the second and subsequent tokens is that a tool might create one token and only later discover that additional tokens are required. With this recommendation, the tool does not have to rename the token it has already created.

NOTE 2: Redaction tokens have no special meaning in properties not specified as "redactable."

If for any reason different values are used, they **MAY** be any readily identifiable strings. An example of a situation where a SARIF producer might choose a different redaction token is if the string `"[REDACTED]"` occurs in the value of a redactable property in `theRun`.

EXAMPLE 1: In this example, the leading portion of a full path name has been redacted from the redactable property `invocation.commandLine` to avoid revealing information about the machine's directory layout.

```
{
  "redactionTokens": [
    "[REDACTED]"
  ],
  "invocation": {
    "commandLine": "SourceScanner --input [REDACTED]/src/ui"
  },
  ...
}
```

3.15 externalPropertyFileReferences Object

3.15.1 General

An `externalPropertyFileReferences` object contains information that enables a SARIF consumer to locate the external property files (see §3.15.2) that contain the values of all externalized properties associated with `theRun`.

3.15.2 Rationale

In some engineering environments, a single tool run might analyze hundreds of thousands of files and produce millions of results. This causes problems for both producers and consumers of such large SARIF log files:

- The log file might be too large for a consumer to hold in memory and might take several minutes to read.
- During production, some information (such as the complete set of artifacts that were analyzed, the complete set of rules that were violated, or the end time of the run) cannot be known until the run is complete. Therefore, it is likely to be serialized at the end of the log file. However, consumers might need to access some of that information before reading the entire file. For example, a SARIF viewer might need to display rule metadata along with each result it displays, or to display the start and end times of a set of tool runs.

To mitigate these problems, SARIF allows certain properties of a `run` object and its sub-objects to be stored in separate files. We refer to these files as “external property files”, and we refer to the file containing the `run` object itself as the “root file”. We refer to a property that can be stored in an external property file as an “externalizable property.” We refer to a property that *has* been stored in an external property file as an “externalized property.”

The format of an external property file is described in §4

A SARIF consumer **SHALL** treat the value of an object-valued property stored in an external property file exactly as if it had appeared inline in the root file as the value of the corresponding property.

A SARIF consumer **SHALL** treat the value of an array-valued property stored in an external property file exactly as if its elements had appeared inline in the root file, appended to the existing value, if any, of that property.

NOTE: This allows a SARIF producer to begin writing the elements of an array-valued property to the root file, and then, if the file grows too large, to “spill” the additional elements into one or more external property files.

3.15.3 Properties

The following table lists all the externalizable properties together with their corresponding property names in the `externalPropertyFileReferences` object:

Externalizable property	Property name	Type
<code>run.addresses</code>	<code>addresses</code>	array
<code>run.artifacts</code>	<code>artifacts</code>	array
<code>run.conversion</code>	<code>conversion</code>	object
<code>run.graphs</code>	<code>graphs</code>	array
<code>run.invocations</code>	<code>invocations</code>	array
<code>run.logicalLocations</code>	<code>logicalLocations</code>	array
<code>run.policies</code>	<code>policies</code>	array
<code>run.properties</code>	<code>externalizedProperties</code>	object
<code>run.webRequests</code>	<code>webRequests</code>	array
<code>run.webResponses</code>	<code>webResponses</code>	array
<code>run.results</code>	<code>results</code>	array
<code>run.taxonomies</code>	<code>taxonomies</code>	array
<code>run.threadFlowLocations</code>	<code>threadFlowLocations</code>	array

Externalizable property	Property name	Type
run.translations	translations	array
run.tool.driver	driver	object
run.tool.extensions	extensions	array

NOTE 1: `run.properties` is externalized under the property name `externalizedProperties` to allow this object to have a property bag named `properties`, consistent with all other objects in this document.

NOTE 2: Note that `run.conversion.tool.driver` and `run.conversion.tool.extensions` are not separately externalizable. Rather, the `run.conversion` property as a whole is externalizable.

Every externalizable property whose type is shown in the table as “object” **SHALL**, if externalized, be stored in a single external property file. In that case, the value of the corresponding property in `externalPropertyFileReferences` **SHALL** be an `externalPropertyFileReference` object (§3.16) specifying the location of the external property file.

Every externalizable property whose type is shown in the table as “array” **SHALL**, if externalized, be stored in one or more external property files. In that case, the value of the corresponding property in `externalPropertyFileReferences` **SHALL** be an array of zero or more `externalPropertyFileReference` objects specifying the locations of those external property files.

EXAMPLE 1: In this example, `run.conversion` is stored in the file `C:\logs\scantool.conversion.sarif-external-properties` and `run.results` is divided into the files `C:\logs\scantools.results-1.sarif-external-properties` and `C:\logs\scantools.results-2.sarif-external-properties`.

```
{
  "originalUriBaseIds": { # A run object.
    "LOGSDIR": { # See §3.14.14.
      "uri": "file:///C:/logs/"
    }
  },
  "externalPropertyFileReferences": {
    "conversion": { # An externalPropertyFileReference object (§3.16).
      "location": { # See §3.16.3.
        "uri": "scantool.conversion.sarif-external-properties",
        "uriBaseId": "LOGSDIR"
      },
      "guid": "11111111-1111-1111-8888-111111111111" # See §3.16.4.
    },
    "results": [
      {
        "location": {
          "uri": "scantool.results-1.sarif-external-properties",
          "uriBaseId": "LOGSDIR"
        },
        "guid": "22222222-2222-1111-8888-222222222222",
        "itemCount": 10000
      },
      {
        "location": {
          "uri": "scantool.results-2.sarif-external-properties",
          "uriBaseId": "LOGSDIR"
        },
        "guid": "33333333-3333-1111-8888-333333333333",
        "itemCount": 4277
      }
    ]
  },
  ...
}
```

With one exception described below, if a property appears inline in the root file, its name **SHALL NOT** appear as one of the property names in `externalPropertyFileReferences`. Since an external property file can contain multiple externalized properties, `externalPropertyFileReference` objects belonging to distinct properties **MAY** denote the same external property file. However, if an array-valued externalizable property is divided among multiple external property files, the `externalPropertyFileReference` objects belonging to that property **SHALL** denote distinct external property files.

EXAMPLE 2: In this example, `theRun.conversion` and `theRun.properties` are stored in the same external property file.

```
{
  # A run object (§3.14).
  "originalUriBaseIds": {
    # See §3.14.14.
    "LOGSDIR": {
      "uri": "file:///C:/logs/"
    }
  },
  "externalPropertyFileReferences": {
    "conversion": {
      # An externalPropertyFileReference object (see (#externalpropertyfilereference-object)).
      # See §3.16.3.
      "location": {
        "uri": "scantool.sarif-external-properties",
        "uriBaseId": "LOGSDIR",
        "index": 0
      },
      "guid": "11111111-1111-1111-8888-111111111111" # See §3.16.4.
    },
    "externalizedProperties": {
      "location": {
        "uri": "scantool.sarif-external-properties",
        "uriBaseId": "LOGSDIR",
        "index": 0
      },
      "guid": "11111111-1111-1111-8888-111111111111"
    }
  },
  ...
}
```

EXAMPLE 3: This example represents invalid SARIF because both elements of the array belonging to the `results` property denote the same external property file.

```
{
  # A run object (§3.14).
  "originalUriBaseIds": {
    # See §3.14.14.
    "LOGSDIR": {
      "uri": "file:///C:/logs/"
    }
  },
  "externalPropertyFileReferences": {
    "results": [
      {
        # An externalPropertyFileReference object (see (#externalpropertyfilereference-object)).
        "location": {
          "uri": "scantool.results.sarif-external-properties",
          "uriBaseId": "LOGSDIR",
          "index": 0
        },
        "guid": "22222222-2222-1111-8888-222222222222"
      },
      {
        # INVALID: The two external property files are the same.
        "location": {
          "uri": "scantool.results.sarif-external-properties",
          "uriBaseId": "LOGSDIR",
          "index": 0
        },
        "guid": "22222222-2222-1111-8888-222222222222"
      }
    ]
  },
  ...
}
```

The exception is that if `run.tool.driver` is externalized, it **SHALL** still occur inline in the root file. The inline driver property **SHOULD** contain only properties that identify the tool, such as `name` (§3.19.8) and `semanticVersion` (§3.19.12); it **SHOULD NOT** contain properties such as `globalMessageStrings` (§3.19.22), `rules` (§3.19.23), `notifications` (§3.19.24), and `taxa` (§3.19.25), which take up a large amount of space.

NOTE 3: This makes it possible to identify the tool that produced the log file without locating and opening the external property file, while still getting the benefit of externalizing those properties that take up a large amount of space.

3.16 externalPropertyFileReference Object

3.16.1 General

An `externalPropertyFileReference` object contains information that enables a SARIF consumer to locate the external property file (see §3.15.2) that contains the value of an externalized property associated with `theRun`.

3.16.2 Constraints

At least one of the `location` property (§3.16.3) or the `guid` property (§3.16.4) **SHALL** be present. If both are present, they **SHALL** identify the same set of externalized properties (possibly located inline; see §3.13.5).

NOTE: This constraint ensures that it is possible to locate the externalized properties.

3.16.3 `location` Property

Depending on the circumstances, an `externalPropertyFileReference` object either **SHALL** or **MAY** contain a property named `location` whose value is an `artifactLocation` object (§3.4) that specifies the location of the external property file.

If the externalized properties are persisted in a separate file, `location` **SHALL** be present. In that case, if the `artifactLocation` object's `uri` property (§3.4.3) specifies a relative reference and its `uriBaseId` property (§3.4.4) is absent, then `uri` **SHALL** be interpreted relative to the location of the root file.

Otherwise (that is, if the externalized properties are persisted as an element of `theSarifLog.inlineExternalProperties` (§3.13.5)), then `location` **MAY** be present. If `location` is present, its `uri` property **SHALL** resolve to an absolute URI using the `sarif` scheme (§3.10.3). If `location` is absent, then a SARIF consumer that needs to locate the externalized properties **SHALL** do so using the `guid` property (§3.16.4).

3.16.4 `guid` Property

Depending on the circumstances, an `externalPropertyFileReference` object either **SHALL** or **MAY** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) which provides a unique, stable identifier for the external property file.

If the externalized properties are persisted in an element of `theSarifLog.inlineExternalProperties` (§3.13.5) and `location` (§3.16.3) is absent, then `guid` **SHALL** be present.

Otherwise (that is, if the externalized properties are persisted in a separate file, in which case `location` is required, or if the externalized properties are persisted in an element of `theSarifLog.inlineExternalProperties` but `location` is present), `guid` **MAY** be present.

NOTE: The rationale for these constraints is to ensure that there is enough information to locate the external properties. If the properties are in an external file, then `location` is necessary but `guid` can still be present; if the properties are inline, either `location` or `guid` suffices but both can be present.

If `guid` is present, it **SHALL** equal the `guid` property (§4.3.4) of the `externalProperties` object (§4.3) identified by `guid` and/or `location`.

3.16.5 `itemCount` Property

If an `externalPropertyFileReference` object specifies an external property file that contains all or a portion of an array-valued property, it **MAY** contain a property named `itemCount` whose value is a non-negative integer that specifies the number of items in the externalized property array in that file. If the `externalPropertyFileReference` object specifies an external property file that contains an object-valued property, `itemCount` **SHALL** be absent.

If `itemCount` is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set).

NOTE: This information is useful to a SARIF consumer that needs to locate the item at a specified array index in an externalized array-valued property. Without this information, the consumer would have to open in turn each external property file belonging to that property, counting the number of array elements in each, until it reached the file containing the desired element.

EXAMPLE 1: In EXAMPLE 1 in §3.15.3, the array-valued property `results` is divided into two files, the first containing 10,000 elements and the second containing 4,277 elements. A SARIF consumer that needs to access element 12,000 knows immediately that it is contained in the second file, at index 2,000.

3.17 `runAutomationDetails` Object

3.17.1 General

A `runAutomationDetails` object contains information that specifies theRun's identity and role within an engineering system.

EXAMPLE 1: In this example, a run contains the results from one nightly execution of a single security tool over a specified set of binaries. `theRun.automationDetails` describes the run. Its `id` and `guid` properties both identify the run; the former in human-readable form, the latter in a form that might be more useful in an engineering system's database. Its `correlationGuid` property specifies the set of runs identified by *all but the last component* of `id`'s hierarchical string; that is, it identifies the set of runs "Nightly CredScan run for sarif-sdk/master/x86/debug".

The run in this example is part of an aggregate of runs which together comprise the nightly execution of the engineering system's full suite of security tools. `theRun.runAggregates[0]` describes that aggregate. Its `id` and `guid` properties both identify the aggregate. Its `correlationGuid` property specifies the collection of such aggregates identified by *all but the last component* of `id`'s hierarchical string; that is, it identifies the collection of aggregates "Nightly security tools run for sarif-sdk/master/x86/debug".

```
{
  "automationDetails": {           # A run object (§3.14).
    "description": {               # See §3.14.3.
      "text": "This is the {0} nightly run of the Credential Scanner tool on
               all product binaries in the '{1}' branch of the '{2}' repo. The
               scanned binaries are architecture '{3}' and build type '{4}'.",
      "arguments": [
        "October 10, 2018",
        "master",
        "sarif-sdk",
        "x86",
        "debug"
      ]
    },
    "id": "Nightly CredScan run for sarif-sdk/master/x86/debug/2018-10-05",
    "guid": "11111111-1111-1111-8888-111111111111",
    "correlationGuid": "22222222-2222-1111-8888-222222222222"
  },
  "runAggregates": [              # See §3.14.4.
    {
      "id": "Nightly security tools run for sarif-sdk/master/x86/debug/2018-10-05",
      "guid": "33333333-3333-1111-8888-333333333333",
      "correlationGuid": "44444444-4444-1111-8888-444444444444"
    }
  ]
}
```

3.17.2 `description` Property

A `runAutomationDetails` object **MAY** contain a property named `description` whose value is a message object (§3.11) that describes the role played within the engineering system by theRun.

3.17.3 `id` Property

A `runAutomationDetails` object **MAY** contain a property named `id` whose value is a hierarchical string (§3.5.4) that uniquely identifies theRun within the engineering system.

A result management system or other components of the engineering system **MAY** use `run.automationDetails.id` to associate the information in the log with additional information not provided by the analysis tool that produced it.

An engineering system **MAY** define any number of components and interpret them in any way desired.

NOTE: The intent is to use the components of `id` to group results from similar runs, such as “all nightly Credential Scanner runs.” A SARIF viewer might display a set of runs in a tree view, grouped by the components of `id`.

EXAMPLE 1: A run whose `id` is "My Nightly Run/Debug/x64/2018-10-10" belongs to the category "My Nightly Run/Debug/x64". Presumably, this is the run from October 10, 2018.

The trailing component of `id` **MAY** be empty; note that the grammar for a hierarchical identifier (§3.5.4.1) permits any component to be empty. This **SHALL** be taken to signify that the run belongs to the specified category, but that the run itself has no unique identifier.

EXAMPLE 2: A run whose `id` is "My Nightly Run/Debug/x64/" belongs to the category "My Nightly Run/Debug/x64" but is not distinguished from other runs in that category.

`id` **MAY** consist of a single component. This **SHALL** be taken to specify a unique identifier for the run, withough specifying any category that the run belongs to.

EXAMPLE 3: A run whose `id` is "My Nightly Run Debug x64 2018-10-10" has a unique identifier but cannot be inferred to belong to any category.

3.17.4 `guid` Property

A `runAutomationDetails` object **MAY** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) that provides a unique, stable identifier for `theRun`.

A result management system or other components of the engineering system **MAY** use `run.automationDetails.guid` to associate the information in the log with additional information not provided by the analysis tool that produced it.

3.17.5 `correlationGuid` Property

A `runAutomationDetails` object **MAY** contain a property named `correlationGuid` whose value is a GUID-valued string (§3.5.3) which is shared by all such runs of the same type, and differs between any two runs of different types.

If `id` (§3.17.3) is present, `correlationGuid` **SHALL** identify the category of runs specified by all but the last hierarchical component (which **MAY** be empty according to the grammar (§3.5.4.1) for hierarchical strings) of `id`.

NOTE: Consider an engineering system that allows engineers to define “build definitions”, and that assigns a GUID to each build definition. In such a system, the build definition’s GUID could serve as `run.automationDetails.correlationGuid`. It would be the same for all runs produced by the same build definition, and different between any two runs produced by different build definitions.

3.18 `tool` Object

3.18.1 General

A `tool` object describes the analysis tool or converter that was run. The `tool` object in `run.tool` (§3.14.6) describes an analysis tool; the `tool` object in `run.conversion.tool` (§3.14.12, §3.22.2) describes a converter.

A tool consists of one or more “tool components,” each of which consists of one or more files. We refer to the component that contains the tool’s primary executable file as the “driver.” It controls the tool’s execution and typically defines a set of analysis rules. We refer to all other tool components as “extensions.” Extensions can include:

- Libraries of additional rules, which we refer to as “plugins.”

- Files that affect the behavior of the tool, which we refer to as “configuration files.”

NOTE: Configuration files that affect the analysis output are of particular interest in compliance scenarios, where, for example, it is necessary to demonstrate that a particular set of rules has been evaluated.

Each tool component is represented by a `toolComponent` object (§3.19).

If another tool post-processes the log file (for example, by removing certain results, or by adding information that was not known to the analysis tool), the post-processing tool **SHOULD NOT** alter any part of the tool object.

EXAMPLE 1:

```
{
  "driver": {
    # A tool object.
    # See §3.18.2.
    "name": "CodeScanner",
    "fullName": "CodeScanner 1.1, Developer Preview (en-US)",
    "semanticVersion": "1.1.2-beta.12",
    "version": "1.1.2b12",
    ...
  },
  "extensions": [
    # See §3.18.3.
    {
      "name": "CodeScanner Security Rules",
      "version": "3.1",
      ...
    }
  ]
}
```

3.18.2 driver Property

A `tool` object **SHALL** contain a property named `driver` whose value is a `toolComponent` object (§3.19) that describes the component containing the tool’s primary executable file.

3.18.3 extensions Property

If the tool used any extensions during the run, the `tool` object **SHOULD** contain a property named `extensions` whose value is an array of one or more unique (§3.7.3) `toolComponent` objects (§3.19) that describe those extensions. If the tool did not use any extensions during the run, then `extensions` **SHALL** either be absent or an empty array.

3.19 toolComponent Object

3.19.1 General

A `toolComponent` object represents one of the components which comprise an analysis tool or a converter, either its driver or one of its extensions. For more information, see §3.18.1.

SARIF also uses `toolComponent` objects to represent other components that participate in the analysis, including:

- Taxonomies (§3.19.3)
- Translations (§3.19.4)
- Policies (§3.19.5)

NOTE: SARIF makes this design choice because `toolComponent` objects contain properties that are useful in all of these other types of components: properties that represent the component’s identity, localizable properties (§3.5.1) that label the component and describe its purpose, and properties that define rules and similar items that participate in the analysis. Not every property is useful in every component type; for example, `translationMetadata` (§3.19.27) is useful only in `toolComponent` objects that represent translations.

3.19.2 Constraints

At least one of `version` (§3.19.13) and `semanticVersion` (§3.19.12) **SHOULD** be present.

3.19.3 Taxonomies

A taxonomy is a classification of results into a set of categories. Some taxonomies are defined publicly, without reference to any particular tool; we refer to these as “standard taxonomies.” An example is the Common Weakness Enumeration [CWE]. A tool can also define its own classification (in addition to the classification implied by its rule definitions); we refer to this as a “custom taxonomy.” We refer to a category within a taxonomy as a “taxon” (*pl.* “taxa”).

A taxonomy is represented by a `toolComponent` object. Its taxa are stored in the `taxa` property (§3.19.25).

A taxon is represented by a `reportingDescriptor` object (§3.49); hence `toolComponent.taxa` is an array of `reportingDescriptor` objects. This is the same object that represents rules and notifications, so a taxon can specify identity properties such as `id` (§3.49.3) and `guid` (§3.49.5), localizable (§3.5.1) descriptive properties such as `name` (§3.49.7) and `fullDescription` (§3.49.10), and configuration properties in `defaultConfiguration` (§3.49.14).

Standard taxonomies **SHALL** be stored in the `run.taxonomies` array (§3.14.8). Every `toolComponent` object in this array **SHALL** contain a `taxa` property (§3.19.25), and **SHALL NOT** contain rules (§3.19.23) or notifications (§3.19.24) properties.

A custom taxonomy is represented by providing a `toolComponent` object in `tool.driver` (§3.18.2) or `tool.extensions` (§3.18.3) with a `taxa` property. Such a `toolComponent` object **MAY** still contain rules and/or notifications as usual.

EXAMPLE 1: In this example, the tool driver supports the CWE™ taxonomy, and also supports a custom taxonomy that it defines. Any result that violates the driver’s rule "CA2101" falls into the "MemoryManagement" taxon of its custom taxonomy, as shown by the "superset" relationship from the "MemoryManagement" taxon to the rule (which is interpreted as “The MemoryManagement taxon is a superset of rule CA2101”). For more information on relationships, see §3.49.15 and §3.53.

```
{
  # A run object (§3.14).
  "tool": {
    # See §3.14.6.
    "driver": {
      # See §3.18.2.
      "name": "CodeScanner",
      "semanticVersion": "3.3",
      "guid": "11111111-1111-1111-8888-111111111111",
      ...
      "rules": [
        {
          "id": "CA2101",
          "shortDescription": {
            "text": "Failed to release dynamic memory."
          },
          "relationships": [
            # See §3.49.15.
            {
              # A reportingDescriptorRelationship object (§3.53).
              "target": {
                # See §3.53.2
                "id": "MemoryManagement",
                "guid": "66666666-6666-1111-8888-666666666666",
                "toolComponent": {
                  "name": "CodeScanner",
                  "guid": "11111111-1111-1111-8888-111111111111"
                }
              },
              "kinds": [
                # See §3.53.3.
                "superset"
              ]
            }
          ]
        }
      ],
      ...
    },
    "taxa": [
      {
        "id": "MemoryManagement",
        "guid": "66666666-6666-1111-8888-666666666666",
        "shortDescription": {
          "text": "Improper usage of dynamic memory."
        }
      },
      {
        "id": "Cryptography",
        "guid": "77777777-7777-1111-8888-777777777777",
        "shortDescription": {
          "text": "Insecure use of cryptography."
        }
      }
    ],
    "supportedTaxonomies": [
      {
        "name": "CodeScanner",
        "guid": "11111111-1111-1111-8888-111111111111"
      },
      {
        "name": "CWE",
```

```

        "index": 1,
        "guid": "33333333-0000-1111-8888-000000000000"
    }
  ],
  "taxonomies": [
    {
      "name": "CWE",
      "version": "3.2",
      "releaseDateUtc": "2019-01-03",
      "guid": "33333333-0000-1111-8888-000000000000",
      "informationUri": "https://cwe.mitre.org/data/published/cwe_v3.2.pdf/",
      "downloadUri": "https://cwe.mitre.org/data/xml/cwec_v3.2.xml.zip",
      "organization": "MITRE",
      "shortDescription": {
        "text": "The MITRE Common Weakness Enumeration"
      },
      "contents": [
        "localizedData",
        "nonLocalizedData"
      ],
      "isComprehensive": true,
      "minimumRequiredLocalizedDataSemanticVersion": "3.2",
      "taxa": [
        {
          "id": "327",
          "guid": "33333333-0000-1111-8888-111111111111",
          "name": "BrokenOrRiskyCryptographicAlgorithm",
          "shortDescription": {
            "text": "Use of a Broken or Risky Cryptographic Algorithm."
          },
          "defaultConfiguration": {
            "level": "warning"
          }
        },
        {
          "id": "924",
          "guid": "33333333-0000-1111-8888-222222222222",
          "name": "TransmittedMessageIntegrity",
          "shortDescription": {
            "text": "Improper Enforcement of Message Integrity ..."
          },
          "defaultConfiguration": {
            "level": "warning"
          }
        }
      ],
      ...
    }
  ],
  ...
}

```

3.19.4 Translations

A translation is the rendering of a `toolComponent` object's localizable strings (§3.5.1) into another language.

A translation is itself represented by a `toolComponent` object whose localizable properties are the translated versions of the corresponding properties in the component being translated. A translation specifies the tool component to which it applies by way of its `associatedComponent` property (§3.19.33).

Translations **SHALL** be stored in the `run.translations` array (§3.14.9).

A translation **SHALL** specify the component that it translates by way of its `associatedComponent` property (§3.19.33). `associatedComponent` **SHALL NOT** refer to another translation.

A translation component **SHALL** contain the translations of every localizable string in the translated component, even if the translated string is identical to the original string. It **MAY** contain additional strings that do not appear in the translated component.

To some degree, translations and the components they translate can version independently. The versioning relationship between a translation and the translated component is explained in the sections describing `localizedDataSemanticVersion` (§3.19.31), populated by translations, and `requiredMinimumLocalizedDataSemanticVersion` (§3.19.32), populated by translated components.

A translation **SHOULD** include the value `"localizedData"` in its `contents` array (§3.19.29). It **MAY** also include the value `"nonLocalizedData"`.

To facilitate the identification of translations that are associated with a given component, a `toolComponent` **SHOULD** populate its `guid` property (§3.19.6), and a translation for that component **SHOULD** set its `guid` property to the same value.

In many cases, a new version of a `toolComponent` defines new localizable strings or requires changes to existing ones (for example, when the tool defines new analysis rules). But in some cases, a new version of a `toolComponent` can use existing translations (for example, in the case of a bug fix release). To ensure that new translations are created only when necessary, a translation component **SHOULD** populate `localizedDataSemanticVersion` (§3.19.31), and a translatable component **SHOULD** populate `minimumRequiredLocalizedDataSemanticVersion` (§3.19.32). See the descriptions of those two properties for an explanation of the interaction between them.

EXAMPLE 1: In this example, a French translation is available. It translates localizable component-level properties such as `toolComponent.name` (§3.19.8), as well as rule-level properties such as `reportingDescriptor.shortDescription` (§3.49.9). The translation can be used because its `localizedDataSemanticVersion` property (§3.19.31) is compatible with the translated component's `minimumRequiredLocalizedDataSemanticVersion` property (§3.19.32).

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "semanticVersion": "3.3",
      "minimumRequiredLocalizedDataSemanticVersion": "3.1",
      ...
    },
    "rules": [
      {
        "id": "CA2101",
        "shortDescription": {
          "text": "Do not do dangerous things."
        }
      }
    ]
  },
  "translations": [
    {
      "language": "fr-FR",
      "semanticVersion": "3.1.3",
      "localizedDataSemanticVersion": "3.1.2",
      "contents": [
        "localizedData"
      ],
      "translationMetadata": {
        "name": "French translation for CodeScanner"
      },
      "name": "<The tool name 'CodeScanner' translated into French>",
      ...
      "rules": [
        {
          "id": "CA2101",
          "shortDescription": {
            "text": "<'Do not do dangerous things.' Translated into French>"
          }
        }
      ]
    }
  ],
  ...
}
```

3.19.5 Policies

A policy is a set of rule configurations that specify how results that violate the rules defined by a particular tool component are to be treated.

A policy is represented by a `toolComponent` object. A policy specifies the tool component to which it applies by way of its `associatedComponent` property (§3.19.33).

A policy **SHALL** contain a `rules` property (§3.19.23), each `reportingDescriptor`-valued (§3.49) element of which in turn contains a `defaultConfiguration` property (§3.49.14). Each element of the `rules` array **SHALL** correspond to a rule defined by the associated component. The `rules` array **MAY** contain elements describing any or all of the rules defined by the associated component. The elements of the `rules` array **MAY** alter rule properties such as `level` (§3.50.3), and **MAY** enable or disable rules. In this way, the policy defines the code analysis standard that is expected of the engineering team.

Policies **SHALL** be stored in the `run.policies` array (§3.14.10).

A SARIF consumer **MAY** offer the user the option of treating results according to the associated component's default rule configuration (possibly modified by command line options stored in the `Invocation.ruleConfigurationOverrides` (§3.20.5), by configuration files, by environment variables, or by any other means), or according to the configuration defined by a selected element of `run.policies`. If the user selects a policy, then for any result that violates a rule covered by that policy, the SARIF consumer **SHALL** treat the result according to the policy, regardless of the associated component's default configuration, regardless of any configuration overrides, and regardless of whether the `result` object (§3.27) itself specifies a configuration property such as `level` (§3.27.10).

NOTE: The rationale is that when a user asks to see how a policy views a set of results, they want to see exactly what the policy has to say, regardless of any configuration options that might have been selected when the log was created.

EXAMPLE 1: In this example, the tool driver defines rule CA2101 to be a warning and disables rule CA2551 by default. However, the corporate security policy specifies that a violation of rule CA2101 is an error and requires rule CA2551 to be run. The presence of `run.policies` allows a SARIF viewer to display the results according to the tool's view or the policy's view.

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "CA2101",
          "defaultConfiguration": {
            "level": "warning"
          }
        },
        {
          "id": "CA2551",
          "defaultConfiguration": {
            "level": "warning",
            "enabled": false
          }
        }
      ]
    }
  },
  "policies": [
    {
      "name": "Example Corp. Security Policy",
      "semanticVersion": "7.0",
      "rules": [
        {
          "id": "CA2101",
          "defaultConfiguration": {
            "level": "error"
          }
        },
        {
          "id": "CA2551",
          "defaultConfiguration": {
            "enabled": true
          }
        }
      ]
    }
  ]
}
```

3.19.6 guid Property

A `toolComponent` object **MAY** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) that provides a unique, stable identifier for the component. `guid` **SHALL NOT** vary between versions of a given component.

3.19.7 Product hierarchy properties

The `name` (§3.19.8) or `fullName` (§3.19.9), `product` (§3.19.10), and `productSuite` (§3.19.11) properties establish a hierarchy of related software: the tool component identified by `name` and/or `fullName` is part of the product named by `product`, which in turn is part of the product suite identified by `productSuite`.

3.19.8 **name** Property

A `toolComponent` object **SHALL** contain a property named `name` whose value is a localizable string (§3.5.1) containing the name of the tool component.

EXAMPLE 1: "CodeScanner"

EXAMPLE 2: "CodeScanner Security Rules Plugin"

EXAMPLE 3: "CodeScanner configuration file"

3.19.9 **fullName** Property

A `toolComponent` object **MAY** contain a property named `fullName` whose value is a localizable string (§3.5.1) containing the name of the tool component along with its version and any other useful identifying information, such as its locale.

EXAMPLE 1: "CodeScanner 1.1, Developer Preview (en-US)"

3.19.10 **product** Property

A `toolComponent` object **MAY** contain a property named `product` whose value is a localizable string (§3.5.1) containing the name of the product to which the tool component belongs.

EXAMPLE 1: "product": "Example Software Corp. Security Scanner"

3.19.11 **productSuite** Property

A `toolComponent` object **MAY** contain a property named `productSuite` whose value is a localizable string (§3.5.1) containing the name of the suite of products to which the tool component belongs.

EXAMPLE 1: "productSuite": "Example Software Corp. Quality Tools"

3.19.12 **semanticVersion** Property

A `toolComponent` object **MAY** contain a property named `semanticVersion` whose value is a string containing the tool component's version in a format that conforms to the syntax and semantics specified by Semantic Versioning [SEMVER].

EXAMPLE 1: "semanticVersion": "1.1.2-beta.12"

NOTE 1: Semantic versions are sortable in chronological order of release. The presence of the `semanticVersion` property allows results management systems to (for example) restrict the results they display to versions newer than a specified version, or to restrict the results to a particular major version.

Unless the author of the converter knows that the version number of the tool from which it converts is intended to be interpreted according to Semantic Versioning [SEMVER], the converter **SHALL NOT** emit the `semanticVersion` property in `run.tool` (§3.14.6), although of course it may emit its own `semanticVersion` property (the one in `run.conversion.tool` (§3.22.2)).

3.19.13 **version** Property

A `toolComponent` object **MAY** contain a property named `version` whose value is a string containing the tool component's version in whatever format the component natively provides.

NOTE: Plugins are often binary files whose version can be determined; configuration files are typically text files with no embedded version information.

3.19.14 **dottedQuadFileVersion** Property

If the operating system on which the tool runs provides a value for the file version of the tool component's primary executable file, and if that value logically consists of an ordered set of four non-negative integers, then the `toolComponent` object **MAY** contain a property named `dottedQuadFileVersion` whose value is a string representation of that file version in this syntax:

```
dottedQuadFileVersion = non negative integer, 3*(".", non negative integer);
```

where the `non negative integers` follow the logical order of the components of the file version.

If the operating system does not provide such a value, the `dottedQuadFileVersion` property **SHALL** be absent.

EXAMPLE 1: On the Microsoft Windows® platform, this information is available in the `FILEVERSION` member of the `VERSIONINFO` structure.

3.19.15 **releaseDateUtc** Property

A `toolComponent` object **MAY** contain a property named `releaseDateUtc` whose value is a string in the format specified in §3.9, specifying the UTC date (and optionally, the time) of the component's release.

3.19.16 **downloadUri** Property

A `toolComponent` object **MAY** contain a property named `downloadUri` whose value is a localizable string (§3.5.1) containing the absolute URI [RFC3986] from which this version of the tool component can be downloaded.

NOTE: This property is localizable to allow different language versions of a tool to be downloaded from their own URIs.

3.19.17 **informationUri** Property

A `toolComponent` object **MAY** contain a property named `informationUri` whose value is a localizable string (§3.5.1) containing the absolute URI [RFC3986] at which information about this version of the tool component can be found.

NOTE: This property is localizable to allow tool information in different languages to be found at different URIs.

3.19.18 **organization** Property

A `toolComponent` object **MAY** contain a property named `organization` whose value is a localizable string (§3.5.1) containing the name of the company or organization that produced the tool component.

EXAMPLE 1: "organization": "Example Software Corp."

3.19.19 **shortDescription** Property

A `toolComponent` object **MAY** contain a property named `shortDescription` whose value is a localizable `multiformatMessageString` object (§3.12, §3.12.2) containing a brief description of the tool component.

The `shortDescription` property **SHOULD** be a single sentence that is understandable when visible space is limited to a single line of text.

3.19.20 **fullDescription** Property

A `toolComponent` object **MAY** contain a property named `fullDescription` whose value is a localizable `multiformatMessageString` object (§3.12, §3.12.2) containing a comprehensive description of the tool component.

The beginning of `fullDescription` (for example, its first sentence) **SHOULD** provide a concise description of the tool component, suitable for display in cases where available space is limited. Tools that construct `fullDescription` in this way do not need to provide a value for `shortDescription` (§3.19.19). Tools that do not construct `fullDescription` in this way **SHOULD** provide a value for `shortDescription`.

NOTE: The rationale for this guidance is that in the absence of `shortDescription`, a viewer with limited display space might display a truncated version of `fullDescription`, for example, the first sentence (if a sentence is identifiable), the first paragraph, or the first 100 characters. If this guidance is not followed, that truncated description might not be understandable.

3.19.21 **language** Property

Depending on the circumstances, a `toolComponent` object either **SHALL** or **MAY** contain a property named `language` whose value is a string specifying the language of the localizable strings (§3.5.1) contained in the component (except for those in the `translationMetadata` property (§3.19.27)), in a subset of the format specified by the language tags standard [RFC5646]. The subset consists of strings conforming to the syntax

`language value = language code, "-", country code;`

`language code = ? ISO 2-character language name \[[IS0639-1:2002](#IS0639-1;2002)\] ?;`

`country code = ? ISO country code \[[IS03166-1:2013](#IS03166-1;2013)\] ?;`

If this object represents a translation (see §3.19.4), `language` **SHALL** be present; otherwise it **MAY** be present.

If this property is absent, it **SHALL** default to "en-US".

EXAMPLE 1: The language is region-neutral English:

```
"language": "en"
```

EXAMPLE 2: The language is French as spoken in France:

```
"language": "fr-FR"
```

3.19.22 **globalMessageStrings** Property

A `toolComponent` object **MAY** contain a property named `globalMessageStrings` whose value is an object (§3.6) each of whose property values is a localizable `multiformatMessageString` object (§3.12, §3.12.2). The property names correspond to `id` properties (§3.11.10) within message objects (§3.11).

EXAMPLE 1:

```

"driver": {                                # A toolComponent object (§3.19).
  "globalMessageStrings": {
    "call": {                             # A multiFormatMessageString object (§3.12).
      "text": "Function call",
      "markdown": "Function **call**"
    },
    "return": {
      "text": "Function return",
      "markdown": "Function **return**"
    }
  }
}

```

NOTE: The message strings in this property are not associated with a single rule (hence the “global” in the property name).

3.19.23 rules Property

A `toolComponent` object **MAY** contain a property named `rules` whose value is an array of zero or more unique (§3.7.3) `reportingDescriptor` objects (§3.49) each of which provides information about an analysis rule supported by the tool component.

Some tools use the same identifier to refer to multiple distinct (although logically related) rules. Therefore, the `id` properties (§3.49.3) of the `reportingDescriptor` objects do not need to be unique within the array.

EXAMPLE 1: In this example, two distinct but related rules have the same rule id. They are distinguished by their message strings.

```

"driver": {                                # A toolComponent object (§3.19).
  "name": "CodeScanner",
  "rules": [                               # A reportingDescriptor object (§3.49).
    {
      "id": "CA1711",
      "shortDescription": {
        "text": "Certain type name suffixes should not be used."
      },
      "messageStrings": {
        "default": {
          "text": "Rename type name {0} so that it does not end in '{1}'."
        }
      }
    },
    {
      "id": "CA1711",
      "shortDescription": {
        "text": "Certain type name suffixes have preferred alternatives."
      },
      "messageStrings": {
        "default": {
          "text": "Either replace the suffix '{0}' in member name '{1}' with
                  the suggested numeric alternate or provide
                  a more meaningful suffix."
        }
      }
    }
  ]
}

```

3.19.24 notifications Property

A `toolComponent` object **MAY** contain a property named `notifications` whose value is an array of zero or more unique (§3.7.3) `reportingDescriptor` objects (§3.49) each of which provides information about a notification provided by the tool component.

A tool might use the same identifier to refer to multiple distinct (although logically related) notifications. Therefore, the `id` properties (§3.49.3) of the `reportingDescriptor` objects do not need to be unique within the array.

EXAMPLE 1: In this example, two distinct but related notifications have the same id. They are distinguished by their descriptions and message strings.

```

"driver": {                                # A toolComponent object (§3.19).
  "notifications": [
    {                                       # A reportingDescriptor object (§3.49).
      "id": "ERR0001",

```

```

    "level": "error",
    "shortDescription": {
      "text": "A plugin could not be loaded because it does not exist."
    },
    "messageStrings": {
      "default": "Cannot load plugin '{0}' because it was not found."
    }
  },
  {
    "id": "ERR0001",
    "level": "error",
    "shortDescription": {
      "text": "A plugin could not be loaded because it is not signed."
    },
    "messageStrings": {
      "default": "Cannot load plugin '{0}' because it is not signed."
    }
  }
]
}

```

3.19.25 **taxa** Property

A `toolComponent` object **MAY** contain a property named `taxa` whose value is an array of zero or more unique (§3.7.3) `reportingDescriptor` objects (§3.49) each of which provides information about a taxon defined by the component.

If the `toolComponent` describes a standard taxonomy (for example, the Common Weakness Enumeration [CWE]), it **SHALL NOT** contain rules (§3.19.23) or notifications (§3.19.24).

NOTE: Tool components representing standard taxonomies are stored in `run.taxonomies` (§3.14.8), but will typically be persisted to external property files (see §3.15.2).

If the `toolComponent` describes a tool driver or plugin that defines its own custom taxonomy, it **MAY** contain all of rules, notifications, and taxa.

EXAMPLE 1: In this example, a `toolComponent` object represents the Common Weakness Enumeration.

```

{
  # A toolComponent object.
  "name": "CWE",
  "version": "3.2",
  "guid": "11111111-1111-1111-8888-111111111111",
  "releaseDateUtc": "2019-01-03",
  "informationUri": "https://cwe.mitre.org/data/published/cwe_v3.2.pdf/",
  "downloadUri": "https://cwe.mitre.org/data/xml/cwec_v3.2.xml.zip",
  "organization": "MITRE",
  "shortDescription": {
    "text": "The MITRE Common Weakness Enumeration"
  },
  "taxa": [
    {
      "id": "327",
      "name": "BrokenOrRiskyCryptographicAlgorithm",
      "shortDescription": {
        "text": "Use of a broken or risky cryptographic algorithm."
      },
      "defaultConfiguration": {
        "level": "warning"
      }
    },
    ...
  ]
}

```

3.19.26 **supportedTaxonomies** Property

A `toolComponent` object **MAY** contain a property named `supportedTaxonomies` whose value is an array of zero or more unique (§3.7.3) `toolComponentReference` objects (§3.54) each of which refers to a taxonomy (§3.19.3) that the component uses to classify results.

A `toolComponent` object that contains a `supportedTaxonomies` property **SHALL** declare which taxa (if any) each of its rules falls into by providing the `relationships` property (§3.49.15) as appropriate on each `reportingDescriptor` object (§3.49) in its rules array (§3.19.23).

NOTE: A SARIF consumer could infer the set of taxonomies that a component supports by examining the set of `relationships` properties of each element of `toolComponent.rules`. The `supportedTaxonomies`

property is a convenience, intended to enable consumers to see this information at a glance.

If a `toolComponent` supports a custom taxonomy, it **SHOULD** include a reference to itself in `supportedTaxonomies`.

EXAMPLE 1: In this example, a `toolComponent` claims to support the Common Weakness Enumeration [CWE], and also supports a custom taxonomy.

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "guid": "22222222-2222-1111-8888-222222222222",
      "rules": [
        ...
      ],
      "taxa": [
        ...
      ],
      "supportedTaxonomies": [
        {
          "name": "CWE",
          "index": 0,
          "guid": "11111111-1111-1111-8888-111111111111"
        },
        {
          "name": "CodeScanner",
          "guid": "22222222-2222-1111-8888-222222222222"
        }
      ]
    },
    "taxonomies": [
      {
        "name": "CWE",
        "version": "3.2",
        "guid": "11111111-1111-1111-8888-111111111111",
        ...
        "taxa": [
          ...
        ]
      }
    ]
  },
  ...
}
```

3.19.27 `translationMetadata` Property

If a `toolComponent` object represents a translation (§3.19.4), it **SHALL** contain a property named `translationMetadata` whose value is a `translationMetadata` object (§3.26) that contains descriptive information about the translation itself, as opposed to describing the component whose localizable strings (§3.5.1) it translates. Otherwise, `translationMetadata` **SHALL** be absent.

3.19.28 `locations` Property

A `toolComponent` object **MAY** contain a property named `locations` whose value is an array of zero or more unique (§3.7.3) `artifactLocation` objects (§3.4) each of which specifies the location of one of the files comprising this tool component.

3.19.29 `contents` Property

A `toolComponent` object **SHOULD** contain a property named `contents` whose value is an array of zero or more unique (§3.7.3) strings each of which is one of the following values with the specified meanings:

- `"localizedData"`: The component includes localizable strings (§3.5.1) such as rule messages.
- `"nonLocalizedData"`: The component includes non-localizable properties such as rule severity levels.

If `contents` is absent, it **SHALL** default to [`"localizedData"`, `"nonLocalizedData"`].

NOTE: The purpose of this property is to help protect components from misuse. Within a SARIF file, the component types are all stored in their own properties, so there is no danger of mistaking, for example, a translation (stored in `run.translations` (§3.14.9)) for a policy (stored in `run.policies` (§3.14.10)). But components such as

translations and policies are typically authored independently from a tool and stored separately from its log files. The author of a translation (which contains only "localizedData") can help prevent its misuse as a policy (which requires "nonLocalizedData") by setting contents to ["localizedData"].

For example, a user might specify the path to a policy file on a tool's command line. If the specified file does not claim to contain "nonLocalizedData", the tool could conclude that the file does not contain a policy and warn the user.

3.19.30 isComprehensive Property

A toolComponent object **SHOULD** contain a property named isComprehensive whose value is a Boolean that is true if the component contains complete information for the content types specified by contents (§3.19.29) and false otherwise.

If isComprehensive is absent, it **SHALL** default to false.

NOTE: This property is useful because tools are permitted to emit rules (§3.19.23), notifications (§3.19.24), or taxa (§3.19.25) properties that contain only those items relevant to the current run. For example, a tool might define hundreds of rules, but if a scan detects violations of only two of them, then the rules property (if it is present at all, which it does not need to be) need only contain metadata for those two rules.

So, for example, the author of a translation (§3.19.4) would want to work from a log file whose contents array includes "localizedData" and whose isComprehensive property is set to true. Similarly, the author of a policy (§3.19.5) would want to work from a log file whose contents array contains "nonLocalizedData" and whose isComprehensive property is set to true.

3.19.31 localizedDataSemanticVersion Property

If a toolComponent object represents a translation (§3.19.4), it **SHOULD** contain a property named localizedDataSemanticVersion whose value is a string that specifies the semantic version [SEMVER] of the translated strings. Otherwise, localizedDataSemanticVersion **MAY** be present, in which case it represents the semantic version of the localizable strings (§3.5.1) that are present in this component.

If localizedDataSemanticVersion is absent, it **SHALL** default to thisObject.semanticVersion (§3.19.12).

NOTE 1: See the description of minimumRequiredLocalizedDataSemanticVersion (§3.19.32) for an explanation of how these two properties interact.

NOTE 2: In a translation, localizedDataSemanticVersion will usually be the same as semanticVersion. They will differ only if it is necessary to revise the translation component to correct an error unrelated to the translated strings, for example, an error in its translationMetadata (§3.19.27). In that case, semanticVersion would be incremented but localizedDataSemanticVersion would not.

3.19.32 minimumRequiredLocalizedDataSemanticVersion Property

If a toolComponent object does not represent a translation (§3.19.4), it **SHOULD** contain a property named minimumRequiredLocalizedDataSemanticVersion whose value is a string that specifies the minimum semantic version [SEMVER] of the translated strings that it requires. Otherwise, minimumRequiredLocalizedDataSemanticVersion **SHALL** be absent.

If minimumRequiredLocalizedDataSemanticVersion is absent, it **SHALL** default to thisObject.semanticVersion (§3.19.12).

When a SARIF consumer is seeking a translation for this object, it **SHALL** only accept one whose `localizedDataSemanticVersion` (§3.19.31) is greater than or equal to (in the SEMVER sense) but has the same major version component as `thisObject.minimumRequiredLocalizedDataSemanticVersion`.

NOTE: `minimumRequiredLocalizedDataSemanticVersion` can differ from `semanticVersion` for two reasons. First, successive versions of a translated component (even versions whose minor version component is incremented) might be able to use the same set of translated strings. Second, the translation itself might be versioned if, for example, the translation author discovers a typo or decides to clarify a message string.

EXAMPLE 1: In this example, the tool is at version 3.3, but it only requires strings at version 3.1, because tool versions 3.2 and 3.3 didn't affect any user-facing localizable strings. Therefore, the translation at index 0 in `theRun.translations` (§3.14.9) is acceptable.

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "semanticVersion": "3.3",
      "minimumRequiredLocalizedDataSemanticVersion": "3.1",
      ...
    },
    ...
  },
  "translations": [
    {
      "language": "fr-FR",
      "localizedDataSemanticVersion": "3.1.2",
      ...
    },
    ...
  ],
  ...
}
```

3.19.33 `associatedComponent` Property

If this `toolComponent` object represents a plugin (see §3.18.1), a taxonomy (§3.19.3), a translation (§3.19.4), or a policy (§3.19.5), it **MAY** contain a property named `associatedComponent` whose value is a `toolComponentReference` object (§3.54) which identifies the component (either `theTool.driver` (§3.18.2) or an element of `theTool.extensions` (§3.18.3)) to which this plugin, translation, or policy applies. If `associatedComponent` is absent, it **SHALL** default to a reference to `theTool.driver`.

NOTE: The scenario for a taxonomy component to have an `associatedComponent` property is when a party other than the tool vendor defines a custom taxonomy to categorize the rules defined by a specific tool. In this case, `associatedComponent` would specify the tool's driver. A custom taxonomy defined by the tool vendor would be defined in the `taxa` property (§3.19.25) of the driver itself, so `associatedComponent` would not be necessary.

The associated `toolComponent` object **MAY** itself contain an `associatedComponent` property; for example, a translation might be associated with a plugin which in turn is associated with the driver (see §3.18.1).

3.20 `invocation` Object

3.20.1 General

An `invocation` object describes the invocation of the analysis tool that was run.

3.20.2 `commandLine` Property

An `invocation` object **MAY** contain a property named `commandLine` whose value is a string containing the completely specified command line used to invoke the tool, starting with the name of the tool's executable or script file, optionally qualified by the relative or absolute path to the file.

NOTE 1: The information in the `commandLine` property helps to precisely repeat a run of an analysis tool, and to verify that the results reported in the log file were generated by an appropriate invocation of the tool.

The `commandLine` property is redactable (§3.5.2) because it might contain information which it is not appropriate to disclose, such as passwords, tokens, database connection strings, or in some circumstances even the fully qualified path to the tool's executable or script file.

NOTE 2: Redacting sensitive information from `commandLine` makes it more difficult to precisely reproduce an analysis run. The value of `commandLine` would have to be combined with information from another source to allow the run to be repeated.

EXAMPLE 1: Suppose a tool is invoked with the command line

```
C:\Users\mary\Tools\DbScanner.exe /ConnectionString
"Server=Corp;Db=Accounting;User=Admin;Password=S3cr#t"
/input *.sql
```

Then `commandLine` might contain the redacted string

```
[REDACTED]\DbScanner.exe /connectionString=[REDACTED] /input=*.sql
```

The `commandLine` property might describe a command that would be harmful if it were executed. For this reason, a SARIF consumer that receives a SARIF log file from an untrusted source **SHOULD NOT** execute the command line without first examining it carefully. In particular, an automated SARIF consumer **SHALL NOT** execute a command line in a SARIF log file from an untrusted source.

EXAMPLE 2: An example of a harmful command line:

```
{                                     # An invocation object
  "commandLine": "rm -rf /"
}
```

3.20.3 arguments property

An `invocation` object **MAY** contain a property named `arguments` whose value is either `null` or an array of zero or more strings, containing in order the command line arguments passed to the tool from the operating system.

If `arguments` is absent, it **SHALL** default to `null`.

An empty array **SHALL** mean that the tool was invoked with no command line arguments. `null` **SHALL** mean that the command line arguments, if any, are not known.

EXAMPLE 1: If the tool is implemented as a C# or Java program, `arguments` would contain the contents of the `args` array passed to the entry point method.

NOTE: Although the `commandLine` property (§3.20.2) contains the same information, parsing it is error prone even if one understands the command shell's quoting and escaping conventions. SARIF consumers might find the pre-parsed `arguments` property easier to use.

3.20.4 responseFiles Property

An `invocation` object **MAY** contain a property named `responseFiles` whose value is either `null` or an array of zero or more unique (§3.7.3) `artifactLocation` objects (§3.4) each of which represents a response file specified on the tool's command line.

If `responseFiles` is absent, it **SHALL** default to `null`.

An empty array **SHALL** mean that the tool was invoked with no command line arguments that specified response files. `null` **SHALL** mean that it is not known whether any command line arguments specified a response file.

A SARIF producer **MAY** embed the contents of a response file in the SARIF log file by mentioning the response file in `theRun.artifacts` (§3.14.15) and providing a value for `artifact.contents` (§3.24.8).

EXAMPLE 1:

```
{
  # An invocation object.
  "commandLine": "/quiet @analyzer.rsp @strict.rsp @options.rsp",
  "responseFiles": [
    {
      # An artifactLocation object (§3.4).
      "uri": "analyzer.rsp",
      "uriBaseId": "RESPONSEFILEDIR"
    },
    {
      "uri": "strict.rsp",
      "uriBaseId": "RESPONSEFILEDIR"
    },
    {
      "uri": "options.rsp",
      "uriBaseId": "RESPONSEFILEDIR"
    }
  ],
  ...
}
```

3.20.5 `ruleConfigurationOverrides` Property

An invocation object **MAY** contain a property named `ruleConfigurationOverrides` whose value is an array of zero or more unique (§3.7.3) `configurationOverride` objects (§3.51) each of which overrides the `defaultConfiguration` property (§3.49.14) of a `reportingDescriptor` object (§3.48.7) that describes a rule (that is, a `reportingDescriptor` object that is an array element of the `rules` property (§3.19.23) of some `toolComponent` object (§3.19)).

3.20.6 `notificationConfigurationOverrides` Property

An invocation object **MAY** contain a property named `notificationConfigurationOverrides` whose value is an array of zero or more unique (§3.7.3) `configurationOverride` objects (§3.51) each of which overrides the `defaultConfiguration` property (§3.49.14) of a `reportingDescriptor` object (§3.49) that describes a notification (that is, a `reportingDescriptor` object that is an array element of the `notifications` property (§3.19.24) of some `toolComponent` object (§3.19)).

3.20.7 `startTimeUtc` Property

An invocation object **MAY** contain a property named `startTimeUtc` whose value is a string in the format specified in §3.9, specifying the UTC date and time at which the invocation started.

3.20.8 `endTimeUtc` Property

An invocation object **MAY** contain a property named `endTimeUtc` whose value is a string in the format specified in §3.9, specifying the UTC date and time at which the invocation ended.

3.20.9 `exitCode` Property

If the SARIF producer process did not exit due to a signal, an invocation object **SHOULD** contain a property named `exitCode` whose value is an integer specifying the process exit code.

If the SARIF producer process exited due to a signal, the `exitCode` property **SHALL** be absent.

For examples, see §3.20.10.

3.20.10 `exitCodeDescription` Property

If the SARIF producer process did not exit due to a signal, an `invocation` object **MAY** contain a property named `exitCodeDescription` whose value is a string describing the reason for the process exit.

EXAMPLE 1:

```
{
  # An invocation object
  "exitCode": 0,
  "exitCodeDescription": "Normal successful completion"
}
```

EXAMPLE 2:

```
{
  # An invocation object
  "exitCode": 2,
  "exitCodeDescription": "File not found"
}
```

3.20.11 `exitSignalName` Property

If the SARIF producer process exited due to a signal, an `invocation` object **SHOULD** contain a property named `exitSignalName` whose value is a string containing the name of the signal that caused the process to exit.

If the SARIF producer process did not exit due to a signal, the `exitSignalName` property **SHALL** be absent.

For an example, see §3.20.12.

3.20.12 `exitSignalNumber` Property

If the SARIF producer process exited due to a signal, an `invocation` object **MAY** contain a property named `exitSignalNumber` whose value is an integer specifying the numeric value of the signal that caused the process to exit.

If the SARIF producer process did not exit due to a signal, the `exitSignalNumber` property **SHALL** be absent.

EXAMPLE 1:

```
{
  # An invocation object
  "exitSignalNumber": 3,
  "exitSignalName": "SIGQUIT"
}
```

3.20.13 `processStartFailureMessage` Property

If the analysis tool process failed to start, an `invocation` object **MAY** contain a property named `processStartFailureMessage` whose value is a string containing the operating system's message describing the failure.

NOTE: In this case, the SARIF file would not be produced by the analysis tool (since it failed to start), but rather by some other component of the user's engineering system which is responsible for monitoring the operation of the analysis tool.

If the analysis tool process started successfully (regardless of whether or how it subsequently failed), the `processStartFailureMessage` property **SHALL** be absent.

EXAMPLE 1:

```
{
  # An invocation object
  "processStartFailureMessage": "WebScan.exe is not recognized as a command."
}
```

3.20.14 **executionSuccessful** Property

An invocation object **SHALL** contain a property named `executionSuccessful` whose value is a Boolean that is `true` if the engineering system that started the process knows that the analysis tool succeeded, and `false` if the engineering system knows that the tool failed.

NOTE: This property is needed because not all programs exit with an exit code of 0 on success and non-0 on failure.

EXAMPLE 1:

```
{
  "exitCode": 1,
  "exitCodeDescription": "Scan successful; warnings detected.",
  "executionSuccessful": true
}
```

3.20.15 **machine** Property

An invocation object **MAY** contain a property named `machine` whose value is a redactable (§3.5.2) string containing the name of the machine on which the invocation occurred.

3.20.16 **account** Property

An invocation object **MAY** contain a property named `account` whose value is a redactable (§3.5.2) string containing the name of the account under which the invocation occurred.

3.20.17 **processId** Property

An invocation object **MAY** contain a property named `processId` whose value is an integer containing the id of the process in which the invocation occurred.

3.20.18 **executableLocation** Property

An invocation object **MAY** contain a property named `executableLocation` whose value is an `artifactLocation` object (§3.4) specifying the location of the primary executable file for the program or script that was invoked.

NOTE 1: This property is defined in the `invocation` object rather than in the `toolComponent` object (§3.19) because the identical tool might be invoked from different paths on different machines.

NOTE 2: This property might duplicate information in the `commandLine` property (§3.20.2). It is necessary because the command line might not explicitly specify the path to the tool (for example, if the tool directory is on the execution path), and this information is important for troubleshooting.

NOTE 3: Absolute path names can reveal information that might be sensitive.

3.20.19 **workingDirectory** Property

An invocation object **MAY** contain a property named `workingDirectory` whose value is an `artifactLocation` object (§3.4) specifying the fully qualified path name of the process's working directory (a directory that the operating system associates with the process, with respect to which the operating system interprets relative file paths).

NOTE: Absolute path names can reveal information that might be sensitive.

3.20.20 `environmentVariables` Property

An invocation object **MAY** contain a property named `environmentVariables` whose value is an object. The property names in this object **SHALL** contain the names of all the environment variables in the tool's execution environment. The value of each property **SHALL** be a string containing the value of the specified environment variable. If the value of the environment variable is an empty string, the corresponding property value **SHALL** be an empty string.

NOTE 1: Environment variables might be useful to include in a log file because they might affect the tool's analysis output, for example, by specifying the location of a directory containing plugins (see §3.18.1). However, environment variable names and values are likely to reveal highly sensitive information. For example, on a machine running Microsoft Windows®, environment variables reveal the directories on the execution path, user account name, machine name, logon domain controller, *etc.*

NOTE 2: The result of setting an environment variable to an empty string is operating system dependent. On Microsoft Windows®, it removes the variable from the environment. In UNIX®, an environment variable can have an empty value.

Both the property names and their values are redactable (§3.5.2). A distinct redaction token (§3.14.28) **SHALL** be used for each redacted property name.

NOTE 3: This is necessary to prevent the creation of an object with identical property names, which is invalid in the JSON serialization.

3.20.21 `toolExecutionNotifications` Property

An invocation object **MAY** contain a property named `toolExecutionNotifications` whose value is an array of zero or more `notification` objects (§3.58). Each element of the array represents a runtime condition detected by the invoked process, either by the tool's driver or by one of its extensions. The presence within this array of any `notification` object whose `level` property (§3.58.6) is "error" **SHALL** mean that the run failed. A SARIF consumer **SHALL NOT** assume that a failed run contains a complete set of analysis results.

NOTE: This is important in compliance scenarios, where, for example, a corporate policy might require that a project's entire code base be analyzed with a specified set of rules.

The information in `toolExecutionNotifications` is primarily intended for the developers of the analysis tool, to aid them in diagnosing bugs in the tool. This contrasts with the information in `results`, which is intended for the developers of the code being analyzed. However, viewers **MAY** still present tool notifications to users, so users are aware of any tool problems. At a minimum, viewers **SHOULD** make users aware of tool notifications whose `level` property is "error".

NOTE: Depending on the nature of the error, a tool that encounters a runtime error might or might not be able to continue running.

If the error occurs in the course of evaluating a rule, the tool might report the error in `toolExecutionNotifications`, disable the rule, and continue to execute the remaining rules.

If the error occurs outside of the evaluation of a rule, the tool might report the error in `toolExecutionNotifications` and then halt. If the tool exits abnormally, it might not have the opportunity to report the error. But if the tool is running under the control of an orchestration process that can detect the error, that process might add a notification for the error to the log file, or even synthesize a log file to hold the error, if the tool did not have the opportunity to create one.

3.20.22 `toolConfigurationNotifications` Property

An invocation object **MAY** contain a property named `toolConfigurationNotifications` whose value is an array of zero or more `notification` objects (§3.58). Each element of the array represents a condition relevant to the

configuration of the tool's driver or one of its extensions. The presence within this array of any notification object whose `level` property (§3.58.6) is "error" **SHALL** mean that the run failed.

The information in `toolConfigurationNotifications` is primarily intended for the engineers who configure the analysis tool, to aid them in diagnosing errors in the configuration. This contrasts with the information in `results`, which is intended for the developers of the code being analyzed. However, viewers **MAY** still present configuration notifications to users, so users are aware of any configuration problems. At a minimum, viewers **SHOULD** make users aware of configuration notifications whose `level` property is "error".

NOTE: Many tools can be parameterized with information about which rules to run, and how those rules should be configured. In some cases, if the configuration information is invalid, the tool can ignore the invalid information and continue to run.

EXAMPLE 1: A tool is invoked with a configuration file which specifies that the tool should disable rule ABC0001, but there is no rule whose id is ABC0001. The tool reports the problem in `toolConfigurationNotifications`. The tool might continue to run, reporting results for the rules that are correctly configured.

```
"toolConfigurationNotifications": [
  {
    "descriptor": {
      "id": "UnknownRule"
    },
    "associatedRule": {
      "ruleId": "ABC0001"
    },
    "level": "warning",
    "message": {
      "text": "Could not disable rule \"ABC0001\"
              because there is no rule with that id."
    }
  }
]
```

EXAMPLE 2: A tool is invoked with an unknown command-line argument. The tool reports the problem in `toolConfigurationNotifications`. The tool might report the problem as a warning and continue to run, or it might report the problem as an error and terminate.

```
"toolConfigurationNotifications": [
  {
    "descriptor": {
      "id": "UnknownCommandLineArgument"
    },
    "level": "error",
    "message": {
      "text": "Command line argument \"/X\" is unknown."
    }
  }
]
```

EXAMPLE 3: A tool is invoked with a command-line argument that specifies the name of a directory containing files to analyze, but the user who invoked the tool does not have read access to that directory. The tool reports the problem as an error in `toolConfigurationNotifications` and then terminates.

```
"toolConfigurationNotifications": [
  {
    "descriptor": {
      "id": "AccessDenied"
    },
    "level": "error",
    "message": {
      "text": "Cannot read from directory \"C:\\code\"."
    }
  }
]
```

3.20.23 `stdin`, `stdout`, `stderr`, and `stdoutStderr` Properties

An invocation object **MAY** contain any or all of the properties `stdin`, `stdout`, `stderr`, and `stdoutStderr`, whose values are `artifactLocation` objects (§3.4) referring to files that contain the input to and output from the SARIF producer process. `stdin`, `stdout`, and `stderr` refer, respectively, to files containing the contents of the standard input, standard output, and standard error streams. `stdoutStderr` refers to a file containing the interleaved contents of the standard output and standard error streams. This is useful when the output of those two streams was written to the

same file by means of command shell redirection syntax such as "`> output.txt 2>&1`".

A SARIF producer **MAY** embed the stream contents in the log file by mentioning the corresponding file in `theRun.artifacts` (§3.14.15) and providing a value for `artifact.contents` (§3.24.8).

3.21 attachment Object

3.21.1 General

An attachment object describes an artifact relevant to the detection of a result (see §3.27.26).

A SARIF producer **MAY** embed the contents of an attachment in the log file by mentioning the attachment in `theRun.artifacts` (§3.14.15) and providing a value for `artifact.contents` (§3.24.8).

EXAMPLE 1: In this example, `image001.png` is a screen shot of the program being analyzed at the point where the result was detected. Note that this example is more appropriate to a dynamic analysis tool than to a static analysis tool.

```
{
  ...                                     # A result object (§3.27).
  "attachments": [                       # See §3.27.26.
    {                                     # An attachment object.
      "description": {                  # See §3.21.2.
        "text": "Screen shot"
      },
      "location": {                     # See §3.21.3.
        "uri": "file:///C:/ScanOutput/image001.png"
      }
    }
  ]
}
```

3.21.2 description Property

An attachment object **SHOULD** contain a property named `description` whose value is a message object (§3.11) describing the role played by the attachment.

3.21.3 location Property

An attachment object **SHALL** contain a property named `location` whose value is an `artifactLocation` object (§3.4) that specifies the location of the attachment.

3.21.4 regions Property

An attachment object **MAY** contain a property named `regions` whose value is an array of zero or more unique (§3.7.3) `region` objects (§3.30) each of which **SHALL** specify a region of interest within the attachment, and **SHOULD** contain a message property (§3.30.14) so a user can understand its relevance.

3.21.5 rectangles Property

An attachment object **MAY** contain a property named `rectangles` whose value is an array of zero or more unique (§3.7.3) `rectangle` objects (§3.31). If the attachment is an image (for example `.png` or `.svg`), each `rectangle` object **SHALL** specify an area of interest within the image, and **SHOULD** contain a message property (§3.31.3) so a user can understand its relevance.

If the attachment is not an image, and `rectangles` is present, its value **SHALL** be an empty array.

3.22 conversion Object

3.22.1 General

A `conversion` object describes how a converter transformed the output of an analysis tool from the analysis tool's native output format into the SARIF format.

EXAMPLE 1: In this example, a converter has converted an AndroidStudio output file into a SARIF log file:

```
{
  ...
  "runs": [
    {
      "tool": {
        "driver": {
          "name": "AndroidStudio"
        }
      },
      "conversion": {
        "tool": {
          "driver": {
            "name": "SARIF SDK Multitool"
          }
        },
        "invocation":
          "Sarif.Multitool.exe convert -t AndroidStudio northwind.log",
        "analysisToolLogFileLocation": {
          "uri": "northwind.log",
          "uriBaseId": "$LOG_DIR$"
        }
      },
      "results": [
        ...
      ]
    }
  ]
}
```

3.22.2 tool Property

A `conversion` object **SHALL** contain a property named `tool` whose value is a `tool` object (§3.18) that describes the converter.

3.22.3 invocation Property

A `conversion` object **MAY** contain a property named `invocation` whose value is an `invocation` object (§3.20) that describes the invocation of the converter.

3.22.4 analysisToolLogFiles Property

Some analysis tools produce one or more output files that describe the analysis run as a whole; we refer to these as “per-run” files. Some tools produce one or more output files for each result; we refer to these as “per-result” files. Some tools produce both per-run and per-result files.

A `conversion` object **MAY** contain a property named `analysisToolLogFiles` whose value is an array of zero or more unique (§3.7.3) `artifactLocation` objects (§3.4) that specify the locations of the per-run files.

If the analysis tool did not produce any per-run files, and `analysisToolLogFiles` is present, its value **SHALL** be an empty array.

Per-result files are handled by the `resultProvenance.conversionSources` property (§3.48.7).

3.23 **versionControlDetails** Object

3.23.1 General

A `versionControlDetails` object specifies the information necessary to retrieve from a version control system (VCS) the correct revision of the files that were scanned during the run.

For an example, see §3.14.13.

3.23.2 Constraints

A `versionControlDetails` object **SHOULD** contain sufficient information to uniquely and permanently identify the revision of the files that were scanned.

NOTE: The required set of properties depends on the VCS and on the engineering system within which it is used. Consider Git as an example. The `revisionId` property (containing a commit id) would suffice. The `branch` property (§3.23.5) might not suffice because a Git branch is a pointer to the latest commit along a line of development; however, `branch` together with `asOfTimeUtc` (§3.23.7) might suffice (although that is not an idiomatic use of Git). Similarly, `revisionTag` (§3.23.6) might not suffice because a Git tag can be removed, but if the engineering system guaranteed that certain tags (such as those specifying public releases) were stable, then `revisionTag` might suffice.

3.23.3 **repositoryUri** Property

A `versionControlDetails` object **SHALL** contain a property named `repositoryUri` whose value is a string containing an absolute URI [RFC3986] that specifies the location of the repository containing the scanned files.

3.23.4 **revisionId** Property

A `versionControlDetails` object **SHOULD** contain a property named `revisionId` whose value is a redactable (§3.5.2) string that uniquely and permanently identifies the appropriate revision of the scanned files.

3.23.5 **branch** Property

A `versionControlDetails` object **MAY** contain a property named `branch` whose value is a redactable (§3.5.2) string containing the name of a branch containing the correct revision of the scanned files.

3.23.6 **revisionTag** Property

A `versionControlDetails` object **MAY** contain a property named `revisionTag` whose value is a redactable (§3.5.2) string containing a tag that has been applied to the revision in the VCS.

NOTE 1: This document refers to an identifier for a revision in a VCS as a “tag”. Different VCSs use different terms; for example, Visual Studio Team Services Version Control calls it a “label”.

NOTE 2: Although VCSs generally allow a revision to have more than one tag, the `revisionTag` property is not an array. The purpose of `revisionTag` is to aid in identifying a revision so that a scan can be reproduced, not to exhaustively describe the revision.

3.23.7 asOfTimeUtc Property

A `versionControlDetails` object **MAY** contain a property named `asOfTimeUtc` whose value is a string in the format specified in §3.9, specifying a UTC date and time that can be used to synchronize an enlistment to the state of the repository as of that time.

NOTE: In some VCSs, the “synchronize by date” feature requires the time to be expressed in the server’s time zone. In such a case, the SARIF producer would need to know the server’s time zone to correctly populate `asOfTimeUtc`.

3.23.8 mappedTo Property

A `versionControlDetails` object **MAY** contain a property named `mappedTo` whose value is an `artifactLocation` object (§3.4) that specifies the location in the local file system to which the root of the repository was mapped at the time of the analysis.

This property makes it possible to map any `artifactLocation` to the repository, if any, to which the file belongs. The mapping algorithm **SHALL** be as follows, or any algorithm with the same result (a clarifying example follows):

1. Resolve the `artifactLocation` as far as possible using the procedure specified in §3.14.14. Denote the resolved `artifactLocation` by `a`.
2. For every `versionControlDetails` object `vcd` in `theRun.versionControlProvenance` (§3.14.13), resolve the `artifactLocation` object specified by `vcd.mappedTo`, again using the procedure specified in §3.14.14. Denote each such resolved `artifactLocation` object by `v`.
3. Let `S` be the set of all `versionControlDetails` objects `vcd` for which `v.uriBaseId` equals `a.uriBaseId` and `v.uri` is a prefix of `a.uri`.
4. If `S` is the empty set, then the file specified by `artifactLocation` does not belong to any repository.
5. Otherwise, the file specified by `artifactLocation` belongs to the repository specified by the member of `S` with the longest `v.uri`.

EXAMPLE 1: This example illustrates the mapping algorithm. Consider this SARIF file:

```
{
  "originalUriBaseIds": {
    "HOME": {
      "uri": "file:///home/user/"
    },
    "PACKAGE_ROOT": {
      "uri": "package/",
      "uriBaseId": "HOME"
    }
  },
  "versionControlProvenance": [
    {
      "repositoryUri": "https://github.com/example-corp/package",
      "revisionId": "b87c4e9",
      "mappedTo": {
        "uriBaseId": "PACKAGE_ROOT"
      }
    },
    {
      "repositoryUri": "https://github.com/example-corp/plugin1",
      "revisionId": "cafdac7",
      "mappedTo": {
        "uriBaseId": "PACKAGE_ROOT",
        "uri": "plugin1"
      }
    },
    {
      "repositoryUri": "https://github.com/example-corp/plugin2",
      "revisionId": "d0dc2c0",
      "mappedTo": {
        "uriBaseId": "PACKAGE_ROOT",
        "uri": "plugin2"
      }
    }
  ],
  "results": [
    {
      "ruleId": "CA1000",
      "locations": [
```

```

{
  "physicalLocation": {
    "artifactLocation": {
      "uri": "plugin1/x.c",
      "uriBaseId": "PACKAGE_ROOT"
    }
  }
}
]
}

```

The object is to determine to which repository, if any, the file `plugin1/x.c` specified by the result location belongs. The algorithm proceeds as follows, using a simplified notation (*uriBaseId*, *uri*) to denote an `artifactLocation`:

1. Use the information in `originalUriBaseIds` and the procedure specified in §3.14.14 to calculate the “resolved artifact location” *a*:
 $(\text{PACKAGE_ROOT}, \text{plugin1/x.c}) \rightarrow (\text{HOME}, \text{package/plugin1/x.c}) \rightarrow (\text{null}, \text{file:///home/user/package/plugin1/x.c}).$
2. In the same way, calculate the resolved artifact location *v* from the `mappedTo` property of each element *vcd* of the `versionControlProvenance` array:
 - $(\text{PACKAGE_ROOT}, \text{null}) \rightarrow (\text{HOME}, \text{package}) \rightarrow (\text{null}, \text{file:///home/user/package})$
 - $(\text{PACKAGE_ROOT}, \text{plugin1}) \rightarrow (\text{HOME}, \text{package/plugin1}) \rightarrow (\text{null}, \text{file:///home/user/package/plugin1})$
 - $(\text{PACKAGE_ROOT}, \text{plugin2}) \rightarrow (\text{HOME}, \text{package/plugin2}) \rightarrow (\text{null}, \text{file:///home/user/package/plugin2})$
3. The set of *vcd* for which *v.uriBaseId* equals *a.uriBaseId* (which is `null`) and for which *v.uri* is a *prefix* of *a.uri* (which is `file:///home/user/package/plugin1/x.c`) contains the objects at indices 0 and 1. It does not contain the object at index 2 because `file:///home/user/package/plugin2` is not a prefix of `file:///home/user/package/plugin1/x.c`.
4. The set is not empty (it contains indices 0 and 1).
5. The member of the set for with the longest *v.uri* is the object at index 1, because `file:///home/user/package/plugin1` is longer than `file:///home/user/package`.

Therefore, the specified file belongs to the repository specified by the `versionControlDetails` object at index 1, namely `https://github.com/example-corp/plugin1`.

3.24 artifact Object

3.24.1 General

An `artifact` object represents a single artifact.

3.24.2 ‘location’ property

Depending on the circumstances, an `artifact` object either **SHALL**, **MAY**, or **SHALL NOT** contain a property named `location` whose value is an `artifactLocation` object (§3.4).

If the `artifact` object represents a top-level artifact, then `location` **SHALL** be present.

If the `artifact` object represents a nested artifact whose location relative to the root of its parent can be expressed only by means of a path, then `location` **SHALL** be present, and the value of its `uri` property **SHALL** be a relative reference [RFC3986] beginning with `" / "` expressing that path.

If the `artifact` object represents a nested artifact whose location within its parent can be expressed only by a byte offset from the start of the parent, and not by means of a path, then `location` **SHALL NOT** be present.

If the `artifact` object represents a nested artifact whose location within its parent can be expressed either by means of a path or by means of a byte offset from the start of the parent, then `location` **MAY** be present; if it is absent, then `offset` (§3.24.4) **SHALL** be present. If `location` is present, the value of its `uri` property **SHALL** be a relative reference expressing the path of the nested artifact within the parent.

For an example, see §3.24.3.

3.24.3 `parentIndex` Property

If this `artifact` object represents a nested artifact, then it **SHALL** contain a property named `parentIndex` whose value is the array index (§3.7.4) of the parent artifact's `artifact` object within `theRun.artifacts` (§3.14.15).

If this `artifact` object represents a top-level artifact, then `parentIndex` **SHALL** be absent.

NOTE: `parentIndex` makes it possible to navigate from the `artifact` object representing a nested artifact to the `artifact` objects representing each of its parent artifacts in turn, up to the top-level artifact.

EXAMPLE 1: This example demonstrates two levels of artifact nesting. The top-level artifact is a ZIP archive represented by the `artifact` object at index 0 in the `artifacts` array. The archive contains a word processing document at the specified absolute path from its root; the document is represented by the `artifact` object at index 1. Finally, the document contains an embedded media object of the specified length at the specified offset from its beginning; the media object is represented by the `artifact` object at index 2. The media object's `parentIndex` property refers to its parent document; the document's `parentIndex` property refers to its parent ZIP archive, and the ZIP archive does not have a `parentIndex` property.

```
"artifacts": [
  {
    "location": {
      "uri": "file:///C:/Code/app.zip"
    },
    "mimeType": "application/zip"
  },
  {
    "location": {
      "uri": "/docs/intro.docx"
    },
    "mimeType":
      "application/vnd.openxmlformats-officedocument.wordprocessingml.document",
    "parentIndex": 0
  },
  {
    "offset": 17522,
    "length": 4050,
    "mimeType": "application/x-contoso-animation",
    "parentIndex": 1
  }
]
```

3.24.4 `offset` Property

Depending on the circumstances, an `artifact` object either **SHALL**, **MAY**, or **SHALL NOT** contain a property named `offset` whose value is a non-negative integer.

If the `artifact` object represents a top-level artifact, then `offset` **SHALL NOT** be present.

If the `artifact` object represents a nested artifact whose location relative to its parent can be expressed only by means of a byte offset from the start of its parent artifact, then `offset` **SHALL** be present, and its value **SHALL** be that byte offset.

If the `artifact` object represents a nested artifact whose location within its parent can only be expressed by means of a path, and not by means of a byte offset from the start of the parent, then `offset` **SHALL NOT** be present.

If the `artifact` object represents a nested artifact whose location within its parent can be expressed either by means of a path or by means of a byte offset from the start of the parent, then `offset` **MAY** be present; if it is absent, then `location` (§3.24.2) **SHALL** be present. If `offset` is present, its value **SHALL** be that byte offset.

3.24.5 **length** Property

An artifact object **MAY** contain a property named `length` whose value is a non-negative integer specifying the length of the artifact in bytes.

If `length` is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set).

3.24.6 **roles** Property

An artifact object **MAY** contain a property named `roles` whose value is an array of zero or more unique (§3.7.3) strings each of which specifies a role that this artifact played in the analysis.

Each array element **SHALL** have one of the following values, with the specified meanings:

- `"analysisTarget"`: The analysis tool was instructed to scan this artifact.
- `"attachment"`: The artifact is an attachment mentioned in `result.attachments` (§3.27.26).
- `"conversionSource"`: The artifact is an output from an analysis tool in a non-SARIF format that was converted to SARIF.
- `"debugOutputFile"`: The artifact contains debug output from the tool.
- `"directory"`: The artifact is a directory (a container for other files and directories) rather than a file.

NOTE 1: URIs do not represent “directories” in the file system sense. Even if the URI `https://www.example.com/dir/` addresses a resource, the URI `https://www.example.com/dir` might also address a resource. Nonetheless, if the analysis tool knows that `https://www.example.com/dir` is not itself a resource, but only a prefix for other URIs that *are* resources, it is appropriate for the tool to mark `https://www.example.com/dir` with the `"directory"` role.

- `"driver"`: The file belongs to the analysis tool’s driver (§3.18.2).
- `"extension"`: The file belongs to one of the analysis tool’s extensions (§3.18.3).
- `"externalPropertyFile"`: The artifact is an external property file (§4).
- `"memoryContents"`: The artifact contains the contents of a portion of memory.
- `"policy"`: The file belongs to a policy (§3.19.5).
- `"referencedOnCommandLine"`: The artifact was referenced on the command line.
- `"repositoryRoot"`: The artifact is the root directory of a source control repository containing files that were analyzed

NOTE 2: A single run might analyze files from multiple repositories.

- `"responseFile"`: The artifact contains command line arguments to a program, as specified in `invocation.response` (§3.20.4).
- `"resultFile"`: A result was detected in this artifact (which the analysis tool was not explicitly instructed to scan).
- `"scannedFile"`: An “indirect” artifact (not directly requested for scanning, but scanned as a result of analyzing another “linked” artifact) in which no result was found.

NOTE 3: For example, a scanner might be configured to analyze a C source file and find a result in a header file that it includes. The header file may be marked with the `"resultFile"` role. The C file should be marked with the `"analysisTarget"` role, however, as it was explicitly configured as a scan target.

- `"standardStream"`: The artifact contains the contents of one of the standard input or output streams, as specified in `invocation.stdin`, `invocation.stdout`, `invocation.stderr`, or `invocation.stdoutStderr` (§3.20.23).

- "taxonomy": The file belongs to a taxonomy (§3.19.3).
- "toolSpecifiedConfiguration": The artifact is a configuration file provided by the tool.
- "tracedFile": The analysis tool traced through this artifact while executing or simulating the execution of the code under test.
- "translation": The file belongs to a translation (§3.19.4).
- "userSpecifiedConfiguration": The artifact is a configuration file provided by the user.

The following role values denote artifacts that have changed since some previous time which we refer to as the "baseline time."

A SARIF producer **MAY** determine the baseline time in any way. (For example, if `theRun.baselineGuid` (§3.14.5) is present, the tool might use its start time as the baseline time. Alternatively, the tool might use version control information, such as the time of some commit before the one being analyzed.)

- "added": The artifact was added after the baseline time.
- "deleted": The artifact was deleted after the baseline time.
- "modified": The artifact was modified after the baseline time.
- "renamed": The artifact was renamed after the baseline time. In this case, the `artifact` object specifies the new name.
- "uncontrolled": The artifact is not under version control.
- "unmodified": The artifact has not been modified since the baseline time.

NOTE 4: The information conveyed by these values could be extracted from a VCS. These properties exist so SARIF consumers can have this information without needing access to the VCS.

3.24.7 mimeType Property

An `artifact` object **MAY** contain a property named `mimeType` whose value is a string that specifies the artifact's MIME type [RFC2045]. For information about the use of `mimeType` by SARIF viewers, see Appendix C.

3.24.8 contents Property

An `artifact` object **MAY** contain a property named `contents` whose value is an `artifactContent` object (§3.3) representing the entire contents of the artifact.

3.24.9 encoding Property

If an `artifact` object represents a text artifact, it **MAY** contain a property named `encoding` whose value is a case-sensitive string that specifies the artifact's text encoding. The string **SHALL** be one of the character set names defined by IANA [IANA-ENC].

If the `artifact` object represents a text artifact and this property is absent, it **SHALL** default to the value of `theRun.defaultEncoding` (§3.14.24), if that property is present; otherwise, the artifact's encoding **SHALL** be taken to be unknown.

If the `artifact` object represents a binary artifact, `encoding` **SHALL** be absent.

EXAMPLE 1: In this example, the encoding of `output.txt` is UTF-16BE (obtained from the default), but the encoding of `data.txt` is UTF-16LE:

```
{
  "defaultEncoding": "UTF-16BE",      # A run object (§3.14).
                                     # See §3.14.24.
}
```

```

"artifacts": [
  {
    "location": {
      "uri": "output.txt"
    }
    # encoding property omitted
  },
  {
    "location": {
      "uri": "data.txt"
    },
    "encoding": "UTF-16LE"
  }
]
}

```

3.24.10 sourceLanguage Property

3.24.10.1 General

If an `artifact` object represents a text artifact that contains source code, it **MAY** contain a property named `sourceLanguage` whose value is a hierarchical string (§3.5.4) that specifies the programming language in which the source code is written. If the `artifact` object does not represent a text artifact containing source code, `sourceLanguage` **SHALL** be absent.

For the remainder of this section, we assume that the `artifact` object represents a text artifact that contains source code.

NOTE 1: This property is intended to help SARIF viewers to render code snippets (§3.30.13) with appropriate syntax coloring.

If the artifact contains source code in a mix of languages, and if it is possible to identify one of those languages as the “primary” language of the artifact, then `sourceLanguage` **SHALL** specify that language.

NOTE 2: Typically, this is the language implied by the file name extension.

EXAMPLE 1: In an HTML file that contains embedded JavaScript™, `sourceLanguage` would be `"html"`.

If it is not possible to identify a primary language, `sourceLanguage` **MAY** specify any language used in the artifact, or it **MAY** be absent.

NOTE 3: In either case, it is possible to specify a source language for any region by using `region.sourceLanguage` (see §3.30.15).

If `sourceLanguage` is absent, it **SHALL** default to the value of `theRun.defaultSourceLanguage` (§3.14.25). If both `artifact.sourceLanguage` and `theRun.defaultSourceLanguage` are absent, the artifact’s source language **SHALL** be taken to be unknown. In that case, a SARIF viewer **MAY** use any method or heuristic to determine the artifact’s source language, for example, by examining its file name extension or MIME type, or by prompting the user.

3.24.10.2 Source language identifier conventions and practices

To maximize interoperability, SARIF producers and consumers **SHOULD** conform to the following conventions and practices with respect to the value of this property:

- Producers:
 - Use only lower-case letters, and numbers (for example, `"c"` rather than `"C"`).
 - Spell out symbols (for example, `"csharp"` rather than `"c#"`).
 - To denote a language variant, use the hierarchical string mechanism (for example, `"csharp/7"`).
 - Do not abbreviate (for example, `"visualbasic"`™ rather than `"vb"`).

- Consumers

- Accept source language identifiers that conform to the above producer conventions.
- In addition, accept a variety of common industry forms, for example, {"cplusplus", "c++", "cpp"}, or {"javascript", "js"}.
- Compare source language identifiers case-insensitively.

§Appendix J., “Sample sourceLanguage values,” provides sample values for common programming languages.

3.24.11 hashes Property

An `artifact` object **MAY** contain a property named `hashes` whose value is a non-empty object (§3.6) each of whose property names specifies the name of a hash function, and each of whose property values represents the value produced by that hash function.

EXAMPLE 1: In this example, each of the hash functions SHA-256 and SHA-512 were used to compute hash values for the file.

```
{
  "hashes": {
    "sha-256": "...",
    "sha-512": "..."
  }
}
```

A file object.

To maximize interoperability, the property names **SHOULD** appear in the IANA registry of hash function textual names [IANA-HASH]. SARIF consumers that need to verify hash values **SHOULD** be able to compute any hash function whose name appears in the IANA registry.

The object **SHOULD** contain a property named `"sha-256"`. SARIF consumers that need to verify hash values **SHALL** be able to compute a SHA-256 hash.

The object **MAY** contain properties whose names do not appear in the IANA registry, but at the expense of interoperability. A SARIF consumer **MAY** implement any hash function, but it does not have to implement any hash function that does not appear in the IANA registry.

If the hash function is one whose name appears in the IANA registry, the property name **SHALL** equal the name as it appears in the registry (for example, `"sha-256"` rather than `"sha256"`); otherwise the property name **MAY** be any suitable name, but it **SHALL NOT** equal any name defined in the IANA registry.

SARIF consumers **SHALL** treat the property name as case insensitive (even when comparing to hash function names in the IANA registry).

Each property value **SHALL** be a string representation of the hash digest of the artifact, computed by the hash function specified by the property name. The string **SHALL** conform to the format produced by the hash algorithm (for example, if the hash algorithm produces a string of hexadecimal digits, the producer would not prepend `"0x"` to it).

NOTE 1: The value is represented as a string because hash values are typically represented in hexadecimal notation, and JSON integer values must be decimal.

NOTE 2: A hash value for an analysis target can be useful when a log file is processed by a result management system. The value can be used as a key when persisting results in a database. This allows a build system to use cached results, rather than repeating the analysis, when a target has not changed. A file hash can also be useful for validating results in a policy compliance system, allowing an auditor to validate that rerunning analysis against a target that hashes to a specific value reproduces the provided results.

The `artifact` object defines a set of hash values, rather than a single hash value, to allow a log file to be consumed by multiple tool chains that might expect hash values produced by differing hash function. Compliance systems, for example, will favor the use of more secure hash functions (such as SHA-256) that minimize the possibility that

two different targets will produce the same hash (at the expense of speed to produce the hash). In situations where compliance and security are not a concern, a system might prefer to use a fast hash function (such as MD5 or SHA-1) even though they have known weaknesses that allow adversaries to more easily generate hash collisions. To populate the `hashes` property, an analysis tool needs the ability to produce hashes for its analysis targets. Alternatively, the hashes could be added to the log file as a post-processing step. To make the best use of such an analysis tool, a user (such as a build engineer) would determine what systems in their build environment will consume the log file. The user would then configure the tool to produce hashes using the hash functions required by those systems. Analysis tools that are configurable to produce hashes with a variety of commonly used hash functions will interoperate most easily with such systems.

3.24.12 `lastModifiedTimeUtc` Property

An `artifact` object **MAY** contain a property named `lastModifiedTimeUtc` whose value is a string in the format specified in §3.9, specifying the UTC date and time at which the artifact was most recently modified.

NOTE: In scenarios where a tool has analyzed files on a network file share or on a local disk, an engineering system might use this property, rather than `hashes` (§3.24.11), as the most lightweight mechanism to determine whether the analysis needs to be repeated.

3.24.13 `description` Property

An `artifact` object **MAY** have a property named `description` whose value is a `message` object (§3.11) that describes the artifact.

3.25 `specialLocations` Object

3.25.1 General

A `specialLocations` object defines locations of special significance to SARIF consumers.

NOTE: This version of SARIF defines only one such location, `displayBase` (§3.25.2). In the future, other specially treated locations might be defined.

3.25.2 `displayBase` Property

A `specialLocations` object **MAY** contain a property named `displayBase` whose value is an `artifactLocation` object (§3.4) which provides a suggestion to consumers to display file paths relative to the specified location.

A consumer **MAY** act on this hint as follows:

1. Resolve `displayBase` to a URI (the “base URI”) by the procedure defined in §3.14.14 or any procedure with the same result. If the result is not an absolute URI, the procedure fails.
2. Normalize the base URI and the displayed URI by the procedures defined in §3.10.1 and §3.10.2 or any procedures with the same result.
3. If the base URI and the displayed URI have the identical scheme, authority, and initial path segments, then display only the remaining path segments of the displayed URI, or “.” if there are no remaining path segments.
4. Otherwise, render the displayed URI as an absolute URI (or in some other appropriate form, such as a (`uriBaseId`, `uri`) pair.

EXAMPLE 1: Given the following:

```
{
  "originalUriBaseIds": {
    # A run object (§3.14).
    # See §3.14.14.
  }
}
```

```

"WEBHOST": {
  "uri": "http://www.example.com/"
},
"ROOT": {
  "uri": "file:/"
},
"HOME": {
  "uri": "/home/user/",
  "uriBaseId": "ROOT"
},
"PACKAGE": {
  "uri": "mySoftware/",
  "uriBaseId": "HOME"
},
"SRC": {
  "uri": "src/",
  "uriBaseId": "PACKAGE"
}
},
"specialLocations": {
  "displayBase": {
    "uri": "",
    "uriBaseId": "PACKAGE"
  }
}
}

```

These equivalent locations would display as `src/f.c` because the scheme, authority, and initial path segments match:

```

{
  "uri": "f.c",
  "uriBaseId": "SRC"
}

{
  "uri": "src/f.c",
  "uriBaseId": "PACKAGE"
}

{
  "uri": "file:///home/user/mySoftware/src/f.c"
}

```

These equivalent locations would display as `/usr/include/stdio.h` because the scheme and authority match, but not the path:

```

{
  "uri": "/usr/include/stdio.h",
  "uriBaseId": "ROOT"
}

{
  "uri": "file:///usr/include/stdio.h"
}

```

These equivalent locations would display as `http://www.example.com/hello` because the scheme and authority do not match:

```

{
  "uri": "hello",
  "uriBaseId": "WEBHOST"
}

{
  "uri": "http://www.example.com/hello"
}

```

If `displayBase` were changed to

```

"displayBase": {
  "uri": "",
  "uriBaseId": "HOME"
}

```

the URIs displayed as `src/f.c` would instead be displayed as `mySoftware/src/f.c`. All other display values would be unchanged.

3.26 translationMetadata Object

3.26.1 General

A `translationMetadata` object describes a translation. It is necessary because in a `toolComponent` object that represents a translation, the usual descriptive properties `name` (§3.19.8), `fullName` (§3.19.9), *etc.* contain the translations of the corresponding strings in the `toolComponent` being translated; therefore, they are not available to hold descriptive information for the translation itself.

Because they occur only in `toolComponent` objects that represent translations, the properties of a `translationMetadata` object are not themselves localized (§3.5.1).

EXAMPLE 1:

```
{
  "language": "fr-FR",          # A toolComponent object (§3.19).
                               # The language of the translation (see (#language-property)).

  "translationMetadata": { # A translation metadata object.
    "name": "CodeScanner translation for fr-FR ",
    "fullName": "CodeScanner translation for fr-FR by Example Corp.",
    "shortDescription": {
      "text": "A good translation"
    },
    "fullDescription": {
      "text": "A good translation performed by native en-US speakers."
    }
  },

  "name": "(fr-FR translation of translated component's name)",
  "fullName": "(fr-FR translation of translated component's full name)",
  ...
}
```

3.26.2 name Property

A `translationMetadata` object **SHALL** contain a property named `name` whose value is a string containing a name for the translation.

3.26.3 fullName Property

A `translationMetadata` object **MAY** contain a property named `fullName` whose value is a string containing the name of the translation along with any other useful identifying information.

3.26.4 shortDescription Property

A `translationMetadata` object **MAY** contain a property named `shortDescription` whose value is a `multiformatMessageString` object (§3.12) containing a brief description of the translation.

3.26.5 fullDescription Property

A `translationMetadata` object **MAY** contain a property named `fullDescription` whose value is a `multiformatMessageString` object (§3.12) containing a comprehensive description of the translation.

3.26.6 downloadUri Property

A `translationMetadata` object **MAY** contain a property named `downloadUri` whose value is a string containing the absolute URI [RFC3986] from which the translation can be downloaded.

3.26.7 informationUri Property

A `translationMetadata` object **MAY** contain a property named `informationUri` whose value is a string containing the absolute URI [RFC3986] at which information about the translation can be found.

3.27 **result** Object

3.27.1 General

A **result** object describes a single result detected by an analysis tool.

Each result is produced by the evaluation of a rule. If the **tool** contains a **reportingDescriptor** object (§3.49) that describes that rule, we refer to that object as the **descriptor**, and we refer to the **toolComponent** object (§3.19) that defines the **descriptor** as the **component**.

3.27.2 Distinguishing logically identical from logically distinct results

Successive runs might detect the same condition in the code. When two **result** objects represent the same condition, we say that the results are “logically identical;” when they represent different conditions, we say that the results are “logically distinct.” Two results can be logically identical even if the **result** objects are not identical. For example, if code is inserted into a file between runs, the same condition might be reported on two different lines.

To avoid reporting the same condition repeatedly, result management systems typically group results into equivalence classes such that results in any one class are logically identical and results in different classes are logically distinct.

Some result management systems do this by calculating a “fingerprint” for each result and considering results with the same fingerprint to be logically identical. A fingerprint is calculated from information contained in the result and might contain readable information from the result.

Other result management systems group results into equivalence classes *without* associating a computed fingerprint with each result, and they denote each equivalence class with an arbitrary unique identifier. This identifier is opaque: it is *not* calculated from information stored in the result, and hence contains no readable information about the result.

Still other result management systems compute a fingerprint, associate an arbitrary unique identifier with the fingerprint, and use that identifier rather than the fingerprint to identify the equivalence class of results.

SARIF accommodates all these types of result management systems. Result management systems that compute fingerprints **SHOULD** populate the **fingerprints** property (§3.27.16). Result management systems that group results into equivalence classes based on an arbitrary unique identifier **SHOULD** populate the **correlationGuid** property (§3.27.4), regardless of whether they also compute a fingerprint.

3.27.3 **guid** property

A **result** object **MAY** contain a property named **guid** whose value is a GUID-valued string (§3.5.3) defining a unique, stable identifier for the result.

Direct SARIF producers and SARIF converters **MAY** but do not need to set this property. A result management system **SHOULD** set this property when it ingests a SARIF log file. If it does so, then later, when a SARIF consumer retrieves results in SARIF format from the result management system, the result management system **SHALL** set this property to the value it assigned.

A result management system **MAY** store multiple results with identical fingerprints (see §3.27.16 and §Appendix B.), but the **guid** properties for those results **SHALL** be distinct.

3.27.4 `correlationGuid` Property

A `result` object **MAY** contain a property named `correlationGuid` whose value is a GUID-valued string (§3.5.3) that is shared by all results that are considered logically identical, and that is different between any two results that are considered logically distinct.

Direct SARIF producers and SARIF converters **SHOULD NOT** set this property. A result management system **MAY** set this property when it ingests a SARIF log file. If it does so, then later, when a SARIF consumer retrieves results in SARIF format from the result management system, the result management system **MAY** set this property to the value it assigned.

NOTE: `correlationGuid` and fingerprints (§3.27.16) provide two different ways for result management systems to associate results that are logically identical. See §3.27.2 for more information.

3.27.5 `ruleId` Property

Depending on the circumstances, a `result` object either **SHALL**, **MAY**, or **SHALL NOT** contain a property named `ruleId` whose value is a hierarchical string (§3.5.4) whose leading components specify the stable identifier of the rule that was evaluated to produce the result. In addition to being stable, `ruleId` **SHOULD** be opaque.

NOTE: `ruleId` will usually consist entirely of the rule's stable opaque identifier. In some cases, it might be helpful to specify additional hierarchical components to more precisely describe the rule violation.

A SARIF viewer or result management system **MAY** use the additional hierarchical components to allow a user to suppress a subset of the violations of a given rule. A result management system **MAY** also use the additional components to more precisely match results between runs.

EXAMPLE 1: In this example, the first result describes a violation of rule CA2101. Its `ruleId` consists entirely of the rule's identifier. The second and third results both describe violations of rule CA5350. Each of their `ruleIds` specifies an additional hierarchical component that more precisely describes the rule violation. Note that `rule.index` (§3.27.7, §3.52.5) for both those results is 1; despite the additional hierarchical components in `ruleId`, both results describe violations of the same rule.

A SARIF viewer or result management system might allow a user to suppress, for example, only those violations of rule CA5350 which specify `md5` as the second hierarchical component of `ruleId`; that is, to allow the use of MD5 but still warn about the uses of other weak cryptographic algorithms.

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "CA2101",
          "shortDescription": {
            "text": "Specify marshaling for P/Invoke string arguments."
          }
        },
        {
          "id": "CA5350",
          "shortDescription": {
            "text": "Do not use weak cryptographic algorithms."
          }
        }
      ]
    }
  },
  "results": [
    {
      "ruleId": "CA2101",
      "rule": {
        "index": 0
      }
    },
    {
      "ruleId": "CA5350/md5",
      "rule": {
        "index": 1
      }
    },
    {
      "ruleId": "CA5350/sha-1",
      "rule": {
        "index": 1
      }
    }
  ]
}
```

```

    }
  ]
}

```

Direct producers **SHALL** emit either or both of `ruleId` and `rule.id` (§3.27.7, §3.52.4). If `rule.id` is absent, `ruleId` **SHALL** be present. If `rule.id` is present, `ruleId` **MAY** be present. If `ruleId` and `rule.id` are both present, they **SHALL** be equal.

For an example of the interaction between `ruleId` and `rule.id`, see §3.52.4.

Not all existing analysis tools emit the equivalent of a `ruleId` in their output. A SARIF converter which converts the output of such an analysis tool to the SARIF format **SHOULD** synthesize `ruleId` from other information available in the analysis tool's output.

Each SARIF converter might synthesize `ruleId` in a different way. Therefore, a SARIF consumer **SHOULD NOT** attempt to compare or combine the output from different converters for the same analysis tool. See Appendix D for more information about production of SARIF by converters.

3.27.6 `ruleIndex` Property

If `theDescriptor` exists (that is, if `theTool` contains a `reportingDescriptor` object (§3.49) that describes the rule that was violated), a `result` object **MAY** contain a property named `ruleIndex` whose value is the array index (§3.7.4) of the `Descriptor` within the `Component.ruleDescriptors` (§3.19.23). Otherwise, `ruleIndex` **SHALL** be absent.

The semantics of `ruleIndex` are identical to the semantics of `reportingDescriptorReference.index` (§3.52.5), and are described there.

If `ruleIndex` and `rule.index` (§3.27.7, §3.52.5) are both present, they **SHALL** be equal.

3.27.7 `rule` Property

Depending on the circumstances, a `result` object either **SHALL NOT**, **SHOULD**, or **MAY** contain a property named `rule` whose value is a `reportingDescriptorReference` object (§3.52) that identifies the `Descriptor`. The procedure for looking up a `reportingDescriptor` from a `reportingDescriptorReference` is described in §3.52.3.

If the `Descriptor` does not exist (that is, if `theTool` does not contain a `reportingDescriptor` object (§3.49) that describes the rule that was violated), then `rule` **SHALL NOT** be present.

If the `Descriptor` occurs in the `Tool.extensions` (§3.18.3), then `rule` **SHOULD** be present.

NOTE 1: If the `Descriptor` occurs in the `Tool.extensions` and `rule` is absent, the SARIF consumer will not be able to locate the rule metadata, even if `ruleIndex` (§3.27.6) is present, because `ruleIndex` alone does not specify which extension contains the `Descriptor`.

If the `Descriptor` occurs in the `Tool.driver` (§3.18.2) and `ruleIndex` is absent, then again `rule` **SHOULD** be present.

NOTE 2: If the `Descriptor` occurs in the `Tool.driver` and `ruleIndex` is absent, the SARIF consumer will not be able to locate the rule metadata within the `Tool.driver.ruleDescriptors`.

If the `Descriptor` occurs in the `Tool.driver` and `ruleIndex` is present, then `rule` **MAY** be present.

NOTE 3: If the `Descriptor` occurs in the `Tool.driver`, then `ruleIndex` suffices to locate the rule metadata within the `Tool.driver.ruleDescriptors`.

If `rule.id` (§3.52.4) is absent, it **SHALL** default to `thisObject.ruleId`. If `rule.id` and `thisObject.ruleId` are both present, they **SHALL** be equal.

If `rule.index` (§3.52.5) is absent, it **SHALL** default to `thisObject.ruleIndex`. If `rule.index` and `thisObject.ruleIndex` are both present, they **SHALL** be equal.

If `rule` is absent, it **SHALL** default to a `reportingDescriptorReference` object whose `id` property is set to `thisObject.ruleId` and whose `index` property is set to `thisObject.ruleIndex`.

NOTE: If the relevant rule is defined by the driver (see §3.18.1), which is likely to be the most common case, then `ruleId` and/or `ruleIndex` suffice to identify the rule, and take up less space in the log file than `rule`.

3.27.8 `taxa` Property

A result object **MAY** contain a property named `taxa` whose value is an array of zero or more unique (§3.7.3) `reportingDescriptorReference` objects (§3.52) each of which refers to a taxon (see §3.19.3) into which this result falls.

If the `toolComponent` object (§3.19) `theComponent` that defines the rule that was violated contains a `reportingDescriptor` object (§3.49) `theDescriptor` (a member of `toolComponent.rules` (§3.19.23)) that describes that rule, then `thisObject.taxa` **SHALL** contain elements corresponding to those elements of `theDescriptor.relationships` (§3.49.15) that describe taxa into which this result falls. `thisObject.taxa` does not need to contain elements which correspond to `superset` or `equals` relationships; rather, the result **SHALL** implicitly be taken to fall into all the taxa described by those relationships.

NOTE 1: See the example below for an illustration of this point. See §3.53.3 for descriptions of the various types of relationships.

Otherwise (that is, if `theDescriptor` does not exist), `thisObject.taxa` **SHALL** contain elements that describe all taxa into which the result falls.

In either case, if there is no `toolComponent` that defines the taxonomy to which an element of `thisObject.taxa` refers, then that element (a `reportingDescriptorReference` object) **SHALL NOT** contain `index` (§3.52.5) or `toolComponent.index` (§3.52.7, §3.54.4).

NOTE 2: The rationale for this restriction is that `toolComponent.index` serves to locate the `toolComponent` object defining the rule, and `index` serves to locate the rule within that `toolComponent`. If there is no relevant `toolComponent` object, neither of those properties is meaningful. On the other hand, properties such as `id` (§3.52.4), `guid` (§3.52.6), `toolComponent.name` (§3.54.3), and `toolComponent.guid` (§3.54.5) are useful for readability and for identification, even if the `toolComponent` itself is absent, so they are permitted.

EXAMPLE 1: In this example, a tool defines a custom taxonomy (see §3.19.3) consisting of three taxa with ids "SUP", "INC1", and "INC2". The tool emits a result that falls into the taxa "SUP" and "INC2", but not into "INC1". According to `relationships[0]`, "SUP" is a superset of "CA2101"; that is, every result that violates "CA2101" falls into the taxon "SUP". Therefore, it is not necessary to mention "SUP" in `theResult.taxa`. On the other hand, according to `relationships[2]`, "INC2" is incomparable to "CA2101"; that is, the set of results that violate "CA2101" intersects with but is neither a superset nor a subset of the set of results that fall into the taxon "INC2". Therefore, it is necessary to mention "INC2" in `theResult.taxa`.

```
# A run object (§3.14).
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      ...
      "rules": [
        {
          "id": "CA2101",
          ...
          "relationships": [
            {
              "target": {
                "id": "SUP",
                "guid": "11111111-1111-1111-8888-111111111111"
              },
              "kinds": [
                "superset"
              ]
            }
          ]
        }
      ]
    }
  }
}
```



```

    ],
    {
      "target": {
        "id": "INC1",
        "guid": "22222222-2222-1111-8888-222222222222"
      },
      "kinds": [
        "incomparable"
      ]
    },
    {
      "target": {
        "id": "INC2",
        "guid": "33333333-3333-1111-8888-333333333333"
      },
      "kinds": [
        "incomparable"
      ]
    }
  ],
  "taxa": [
    {
      "id": "SUP",
      "guid": "11111111-1111-1111-8888-111111111111",
      ...
    },
    {
      "id": "INC1",
      "guid": "22222222-2222-1111-8888-222222222222",
      ...
    },
    {
      "id": "INC2",
      "guid": "33333333-3333-1111-8888-333333333333",
      ...
    }
  ]
},
"results": [
  {
    "ruleId": "CA2101",
    "rule": {
      "index": 0
    },
    "taxa": [
      {
        "id": "INC2",
        "guid": "33333333-3333-1111-8888-333333333333"
      }
    ]
  }
]
}

```

3.27.9 **kind** Property

A **result** object **MAY** contain a property named **kind** whose value is one of a fixed set of strings that specify the nature of the result.

If present, the **kind** property **SHALL** have one of the following values, with the specified meanings:

- **"pass"**: The rule specified by **ruleId** (§3.27.5), **ruleIndex** (§3.27.6), and/or **rule** (§3.27.7) was evaluated, and no problem was found.
- **"open"**: The specified rule was evaluated, and the tool concluded that there was insufficient information to decide whether a problem exists.

NOTE 1: This value is used by proof-based tools. Sometimes such a tool can prove that there is no violation (**kind** = **"pass"**), sometimes it can prove that there is a violation (**kind** = **"fail"**), and sometimes it does not detect a violation but is unable to prove that there is none (**kind** = **"open"**). In such a tool, a **kind** value of **"open"** might be an indication that the user should add additional assertions to enable the tool to determine if there is a violation.

- **"informational"**: The specified rule was evaluated and produced a purely informational result that does not indicate the presence of a problem. (See the example below.)
- **"notApplicable"**: The rule specified by **ruleId** was not evaluated, because it does not apply to the analysis target.

EXAMPLE 1: In this example, a binary checker has a rule that applies to 32-bit binaries only. It produces a "notApplicable" result if it is run on a 64-bit binary. It also has a rule that checks the compiler version and produces an informational result:

```
"results": [
  {
    "ruleId": "ABC0001",
    "kind": "notApplicable",
    "message": {
      "text": "\"MyTool64.exe\" was not evaluated for rule ABC0001
              because it is not a 32-bit binary."
    },
    "locations": [
      {
        "physicalLocation": {
          "uri": "file://C:/bin/MyTool64.exe"
        }
      }
    ]
  },
  {
    "ruleId": "ABC0002",
    "kind": "informational",
    "message": {
      "text": "\"MyTool64.exe\" was compiled with Example Corporation
              Compiler version 10.2.2."
    },
    "locations": [
      {
        "physicalLocation": {
          "uri": "file://C:/bin/MyTool64.exe"
        }
      }
    ]
  }
]
```

- "review": The result requires review by a human user to decide if it represents a problem.

NOTE 2: This value is used by tools that are unable to check for certain conditions, but that wish to bring to the user's attention the possibility that there might be a problem. For example, an accessibility checker might produce a result with the message "Do not use color alone to highlight important information," with `kind = "review"`. A user might address this issue by visually inspecting the UI.

- "fail": The result represents a problem whose severity is specified by the `level` property (§3.27.10).

If `kind` is absent, it SHALL default to "fail".

If `level` has any value other than "none" and `kind` is present, then `kind` SHALL have the value "fail".

3.27.10 `level` Property

A `result` object MAY contain a property named `level` whose value is one of a fixed set of strings that specify the severity level of the result.

If present, the `level` property SHALL have one of the following values, with the specified meanings:

- "warning": The rule specified by `ruleId` was evaluated and a problem was found.
- "error": The rule specified by `ruleId` was evaluated and a serious problem was found.
- "note": The rule specified by `ruleId` was evaluated and a minor problem or an opportunity to improve the code was found.
- "none": The concept of "severity" does not apply to this result because the `kind` property (§3.27.9) has a value other than "fail".

EXAMPLE 1: In this example, the tool reports an opportunity to improve the code.

```
"results": [
  {
    "ruleId": "ABC0003",
    "kind": "fail",
    "level": "note",
    "message": {
      "text": "Consider using 'nameof(start)' instead of hard-coding
              the parameter name 'start'."
    }
  }
]
```

```

    },
    "locations": [
      {
        "physicalLocation": {
          "uri": "file:///C:/code/a.cs",
          "region": {
            "startLine": 6
          }
        }
      }
    ]
  }
]

```

If `kind` (§3.27.9) has any value other than "fail", then if `level` is absent, it **SHALL** default to "none", and if it is present, it **SHALL** have the value "none".

If `kind` has the value "fail" and `level` is absent, then `level` **SHALL** be determined by the following procedure:

If rule (§3.27.7) is present THEN

LET `theDescriptor` be the `reportingDescriptor` object (§3.49) that it specifies.

Is there a configuration override for the `level` property?

IF `result.provenance.invocationIndex` (§3.27.29, §3.48.6) is ≥ 0 THEN

LET `theInvocation` be the `invocation` object (§3.20) that it specifies.

IF `theInvocation.ruleConfigurationOverrides` (§3.20.5) is present

AND it contains a `configurationOverride` object (§3.51) whose

`descriptor` property (§3.51.2) specifies `theDescriptor` THEN

LET `theOverride` be that `configurationOverride` object.

IF `theOverride.configuration.level` (§3.51.3, §3.50.3) is present THEN

Set `level` to `theConfiguration.level`.

ELSE

There is no configuration override for `level`. Is there a default configuration for it?

IF `theDescriptor.defaultConfiguration.level` (§3.49.14, §3.50.3) is present THEN

SET `level` to `theDescriptor.defaultConfiguration.level`.

IF `level` has not yet been set and `result.level` is absent THEN

SET `level` to "warning".

ELSE IF `result.level` is present then

USE that value regardless of what the rule metadata states.

3.27.11 message Property

A `result` object **SHALL** contain a property named `message` whose value is a `message` object (§3.11) that describes the result.

The `message` property **SHOULD** provide sufficient details to allow an end user to resolve any problem that the result might indicate. In particular, it **SHALL** include all of the following information that is available and relevant to the result:

- Information sufficient to identify the analysis target, and the location within the target where the problem occurred.
- The condition within the analysis target that led to the problem being reported.
- The risks potentially associated with not fixing the problem.
- The full range of responses to the problem that the end user could take (including the definition of conditions where it might be appropriate not to fix the problem, or to conclude that the result is a false positive).

EXAMPLE 1: This is an example of a message:

```
"results": [
  {
    "message": {
      "text": "Deleting member 'x' of variable 'y' may compromise
performance on subsequent accesses of 'y'. Consider
setting object member 'x' to null instead, unless this
object is a dictionary or if runtime semantics otherwise
dictate that the existence of a null member is distinct
from one that is not present at all. This violation can
also be ignored for infrequently called code paths."
    }
  }
]
```

See §3.11.7 for the procedure for looking up a message string from a `message` object, in particular, for the case where the `message` object occurs as the value of `result.message`.

EXAMPLE 2: In this example, `message.id` refers to the property named `default` defined in the `messageStrings` property of the `reportingDescriptor` object identified by "CA2101".

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "CA2101",
          "messageStrings": {
            "default": {
              "text": "The default message for this rule.",
              "markdown": "The default message for *this* rule."
            },
            "special": {
              "text": "Another message, for special cases.",
              "markdown": "Another message, for *special* cases."
            }
          }
        }
      ]
    }
  },
  "results": [
    {
      "ruleId": "CA2101",
      "rule": {
        "index": 0
      },
      "message": {
        "id": "default"
      },
      ...
    }
  ]
}
```

3.27.12 locations Property

A `result` object **SHOULD** contain a property named `locations` whose value is an array of zero or more `location` objects (§3.28) each of which specifies a location where the result occurred.

NOTE 1: In rare circumstances, it might not be possible to specify a location for a result. However, the `locations` property contains very valuable information for anyone who needs to diagnose and correct the condition described by the result, so the authors of analysis tools should make every effort to provide it.

EXAMPLE 1: If a C++ analyzer detects that no file defines a global function `main`, then that result cannot be associated with a file.

NOTE 2: The `locations` array is not defined to contain unique (§3.7.3) elements because some tools report a line number but not a column number for a result's location. Such a tool might report the same result twice on the same line, in some cases producing multiple identical `location` objects.

The `locations` array **SHALL NOT** contain more than one element unless the condition indicated by the result, if any, can only be corrected by making a change at every location specified in the array.

EXAMPLE 2: In C#, which supports “partial” classes, portions of the declaration of a single class can occur at multiple locations in the source code. If an analysis tool reports that the name of such a class does not conform to a specified convention, then the resulting log file might contain a single result object, which would contain a `locations` array each of whose elements specifies a location in the source code where the class name occurs.

The `locations` array **SHALL NOT** be used to specify distinct occurrences of the same result which can be corrected independently.

EXAMPLE 3: Consider an analysis tool which locates misspelled words in documentation, and suppose this tool scans a document in which the same word is misspelled in two distinct locations. Then the resulting log file must contain two distinct `result` objects each of which contains a `locations` array containing a single `location` object specifying the location of one instance of the misspelled word.

EXAMPLE 4: In contrast, consider a tool which locates misspelled words in variable names. If the tool detects a misspelled variable name, it might produce a single `result` object whose `locations` array contains the location of every reference to the variable, since fixing some but not all of the references would cause a compilation error.

3.27.13 `analysisTarget` Property

If the analysis target differs from the result file, a `result` object **SHOULD** contain a property named `analysisTarget` whose value is an `artifactLocation` object (§3.4) that specifies the analysis target.

If the analysis target and the result file are the same, the `analysisTarget` property **SHOULD** be absent.

EXAMPLE 1: In this example, the tool's analysis target was the file `mouse.c`. During the scan, the tool detected a result in the included file `mouse.h`.

```
{
  "analysisTarget": {           # A result object (§3.27).
    "uri": "input/mouse.c",     # An artifactLocation object (§3.4).
    "uriBaseId": "SRCROOT"
  },
  "locations": [               # See §3.27.12.
    {                           # A location object (§3.28).
      "physicalLocation": {     # See §3.28.3.
        "artifactLocation": {   # An artifactLocation object.
          "uri": "input/mouse.h",
          "uriBaseId": "SRCROOT"
        },
        "region": {
          "startLine": 42
        }
      }
    ]
  }
}
```

3.27.14 **webRequest** Property

A `result` object **MAY** contain a property named `webRequest` whose value is a `webRequest` object (§3.46) that describes the HTTP request which led to this result.

NOTE: This property is primarily useful to web analysis tools.

3.27.15 **webResponse** Property

A `result` object **MAY** contain a property named `webResponse` whose value is a `webResponse` object (§3.47) that describes the response to the HTTP request which led to this result.

NOTE: This property is primarily useful to web analysis tools.

3.27.16 **fingerprints** Property

A `result` object **MAY** contain a property named `fingerprints` whose value is an object (§3.6).

Each property value in this object **SHALL** be a string that provides a stable identifier for the result. This identifier **SHALL**, to the extent that it is feasible, be the same for all results that are logically identical, and different for any two results that are logically distinct. This requirement is intended to ensure that a fingerprint is resistant to changes that do not affect the logical identity of the result, such as the location of the root of a source code enlistment, or the line number where a result appears in a source file.

Each property name in this object **SHALL** be a versioned hierarchical string (§3.5.4.2). A result management system **MAY** use the property names to identify the method used to calculate the fingerprint.

EXAMPLE 1: In this example, the producer has calculated a fingerprint using version 2 of a fingerprinting method it refers to as "stableResultHash":

```
{
  "fingerprints": {
    "stableResultHash/v2": "097886bc876fe"
  }
}
```

When a result management system uses fingerprint information to determine whether two results are logically identical, it **SHOULD** use the latest version of the fingerprint available in both results.

EXAMPLE 2: In this example, one result has values for versions 1 and 2 of the “context region hash” fingerprint. Another result has values for versions 2 and 3. A result management system would use version 2 (the greatest common version) to compare the two results.

```
{
  "results": [
    {
      "fingerprints": {
        "stableResultHash/v1": "1234567900abcd",
        "stableResultHash/v2": "234567900abcd"
      }
    },
    {
      "fingerprints": {
        "stableResultHash/v2": "234567900abcd",
        "stableResultHash/v3": "34567900abcde"
      }
    }
  ]
}
```

NOTE: This property is an array, rather than a single string, for two reasons:

- To allow a result management system to continue to support outdated fingerprinting algorithms while upgrading to a newer, more reliable algorithm.
- Less likely but possible, to allow multiple result management systems to record their final fingerprints.

A direct SARIF producer **SHOULD NOT** populate this property. A SARIF converter **MAY** populate this property if the analysis tool's native output format provides a value that qualifies as a fingerprint (a stable identifier for the result). A result management system **MAY** populate this property when it ingests a SARIF file. If it does so, then later, when a SARIF consumer retrieves results in SARIF format from the result management system, the result management system **MAY** set this property to the value it assigned.

§Appendix B. provides requirements for how a result management system computes fingerprints.

NOTE: `fingerprints` and `correlationGuid` (§3.27.4) provide two different ways for result management systems to associate results that are logically identical. See §3.27.2 for more information.

3.27.17 `partialFingerprints` Property

A `result` object **MAY** contain a property named `partialFingerprints` whose value is an object (§3.6).

Each property value in this object **SHALL** be a string that contributes to the stable, unique identity, or “fingerprint,” of the result (see §3.27.16). Appendix B explains how a result management system can compute these fingerprints.

Each property name in this object **SHALL** be a versioned hierarchical string (§3.5.4.2). A SARIF producer **MAY** use the property name to identify the nature of the information used to compute the partial fingerprint.

EXAMPLE 1: In this example, the producer has calculated a partial fingerprint using version 3 of a partial fingerprint value it refers to as “prohibitedWordHash”:

```
{
  "partialFingerprints": {
    "prohibitedWordHash/v3": "097886bc876fe"
  }
}
```

A result object (§3.27).

When a result management system uses partial fingerprint information to determine whether two results are logically identical, it **SHOULD** use the latest version of the partial fingerprint available in both results.

EXAMPLE 2: In this example, one result has values for versions 1 and 2 of the “prohibited word hash” partial fingerprint. Another result has values for versions 2 and 3. A result management system would use version 2 (the greatest common version) to compare the two results.

```
{
  "results": [
    {
      "partialFingerprints": {
        "prohibitedWordHash/v1": "1234567900abcd",
        "prohibitedWordHash/v2": "234567900abcd"
      }
    },
    {
      "partialFingerprints": {
        "prohibitedWordHash/v2": "234567900abcd",
        "prohibitedWordHash/v3": "34567900abcde"
      }
    }
  ]
}
```

A run object (§3.14).
See §3.14.23.
A result object.

A result management system **MAY** use any algorithm to combine the information contained in the various partial fingerprints. (For example, it might decide that two results are logically identical if any one of their partial fingerprints match, or only if a majority of them match, or only if all of them match.)

To make use of the information, if any, embodied in the property names, a result management system requires knowledge of the naming convention used by the SARIF producer. A result management system with that knowledge **MAY** use the property names to decide which partial fingerprints to include in its fingerprint computation. A result management system lacking that knowledge **SHOULD NOT** attempt to interpret the information embodied in the partial fingerprint names.

Because result management systems might come to depend on the choice of property names, SARIF producers that use property names to identify the nature of the information used to compute the partial fingerprint **SHOULD** adhere to the following guidelines:

- Choose meaningful property names that describe the information used to compute the partial fingerprint.
- Document the property names.
- When introducing a partial fingerprint computed with a different approach, associate it with a new property name.
- Avoid removing existing property names and partial fingerprints, since existing result management systems might rely on them.

EXAMPLE 3: In this example, a SARIF-producing document checker has computed a partial fingerprint that hashes a word that should not appear in a document together with the document's language.

```
{
    ...
    # A result object.
    "partialFingerprints": {
        "wordPlusLangHash":
            "2c26b46b68ffc68ff99b453cd30413413422d706483bfa0f98a5e886266e7ae"
    }
}
```

EXAMPLE 4. In this example, the SARIF producer has chosen an arbitrary value for the property name.

```
{
    ...
    # A result object
    "partialFingerprints": {
        "1": "56eaf900cc8f6"
    }
}
```

3.27.18 codeFlows Property

A `result` object **MAY** contain a property named `codeFlows` whose value is an array of zero or more `codeFlow` objects (§3.36). The `codeFlows` property is intended for use by analysis tools that provide execution path details that illustrate a possible problem in the code.

NOTE: The SARIF file format allows multiple `codeFlow` objects within a single `result` object to allow for the possibility that more than one code flow might be relevant to a single result.

3.27.19 graphs Property

A `result` object **MAY** contain a property named `graphs` whose value is an array of zero or more unique (§3.7.3) `graph` objects (§3.39). A `graph` object represents a directed graph: a network of nodes and directed edges that describes some aspect of the structure of the code (for example, a call graph).

A `graph` object defined at the `result` level **SHALL** be referenced only by `graphTraversal` objects (§3.42) defined in the `graphTraversals` property (§3.27.20) of the `result` object in which it is defined. This contrasts with `graph` objects defined at the `run` level (§3.14.20), which **MAY** be referenced by `graphTraversal` objects defined in the `graphTraversals` property of any `result` object in the `run`.

3.27.20 graphTraversals Property

If a `result` object contains a `graphs` property (§3.27.19), or if the `run` contains a `graphs` property (§3.14.20), then the `result` object **MAY** contain a property named `graphTraversals` whose value is an array of zero or more unique (§3.7.3) `graphTraversal` objects (§3.42). If neither the `result` object nor the `run` contains a `graphs` property, the `graphTraversals` property **SHALL** be absent. A `graph traversal` is a path through the code that visits one or more nodes in a specified graph.

3.27.21 **stacks** Property

A **result** object **MAY** contain a property named **stacks** whose value is an array of zero or more unique (§3.7.3) **stack** objects (§3.44). The **stacks** property is intended for use by analysis tools that compute or collect call stack information in the process of producing results.

NOTE: The SARIF file format allows multiple **stack** objects within a single **result** object to allow for the possibility that more than one call stack might be relevant to a single result.

3.27.22 **relatedLocations** Property

A **result** object **MAY** contain a property named **relatedLocations** whose value is an array of zero or more unique (§3.7.3) **location** objects (§3.28) each of which represents a location relevant to understanding the result.

EXAMPLE 1: Suppose that a tool for analyzing JavaScript™ has a rule that reports a problem when a variable declared in an inner scope hides a variable with the same name in an enclosing scope. The tool would report the problem on the line where the inner variable is declared. The tool could choose to add an element to the **relatedLocations** array, specifying the location where the outer variable was declared.

The result might appear in the log file like this:

```
"results": [
  {
    "ruleId": "JS3056",
    "level": "error",
    "message": {
      "text": "Name 'index' cannot be used in this scope because
              it would give a different meaning to 'index'
              ([declared here](0))."
    },
    "locations": [
      {
        "physicalLocation": {
          "uri": "file:///C:/Code/a.js",
          "region": {
            "startLine": "6",
            "startColumn": "10"
          }
        }
      }
    ],
    "relatedLocations": [ # An array of location objects (§3.28).
      { # A location object.
        "id": 0,
        "message": {
          "text": "The previous declaration of 'index' was here."
        },
        "physicalLocation": {
          "uri": "file:///C:/Code/a.js",
          "region": {
            "startLine": "2",
            "startColumn": "6"
          }
        }
      }
    ]
  },
  ...
]
```

The tool might write messages to the console like this:

```
C:\Code\a.js(6,10-10): error : JS3056: Name 'index' cannot be used in this scope because it would give a different meaning to 'index'.
C:\Code\a.js(2,6-6): info : JS3056: The previous declaration of 'index' was here.
```

3.27.23 **suppressions** Property

A **result** object **MAY** contain a property named **suppressions** whose value is an array of zero or more unique (§3.7.3) **suppression** objects (§3.35) each of which describes a request to “suppress” a result (that is, to exclude it from result lists, bug counts, *etc.*).

If **suppressions** is absent, it **SHALL** default to **null**.

The presence of an array value, whether or not the array is empty, **SHALL** mean that suppression information is available for the result. In this case, if the array is empty, a consumer **SHALL** treat the result as not suppressed. If the array is non-empty,

a consumer that needs to determine the result's suppression state **SHALL** examine the `status` properties (§3.35.3) of the suppression objects in the array.

The absence of an array value, or the presence of a `null` value, **SHALL** mean that suppression information is not available for the result. A SARIF consumer **SHALL** treat such a result as not suppressed.

The `Suppressions` values for all `result` objects in `theRun` **SHALL** be either all `null` or all non-`null`.

NOTE: The rationale is that an engineering system will generally evaluate all results for suppression, or none of them. Requiring that the `Suppressions` values be either all `null` or all non-`null` enables a consumer to determine whether suppression information is available for the run by examining a single `result` object.

3.27.24 `baselineState` Property

A `result` object **MAY** contain a property named `baselineState` whose value is a string that specifies the state of this result with respect to some previous run, which we refer to as the “baseline run.”

If `theRun.baselineGuid` (§3.14.5) is present, its value **SHALL** specify the baseline run.

This property **SHALL** have one of the following values, with the specified meanings:

- `"new"`: This result was detected in the current run but was not detected in the baseline run.
- `"unchanged"`: This result was detected both in the current run and in the baseline run, and it did not change between those two runs in any way that the tool considers significant.
- `"updated"`: This result was detected both in the current run and in the baseline run, but it changed between those two runs in a way that the tool considers significant.
- `"absent"`: This result was detected in the baseline run but was not detected in the current run.

NOTE 1: The purpose of `baselineState` is to allow (for example) a measurement of how many new results were introduced in the run, and how many previously existing results no longer appear.

To assign a value to `baselineState`, a tool needs a way to determine whether a result is logically “the same”, in some sense, as a result that appeared in the baseline. §Appendix B. discusses how a result management system can assign a “fingerprint” to each result. See also the description of the `fingerprints` (§3.27.16) and `partialFingerprints` (§3.27.17) properties.

An analysis tool that works together with such a result management system can use the fingerprint to determine whether two results are logically the same; two results with the same fingerprint are considered logically the same.

NOTE 2: A result management system might respond to a “new” result by filing an issue in a bug tracking system. It might respond to an “updated” result by editing the details of an existing issue in the bug tracking system, or by attaching an updated SARIF log to the issue. It might respond to an “absent” result by resolving the issue. It might take no action at all for an “unchanged” issue, or it might simply update its internal information about the range of runs that contained the result.

If `baselineState` is present on any `result` object in `theRun`, it **SHALL** be present on every such `result` object.

NOTE 3: The presence of `baselineState` on any `result` implies that the SARIF producer performed a comprehensive comparison between the results in the current run and those in some previous run. A SARIF consumer is entitled to expect that the differencing operation produced a `baselineState` value for every result.

This is conceptually similar to a tool that compares two text files, and for every line, concludes that it exists in the left-hand file, the right-hand file, or both. The tool must provide this information for every line in both files; it cannot leave some lines “undetermined.”

3.27.25 **rank** Property

A **result** object **MAY** contain a property named **rank** whose value is a number between 0.0 and 100.0 inclusive, representing the priority or importance of the result. 0.0 is the lowest priority and 100.0 is the highest.

rank is only meaningful if **kind** (§3.27.9) has the value "fail".

If **kind** has the value "fail", then if **rank** is absent, it **SHALL** default to the value determined by the procedure defined for **level** (§3.27.10), except throughout the procedure, replace "level" with "rank" and replace "warning" with -1.0.

If **kind** has any other value, then **rank** **SHALL** be absent.

If **rank** is absent, it **SHALL** default to -1.0, which indicates that the value is unknown (not set).

NOTE: rank values produced by different tools are in general not commensurable. If Tool A produces one result with rank 0.65 and a second result with rank 0.70, the consumer is entitled to assume that the second result is of higher priority than the first. But if Tool A produces a result with rank 0.65 and Tool B produces a result with rank 0.70, the result produced by Tool B might or might not be of higher priority than the result produced by Tool A. In an engineering system that aggregates results from multiple tools, rank values might need to be adjusted, either automatically or by end users, so that rank values from different tools can be interleaved in a meaningful way.

3.27.26 **attachments** Property

A **result** object **MAY** contain a property named **attachments** whose value is an array of zero or more unique (§3.7.3) attachment objects (§3.21) each of which describes an artifact relevant to the detection of the result.

3.27.27 **workItemUri** Property

A **result** object **MAY** contain a property named **workItemUri** whose value is either **null** or an array of zero or more unique (§3.7.3) strings each of which contains the absolute URI [RFC3986] of a work item associated with this result.

If **workItemUri** is absent, it **SHALL** default to **null**.

An empty array **SHALL** mean that there are no work items associated with this result. **null** **SHALL** mean that the set of work items associated with this result, if any, is not known.

The **workItemUri** values for all **result** objects in theRun **SHALL** be either all **null** or all non-**null**.

NOTE 1: The rationale is that an engineering system will generally track work item status for all results or for none of them. Requiring that the **workItemUri** values be either all **null** or all non-**null** enables a consumer to determine whether work item information is available for the run by examining a single **result** object.

NOTE 2: Result management systems are likely to generate work items from at least some of the results in a SARIF log file. Depending on the engineering system, these work items might take the form of Git issues, Jira tickets, TFS work items, or the equivalent in other work item tracking systems.

3.27.28 **hostedViewerUri** Property

A **result** object **MAY** contain a property named **hostedViewerUri** whose value is a string containing an absolute URI [RFC3986] at which the result can be viewed. The URI **SHALL** be valid as of the time the tool generated this result. It is not guaranteed to be valid at later times (for example, the hosting environment might not keep results older than a specified age).

NOTE: This property can be used by tools that provide an online viewing experience for the results they generate. This experience might be specifically designed to display the results from that tool, as opposed to a generic SARIF

viewer that displays results from any tool that produces SARIF.

3.27.29 **provenance** Property

A `result` object **MAY** contain a property named `provenance` whose value is a `resultProvenance` object (§3.48) that contains information about how and when the result was detected.

3.27.30 **fixes** Property

A `result` object **MAY** contain a property named `fixes` whose value is an array of zero or more unique (§3.7.3) `fix` objects (§3.55).

3.27.31 **occurrenceCount** Property

A `result` object **MAY** contain a property named `occurrenceCount` whose value is a positive integer specifying the number of times a result with `theResult.correlationGuid` (§3.27.4) has been observed.

NOTE: This property is intended for the scenario where multiple SARIF files are being merged into a single SARIF file, with the intent that each logically distinct result (see §3.27.2) occurs only once in the merged file. In that case, the system performing the merge would select one occurrence of each logically distinct result to serve as the exemplar for that class of results, and it would set `occurrenceCount` on that instance to the number of times a result with that `correlationGuid` occurred in the input files.

This property can also be useful even in the context of a single log file. Consider an accessibility checker that detects an accessibility problem at a particular location. Suppose the checker has access to activity logs that trace user paths through the application. The checker could use those logs to determine how many times users encountered the location with the accessibility problem, and store that information in `occurrenceCount`.

3.28 **location** Object

3.28.1 **General**

A `location` object describes a location. Depending on the circumstances, a `location` object is described by physical location (§3.29), a logical location (§3.33), both, or in rare circumstances, neither (see below).

A logical location specifies a programmatic construct, for example, a class name or a function name, without specifying the artifact within which that construct occurs.

NOTE: Among the reasons for including logical locations in the SARIF format in addition to physical locations are the following:

- In the absence of symbol information, binary analysis tools might not have source code locations available, so information about line and column numbers might not be present in the log file. In this case, code editors, other programs, or end users can use logical location to navigate from a result to the correct source code location.
- Logical location information is an important contributor to fingerprinting scenarios because it is typically more resilient to changes in source code than are the line numbers included in physical locations. See §Appendix B. for more information about fingerprinting. The `logicalLocation.fullyQualifiedName` property (§3.33.5) is particularly convenient for fingerprinting.
- In the analysis of structured data files such as XML or JSON, internal structural information (such as an XML path like `"/orders[2]/customers/lastName"`) might be helpful.

In rare circumstances, there might be neither physical nor logical location information available for a `location` object. See §3.38 for an example. In that case, the location object **SHOULD** contain a message property (§3.28.5) explaining the significance of this “location.”

3.28.2 `id` Property

A `location` object **MAY** contain a property named `id` whose value is a non-negative integer that is unique among all `location` objects belonging to the `LocationOwner`. The value does not need to be unique across all `result` (§3.27) or `notification` (§3.58) objects in the `Run`.

If `id` is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set).

NOTE: Negative values are forbidden because their use would suggest some non-obvious semantic difference between positive and negative values.

EXAMPLE 1: Within a `result` object, the following property values (among others) are `location` objects, and no two of them can have the same value for `id`:

```
result.relatedLocations[0]
result.codeFlows[0].threadFlows[0].locations[0].location
result.stacks[0].frames[0].location
```

The `id` property has two purposes: to enable an embedded link (§3.11.6) within a message object (§3.11) to refer to `thisObject`, and to identify `thisObject` as the target of a `locationRelationship` (§3.34). If no message object within the `LocationOwner` refers to `thisObject` *via* an embedded link and no `locationRelationship` object within the `LocationOwner` specifies `thisObject` as its target, the `id` property does not need to appear.

3.28.3 `physicalLocation` Property

Depending on the circumstances, a `location` object either **SHALL**, **MAY**, or **SHALL NOT** contain a property named `physicalLocation` whose value is a `physicalLocation` object (§3.29) that identifies the file within which the location lies. If physical location information is available and the `logicalLocations` property (§3.28.4) is absent or empty, `physicalLocation` **SHALL** be present. If physical location is available and `logicalLocations` is present and non-empty, `physicalLocation` **MAY** be present. If physical location information is not available, `physicalLocation` **SHALL NOT** be present.

3.28.4 `logicalLocations` Property

Depending on the circumstances, a `location` object either **SHALL**, **MAY**, or **SHALL NOT** contain a property named `logicalLocations` whose value is an array of zero or more unique (§3.7.3) `logicalLocation` objects (§3.33) that identify the programmatic construct within which the location lies. If logical location information is available and the `physicalLocation` property (§3.28.3) is absent, `logicalLocations` **SHALL** be present and non-empty. If logical location information is available and `physicalLocation` is present, `logicalLocations` **MAY** be present. If logical location information is not available, `logicalLocations` **SHALL NOT** be present.

NOTE: `logicalLocations` is an array because some logical locations can be expressed in more than one way. For example, the logical location of an element in an HTML document might be expressed by an XML Path expression such as `/html/body/img[1]` or by a CSS selector such as `#logo`.

3.28.5 `message` Property

A `location` object **MAY** contain a property named `message` whose value is a message object (§3.11) relevant to the location.

3.28.6 annotations Property

A `location` object **MAY** contain a property named `annotations` whose value is an array of zero or more unique (§3.7.3) `region` objects (§3.30) each of which describes a region within the artifact specified by the `location` object that is relevant to the location. Each of these `region` objects **SHOULD** contain a `message` property (§3.30.14) that explains the relevance of the region to the location.

EXAMPLE 1: Consider a `location` object which describes the declaration statement

```
int x = (y + z) * q;
```

If the analysis tool wanted to emphasize the expression `(y + z)`, it might set the `annotations` property to:

```
"annotations": [
  {
    "startLine": 12,
    "startColumn": 9,
    "endColumn": 16,
    "message": {
      "text": "(y + z) = 42"
    }
  }
]
```

An array of region objects.
A region object (§3.30).

3.28.7 relationships Property

A `location` object **MAY** contain a property named `relationships` whose value is an array of zero or more unique (§3.7.3) `locationRelationship` objects (§3.34) each of which declares one or more directed relationship from `thisObject` to another `location` object, which we refer to as `theTarget`, specified by `locationRelationship.target` (§3.34.2). The natures of the relationships between `thisObject` and `theTarget` are specified by `locationRelationship.kinds` (§3.34.3).

3.29 physicalLocation Object

3.29.1 General

A `physicalLocation` object represents the physical location where a result was detected. A physical location specifies a reference to an artifact together with a region within that artifact.

3.29.2 Constraints

Either the `artifactLocation` property (§3.29.3), the `address` property (§3.29.6), or both **SHALL** be present.

If `region.byteLength` (§3.29.4, §3.30.12) and `address.length` (§3.29.6, §3.32.9) are both present, then `region.byteLength` **SHALL** equal the absolute value of `address.length`.

3.29.3 artifactLocation Property

A `physicalLocation` object **MAY** contain a property named `artifactLocation` whose value is an `artifactLocation` object (§3.4) that represents the location of the artifact. If `artifactLocation` is absent, then `address` (§3.29.6) **SHALL** be present.

3.29.4 region Property

A `physicalLocation` object **MAY** contain a property named `region` whose value is a `region` object (§3.30) that represents a relevant portion of the artifact. In particular, if the `physicalLocation` object occurs within the `locations` property (§3.27.12) of a `result` object (§3.27), the `region` property **SHALL** specify the region within the artifact where the result was detected.

EXAMPLE 1: In this example, a `physicalLocation` object specifies the location where a result was detected. Its `region` property specifies the portion of the file where the result was detected.

```
{
  "locations": [
    {
      "physicalLocation": {
        "artifactLocation": {
          "uri": "ui/window.c",
          "uriBaseId": "SRCROOT"
        },
        "region": {
          "startLine": 42
        }
      }
    }
  ]
}
```

A result object (§3.27).
 # See §3.27.12.
 # A location object (§3.28).
 # See §3.28.3.
 # A artifactLocation object.
 # The region specifies the portion of the file
 # where the result was detected.

If the `physicalLocation` object specifies a location in a nested artifact, then the `region` property **SHALL** specify the location with respect to the innermost nested artifact.

EXAMPLE 2: If a result occurs in a C++ file contained in a compressed archive, then the region would represent the line and column number of the result with the C++ file. It would not represent (for example) the offset of the C++ file from the start of the archive.

If the `region` property is absent, the `physicalLocation` object refers to the entire artifact.

3.29.5 contextRegion Property

If a `physicalLocation` object contains a `region` property (§3.29.4), it **MAY** also contain a property named `contextRegion` whose value is a `region` object (§3.30) which specifies a region that is a proper superset of the region specified by the `region` property. If `region` is absent, `contextRegion` **SHALL** be absent.

NOTE: `contextRegion` enables a viewer to provide visual context when displaying a portion of an artifact. It can also be used to improve result matching.

EXAMPLE In this example, an analysis tool detected a result on line 42. The tool provides additional context for SARIF viewers by specifying a range of content surrounding the result line.

```
{
  "locations": [
    {
      "physicalLocation": {
        "artifactLocation": {
          "uri": "ui/window.c",
          "uriBaseId": "SRCROOT"
        },
        "region": {
          "startLine": 42,
          "snippet": {
            "text": "int n = m + 1;"
          }
        },
        "contextRegion": {
          "startLine": 41,
          "endLine": 43,
          "snippet": {
            "text": "int m;\nint n = m + 1\n\n"
          }
        }
      }
    }
  ]
}
```

A result object (§3.27).
 # See §3.27.12.
 # A location object (§3.28).
 # A physicalLocation object (§3.29).
 # An artifactLocation object (§3.4).
 # See §3.29.4.

3.29.6 address Property

A `physicalLocation` object **MAY** contain a property named `address` whose value is an `address` object (§3.32) that represents the physical or virtual address of this location. If `address` is absent, then `artifactLocation` (§3.29.3) **SHALL** be present.

3.30 region Object

3.30.1 General

A **region** object represents a region, that is, a contiguous portion of an artifact.

The **region** object defines both “text properties” and “binary properties.” The text properties represent a region as a contiguous range of zero or more characters (a “text region”). The binary properties represent a region as a contiguous range of zero or more bytes (a “binary region”).

A region **SHALL** contain at least one of `startLine`, `charOffset`, or `byteOffset`.

If `startLine` (§3.30.5) > 0 or `charOffset` (§3.30.10) ≥ 0 , this **region** object **SHALL** define a text region. If `byteOffset` (§3.30.11) ≥ 0 , this **region** object **SHALL** define a binary region. If a **region** object defines both a text region and a binary region, the text region and the binary region **SHALL** specify the identical range of bytes in the artifact, as determined by the artifact’s character encoding.

For regions in text artifacts, a **region** object **SHOULD** define a text region and **MAY** also define a binary region; it **SHALL** define either a text region or a binary region or both.

For regions in binary artifacts, a **region** object **SHALL** define a binary region and **SHALL NOT** define a text region.

If any text properties are present, enough text properties **SHALL** be present to fully specify a text region (see §3.30.2). If any binary properties are present, then enough binary properties **SHALL** be present to fully specify a binary region (see §3.30.3).

3.30.2 Text Regions

NOTE 1: The examples in this section assume a text file with the following contents:

```
abcd\r\nefg\r\nhijk\r\nlmn\r\n
```

Breaking the lines for the sake of readability, the contents are:

```
abcd\r\n
efg\r\n
hijk\r\n
lmn\r\n
```

The file contains four lines, each of which ends with the two-character newline sequence “`\r\n`”, which is explicitly displayed for clarity.

The line number of the first line in a text artifact **SHALL** be 1. The column number of the first character in each line **SHALL** be 1. The character offset of the first character in the artifact **SHALL** be 0.

The values of text properties **SHALL NOT** depend on the presence or absence of a byte order mark (BOM) at the start of the artifact.

Column numbers are expressed in the measurement unit specified by `theRun.columnKind` (§3.14.27).

A SARIF viewer **MAY** choose to present column numbers that match the visual offset of each character from the beginning of the line. These “visual” column numbers might not match the column numbers contained in the SARIF file.

NOTE 2: Such a mismatch might occur if, for example, the line contains a tab character, or an accented character represented by a base character plus a combining character.

A text artifact’s character encoding determines the number of bytes that represent each character, and therefore determines the range of bytes represented by a text region. A SARIF consumer **SHALL** consider an artifact to have the encoding specified by `artifact.encoding` (§3.24.9), if present, or else by `theRun.defaultEncoding` (§3.14.24), if present.

If neither is present, the consumer **MAY** use any heuristic or procedure to determine the encoding, including (for example) prompting the user.

NOTE 3: If a consumer incorrectly determines an artifact's encoding, it might not display the artifact correctly. For example, when it attempts to highlight a region, it might highlight an incorrect range of characters.

A text region **MAY** be specified in two ways:

- By means of the “line/column” properties `startLine` (§3.30.5), `startColumn` (§3.30.6), `endLine` (§3.30.7), and `endColumn` (§3.30.8).
- By means of the “offset/length” properties `charOffset` (§3.30.9) and `charLength` (§3.30.10).

A text region **SHALL** specify both its start (the location of its first character) and its end (the location of its last character).

NOTE 4: The end of a text region does not have to be specified explicitly if the default values for `endLine`, `endColumn`, and/or `charLength` correctly describe the region.

A text region does not include the character specified by `endColumn` (see §3.30.8).

EXAMPLE 1: The following regions (among others) all specify the range of characters "bc".

```
{
  "startLine": 1,
  "startColumn": 2,
  "endLine": 1,
  "endColumn": 4    # The region excludes the character at endColumn.
}

{
  "charOffset": 1,
  "charLength": 2
}

{
  "startLine": 1,
  "startColumn": 2,
  "endLine": 1,
  "endColumn": 4,
  "charOffset": 1,
  "charLength": 2
}
```

EXAMPLE 2: The following region is invalid, even though it might appear to specify the same range of characters "bc" as in > EXAMPLE 1:

```
{
  "startLine": 1,
  "charOffset": 1,    # Specifies the "b"
  "endColumn": 4      # Specifies the column one past the "c"
}
```

This is because the line/column properties and the offset/length properties, taken independently, specify different regions:

- "startColumn" is absent, and so defaults to 1 (see §3.30.6).
- "endLine" is absent, and so defaults to "startLine", which in this example is 1 (see §3.30.7).
- "charLength" is absent, and so defaults to 0 (see §3.30.10).

In summary, the above region is equivalent to the region

```
{
  "startLine": 1,
  "startColumn": 1,
  "endLine": 1,
  "endColumn": 4,

  "charOffset": 1,
  "charLength": 0
}
```


Now we can see that the line/column properties represent the range of characters "abc", while the offset/length properties represent an insertion point before the character "b" (see §3.30.10). Those two regions are not the same, and so the region is invalid.

If a region spans one or more lines, it **SHALL** include the newline sequences of all but the last line in the region.

NOTE 5: This is not an independent requirement; it is a consequence of the specification for the default value of `endColumn`.

EXAMPLE 3: The region

```
{ "startLine": 2 }
```

includes the characters "efg".

EXAMPLE 4: The region

```
{ "startLine": 2, "endLine": 3 }
```

includes the characters "efg\r\nhijk".

To specify an entire line together with its trailing newline sequence, specify the region's end point to be column 1 on the next line.

NOTE 6: This is again a consequence of the specification of `endColumn`, which states that it specifies the character one past the end of the region.

EXAMPLE 5: The region

```
{ "startLine": 2, "endLine": 3, "endColumn": 1 }
```

includes the characters "efg\r\n".

A region of length 0 is referred to as an "insertion point." An insertion point **MAY** be specified either by specifying `charLength` as 0, or by specifying the same values for `startColumn` and `endColumn`.

NOTE 7: Once more, this is again a consequence of the specification of `endColumn`.

EXAMPLE 6: These regions (among others) specify an insertion point before the "b" on line 1.

```
{ "startLine": 1, "startColumn": 2, "endColumn": 2 }
{ "charOffset": 1, "charLength": 0 }
```

EXAMPLE 7: These regions (among others) specify an insertion point at the beginning of the file:

```
{ "startLine": 1, "startColumn": 1, "endColumn": 1 }
{ "charOffset": 0, "charLength": 0 }
```

To specify an insertion point after the last character in an artifact, set `endLine` to the number of the last line in the artifact, and set `endColumn` to a value one greater than the number of characters on the line, *including* any trailing newline sequence.

EXAMPLE 8: These regions (among others) specify an insertion point at the very end of the file. Note that the last line contains the five characters (including the newline sequence) "lmn\r\n".

```
{ "startLine": 4, "startColumn": 6, "endColumn": 6 }
```

```
{ "charOffset": 22, "charLength": 0 }
```

3.30.3 Binary Regions

The byte offset of the first byte in an artifact **SHALL** be 0.

To specify a byte region, at least `byteOffset` (§3.30.11) **SHALL** be present. `byteLength` (§3.30.12) **MAY** also be present. `byteOffset` specifies the start of the region. `byteLength` specifies the region's length and thereby, indirectly, its end. A `byteLength` value of 0 represents an insertion point before the byte specified by `byteOffset`.

3.30.4 Independence of Text and Binary Regions

The text-related and binary-related properties in a `region` object **SHALL** be treated independently. That is, the value of a text-related property **SHALL NOT** be inferred from the value of any set of binary-related properties, and *vice versa*.

EXAMPLE 1: This example is based on the sample text file shown in NOTE 1 of §3.30.2. It represents invalid SARIF because the text-related and binary-related properties are inconsistent. At first glance they appear to be consistent because the byte at offset 2 is indeed on line 1:

```
{ "startLine": 1, "byteOffset": 2, "byteLength": 6 }
```

However, because the default values for the missing text-related properties are determined entirely from the existing text-related properties, and independently of any binary-related properties, this region is in fact equivalent to this one:

```
{
  "startLine": 1,
  "startColumn": 1, // Missing startColumn defaults to 1.
  "endLine": 1,    // Missing endLine defaults to startLine.
  "endColumn": 5,  // Missing endColumn defaults to (length of endLine + 1),
                  // exclusive of newline sequence.
  "byteOffset": 2,
  "byteLength": 6
}
```

This makes it clear that the text-related and binary-related properties represent different ranges of bytes, and therefore the region is invalid.

3.30.5 startLine Property

When a `region` object represents a text region specified by line/column properties, it **SHALL** contain a property named `startLine` whose value is a positive integer equal to the line number of the line containing the first character in the region.

3.30.6 startColumn Property

When a `region` object represents a text region specified by line/column properties, it **MAY** contain a property named `startColumn` whose value is a positive integer equal to the column number of the first character in the region.

If `startColumn` is absent, it **SHALL** default to 1.

3.30.7 endLine Property

When a `region` object represents a text region specified by line/column properties, it **MAY** contain a property named `endLine` whose value is a positive integer equal to the line number of the line containing the last character in the region.

If `endLine` is absent, its value **SHALL** default to `startLine`.

3.30.8 **endColumn** Property

When a **region** object represents a text region specified by line/column properties, it **MAY** contain a property named **endColumn** whose value is an integer whose value is one greater than the column number of the last character in the region.

If **endColumn** is absent, it **SHALL** default to a value one greater than the column number of the last character on the line, excluding any newline sequence.

3.30.9 **charOffset** Property

When a **region** object represents a text region specified by offset/length properties, it **SHALL** contain a property named **charOffset** whose value is an integer equal to the zero-based character offset of the first character in the region from the beginning of the artifact. If **charOffset** is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set).

3.30.10 **charLength** Property

When a **region** object represents a text region specified by offset/length properties, it **MAY** contain a property named **charLength** whose value is a non-negative integer equal to the number of characters in the region.

If **charLength** is absent, it **SHALL** default to 0, which **SHALL** be interpreted as an insertion point at the position specified by **charOffset** (§3.30.9)

The sum of **charOffset** and **charLength** **SHALL** be greater than or equal to 0 and less than or equal to the number of characters in the artifact.

A region whose **charOffset** is equal to the number of characters in the artifact and whose **charLength** is 0 is permitted and **SHALL** represent an insertion point at the end of the artifact.

3.30.11 **byteOffset** Property

When a **region** object represents a binary region, it **SHALL** contain a property named **byteOffset** whose value is an integer equal to the zero-based byte offset of the first byte in the region from the beginning of the artifact. If **byteOffset** is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set).

3.30.12 **byteLength** Property

When a **region** object represents a binary region, it **MAY** contain a property named **byteLength** whose value is an integer equal to the number of bytes in the region. If **byteLength** is absent, it **SHALL** default to 0, which **SHALL** be interpreted as an insertion point at the position specified by **byteOffset** (§3.30.11).

The sum of **byteOffset** and **byteLength** **SHALL** be greater than or equal to 0 and less than or equal to the number of bytes in the artifact.

A region object whose **byteOffset** equals the number of bytes in the artifact and whose **byteLength** is 0 is permitted, and **SHALL** represent an insertion point at the end of the artifact.

3.30.13 **snippet** Property

A region object **MAY** contain a property named **snippet** whose value is an **artifactContent** object (§3.3) representing the portion of the artifact specified by the **region** object.

NOTE: The **snippet** property has various uses:

- It allows a SARIF viewer to present the contents of the region even if the artifact from which it was taken is

not available.

- It also allows an end user examining a SARIF log file to see the relevant content without opening another file.
- It can be used to improve result matching.

3.30.14 **message** Property

A **region** object **MAY** contain a property named **message** whose value is a **message** object (§3.11) containing a message relevant to the region.

A SARIF viewer **MAY** display this message when the user interacts with the region. (For example, if the user hovers over the region with the mouse, the viewer might present the message as hover text.)

3.30.15 **sourceLanguage** Property

If the **region** object represents a portion of a text artifact that contains source code, it **MAY** contain a property named **sourceLanguage** whose value is a hierarchical string (§3.5.4) that specifies the programming language in which this portion of the source code is written. If the **region** object does not represent a portion of a text artifact containing source code, then **sourceLanguage** **SHALL** be absent.

For the remainder of this section, we assume that the **region** object represents a portion of a text artifact that contains source code.

NOTE: This property is intended to help SARIF viewers to render code snippets (§3.30.13) with appropriate syntax coloring. It is intended for use in mixed-language files, such as HTML files that contain JavaScript™. For more information about this usage, see §3.24.10.

if **sourceLanguage** is absent, it **SHALL** default to the value of the **sourceLanguage** property (§3.24.10) of the **artifact** object (§3.24) which describes the artifact that contains the region. **artifact.sourceLanguage** in turn defaults to **theRun.defaultSourceLanguage** (§3.14.25). If all three of **region.sourceLanguage**, **artifact.sourceLanguage**, and **theRun.defaultSourceLanguage** are absent, the source language of the region object **SHALL** be taken to be unknown. In that case, a SARIF viewer **MAY** use any method or heuristic to determine the region's source language, for example, by examining the file's file name extension or MIME type, or by prompting the user.

For conventions and practices regarding the value of this property, see §3.24.10.2.

3.31 **rectangle** Object

3.31.1 **General**

A **rectangle** object specifies a rectangular area within an image. When a SARIF viewer displays an image, it **MAY** indicate the presence of these areas, for example, by highlighting them or surrounding them with a border.

3.31.2 **top**, **left**, **bottom**, and **right** Properties

A **rectangle** object **SHALL** contain properties named **top**, **left**, **bottom**, and **right**, each of which contains a number (as defined by the JSON Schema standard [JSONSchema01]) specifying one of the coordinates of the rectangle within the image. These properties **SHALL** be measured in the image format's natural units (for example, pixels for raster-based image formats). These values **MAY** be positive or negative, depending on the natural coordinate system of the image format. They **MAY** increase either from left to right or from right to left, and either from top to bottom or from bottom to top, again depending on the natural coordinate system of the image format.

NOTE: A number in JSON schema can take a variety of forms, including simple integers (42) and floating-point numbers (3.14).

3.31.3 message Property

A `rectangle` object **SHOULD** contain a property named `message` whose value is a `message` object (§3.11) containing a message relevant to this area of the image.

A SARIF viewer **MAY** display this message when the user interacts with the area. For example, if the user hovers over the area with the mouse, the viewer might present the message as hover text.

3.32 address Object

3.32.1 General

An address object describes a physical or virtual address, or a range of addresses, in an “addressable region” (memory or a binary file).

3.32.2 Parent-child Relationships

address objects can be linked by their `parentIndex` properties (§3.32.13) to form a chain in which each address is specified as an offset from a “parent” object which we refer to as `theParent`.

EXAMPLE 1: In this example, the location of the Sections region of a Windows® Portable Executable file [PE] is expressed as an offset from the start of the module. The location of the `.text` section is in turn expressed as an offset from Sections.

```
{
  "addresses": [
    {
      "name": "Multitool.exe",
      "kind": "module",
      "absoluteAddress": 1024
    },
    {
      "name": "Sections",
      "kind": "header",
      "parentIndex": 0,
      "offsetFromParent": 376,
      "absoluteAddress": 1400,
      "relativeAddress": 376
    },
    {
      "name": ".text",
      "kind": "section",
      "parentIndex": 1,
      "offsetFromParent": 136,
      "absoluteAddress": 1536,
      "relativeAddress": 512
    }
  ],
  ...
}
```

3.32.3 Absolute Address Calculation

Each address object has an associated value called its “absolute address” which is the offset of the address from the start of the addressable region. The absolute address is calculated by executing the function `CalculateAbsoluteAddress` defined below on `thisObject` or by any procedure with the same result.

This procedure assumes that the `offsetFromParent` (§3.32.8) and `parentIndex` (§3.32.13) properties are either both present or both absent; if this is not the case, the SARIF file is invalid.

FUNCTION `CalculateAbsoluteAddress(addr)`

IF `addr.absoluteAddress` exists THEN

RETURN `addr.absoluteAddress`

ELSE IF `addr.parentIndex` exists THEN

 LET `theParent` = the parent object (see §3.32.2) of `addr`

 RETURN `addr.offsetFromParent + CalculateAbsoluteAddress(theParent)`

ELSE

 ERROR “Absolute address cannot be determined”.

If `CalculateAbsoluteAddress(thisObject)` or any of its recursive invocations encounters an ERROR, the absolute address cannot be determined.

If both `absoluteAddress` and `offsetFromParent` exist, then `absoluteAddress` **SHALL** equal the value that `CalculateAbsoluteAddress` would have returned if `absoluteAddress` were absent, if `CalculateAbsoluteAddress` would have returned successfully in that circumstance.

3.32.4 Relative Address Calculation

Each address object has an associated value called its “relative address” which is the offset of the address from the address of the top-most object in its parent chain. The relative address is calculated by executing the function `CalculateRelativeAddress` defined below on `thisObject` or by any procedure with the same result.

This procedure assumes that the `offsetFromParent` (§3.32.8) and `parentIndex` (§3.32.13) properties are either both present or both absent; if this is not the case, the SARIF file is invalid.

FUNCTION `CalculateRelativeAddress(addr)`

 IF `addr.relativeAddress` exists THEN

 RETURN `addr.relativeAddress`

 ELSE IF `addr.parentIndex` exists THEN

 LET `theParent` = the parent object (see §3.32.2) of `addr`

 RETURN `addr.offsetFromParent + CalculateRelativeAddress(theParent)`

 ELSE

 RETURN 0

If `CalculateRelativeAddress(thisObject)` or any of its recursive invocations encounters an ERROR, the relative address cannot be determined.

If both `relativeAddress` and `offsetFromParent` exist, then `relativeAddress` **SHALL** equal the value that `CalculateRelativeAddress` would have returned if `relativeAddress` were absent, if `CalculateRelativeAddress` would have returned successfully in that circumstance.

3.32.5 **index** Property

Depending on the circumstances, an address object either **MAY**, **SHALL NOT**, or **SHALL** contain a property named `index` whose value is the array index (§3.7.4) within `theRun.addresses` (§3.14.18) of an address object that provides the properties for `thisObject`. We refer to the object in `theRun.addresses` as the “cached object.”

If `thisObject` is an element of `theRun.addresses`, then `index` **MAY** be present. If present, its value **SHALL** be the index of `thisObject` within `theRun.addresses`.

Otherwise, if `theRun.addresses` is absent, or if it does not contain a cached object for `thisObject`, then `index` **SHALL NOT** be present.

Otherwise (that is, if `thisObject` belongs to a result, and `theRun.addresses` contains a cached object for `thisObject`), then `index` **SHALL** be present, and its value **SHALL** be the array index within `theRun.addresses` of the cached object.

If `index` is present, `thisObject` **SHALL** take all properties present on the cached object. If `thisObject` contains any properties other than `index`, they **SHALL** equal the corresponding properties of the cached object.

NOTE 1: This allows a SARIF producer to reduce the size of the log file by reusing the same address object in multiple results.

NOTE 2: For examples of the use of an `index` property to locate a cached object, see §3.38.2.

3.32.6 **absoluteAddress** Property

An address object **MAY** contain a property named `absoluteAddress` whose value is a non-negative integer containing the absolute address (see §3.32.3) of `thisObject`.

If `absoluteAddress` is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set).

3.32.7 **relativeAddress** Property

If `parentIndex` (§3.32.13) is present, an address object **MAY** contain a property named `relativeAddress` whose value, if present, is an integer containing the relative address (see §3.32.4) of `thisObject`.

If `parentIndex` is absent, `relativeAddress` **SHALL** be absent.

If `relativeAddress` is absent, it **SHALL** default to `null`, which indicates that the value is unknown (not set).

3.32.8 **offsetFromParent** Property

If `parentIndex` (§3.32.13) is present, an address object **MAY** contain a property named `offsetFromParent` whose value, if present, is an integer containing the offset of this address from the absolute address of `theParent` (see §3.32.2). This is the case even if the absolute address of the parent cannot be determined by the procedure in §3.32.3.

NOTE 1: The rationale is that the absolute address always exists, even if the log file does not contain enough information to determine it, so it is always sensible to talk about an offset from that address.

If `parentIndex` is absent, `offsetFromParent` **SHALL** be absent.

If `offsetFromParent` is absent, it **SHALL** default to `null`, which indicates that the value is unknown (not set).

3.32.9 **length** Property

An address object **MAY** contain a property named **length** whose value, if present, is an integer whose absolute value specifies the number of bytes in the range of addresses specified by this object.

A negative value for **length** **SHALL** mean that the data structure being described grows from higher addresses towards lower addresses (as, for example, is often the case for a stack).

If **length** is absent, it **SHALL** default to **null**, which indicates that the value is unknown (not set).

3.32.10 **name** Property

An address object **MAY** contain a property named **name** whose value is a string containing the name of this address.

3.32.11 **fullyQualifiedName** Property

An address object **MAY** contain a property named **fullyQualifiedName** whose value is a string containing the fully qualified name of this address.

EXAMPLE 1: "fullyQualifiedName": "MyDll.dll+0x47"

This name consists of two components. The first component is the name of the address at which the module was loaded into memory. The second component represents an offset from that address.

3.32.12 **kind** Property

An address object **MAY** contain a property named **kind** whose value is a string that specifies the kind of addressable region in which this address is located.

When possible, SARIF producers **SHOULD** use the following values, with the specified meanings.

- "data": An addressable location containing non-executable data.
- "header": A data structure that precedes one or more addressable regions and specifies the layout and location of objects within the address space.
- "function": An addressable region, possibly named, containing a sequence of instructions that perform a specified task.
- "instruction": An addressable location containing executable code.
- "page": An addressable region whose contents can be moved between primary and secondary storage.
- "section": A named region of a file containing executable code or data, which in some circumstances is loaded into memory.
- "segment": 1. A data structure in a binary that describes a region of memory, specifying its addressing and permissions information, as well as information about which sections are to be loaded into the segment. 2. A region of memory whose contents are specified by the information in a segment defined in a binary, or by the operating system.
- "stack": An addressable region containing a call stack.
- "stackFrame": An addressable region containing a single frame from within a call stack.
- "module": The location at which a module was loaded.
- "table": An addressable region with a distinct purpose and a specified internal organization

The definitions of some of these "kind" values vary across operating systems. A SARIF producer **SHOULD** use the term most appropriate for the target operating system.

Although a function does contain executable code, the value "function" **SHOULD** be used for the address of the start of a function, because it is more specific. The value "instruction" **SHOULD** be used for an address within the body of a function.

If none of these values are appropriate, a SARIF producer **MAY** use any value.

3.32.13 `parentIndex` Property

If `theParent` exists (that is, if `thisObject` is expressed as an offset from some other address), then an `address` object **SHALL** contain a property named `parentIndex` whose value is the array index (§3.7.4) of `theParent` within `theRun.addresses` (§3.14.18).

If `theParent` does not exist, then `parentIndex` **SHALL** be absent.

3.33 `logicalLocation` Object

3.33.1 General

A `logicalLocation` object describes a logical location. A logical location is a location specified by a programmatic construct such as a namespace, a type, or a method, without regard to the physical location where the construct occurs.

`logicalLocation` objects occur in two places: as array elements of `run.logicalLocations` (§3.14.17) and as array elements of `location.logicalLocations` (§3.28.4).

3.33.2 Logical Location Naming Rules

Every logical location has a “fully qualified logical name” (more briefly, a “fully qualified name”) that fully specifies the programmatic construct to which it refers. When programmatic constructs are nested (such as a method within a class within a namespace), the fully qualified name is typically a hierarchical identifier such as `"N.C.F(void)"` or `"N::C::F(void)"`. We refer to the rightmost component of this hierarchical identifier as the “logical name” (more briefly, the “name”) of the logical location.

Whenever possible, logical names and fully qualified logical names **SHOULD** conform to the syntax of the programming language in which the programmatic construct specified by the logical location was expressed.

EXAMPLE 1: The fully qualified logical name of the C++ method `f(void)` in class `C` in namespace `N` is `"N::C::f(void)"`. Its logical name is `"f(void)"`.

This is not always possible, for two reasons:

- For certain values of `logicalLocation.kind` (§3.33.7), there is no language syntax to specify the fully qualified name.

EXAMPLE 2: Suppose the logical location is the local variable `pBuffer` in the C++ method `"N::C::f(void)"`. `logicalLocation.kind` is `"variable"`. There is no way to express the fully qualified name in C++. The SARIF producer might choose a fully qualified name such as `"N::C::f(void)?pBuffer"`.

- For other values of `logicalLocation.kind`, it is sometimes but not always possible to express the logical location in language syntax.

EXAMPLE 3: Suppose the logical location is the anonymous callback function in this JavaScript™ function:

```
function click_it() {
  $("button").click(function(){
    alert("Clicked");
  });
}
```

```
}
```

`logicalLocation.kind` is "function", for which it is sometimes possible to specify a fully qualified name. But there is no language syntax to express the name of an anonymous callback. The SARIF producer might choose a fully qualified name such as "click_it?anon-1".

3.33.3 `index` Property

Depending on the circumstances, a `logicalLocation` object either **MAY**, **SHALL NOT**, or **SHALL** contain a property named `index` whose value is the array index (§3.7.4) within `theRun.logicalLocations` (§3.14.17) of a `logicalLocation` object that provides the properties for `thisObject`. We refer to the object in `theRun.logicalLocations` as the “cached object.”

If `thisObject` is an element of `theRun.logicalLocations`, then `index` **MAY** be present. If present, its value **SHALL** be the index of `thisObject` within `theRun.logicalLocations`.

Otherwise, if `theRun.logicalLocations` is absent, or if it does not contain a cached object for `thisObject`, then `index` **SHALL NOT** be present.

Otherwise (that is, if `thisObject` belongs to a result, and `theRun.logicalLocations` contains a cached object for `thisObject`), then `index` **SHALL** be present, and its value **SHALL** be the array index within `theRun.logicalLocations` of the cached object.

If `index` is present, `thisObject` **SHALL** take all properties present on the cached object. If `thisObject` contains any properties other than `index`, they **SHALL** equal the corresponding properties of the cached object.

NOTE 1: This allows a SARIF producer to reduce the size of the log file by reusing the same `logicalLocation` object in multiple results.

NOTE 2: For examples of the use of an `index` property to locate a cached object, see §3.38.2.

3.33.4 `name` Property

A `logicalLocation` object **SHOULD** contain a property named `name` whose value is the logical name of the programmatic construct specified by this object. For example, this property might contain the name of a class or a method.

The `name` property **SHALL** be suitable for display and **SHALL** follow the naming rules for logical names described in §3.33.2.

NOTE: A C++ analysis tool might have available both the source code form of a function name and the compiler’s “decorated” function name (which encodes the function signature in a manner that is compiler-dependent and not easily readable). The tool would place the source code form of the function name in the `name` property, and the decorated name in the `decoratedName` property (§3.33.6).

EXAMPLE 1: In this C++ example, the fully qualified name is "b::c(float)", so "name" is the rightmost component, "c(float)".

```
{
    # A logicalLocation object.
    "name": "c(float)",
    "fullyQualifiedName": "b::c(float)", # See §3.33.5.
    "kind": "function"                 # See §3.33.7
}
```

3.33.5 `fullyQualifiedName` Property

Depending on the circumstances, a `logicalLocation` object either **SHOULD** or **MAY** contain a property named `fullyQualifiedName` whose value is the fully qualified name of the logical location. This name **SHALL** follow the naming rules for fully qualified names described in §3.33.2.

If this `logicalLocation` object represents a top-level logical location, then `fullyQualifiedName` **MAY** be present. If present, it **SHALL** equal `name`; if absent, it **SHALL** default to `name`. If this object does not represent a top-level logical location, `fullyQualifiedName` **SHOULD** be present.

It is possible for two or more distinct logical locations to have the same fully qualified name.

NOTE: This is an extremely rare corner case.

EXAMPLE 1: Suppose a tool analyzes two C++ source files:

```
// file1.cpp
namespace A {
  class B {
  }
}

// file2.cpp
namespace A {
  namespace B {
    class C {
    }
  }
}
```

These could not coexist in the same compilation, but there is no reason two such source files could not exist. If the tool detected one result in `class B` in *file1.cpp*, and another result in `namespace B` in *file2.cpp*, the `fullyQualifiedName` for both would be `A::B`. However, they would be distinguished by their `parentIndex` properties:

```
"logicalLocations": [
  {
    "name": "B",
    "fullyQualifiedName": "A::B",
    "kind": "namespace",
    "parentIndex": 1
  },
  {
    "name": "A",
    "kind": "namespace"
  },
  {
    "name": "B",
    "fullyQualifiedName": "A::B",
    "kind": "type",
    "parentIndex": 3
  },
  {
    "name": "A",
    "kind": "namespace"
  }
]
```

NOTE: There are a few reasons the `fullyQualifiedName` property exists, even though the information it contains can be reconstructed from the `name` properties of this object and its parent objects in `run.logicalLocations`:

- `run.logicalLocations` might not be present.
- It allows a SARIF viewer to display the logical location in a way that is easily understood by users.
- As mentioned in §3.28.1, `fullyQualifiedName` is also particularly convenient for fingerprinting, although the more detailed information in `run.logicalLocations` could be used instead.
- It relieves viewers from having to format the logical location from the more detailed information in `run.logicalLocations`.
- It is useful for producing readable in-source suppressions (for example, “suppress all instance of rule CA2101 in the class `NamespaceA.NamespaceB.ClassC`”).

3.33.6 `decoratedName` Property

A `LogicalLocation` object **MAY** contain a property named `decoratedName` whose value is a string containing the compiler's internal representation of the logical location associated with this `location` object.

NOTE: Some compilers refer to this representation as a “mangled name.” It typically encodes the function's name, signature, return type, and the class and namespace (if any) to which it belongs.

EXAMPLE 1: In this example, the `decoratedName` property contains a “mangled” name emitted by a C++ compiler:

```
{                                     # A LogicalLocation object
  "name": "c(float)",
  "fullyQualifiedName": "b::c(float)",
  "decoratedName": "?c@b@AAGXM@Z"
}
```

3.33.7 `kind` Property

A `LogicalLocation` object **SHOULD** contain a property named `kind` whose value is one of the following strings, if any of those strings accurately describes the construct identified by this object.

Although the values suggested here are useful in the specified categories (for example, "member" is useful in describing executable code), they **MAY** be used in other contexts as appropriate.

- Values for locations within executable code:
 - "function"
 - "member"
 - "module"
 - "namespace"
 - "resource"
 - "type"
 - "returnType"
 - "parameter"
 - "variable"
- Values for locations within XML or HTML documents:
 - "element"
 - "attribute"
 - "text"
 - "comment"
 - "processingInstruction"
 - "dtd"
 - "declaration"

EXAMPLE 1: Consider the following XML document:

```
1. <?xml version="1.0"?>
2. <orders>
3.   <order number="">
4.     <total>-$3.25</total>
5.   </order>
6. </orders>
```

Suppose that an analysis tool detects errors on line 3 (the order number is blank) and line 4 (the total is negative). It might represent the logical locations of these errors as XML Paths (although this is not required), as follows:

```
{
  "results": [
    {
      "locations": [
        {
          "logicalLocations": [
            {
              "fullyQualifiedName": "/orders/order[1]/@number",
              "index": 2
            }
          ]
        }
      ],
      ...
    },
    {
      "locations": [
        {
          "logicalLocations": [
            {
              "fullyQualifiedName": "/orders/order[1]/total/text()",
              "index": 3
            }
          ]
        }
      ],
      ...
    }
  ],
  "logicalLocations": [
    {
      "name": "orders",
      "fullyQualifiedName": "/orders",
      "kind": "element"
    },
    {
      "name": "order[1]",
      "fullyQualifiedName": "/orders/order[1]",
      "kind": "element",
      "parentIndex": 0
    },
    {
      "name": "number",
      "fullyQualifiedName": "/orders/order[1]/@number",
      "kind": "attribute",
      "parentIndex": 1
    },
    {
      "name": "text",
      "fullyQualifiedName": "/orders/order[1]/total/text()",
      "kind": "text",
      "parentIndex": 1
    }
  ]
}
```

- Values for locations within JSON documents:

- "object"
- "array"
- "property"
- "value"

EXAMPLE 2: Consider the following JSON document:

```
1. {
2.   "orders": [
3.     {
4.       "productIds": [ "A-101", "", "A-223" ],
5.       "total": "-$3.25"
6.     }
7.   ]
8. }
```

Suppose that an analysis tool detects errors on line 4 (one of the product ids blank) and line 5 (the total is negative). It might represent the logical locations of these errors as JSON Pointers (although this is not required), as follows:

```
{
  "results": [
    {
      "locations": [
        {
          "logicalLocations": [
            {
              "fullyQualifiedName": "/orders/order[1]/@number",
              "index": 2
            }
          ]
        }
      ],
      ...
    },
    {
      "locations": [
        {
          "logicalLocations": [
            {
              "fullyQualifiedName": "/orders/order[1]/total/text()",
              "index": 3
            }
          ]
        }
      ],
      ...
    }
  ],
  "logicalLocations": [
    {
      "name": "orders",
      "fullyQualifiedName": "/orders",
      "kind": "element"
    },
    {
      "name": "order[1]",
      "fullyQualifiedName": "/orders/order[1]",
      "kind": "element",
      "parentIndex": 0
    },
    {
      "name": "number",
      "fullyQualifiedName": "/orders/order[1]/@number",
      "kind": "attribute",
      "parentIndex": 1
    },
    {
      "name": "text",
      "fullyQualifiedName": "/orders/order[1]/total/text()",
      "kind": "text",
      "parentIndex": 1
    }
  ]
}
```

```

    {
      # A logicallocation object (§3.33).
      "fullyQualifiedName": "/orders/0/productIds/1",
      "index": 3
    }
  ]
},
{
  "locations": [
    {
      "logicalLocations": [
        {
          "fullyQualifiedName": "/orders/0/total",
          "index": 4
        }
      ]
    }
  ]
},
{
  "logicalLocations": [
    # See §3.14.17.
    # A logicalLocation object (§3.33).
    {
      "name": "orders",
      "fullyQualifiedName": "/orders",
      "kind": "array"
    },
    {
      "name": "0",
      "fullyQualifiedName": "/orders/0",
      "kind": "object",
      "parentIndex": 0
    },
    {
      "name": "productIds",
      "fullyQualifiedName": "/orders/0/productIds",
      "kind": "array",
      "parentIndex": 1
    },
    {
      "name": "1",
      "fullyQualifiedName": "/orders/0/productIds/1",
      "kind": "value",
      "parentIndex": 2
    },
    {
      "name": "total",
      "fullyQualifiedName": "/orders/0/total",
      "kind": "property",
      "parentIndex": 1
    }
  ]
}
}

```

If none of those strings accurately describes the construct, `kind` **MAY** contain any value specified by the analysis tool.

If a logical location is both a member and a type (for example, a nested class in C++ or C#), the value of `kind`, if present, **SHALL** be `"type"`.

NOTE: The purpose of this property is to help result management systems group results that occur in the same logical location. If one result specifies the logical location “namespace A”, and another result specifies the logical location “class A”, the difference in the `kind` property between the two results tells the result management system to sort them into different groups.

3.33.8 `parentIndex` Property

If this `logicalLocation` object represents a nested logical location, then it **SHALL** contain a property named `parentIndex` whose value is the array index (§3.7.4) of the parent `logicalLocation` object within `theRun.logicalLocations` (§3.14.17).

If this `Object` represents a top-level logical location, then `parentIndex` **SHALL** be absent.

NOTE: `parentIndex` makes it possible to navigate from the `logicalLocation` object representing a nested logical location to the `logicalLocation` objects representing each of its parent logical locations in turn, up to the top-level logical location.

EXAMPLE 1: In this example, the logical location `n::f(void)` is nested within the top-level logical location `n`. The `logicalLocation` object representing `n::f(void)` contains a `parentIndex` property that points to the

object representing `n`; the object representing `n` does not contain a `parentIndex` property.

```
{
  "logicalLocations": [
    {
      "name": "f(void)",
      "fullyQualifiedName": "n::f(void)",
      "kind": "function",
      "parentIndex": 1
    },
    {
      "name": "n",
      "kind": "namespace"
    }
  ]
}
```

A run object (§3.14).
See §3.14.17.
See §3.33.4.
See §3.33.5.
See §3.33.7.

3.34 locationRelationship Object

3.34.1 General

A `locationRelationship` object specifies one or more directed relationships from one `location` object (§3.28), which we refer to as the `Source`, to another one, which we refer to as the `Target`.

`locationRelationship` objects appear as elements of the `location.relationships` array (§3.28.7). The `location` object containing this property is the `Source`.

EXAMPLE 1: In this example, the location relationships specify that the file `f.h` in which the result was found is included by `g.h`, which is in turn included by `g.c`. Depending on the circumstances, it might or might not be useful to include both the `"includes"` and `"isIncludedBy"` relationships, as this example does for `g.h`.

```
{
  "locations": [
    {
      "id": 0,
      "physicalLocation": {
        "artifactLocation": {
          "uri": "f.h"
        },
        "region": {
          "startLine": 42
        }
      },
      "relationships": [
        {
          "target": 1,
          "kinds": [ "isIncludedBy" ]
        }
      ]
    },
    {
      "id": 1,
      "physicalLocation": {
        "artifactLocation": {
          "uri": "g.h"
        },
        "region": {
          "startLine": 17
        }
      },
      "relationships": [
        {
          "target": 0,
          "kinds": [ "includes" ]
        },
        {
          "target": 2,
          "kinds": [ "isIncludedBy" ]
        }
      ]
    },
    {
      "id": 2,
      "physicalLocation": {
        "artifactLocation": {
          "uri": "g.c"
        },
        "region": {
          "startLine": 8
        }
      },
      "relationships": [
        {
          "target": 1,
          "kinds": [ "includes" ]
        }
      ]
    }
  ],
  "relatedLocations": [
    {
      "id": 1,
      "physicalLocation": {
        "artifactLocation": {
          "uri": "g.h"
        },
        "region": {
          "startLine": 17
        }
      },
      "relationships": [
        {
          "target": 0,
          "kinds": [ "includes" ]
        },
        {
          "target": 2,
          "kinds": [ "isIncludedBy" ]
        }
      ]
    },
    {
      "id": 2,
      "physicalLocation": {
        "artifactLocation": {
          "uri": "g.c"
        },
        "region": {
          "startLine": 8
        }
      },
      "relationships": [
        {
          "target": 1,
          "kinds": [ "includes" ]
        }
      ]
    }
  ]
}
```

A result object (§3.27).
See §3.27.12.
A location object (§3.28).
See §3.28.2.
See §3.28.7
A locationRelationship object.
See §3.34.2.
See §3.34.3.
See §3.27.22.
The line that includes f.h.
The line that includes g.h.

```

    ]
  }
}

```

3.34.2 **target** Property

A `locationRelationship` object **SHALL** contain a property named `target` whose value is a non-negative integer which identifies the `target` (see §3.34.1) among all `location` objects (§3.28) in the `result` by virtue of being equal to `target.id` (§3.28.2).

NOTE: Negative values are forbidden because their use might suggest some non-obvious semantic difference between positive and negative values.

3.34.3 **kinds** Property

A `locationRelationship` object **MAY** contain a property named `kinds` whose value is an array of one or more unique (§3.7.3) strings each of which specifies a relationship between the `source` and the `target` (see §3.34.1). If `kinds` is absent, it **SHALL** default to [`"relevant"`] (see below for the meaning of `"relevant"`).

When possible, SARIF producers **SHOULD** use the following values, with the specified meanings.

- `"includes"`: The artifact identified by the `source` includes the artifact identified by the `target`.
- `"isIncludedBy"`: The artifact identified by the `source` is included by the artifact identified by the `target`.
- `"relevant"`: the `target` is relevant to the `source` in a way not covered by other relationship kinds.

If none of these values are appropriate, a SARIF producer **MAY** use any value.

NOTE: Although `"relevant"` is a catch-all for any relationship not described by the other values, a producer might still wish to define its own more specific values.

In particular, the values defined for `logicalLocation.kind` (§3.33.7) and `threadFlowLocation.kinds` (§3.38.8) might prove useful.

3.34.4 **description** Property

A `locationRelationship` object **MAY** contain a property named `description` whose value is a message object (§3.11) that describes the relationship.

3.35 **suppression** Object

3.35.1 **General**

A `suppression` object describes a request to suppress a result.

NOTE 1: The `suppression` object is valuable in compliance scenarios, where teams must show an auditor that they have looked at all results that corporate policy requires, and either fixed them or explicitly decided not to fix them. The `kind` property (§3.35.2) enables a review process that ensures that the engineering team agrees with the suppression, and makes the agreement explicit in the log file.

NOTE 2: The treatment of suppressed results depends on the development environment within which the log file is used, for example, a build system, an integrated development environment (IDE), or a result management system. Typically, development environments do not expose suppressed results to the user. For example, they do not include them in build log files, display them in error lists, or include them in bug counts.

3.35.2 **kind** Property

A `suppression` object **SHALL** contain a property named `kind` whose value is a string with one of the following values, with the specified meanings:

- `"inSource"`: The result is suppressed by a syntactic construct offered by the programming language.

EXAMPLE 1: The `SuppressMessage` attribute in the .NET Framework.

- `"external"`: The result is suppressed in an external, persistent store.

EXAMPLE 1: A database containing historical information about the results from analysis tools. Such a store might offer the ability to mark a result as “suppressed,” meaning that if the result is encountered again, it is to be ignored.

3.35.3 **status** Property

A `suppression` object **MAY** contain a property named `status` whose value is a string with one of the following values, with the specified meanings:

- `"accepted"`: The suppression is accepted.
- `"underReview"`: The engineering team is discussing the result to decide if they will suppress it.
- `"rejected"`: The engineering team decided not to suppress the result.

3.35.4 **location** Property

A `suppression` object **MAY** contain a property named `location` whose value is a `location` object (§3.28) that specifies the location where the suppression is persisted.

NOTE: In the common scenario, a suppression is represented by a source code construct (which we will refer to as a “suppression construct”) such as an attribute or a specially formatted comment at the location where the result was detected. In this scenario, `location` is unnecessary, although it is permitted, because an end user who navigates from the result to the source code location will see the suppression attribute or comment near the relevant code. Nevertheless, there are several scenarios where `location` is useful. Here are some examples:

When the suppression construct is placed in a separate compiled source file, `kind` (§3.35.2) is `"inSource"`, and `location.physicalLocation` (§3.28.3) specifies the location of the suppression attribute in that separate file.

Even when the suppression construct is adjacent to the result line, `location.physicalLocation` can be useful because it allows you to include in the log file a source code snippet containing the suppression construct, using `location.physicalLocation.region.snippet` (§3.29.4, §3.30.13).

When a tool detects a result within a method, but the suppression construct is applied to some higher-level construct such as the enclosing class, then `kind` is again `"inSource"`, `location.logicalLocation` (§3.28.4) can specify the construct to which the suppression was applied, and `location.physicalLocation` can still usefully specify the location of the suppression construct in the source file, since it is distant from the result.

In a similar case, a binary analysis tool that detected the suppression within an executable file’s metadata could provide `location.logicalLocation` even if it could not provide `location.physicalLocation`.

If a suppression is stored in a separate, non-compiled file, sometimes called a “sidecar file,” `kind` is `"external"`, and `location.physicalLocation` specifies the location of the suppression within the sidecar file. The sidecar file might even be another SARIF file.

If a suppression is stored in a database, `kind` is again `"external"`, and `location.physicalLocation` might specify the URI of a query that returns the database information that describes the suppression.

3.35.5 **guid** Property

A `suppression` object **MAY** contain a property named `guid` whose value is a GUID-valued string (§3.5.3).

NOTE: This can be used, for example, to link a `suppression` object in a SARIF file to suppression information in a result management system's database.

3.35.6 `justification` Property

A `suppression` object **MAY** contain a property named `justification` whose value is a user-supplied string that explains why the result was suppressed.

This is one of the few properties that contain textual content supplied by a user rather than by a tool or taxonomy (see §3.19.3) vendor. As such, it might contain undesirable content. Therefore, SARIF consumers **SHOULD** exercise appropriate caution when displaying, sharing, or publishing this information.

NOTE: This property exists because the information it contains is commonly made available by existing suppression mechanisms such as the `SuppressMessage` attribute in the .NET Framework.

3.35.7 `justificationType` Property

A suppression is a filter on an existing result. The free-form `justification` field for arbitrary textual descriptions of a suppression is not easy to parse or to map to finite states. The `justificationType` property is an enumeration providing a useful set of tags to help sort and differentiate suppressions. As with other areas of SARIF design, such buckets assist in routing information to specific actors in end-to-end result management systems.

The `justificationType` property is an enumeration with the following five values:

`FixDeferred`
`NotForRelease`
`RiskAccepted`
`ToolNoise`
`VulnerabilityNotFeasible`

The suggested situations represented by these five enumeration values are the following:

`ToolNoise` for example filters a result because it comprises a false positive. The primary responder to this class of suppression is a tool vendor (with other actual code owners in a secondary role to guarantee the finding is, in fact, incorrect).

`VulnerabilityNotFeasible` designates a vulnerability that looks accurate on surface which cannot be realized or exploited in production due to factors or contexts that are not (or cannot be) considered by the quality tool. The appropriate responders are other code owners to confirm a vulnerability does not impact production (with tool vendors in a secondary review role to look for opportunities to improve or refine analysis).

`NotForRelease` filters a result because it fired against code that does not ship (and therefore affords no quality or security risk). The appropriate responder/reviewer for this class of suppression might be an automation owner who can adjust tool configuration to not scan non-shipping code.

`FixDeferred` acknowledges a result as a true positive but simply requests time to resolve. The appropriate responders are security reviewers and leads accountable for prioritizing or scheduling work items.

`RiskAccepted` acknowledges a result as a true positive but definitively proposes not to act on it. Appropriate responders include security reviewers and leads accountable for signing off on quality and risk.

The buckets represented through the enumeration values aim to be a clear, minimal set that together handle prominent routing and response use cases. It is possible, for example, that `ToolNoise` and `VulnerabilityNotFeasible` could be collapsed into a single `FalsePositive` designation. The rationale for preserving both is the distinction between the primary responder for the two cases (tool vendor and code owner).

3.36 codeFlow Object

3.36.1 General

A `codeFlow` object describes the progress of one or more programs through one or more thread flows, which together lead to the detection of a problem in the system being analyzed. We define a thread flow as a temporally ordered sequence of code locations occurring within a single thread of execution, typically an operating system thread or a fiber. The thread flows in a code flow **MAY** lie within a single process, within multiple processes on the same machine, or within multiple processes on multiple machines.

EXAMPLE

```
{
  "codeFlows": [
    {
      "message": {
        "text": "...",
      },
      "threadFlows": [
        {
          "id": "thread-123",
          "message": {
            "text": "...",
          },
          "locations": [
            {
              "location": {
                "physicalLocation": {
                  "artifactLocation": {
                    "uri": "ui/window.c",
                    "uriBaseId": "SRCROOT"
                  },
                  "region": {
                    "startLine": 42
                  }
                },
                "state": {
                  "x": {
                    "text": "42"
                  },
                  "y": {
                    "text": "54"
                  },
                  "x + y": {
                    "text": "96"
                  }
                },
                "nestingLevel": 0,
                "executionOrder": 2
              },
            }
          ],
        }
      ],
    }
  ],
}
```

A result object (§3.27).
 # See §3.27.18.
 # A codeFlow object.
 # See §3.36.2.
 # See §3.36.3.
 # A threadFlow object (§3.37).
 # See §3.37.2.
 # See §3.37.3.
 # See §3.37.6.
 # A threadFlowLocation object (§3.38).
 # See §3.38.3.
 # See §3.28.3.
 # See §3.38.9.
 # See §3.38.10.
 # See §3.38.11.

3.36.2 message Property

A `codeFlow` object **MAY** contain a property named `message` whose value is a `message` object (§3.11) relevant to the code flow.

3.36.3 threadFlows Property

A `codeFlow` object **SHALL** contain a property named `threadFlows` whose value is an array of one or more `threadFlow` objects (§3.37) each of which describes the progress of a program through a single thread of execution such as an operating system thread or a fiber.

3.37 **threadFlow** Object

3.37.1 General

A thread flow is a sequence of code locations that specify a possible path through a single thread of execution such as an operating system thread or a fiber.

For an example, see §3.36.1.

3.37.2 **id** Property

A **threadFlow** object **MAY** contain a property named **id** whose value is a string that uniquely identifies this **threadFlow** within its containing **codeFlow** object (§3.36).

NOTE: A tool might choose to use an operating system thread id for this purpose. However, if thread ids are reused on a single machine, or if the code flow includes thread flows from more than one machine, the thread id might not be unique.

3.37.3 **message** Property

A **threadFlow** object **MAY** contain a property named **message** whose value is a **message** object (§3.11) relevant to the thread flow.

3.37.4 **initialState** Property

A **threadFlow** object **MAY** contain a property named **initialState** whose value is an object (§3.6) each of whose property values is a **multiFormatMessageString** object (§3.12) that represents the initial value of a relevant item prior to the first location in the thread flow. This property, together with **threadFlowLocation.state** (§3.38.9), enables a SARIF viewer to present a debugger-like “watch window” experience as the user traverses a thread flow.

This property **SHOULD NOT** include items whose values remain constant throughout the thread flow. Such items **SHOULD** be stored in the **immutableState** property (§3.37.5).

For details of how properties within a “state” object are represented, see EXAMPLE 1 in §3.38.9.

3.37.5 **immutableState** Property

A **threadFlow** object **MAY** contain a property named **immutableState** whose value is an object (§3.6) each of whose property values is a **multiFormatMessageString** object (§3.12) that represents the value of a relevant item that remains constant throughout the thread flow.

EXAMPLE 1: In this example, **immutableState** holds the value of a global variable that remains constant throughout the thread flow.

```
{
  # A threadFlow object.
  "immutableState": {
    "MaxFiles": {
      "text": "1000"
    }
  }
}
```

3.37.6 `locations` Property

A `threadFlow` object **SHALL** contain a property named `locations` whose value is an array of one or more `threadFlowLocation` objects (§3.38). Each element of the array **SHALL** represent a single location visited by the tool in the course of producing the result. This array does not need to include every location visited by the tool, but the elements that are present **SHALL** occur in the execution order that demonstrates the problem. The elements do not need to be unique within the array.

NOTE: The locations array might include multiple identical elements if, for example, the analysis tool simulated the execution of a loop in the course of producing the result.

3.38 `threadFlowLocation` Object

3.38.1 General

A `threadFlowLocation` object represents a location visited by an analysis tool in the course of simulating or monitoring the execution of a program.

3.38.2 `index` Property

Depending on the circumstances, a `threadFlowLocation` object either **MAY**, **SHALL NOT**, or **SHALL** contain a property named `index` whose value is the array index (§3.7.4) within `theRun.threadFlowLocations` (§3.14.19) of a `threadFlowLocation` object that provides the properties for `thisObject`. We refer to the object in `theRun.threadFlowLocations` as the “cached object.”

If `thisObject` is an element of `theRun.threadFlowLocations`, then `index` **MAY** be present. If present, its value **SHALL** be the index of `thisObject` within `theRun.threadFlowLocations`.

Otherwise, if `theRun.threadFlowLocations` is absent, or if it does not contain a cached object for `thisObject`, then `index` **SHALL NOT** be present.

Otherwise (that is, if `thisObject` belongs to a result, and `theRun.threadFlowLocations` contains a cached object for `thisObject`), then `index` **SHALL** be present, and its value **SHALL** be the index within `theRun.threadFlowLocations` of the cached object.

If `index` is present, `thisObject` **SHALL** take all properties present on the cached object. If `thisObject` contains any properties other than `index`, they **SHALL** equal the corresponding properties of the cached object.

NOTE 1: This allows a SARIF producer to reduce the size of the log file by reusing the same `threadFlowLocation` object in multiple thread flows.

EXAMPLE 1: In this example, `thisObject` is an element of `theRun.threadFlowLocations`. Its array index is known to be 1, so `thisObject.index` does not need to be present, but since it is present, it equals the array index, as required.

```
{
  "threadFlowLocations": [
    ...
    {
      "index": 1,
      "location": {
        ...
      }
    },
    ...
  ],
  ...
}
```

A run object (§3.14).
See §3.14.19.
A threadFlowLocation object: thisObject.
Optional.

EXAMPLE 2: In this example, `thisObject` is not an element of `theRun.threadFlowLocations`; rather, it is an element of `theResult.codeFlows[0].threadFlows[0].locations`. There is no cached object; that is, there is no object in `theRun.threadFlowLocations` that provides the properties for `thisObject`.

Therefore, `thisObject.index` is absent, as required.

```
{
  "results": [
    {
      "codeFlows": [
        {
          "threadFlows": [
            {
              "locations": [
                {
                  "location": {
                    ...
                  }
                }
              ]
            }
          ]
        }
      ]
    },
    ...
  ],
  "threadFlowLocations": [
    ...
  ]
}
```

A run object (§3.14).
 # See §3.14.23.
 # A result object (§3.27).
 # See §3.27.18.
 # A codeFlow object (§3.36).
 # See §3.36.3.
 # A threadFlow object (§3.37).
 # See §3.37.6.
 # A threadFlowLocation object (thisObject).
 # See §3.38.3.

EXAMPLE 3: In this example, `thisObject` is again an element of `theResult.codeFlows[0].threadFlows[0].locations`, not an element of `theRun.threadFlowLocations`. But in this example, there is a cached object, an element of `theRun.threadFlowLocations` that provides the properties for `thisObject`. Therefore, `thisObject.index` is present, as required.

```
{
  "results": [
    {
      "codeFlows": [
        {
          "threadFlows": [
            {
              "locations": [
                {
                  "index": 0
                }
              ]
            }
          ]
        }
      ]
    },
    ...
  ],
  "threadFlowLocations": [
    {
      "location": {
        ...
      }
    },
    ...
  ]
}
```

A run object (§3.14).
 # See §3.14.23.
 # A result object (§3.27).
 # See §3.27.18.
 # A codeFlow object (§3.36).
 # See §3.36.3.
 # A threadFlow object (§3.37).
 # See §3.37.6.
 # An threadFlowLocation object: thisObject.
 # index is present so no other properties.

See §3.14.19.
 # The cached threadFlowLocation object.
 # See §3.38.3.

3.38.3 location Property

If location information is available, a `threadFlowLocation` object **SHALL** contain a property named `location` whose value is a `location` object (§3.28) that specifies the location to which the `threadFlowLocation` object refers. If location information is not available, `location` **SHALL** be absent.

There are analysis tools whose native output format includes the equivalent of a SARIF code flow, but which do not provide location information for every step in the code flow. A SARIF converter for such a format might not be able to populate `location`. However, if the native output format associates a human readable message with such a step, the SARIF converter **SHOULD** create a `location` object and populate only its `message` property (§3.28.5). A SARIF direct producer which creates such code flows **SHOULD** populate `location.message`, even if no actual location information is available.

EXAMPLE 1: In this example, a file is locked by another program before a thread attempts to write to it. The analysis tool has no location information for the other program; in fact, the analysis tool might merely be simulating an

execution sequence in which a *hypothetical* external program locks the file. Nevertheless, it provides a helpful message.

Note the use of `executionOrder` (§3.38.11) to ensure that the location in the external program executes before the location in the program being analyzed.

```
{
  "threadFlows": [
    {
      "message": {
        "text": "An external program."
      },
      "locations": [
        {
          "executionOrder": 1,
          "location": {
            "message": {
              "text": "File is now locked."
            }
          }
        }
      ]
    },
    {
      "message": {
        "text": "The program being analyzed."
      },
      "locations": [
        ...
        {
          "executionOrder": 2,
          "location": {
            "message": {
              "text": "Attempt to write to the file."
            },
            "physicalLocation": {
              "artifactLocation": {
                "uri": "io/logger.c",
                "uriBaseId": "SRCROOT"
              },
              "region": {
                "startLine": 42,
                "snippet": {
                  "text": "    fprintf(fd, \"test\\n\\n\");"
                }
              }
            }
          }
        }
      ]
    }
  ]
}
```

3.38.4 **module** Property

A `threadFlowLocation` object **MAY** contain a property named `module` whose value is a string containing the name of the module that contains the code location specified by this object.

3.38.5 **stack** Property

A `threadFlowLocation` object **MAY** contain a property named `stack` whose value is a `stack` object (§3.44) that represents the call stack leading to this location.

3.38.6 **webRequest** Property

A `threadFlowLocation` object **MAY** contain a property named `webRequest` whose value is a `webRequest` object (§3.46) that describes an HTTP request sent from this location.

NOTE: This property is primarily useful to web analysis tools.

3.38.7 **webResponse** Property

A `threadFlowLocation` object **MAY** contain a property named `webResponse` whose value is a `webResponse` object (§3.47) that describes the response to the HTTP request sent from this location.

NOTE: This property is primarily useful to web analysis tools.

3.38.8 **kinds** Property

A `threadFlowLocation` object **MAY** contain a property named `kinds` whose value is an array of unique (§3.7.3) strings that describe the meaning of this location. The strings **SHOULD** be human-readable (as opposed to, for example, GUIDs or hash values).

When possible, SARIF producers **SHOULD** use the following values, with the specified meanings.

Verbs:

- `"acquire"`: Gain ownership of something.
- `"branch"`: Conditional transfer of control.
- `"call"`: Point of call into a section of the program such as a function.
- `"catch"`: Catch an exception.
- `"enter"`: Entry point to a section of the program such as a function.
- `"exit"`: Exit point from a section of the program such as a function.
- `"expose"`: Exposure of a secret across a trust boundary (e.g. password written to a logfile or an uninitialized stack copied from kernel back to user space).
- `"longjmp"`: Call to `longjmp` that rewinds the program counter/stack to the location of a previous `setjmp` call.
- `"release"`: Relinquish ownership of something.
- `"return"`: Point of return from a section of the program such as a function.
- `"setjmp"`: Call to `setjmp`.
- `"throw"`: Throw an exception.
- `"unwind"`: Unwind stack frame(s) during exception-handling.

NOTE 1: These values are typically combined with nouns from the list below, as in the examples below.

Nouns:

- `"taint"`: Value obtained from user input.
- `"function"`: Section of a program that can be called into and returned from.
- `"handler"`: Code invoked in response to an exception, signal, or event.
- `"lock"`: Limits access to a resource.
- `"memory"`: Portion of computer's internal storage.
- `"resource"`: Anything that can be acquired and released.
- `sensitive`: a value that is known to be secret e.g. a password or a private key.
- `"scope"`: Section of a program that limits the visibility of variables defined within it.
- `uninitialized`: uninitialized memory.
- `"value"`: The value of a variable.

NOTE 2: `"kinds": [ "acquire", "value" ]` can be used to denote a variable assignment or initialization.

Miscellaneous:

- `"implicit"`: Code was invoked implicitly, for example by a garbage collector.
- `"false"`: A condition evaluated to false.
- `"true"`: A condition evaluated to true.
- `"caution"`: Execution of the code at this location in the current circumstance requires care.
- `"danger"`: Execution of the code at this location in the current circumstance is dangerous.
- `"unknown"`: The state of an item is not known.
- `"unreachable"`: Code at this location is unreachable.

NOTE 3: Some analysis tools effectively “uncomment” unreachable code, allowing a simulated execution to flow through it. If such a tool detected a problem in the uncommented code, it could mark the `threadFlowLocation` as `"unreachable"`. An engineering team might then decide to treat this problem with lower priority.

If none of these values are appropriate, a SARIF producer **MAY** use any value.

The interpretations of values other than those above depends on the producer. A SARIF consumer that wishes to act based on such values **SHOULD** examine `theTool` to determine if it (the consumer) knows how to interpret them.

NOTE 4: This might not be necessary if, for example, the consumer has out of band information telling it how to interpret the values.

A SARIF producer **MAY** provide additional kind-dependent information by populating `threadFlowLocation.properties` with properties whose names and values depend on the kind. A SARIF consumer that knows how to interpret kinds for this tool **MAY** use this additional information.

EXAMPLE 1: In this example, tainted data enters the system at this location.

```
"kinds": [
  "acquire",
  "taint"
]
```

EXAMPLE 2: In this example, the “taint” state of a data item at this location is unknown:

```
"kinds": [
  "taint",
  "unknown"
]
```

EXAMPLE 3: In this example, control leaves a function at this location.

```
"kinds": [
  "exit",
  "function"
]
```

EXAMPLE 4: In this example, an uninitialized memory region is created at this location, such as at the point where a local variable is created on the stack, at an `alloca` call, or at a `malloc` call:

```
"kinds": [
  "acquire",
  "uninitialized"
]
```

EXAMPLE 5: In this example, uninitialized data is copied across a security boundary at this location, such as a copy from kernel-space to user-space within an OS kernel, or transmitting the data across a network:

```
"kinds": [
  "expose",
  "uninitialized"
]
```

EXAMPLE 6: In this example, a password or private key is read into memory at this location:

```
"kinds": [
  "acquire",
  "sensitive"
]
```

EXAMPLE 7: In this example, a password or private key is written to a log file at this location:

```
"kinds": [
  "expose",
  "sensitive"
]
```

3.38.9 state Property

A `threadFlowLocation` object **MAY** contain a property named `state` whose value is an object (§3.6) in which each property name represents an item relevant to the location in the context of the code flow, and the corresponding property value is a `multiFormatMessageString` object (§3.12) that specifies either the value of or a constraint on that item.

NOTE: This property enables a SARIF viewer to present a debugger-like “watch window” experience as the user navigates through a code flow.

A SARIF viewer **SHALL NOT** assume that expressions mentioned in previous steps but not mentioned in the current step are still present with unchanged values.

EXAMPLE 1: In this example, the `state` property captures the values of the expressions `"x"`, `"y"`, and `"x + y"`, and a constraint on the expression `"y - x"`.

```
{
  # A threadFlowLocation object.
  "state": {
    "x": {
      "text": "42"
    },
    "y": {
      "text": "54"
    },
    "x + y": {
      "text": "96"
    },
    "y - x": {
      "text": "{expr} > 0"
    }
  }
}
```

EXAMPLE 2: In C++, a property name within the `state` object might be:

- A variable name such as `"index"`.
- An array element reference such as `"names[index]"`.
- An object property reference such as `"names[index]->first"`.
- Any other expression that produces a value.

EXAMPLE 3: In C++, a property value within the `state` object might be:

- An integer such as `"42"` (note that the property value is a string).
- A string such as `"\"John\""` (the double quotes are escaped as they would be in a JSON serialization; other serializations might represent the double quotes differently).
- A Boolean such as `"true"`.

In a property value that represents a constraint, the item being constrained **SHALL BE** represented by the string `"{expr}"`. (See > EXAMPLE 1 above, which shows a constraint on the expression `"y - x"`.)

A constraint which expresses the equality of `"{expr}"` with a literal value **SHALL** be considered equivalent to that literal value.

EXAMPLE 4: In a language where `==` denotes value equality, the property value `"{expr} == 42"`, which represents a constraint, is identical in meaning to the property value `"42"`, which represents a value.

3.38.10 `nestingLevel` Property

A `threadFlowLocation` object **MAY** contain a property named `nestingLevel` whose value is a non-negative integer that represents any type of logical containment hierarchy among the `threadFlowLocation` objects in the `threadFlow`. Typically, it represents function call depth.

A viewer that renders a `threadFlow` **SHOULD** provide a visual representation of the value of `nestingLevel`. Typically, this would be an indentation indicating the depth of each location in the call tree.

3.38.11 `executionOrder` Property

A `threadFlowLocation` object **MAY** contain a property named `executionOrder` whose value is a non-negative integer that represents the temporal order in which execution reached this location, across all `threadFlowLocation` objects within all `threadFlow` objects belonging to a single `codeFlow` (§3.36). `executionOrder` values are assigned in increasing order of time; for example, execution reaches a `threadFlowLocation` whose `executionOrder` is 2 occurs before it reaches a `threadFlowLocation` whose `executionOrder` is 3. If two `threadFlowLocations` in different `threadFlow` objects within the same `codeFlow` have the same value for `executionOrder`, it means that execution reached both of those locations simultaneously. For that reason, values of `executionOrder` within a single `threadFlow` **SHALL** be unique.

It is only necessary to assign a value to `executionOrder` when the temporal ordering of a `threadFlowLocation` relative to a location in a different `threadFlow` is significant to the detection of a result.

If `executionOrder` is absent, it **SHALL** default to -1, which indicates that the value is unknown (not set).

NOTE: Negative values are forbidden because their use would suggest some non-obvious semantic difference between positive and negative values.

3.38.12 `executionTimeUtc` Property

A `threadFlowLocation` object **MAY** contain a property named `executionTimeUtc` whose value is a string in the format specified in §3.9, specifying the UTC date and time at which the thread of execution through the code reached this location.

3.38.13 importance Property

A `threadFlowLocation` MAY contain a property named `importance` whose value is a string that specifies the importance of this `threadFlowLocation` in understanding the code flow.

The `importance` property SHALL have one of the following values, with the specified meanings:

- `"important"`: this location is important for understanding the code flow.
- `"essential"`: this location is essential for understanding the code flow.
- `"unimportant"`: this location contributes to a more detailed understanding of the code flow but is not normally needed.

If this property is absent, it SHALL be considered to have the value `"important"`.

NOTE: A viewer might use this property to offer the user three options for viewing a lengthy code flow:

- A “normal view,” which omits locations whose `importance` property is `"unimportant"`.
- An “abbreviated view,” which displays only those locations whose `importance` property is `"essential"`.
- A “verbose view,” which displays all the locations in the code flow.

3.38.14 taxa Property

A `threadFlowLocation` MAY contain a property named `taxa` whose value is an array of zero or more unique (§3.7.3) `reportingDescriptorReference` objects each of which specifies a category into which this `threadFlowLocation` falls.

NOTE: The motivation for this property is an analysis tool that uses a set of rules to guide its analysis as it traces tainted data from a source to a sink. For example, at one location, the tool might apply a rule that says: “If the input to `String.Substr` is tainted, then so is the return value.” Such a tool can represent these “helper rules” as a custom taxonomy (§3.19.3), an array of `reportingDescriptor` objects (§3.49). Each member of `threadFlowLocation.taxa` can reference one of these helper rules.

EXAMPLE 1: This example illustrates the scenario in the above note.

```
{
  "tool": {
    "driver": {
      "name": "TaintDetector",
      "rules": [
        {
          "id": "TD0001",
          "name": "UntrustedDataStoredInDatabase",
          "shortDescription": {
            "text": "Data from an untrusted source was stored in a database."
          }
        },
        ...
      ],
      "taxa": [
        {
          "id": "HR0001",
          "name": "SubstrPropagatesTaint",
          "shortDescription": {
            "text": "If the input to String.Substr is tainted,
              so is the return value."
          }
        },
        ...
      ]
    },
    "results": [
      {
        "ruleId": "TD0001",
        ...
        "codeFlows": [
          {
            "threadFlows": [
              {
                ...
              }
            ]
          }
        ]
      }
    ]
  }
}
```

A run object (§3.14).
See §3.14.6.

Custom taxonomy (§3.19.3) for helper rules.
A reportingDescriptor object (§3.49).

See §3.14.23.
A result object §3.27.

See §3.27.18.
A codeFlow object (§3.36).
See §3.36.3.
A threadFlow object (§3.37).


```

"locations": [      # See §3.37.6.
  ...
  {                # A threadFlowLocation object.
    "location": {  # See §3.38.3.
      "physicalLocation": {
        "artifactLocation": {
          "uri": "io/input.c",
          "uriBaseId": "SRCROOT"
        },
        "region": {
          "startLine": 32
        }
      }
    },
    "taxa": [
      {            # A reportingDescriptorReference object (§3.52).
        "id": "HR0001",
        "index": 0
      }
    ]
  },
  ...
]
}

```

3.39 graph Object

3.39.1 General

A graph object represents a directed graph, a network of nodes and directed edges that describes some aspect of the structure of the code (for example, a call graph). graph objects **MAY** be defined both at the run level in `run.graphs` (§3.14.20) and at the result level in `result.graphs` (§3.27.19).

A path through a graph, called a “graph traversal,” is represented by a `graphTraversal` object (§3.42).

3.39.2 description Property

A graph object **MAY** contain a property named `description` whose value is a message object (§3.11) that describes the graph.

3.39.3 nodes Property

A graph object **MAY** contain a property named `nodes` whose value is an array of zero or more unique (§3.7.3) node objects (§3.40) which represent the nodes of the graph.

3.39.4 edges Property

A graph object **MAY** contain a property named `edges` whose value is an array of zero or more unique (§3.7.3) edge objects (§3.41) which represent the edges of the graph.

3.40 node Object

3.40.1 General

A node object represents a node in the graph represented by the containing graph object (§3.39), which we refer to as `theGraph`.

3.40.2 id Property

A node object **SHALL** contain a property named `id` whose value is a string that uniquely identifies the node within `theGraph`. `id` **SHALL** be unique among all nodes in `theGraph`, regardless of nesting (see §3.40.5).

EXAMPLE 1: This graph is invalid because two nodes have the same `id`, even though the nodes are within unrelated nested graphs.

```
{
  "nodes": [
    {
      "id": "n1",
      "children": [
        {
          "id": "n3"
        }
      ]
    },
    {
      "id": "n2",
      "children": [
        {
          "id": "n3"
        }
      ]
    }
  ],
  ...
}
```

A graph object (§3.39).
 # See §3.39.3.
 # A node object.
 # See §3.40.5.
 # INVALID: duplicate id.

3.40.3 `label` Property

A node object **MAY** contain a property named `label` whose value is a message object (§3.11) that provides a short description of the node.

3.40.4 `location` Property

A node object **SHOULD** have a property named `location` whose value is a location object (§3.28) that specifies the location associated with the node.

3.40.5 `children` Property

A node object **MAY** contain a property named `children` whose value is an array of zero or more unique (§3.7.3) node objects, referred to as “child nodes.”

Child nodes are logically subordinate to their containing node, and form a “nested graph” within that node.

3.41 `edge` Object

3.41.1 General

An edge object represents a directed edge in the graph represented by `theGraph`.

3.41.2 `id` Property

An edge object **SHALL** contain a property named `id` whose value is a string that uniquely identifies the edge within `theGraph`.

3.41.3 `label` Property

An edge object **MAY** contain a property named `label` whose value is a message object (§3.11) that provides a short description of the edge.

3.41.4 `sourceNodeId` Property

An edge object **SHALL** contain a property named `sourceNodeId` whose value is a string that identifies the source node (the node at which the edge starts). It **SHALL** equal the `id` property (§3.40.2) of one of the node objects (§3.40) in `theGraph`. It **MAY** equal the `id` of any node within `theGraph`, regardless of nesting (see §3.40.5).

EXAMPLE 1: In this example, an edge connects two nodes defined in unrelated nested graphs.

```
{
  "nodes": [
    {
      "id": "n1",
      "children": [
        {
          "id": "n3"
        }
      ]
    },
    {
      "id": "n2",
      "children": [
        {
          "id": "n4"
        }
      ]
    }
  ],
  "edges": [
    {
      "sourceNodeId": "n3",
      "targetNodeId": "n4"
    }
  ],
  ...
}
```

A graph object (§3.39).
 # See §3.39.3.
 # A node object.
 # See §3.40.5.
 # See §3.39.4.
 # Source node and target node are in separate
 # nested graphs: ok.

3.41.5 targetNodeId Property

An edge object **SHALL** contain a property named `targetNodeId` whose value is a string that identifies the target node (the node at which the edge ends). It **SHALL** equal the `id` property (§3.40.2) of one of the node objects (§3.40) in `theGraph`. It **MAY** equal `sourceNodeId` (§3.41.4).

3.42 graphTraversal Object

3.42.1 General

A `graphTraversal` object represents a “graph traversal,” that is, a path through a graph specified by a sequence of connected “edge traversals,” each of which is represented by an `edgeTraversal` object (§3.43). For an example, see §3.42.8.

3.42.2 Constraints

Exactly one of the `resultGraphIndex` property (§3.42.3) and the `runGraphIndex` property (§3.42.4) **SHALL** be present.

3.42.3 resultGraphIndex Property

If a `graphTraversal` object represents the traversal of a graph object (§3.39) that resides in `theResult.graphs` (§3.27.19), the `graphTraversal` object **SHALL** contain a property named `resultGraphIndex` whose value is the array index (§3.7.4) within `theResult.graphs` of that graph object.

3.42.4 runGraphIndex Property

If a `graphTraversal` object represents the traversal of a graph object (§3.39) that resides in `theRun.graphs` (§3.14.20), the `graphTraversal` object **SHALL** contain a property named `runGraphIndex` whose value is the array index (§3.7.4) within `theRun.graphs` of that graph object.

3.42.5 description Property

A `graphTraversal` object **MAY** contain a property named `description` whose value is a message object (§3.11) that describes the graph traversal.

3.42.6 `initialState` Property

A `graphTraversal` object **MAY** contain a property named `initialState` whose value is an object (§3.6) each of whose properties is a `multiformatMessageString` object (§3.12) that represents the value of a relevant item at the point of entry to the graph. This property, together with `edgeTraversal.finalState` (§3.43.4), enables a SARIF viewer to present a debugger-like “watch window” experience as the user traverses a graph.

This property **SHOULD NOT** include items whose value remains constant throughout the traversal. Such items **SHOULD** be stored in the `immutableState` property (§3.42.7).

For details of how properties within a “state” object are represented, see EXAMPLE 1 in §3.38.9.

3.42.7 `immutableState` Property

A `graphTraversal` object **MAY** contain a property named `immutableState` whose value is an object (§3.6) each of whose properties is a `multiformatMessageString` object (§3.12) that represents the value of a relevant item that remains constant throughout the traversal.

EXAMPLE 1: In this example, `immutableState` holds the value of a global variable that remains constant throughout the traversal.

```
{
  # A graphTraversal object.
  "immutableState": {
    "MaxFiles": {
      "text": "1000"
    }
  }
}
```

3.42.8 `edgeTraversals` Property

A `graphTraversal` object **MAY** contain a property named `edgeTraversals` whose value is an array of zero or more `edgeTraversal` objects (§3.43) which together represent the sequence of edges traversed during this graph traversal.

The `edgeTraversal` objects **SHALL** be connected end to end; that is, the target node of every traversed edge except the last **SHALL** equal the source node of the next edge.

EXAMPLE 1: In this example, the `graphTraversal` contains two `edgeTraversal` objects. The id of the first traversed edge is "e1", which connects node "n1" to node "n2". The id of the second traversed edge is "e3", which connects node "n2" to node "n4". This is a valid graph traversal because the target node of each traversed edge is the source node of the next.

This example also demonstrates the usage of `graphTraversal.initialState` (§3.42.6) and `edgeTraversal.finalState` (§3.43.4).

```
{
  # A result object (§3.27).
  # See §3.27.19.
  # A graph object (§3.39).
  # See §3.39.3.
  # A node object (§3.40).
  "graphs": [
    {
      "nodes": [
        { "id": "n1" },
        { "id": "n2" },
        { "id": "n3" },
        { "id": "n4" }
      ],
      "edges": [
        # See §3.39.4.
        # An edge object (§3.41).
        # See §3.41.2.
        # See §3.41.4.
        # See §3.41.5.
        {
          "id": "e1",
          "sourceNodeId": "n1",
          "targetNodeId": "n2"
        },
        {
          "id": "e2",
          "sourceNodeId": "n2",
          "targetNodeId": "n3"
        },
        {
          "id": "e3",
          "sourceNodeId": "n2",
          "targetNodeId": "n4"
        }
      ]
    }
  ]
}
```

```

],
"graphTraversals": [
  {
    "resultGraphIndex": 0,
    "initialState": {
      "x": {
        "text": "1"
      },
      "y": {
        "text": "2"
      },
      "x + y": {
        "text": "3"
      }
    },
    "edgeTraversals": [
      {
        "edgeId": "e1",
        "finalState": {
          "x": {
            "text": "4"
          },
          "y": {
            "text": "2"
          },
          "x + y": {
            "text": "6"
          }
        }
      },
      {
        "edgeId": "e3",
        "finalState": {
          "x": {
            "text": "4"
          },
          "y": {
            "text": "7"
          },
          "x + y": {
            "text": "11"
          }
        }
      }
    ]
  }
]
}

```

See §3.27.20.
A graphTraversal object (§3.42).
See §3.42.3.
See §3.42.6.
See §3.42.8.
An edgeTraversal object (§3.43).
See §3.43.2.
See §3.43.4.

3.43 edgeTraversal Object

3.43.1 General

An `edgeTraversal` object represents the traversal of a single edge during a graph traversal.

3.43.2 `edgeId` Property

An `edgeTraversal` object **SHALL** contain a property named `edgeId` whose value is a string which equals the `id` property (§3.41.2) of one of the `edge` objects (§3.41) in the graph identified by the `resultGraphIndex` property (§3.42.3) or the `runGraphIndex` property (§3.42.4) of the containing `graphTraversal` object (§3.42).

3.43.3 `message` Property

An `edgeTraversal` object **MAY** contain a property named `message` whose value is a `message` object (§3.11) that contains a message to display to the user as the edge is traversed.

3.43.4 `finalState` Property

An `edgeTraversal` object **MAY** contain a property named `finalState` whose value is an object (§3.6) each of whose properties is a `multiFormatMessageString` object (§3.12) that represents the value of a relevant item after the edge has been traversed.

NOTE: This property, together with `graphTraversal.initialState` (§3.42.6), enables a viewer to present a debugger-like “watch window” experience as the user traverses a graph.

A SARIF viewer **SHALL** display only those properties that are explicitly present in the `finalState` property of the current `edgeTraversal`. It **SHALL NOT** assume that properties present in previous steps are still present with unchanged values.

For details of how properties within a “state” object are represented, see §3.38.9.

3.43.5 `stepOverEdgeCount` Property

An `edgeTraversal` object **MAY** contain a property named `stepOverEdgeCount` whose value is a non-negative integer specifying the number of edges a user can step over.

This property is intended to enable a viewing experience in which the user can either step over or step into the traversal of a nested graph (§3.40.5). Therefore, this property **SHOULD** be specified only on an edge that leads from a node to one of its child nodes, and its value **SHOULD** be the number of edges the user would need to traverse to return to the current nesting level.

If this property is present, a SARIF viewer **MAY** provide a visual cue informing the user that they have the option of either stepping over the current edge and into the nested graph, or of stepping over the entire traversal of the nested graph.

EXAMPLE 1: This example defines a graph containing two nested graphs, the first representing code locations in function A and the second representing locations in function B. Node `na2` in function A represents a call to function B.

The example defines a graph traversal consisting of a set of edge traversals which start at node `"na1"` in function A, call into function B, and ultimately return to and continue execution in function A.

Suppose the user executes the first edge traversal, which traverses edge `ea1`. The next edge traversal has a `stepOverEdgeCount` property value of 4. Therefore, the SARIF viewer informs her that she can now choose to either step into function B by traversing edge `"eab"`, or step over the function call by traversing 4 edges, the last of which (edge `"eba"`) returns to function A at node `"na3"`.

If she chooses to enter the nested graph, she will visit the following nodes, in this order:

[`na1`, `na2`, `nb1`, `nb2`, `nb3`, `na3`, `na4`]

If she chooses not to enter the nested graph, the traversal of the edges

[`eab`, `eb1`, `eb2`, `eba`]

will be collapsed into a single “step over.” As a result, she will visit the following nodes, in this order:

[`na1`, `na2`, `na3`, `na4`]

```
{
  "graphs": [
    {
      "nodes": [
        {
          "id": "functionA",
          "children": [
            { "id": "na1" },
            { "id": "na2", "label": "Call functionB" },
            { "id": "na3" },
            { "id": "na4" }
          ]
        },
        {
          "id": "functionB",
          "nodes": [
            { "id": "nb1" },
            { "id": "nb2" },
            { "id": "nb3" }
          ]
        }
      ],
      "edges": [
        { "id": "ea1", "sourceNodeId": "na1", "targetNodeId": "na2" },
        { "id": "ea2", "sourceNodeId": "na2", "targetNodeId": "na3" },
        { "id": "eab", "sourceNodeId": "na2", "targetNodeId": "nb1" },
        { "id": "ea3", "sourceNodeId": "na3", "targetNodeId": "na4" },
        { "id": "eb1", "sourceNodeId": "nb1", "targetNodeId": "nb2" },
        { "id": "eb2", "sourceNodeId": "nb2", "targetNodeId": "nb3" },
        { "id": "eba", "sourceNodeId": "nb3", "targetNodeId": "na3" }
      ]
    }
  ]
}
```

A result object (§3.27).
See §3.27.19.
A graph object (§3.39).

```

],
"graphTraversals": [
  {
    "resultGraphIndex": 0,
    "edgeTraversals": [
      { "edgeId": "ea1" },
      {
        "edgeId": "eab",
        "stepOverEdgeCount": 4
      },
      { "edgeId": "eb1" },
      { "edgeId": "eb2" },
      { "edgeId": "eba" },
      { "edgeId": "ea3" }
    ]
  }
]
}

```

See §3.27.20.
A graphTraversal object (§3.42).
The graph being traversed.

3.44 **stack** Object

3.44.1 General

A **stack** object describes a single call stack. A call stack is a sequence of nested function calls, each of which is referred to as a stack frame.

3.44.2 **message** Property

A **stack** object **MAY** contain a property named **message** whose value is **message** object (§3.11) relevant to this call stack.

3.44.3 **frames** Property

A **stack** object **SHALL** contain a property named **frames** whose value is an array of zero or more **stackFrame** objects (§3.45). This array **SHALL** include every function call in the stack for which the tool has information, and the entries that are present **SHALL** occur in chronological order with the most recent (innermost) call first and the least recent (outermost) call last. The entries in this array do not need to be unique within the array.

NOTE 1: It is possible for the same frame to occur multiple times if the call stack includes a recursion.

NOTE 2: It is possible that the analysis tool will not have location information for every frame in the call stack. This might happen if, for example, application code for which location information is available calls into operating system code for which location information is not available, which in turn calls back into application code.

3.45 **stackFrame** Object

3.45.1 General

A **stackFrame** object describes a single stack frame within a call stack (§3.44).

3.45.2 **location** Property

A **stackFrame** object **MAY** contain a property named **location** whose value is a **location** object (§3.28) specifying the location to which this stack frame refers.

If location information is unavailable (as it might be, for example, when stepping from application code into library code or operating system code), **location** **SHOULD** be present and **SHOULD** contain a **message** property (§3.28) (for example, with a message string "Call into external code").

3.45.3 `module` Property

A `stackFrame` object **MAY** contain a property named `module` whose value is a string containing the name of the module that contains the location to which this stack frame refers.

3.45.4 `threadId` Property

A `stackFrame` object **MAY** contain a property named `threadId` whose value is an integer which identifies the thread on which the code at the location specified by this object was executed.

3.45.5 `parameters` Property

A `stackFrame` object **MAY** contain a property named `parameters` whose value is an array of zero or more strings representing the parameters of the function call represented by this stack frame.

3.46 `webRequest` Object

3.46.1 General

A `webRequest` object describes an HTTP request [RFC7230]. The response to the request is described by a `webResponse` object (§3.47).

NOTE 1: This object is primarily useful to web analysis tools.

A `webRequest` object does not need to represent a valid HTTP request.

NOTE 2: This allows an analysis tool that intentionally sends invalid HTTP requests to use the `webRequest` object.

3.46.2 `index` Property

Depending on the circumstances, a `webRequest` object either **MAY**, **SHALL NOT**, or **SHALL** contain a property named `index` whose value is the array index (§3.7.4) within `theRun.webRequests` (§3.14.21) of a `webRequest` object that provides the properties for `thisObject`. We refer to the object in `theRun.webRequests` as the “cached object.”

If `thisObject` is an element of `theRun.webRequests`, then `index` **MAY** be present. If present, its value **SHALL** be the index of `thisObject` within `theRun.webRequests`.

Otherwise, if `theRun.webRequests` is absent, or if it does not contain a cached object for `thisObject`, then `index` **SHALL NOT** be present.

Otherwise (that is, if `thisObject` belongs to a result, and `theRun.webRequests` contains a cached object for `thisObject`), then `index` **SHALL** be present, and its value **SHALL** be the array index within `theRun.webRequests` of the cached object.

If `index` is present, `thisObject` **SHALL** take all properties present on the cached object. If `thisObject` contains any properties other than `index`, they **SHALL** equal the corresponding properties of the cached object.

NOTE 1: This allows a SARIF producer to reduce the size of the log file by reusing the same `webRequest` object in multiple results.

NOTE 2: For examples of the use of an `index` property to locate a cached object, see §3.38.2.

3.46.3 **protocol** Property

A `webRequest` object **SHOULD** contain a property named `protocol` whose value is a string containing the name of the web protocol used in the request, found on the HTTP request line.

EXAMPLE 1: `"protocol": "HTTP"`

3.46.4 **version** Property

A `webRequest` object **SHOULD** contain a property named `version` whose value is a string containing the version of the web protocol used in the request, found on the HTTP request line.

EXAMPLE 1: `"version": "1.1"`

3.46.5 **target** Property

A `webRequest` object **SHOULD** contain a property named `target` whose value is a string containing the target of the request, found on the HTTP request line, in the form defined by §5.3 (“Request Target”) of the HTTP standard [RFC7230].

3.46.6 **method** Property

A `webRequest` object **SHOULD** contain a property named `method` whose value is a string containing the HTTP method used in the request, found on the HTTP request line. The string **SHOULD** be one of the values "GET", "PUT", "POST", "DELETE", "PATCH", "HEAD", "OPTIONS", "TRACE", or "CONNECT".

3.46.7 **headers** Property

A `webRequest` object **SHOULD** contain a property named `headers` whose value is an object (§3.6) whose property names are the names of the HTTP headers in the request (for example, "Content-Type") and whose corresponding values are the header values (for example, "text/plain; charset=ascii").

3.46.8 **parameters** Property

A `webRequest` object **MAY** contain a property named `parameters` whose value is an object (§3.6) whose property names are the names of the parameters in the request and whose corresponding values are the values of those parameters.

NOTE: The `parameters` property exists as a convenience for the log file consumer. If it is absent, the consumer can parse the parameters from `body` (§3.46.9), in the case of a forms post, or from the query portion of `uri` (§3.46.5).

3.46.9 **body** Property

A `webRequest` object **MAY** contain a property named `body` whose value is an `artifactContent` object (§3.3) containing the body of the request.

If the request body is entirely textual, `body.text` (§3.3.2) **SHOULD** be present. If present, it **SHALL** contain the request body, transcoded to UTF-8 if necessary.

NOTE 1: The transcoding is required because all textual content in a SARIF log file is represented in UTF-8 (see §3.1).

NOTE 2: If necessary, the character encoding actually used in the request can be deduced from the value of the Content-Type header (see §3.46.7), for example, "text/plain; charset=ascii".

If the request body is entirely textual, `body.binary` (§3.3.3) **MAY** be present. If present, it **SHALL** contain the MIME Base64 encoding [RFC2045] of the body as it was actually transmitted.

If the request body consists partially or entirely of binary data, `body.binary` **SHALL** be present and **SHALL** contain the MIME Base64 encoding of the body. In this situation, `body.text` **SHALL** be absent.

3.47 **webResponse** Object

3.47.1 **General**

A `webResponse` object describes the response to an HTTP request [RFC7230]. The request itself is described by a `webRequest` object (§3.46).

NOTE: This object is primarily useful to web analysis tools.

A `webResponse` object does not need to represent a valid HTTP response.

NOTE 2: This allows an analysis tool to describe a situation where a server produces an invalid response.

3.47.2 **index** Property

Depending on the circumstances, a `webResponse` object either **MAY**, **SHALL NOT**, or **SHALL** contain a property named `index` whose value is the array index (§3.7.4) within `theRun.webResponses` (§3.14.22) of a `webResponse` object that provides additional properties for `thisObject`. We refer to the object in `theRun.webResponses` as the “cached object.”

If `thisObject` is an element of `theRun.webResponses`, then `index` **MAY** be present. If present, its value **SHALL** be the index of `thisObject` within `theRun.webResponses`.

Otherwise, if `theRun.webResponses` is absent, or if it does not contain a cached object for `thisObject`, then `index` **SHALL NOT** be present.

Otherwise (that is, if `thisObject` belongs to a result, and `theRun.webResponses` contains a cached object for `thisObject`), then `index` **SHALL** be present, and its value **SHALL** be the array index within `theRun.webResponses` of the cached object.

If `index` is present, `thisObject` **SHALL** take all properties present on the cached object. If `thisObject` contains any properties other than `index`, they **SHALL** equal the corresponding properties of the cached object.

NOTE 1: This allows a SARIF producer to reduce the size of the log file by reusing the same `webResponse` object in multiple results.

NOTE 2: For examples of the use of an `index` property to locate a cached object, see §3.38.2.

3.47.3 **protocol** Property

A `webResponse` object **SHOULD** contain a property named `protocol` whose value is a string containing the name of the web protocol used in the response, found on the HTTP status line.

EXAMPLE 1: `"protocol": "HTTP"`

3.47.4 **version** Property

A `webResponse` object **SHOULD** contain a property named `version` whose value is a string containing the version of the web protocol used in the response, found on the HTTP status line.

EXAMPLE 1: `"version": "1.1"`

3.47.5 **statusCode** Property

A `webResponse` object **SHOULD** contain a property named `statusCode` whose value is an integer containing the status code that describes the result of the request, found on the HTTP status line.

EXAMPLE 1: `"statusCode": 200`

3.47.6 **reasonPhrase** Property

A `webResponse` object **SHOULD** contain a property named `reasonPhrase` whose value is a string containing the textual description of the `statusCode` (§3.47.5) found on the HTTP status line.

EXAMPLE 1: `"reasonPhrase": "OK"`

If `noResponseReceived` (§3.47.9) is `true`, then `reasonPhrase` **SHOULD** instead contain a string describing the reason that no response was received.

3.47.7 **headers** Property

A `webResponse` object **SHOULD** contain a property named `headers` whose value is an object (§3.6) whose property names are the names of the HTTP headers in the response (for example, `"Content-Type"`) and whose corresponding values are the header values (for example, `"text/plain; charset=ascii"`).

3.47.8 **body** Property

A `webResponse` object **MAY** contain a property named `body` whose value is an `artifactContent` object (§3.3) containing the body of the response.

If the response body is entirely textual, `body.text` (§3.3.2) **SHOULD** be present. If present, it **SHALL** contain the response body, transcoded to UTF-8 if necessary.

NOTE 1: The transcoding is required because all textual content in a SARIF log file is represented in UTF-8 (see §3.1).

NOTE 2: If necessary, the character encoding actually used in the response can be deduced from the value of the `Content-Type` header (see §3.47.7), for example, `"text/plain; charset=ascii"`.

If the response body is entirely textual, `body.binary` (§3.3.3) **MAY** be present. If present, it **SHALL** contain the MIME Base64 encoding [RFC2045] of the body as it was actually transmitted.

If the response body consists partially or entirely of binary data, `body.binary` **SHALL** be present and **SHALL** contain the MIME Base64 encoding of the body. In this situation, `body.text` **SHALL** be absent.

3.47.9 **noResponseReceived** Property

If no response to the HTTP request was received (for example, because of a network failure), the `webResponse` object **SHALL** contain a property named `noResponseReceived` whose value is a Boolean `true`. If a response was received, `noResponseReceived` **SHALL** either be present with the value `false`, or absent, in which case it defaults to `false`.

If `noResponseReceived` is `true`, then `reasonPhrase` (§3.47.6), which normally contains the reason phrase from the HTTP response line, **SHOULD** instead contain a string describing the reason that no response was received.

3.48 **resultProvenance** Object

3.48.1 General

A **resultProvenance** object contains information about the how and when the **result** was detected.

NOTE: This information is useful to various human and automated participants in an engineering system. For example:

- A build engineer might use the information to understand the specific tool invocation that produced the result, for example, if the violated rule should not have been configured to run at all.
- A developer reviewing results might use the information to determine how long an issue has existed in the code.
- A result management system might be responsible for associating logically identical results from one run to the next, making it possible for the developer to determine how long the result has existed. Such a result management system might populate this information.

3.48.2 **firstDetectionTimeUtc** Property

A **resultProvenance** object **MAY** contain a property named **firstDetectionTimeUtc** whose value is a string in the format specified in §3.9, specifying the UTC date and time at which the result was first detected. It **SHOULD** specify the start time of the run in which the result was first detected, as opposed to, for example, the time within the run at which the result was actually generated.

NOTE: Using the run's start time makes it possible to group together results that were first detected in the same run.

3.48.3 **lastDetectionTimeUtc** Property

A **resultProvenance** object **MAY** contain a property named **lastDetectionTimeUtc** whose value is a string in the format specified in §3.9, specifying the UTC date and time at which the result was most recently detected. It **SHOULD** specify the start time of the run in which the result was most recently detected, as opposed to, for example, the time within the run at which the result was actually generated.

NOTE: Using the run's start time makes it possible to group together results that were detected in the same run.

If **lastDetectionTimeUtc** is absent, its default value **SHALL** be determined as follows:

1. If **run.invocations** is present, and if the **startTimeUtc** property (§3.20.7) is present on any of the **invocation** objects (§3.20) in that array, then the default is the earliest of those times.
2. Otherwise, there is no default.

3.48.4 **firstDetectionRunGuid** Property

A **resultProvenance** object **MAY** contain a property named **firstDetectionRunGuid** whose value is a GUID-valued string (§3.5.3) which **SHALL** equal the **automationDetails.guid** property (§3.14.3, §3.17.4) of the run in which the **result** was first detected (either the current run or some previous run).

3.48.5 **lastDetectionRunGuid** Property

A **resultProvenance** object **MAY** contain a property named **lastDetectionRunGuid** whose value is a GUID-valued string (§3.5.3) which **SHALL** equal the **automationDetails.guid** property (§3.14.3, §3.17.4) of the run in which the **result** was most recently detected (either the current run or some previous run).

3.48.6 invocationIndex Property

If theRun.invocations (§3.14.11) is present, a resultProvenance object **MAY** contain a property named invocationIndex whose value is the array index (§3.7.4) within the invocations property of the invocation object (§3.20) that describes the tool invocation as a result of which theResult was detected.

If theRun.invocations is absent, invocationIndex **SHALL** be absent.

NOTE 1: The purpose of this property is to allow a result to be associated with the tool invocation that produced it.

If invocationIndex is absent and theRun.invocations is present and contains a single element, it **SHALL** default to 0; otherwise it **SHALL** default to -1, which indicates that the value is unknown (not set).

NOTE 2: This provides a sensible default in the common case where there is only a single tool invocation in the run.

3.48.7 conversionSources Property

Some analysis tools produce output files that describe the analysis run as a whole; we refer to these as “per-run” files. Some tools produce one or more output files for each result; we refer to these as “per-result” files. Some tools produce both per-run and per-result files.

A resultProvenance object **MAY** contain a property named conversionSources whose value is an array of zero or more unique (§3.7.3) physicalLocation objects (§3.29).

If theResult was produced by a converter, and if the analysis tool whose output was converted to SARIF produced any per-result files for this result, then the physicalLocation objects in the array **SHALL** specify the relevant portions of the per-result files for this result.

Otherwise (that is, if the run object was not produced by a converter, or if there were no per-run files for this result), then if conversionSources is present, its value **SHALL** be an empty array.

Per-run files are handled by the conversion.analysisToolLogFiles property (§3.22.4).

NOTE: This property is intended to be useful to developers of converters, to help them debug the conversion from the analysis tool’s native output format to the SARIF format.

EXAMPLE 1: Given this analysis tool’s output file:

```
<?xml version="1.0" encoding="UTF-8"?>
<problems>
  <problem>
    <file></file>
    <line>242</line>
    ...
    <problem_class ...>Assertions</problem_class>
    ...
    <description>Assertions are unreliable. ...</description>
  </problem>
</problems>
```

a SARIF converter might transform it into the following SARIF log file:

```
{
  ...
  "runs": [
    {
      "tool": {
        "driver": {
          "name": "CodeScanner"
        }
      },
      "conversion": { # A conversion object (see (#conversion-object)).
        ...
      },
      "results": [
        {
          "ruleId": "Assertions",
          "message": {
            "text": "Assertions are unreliable. ..."
          },
          ...
          "provenance": { # See §3.27.29.
```

```

"conversionSources": [      # An array of physicalLocation objects
  {                          # ((#physicallocation-object)).
    "artifactLocation": {   # See §3.29.3.
      "uri": "CodeScanner.log",
      "uriBaseId": "$LOGSR00T"
    },
    "region": {             # See §3.29.4.
      "startLine": 3,
      "startColumn": 3,
      "endLine": 12,
      "endColumn": 13,
      "snippet": {
        "text": "<problem>\n ... \n </problem>"
      }
    }
  },
  ...
]
}

```

3.49 reportingDescriptor Object

3.49.1 General

A `reportingDescriptor` object contains information that describes a “reporting item” generated by a tool. A reporting item is either a result produced by the tool’s analysis (see §3.27), or a notification of a condition encountered by the tool (§3.58). We refer to this descriptive information as “reporting item metadata.” When referring to the metadata that describes a result, we use the more specific term “rule metadata.”

Some of the properties of the `reportingDescriptor` object are interpreted differently depending on whether the object represents a rule or a notification. The description of each property will specify any such differences.

3.49.2 Constraints

Either the `shortDescription` property (§3.49.9) or the `fullDescription` property (§3.49.10) or both **SHOULD** be present.

3.49.3 id Property

A `reportingDescriptor` object **SHALL** contain a property named `id` whose value is a string. In the case of a rule, `id` **SHALL** contain a stable identifier for the rule and **SHOULD** be opaque. In the case of a notification, `id` does not need be a stable, opaque identifier; it **MAY** be a user-readable identifier.

EXAMPLE 1: `"id": "CA2101"`

NOTE 1: Rule identifiers must be stable for two reasons:

- So build automation scripts can refer to specific checks, for example, to disable them, without the risk of a script breaking if a rule id changes.
- So result management systems can compare results from one run to the next, without erroneously designating results as “new” because a rule id has changed.

Rule identifiers should be opaque – that is, they should not convey information to a user – because a rule’s implementation might change over time. Suppose a rule id is `"DoNotDoXOrY"`, suppose circumstances change so that “Y” is now acceptable, and suppose the implementation of the rule changes accordingly. Because the rule id must not change, the string `"DoNotDoXOrY"` will continue to be persisted to logs, where it will convey outdated guidance to users in a way that an opaque identifier such as `"CA2101"` would not.

NOTE 2: Despite the fact that the `result.ruleId` property (§3.27.5) is permitted to be a hierarchical string (§3.5.4) whose trailing components denote a subset of the specified rule, SARIF does not support separate metadata for such “sub-rules”. The `id` property of a `reportingDescriptor` object always specifies an entire rule (or notification), not a subset of one.

3.49.4 deprecatedIds Property

A `reportingDescriptor` object **MAY** contain a property named `deprecatedIds` whose value is an array of zero or more unique (§3.7.3) strings each of which contains an id (see §3.49.3) by which this reporting item was known in some previous version of the analysis tool.

NOTE: This property is most useful for rules. It addresses the scenario where rule ids change from one version of a tool to the next. For example, a tool developer might decide that a rule is too general, covering too many concepts. In the next version of the tool, the tool developer might break this rule into a set of more specific rules.

Now the result management system has the problem of matching results between the newer and the older versions of the tool. `deprecatedIds` solves this problem.

EXAMPLE 1: In this example, version 1 of an analysis tool defines rule CA1000. A run of this tool finds two results. The result management system decides that neither result was previously detected, so it marks them as with `"baselineState": "new"` (§3.27.24), producing this log:

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "version": "1",
      "rules": [
        {
          "id": "CA1000",
          ...
        }
      ]
    }
  },
  "results": [
    {
      "ruleId": "CA1000",
      "rule": {
        "index": 0
      },
      "baselineState": "new",
      ...
    },
    {
      "ruleId": "CA1000",
      "rule": {
        "index": 0
      },
      "baselineState": "new",
      ...
    }
  ]
}
```

The engineering team decides that these results are false positive, so they add in-source suppressions, for example (in C#):

```
[SuppressMessage("CA1000", ...)]
...
[SuppressMessage("CA1000", ...)]
```

Now the tool developers decide that rule CA1000 is too broad, so in version 2 of the tool, they divide it into two new rules, CA1001 and CA1002. The engineering team runs the new tool, and the result management system performs result matching, producing this log:

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "version": "2",
      "rules": [
        {
          ...
        }
      ]
    }
  },
  ...
}
```



```

      "id": "CA1001",
      "deprecatedIds": [
        "CA1000"
      ],
      ...
    },
    {
      "id": "CA1002",
      "deprecatedIds": [
        "CA1000"
      ],
      ...
    }
  ]
},
"results": [
  {
    "ruleId": "CA1001",
    "rule": {
      "index": 0
    },
    "baselineState": "unchanged",
    "suppressions": [
      {
        "kind": "inSource"
      }
    ]
  },
  ...
],
{
  "ruleId": "CA1002",
  "rule": {
    "index": 1
  },
  "baselineState": "updated",
  "suppressions": [
    {
      "kind": "inSource"
    }
  ]
},
...
]
}

```

There are a few things to notice:

- In `tool.driver.rules`, each of the new rules is associated with its id from the previous tool version.
- As a result, the analysis tool can determine that the in-source suppressions still apply, even though the rule ids have changed, so it correctly marks each result with `"kind": "inSource"`.
- Furthermore, the result management system can determine that these are the same results it saw in the previous run, so it correctly marks them with `"baselineState": "unchanged"` or `"updated"` as appropriate (see §3.27.24).

3.49.5 **guid** Property

A `reportingDescriptor` object **MAY** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) that uniquely identifies the descriptor.

3.49.6 **deprecatedGuids** Property

A `reportingDescriptor` object **MAY** contain a property named `deprecatedGuids` whose value is an array of zero or more unique (§3.7.3) GUID-valued strings (§3.5.3) each of which was used by a previous version of the tool as the value of the `guid` property (§3.49.5) for this object.

3.49.7 **name** Property

A `reportingDescriptor` object **MAY** contain a property named `name` whose value is a localizable string (§3.5.1) containing an identifier that is understandable to an end user. If the `name` of a rule contains implementation details that change over time, a tool author might alter a rule's name (while leaving the stable `id` property (§3.49.3) unchanged).

NOTE: A rule name is suitable in contexts where a readable identifier is preferable and where the lack of stability is not a concern.

EXAMPLE 1: "name": "`SpecifyMarshalingForPInvokeStringArguments"

3.49.8 deprecatedNames Property

A `reportingDescriptor` object **MAY** contain a property named `deprecatedNames` whose value is an array of zero or more unique (§3.7.3) localizable (§3.5.1) strings each of which was used by a previous version of the tool as the value of the `name` property (§3.49.7) for this object.

The array elements **SHALL** occur in the same order in every translation (§3.19.3).

3.49.9 shortDescription Property

A `reportingDescriptor` object **MAY** contain a property named `shortDescription` whose value is a localizable `multiformatMessageString` object (§3.12, §3.12.2) that provides a concise description of the reporting item. The `shortDescription` property **SHOULD** be a single sentence that is understandable when visible space is limited to a single line of text.

EXAMPLE 1:

```
{
  # A reportingDescriptor object
  "shortDescription": {
    "text": "Specify marshaling for P/Invoke string arguments."
  }
}
```

3.49.10 fullDescription Property

A `reportingDescriptor` object **SHOULD** contain a property named `fullDescription` whose value is a localizable `multiformatMessageString` object (§3.12, §3.12.2) that comprehensively describes the reporting item.

The `fullDescription` property **SHOULD**, as far as possible, provide details sufficient to enable resolution of any problem indicated by the reporting item.

The beginning of `fullDescription` (for example, its first sentence) **SHOULD** provide a concise description of the reporting item, suitable for display in cases where available space is limited. Tools that construct `fullDescription` in this way do not need to provide a value for `shortDescription` (§3.49.9). Tools that do not construct `fullDescription` in this way **SHOULD** provide a value for `shortDescription`.

NOTE: The rationale for this guidance is that in the absence of `shortDescription`, a viewer with limited display space might display a truncated version of `fullDescription`, for example, the first sentence (if a sentence is identifiable), the first paragraph, or the first 100 characters. If this guidance is not followed, that truncated version might not be understandable.

3.49.11 messageStrings Property

A `reportingDescriptor` object **MAY** contain a property named `messageStrings` whose value is an object (§3.6) consisting of a set of properties with arbitrary names, each of whose values is a localizable `multiformatMessageString` object (§3.12, §3.12.2).

If the `reportingDescriptor` object defines a rule, the set of property names appearing in the `messageStrings` property **SHALL** contain at least the set of strings which occur as values of `result.message.id` properties (§3.27.11, §3.11.10) in the current run object. The `messageStrings` property **MAY** contain additional properties whose names do not appear as the value of the `result.message.id` property for any `result` object in the run.

If the `reportingDescriptor` object describes a notification, the set of property names appearing in the `messageStrings` property **SHALL** contain at least the set of strings which occur as values of `notification.message.id` for any notification object in the run.

NOTE: Additional properties are permitted in the `messageStrings` property for the convenience of tool vendors, who might find it easier to emit the entire set of messages defined in the reporting metadata, rather than restricting it to those messages that happen to appear in the log file.

EXAMPLE 1:

```
{
    # A reportingDescriptor object for a rule.
    "messageStrings": {
        "objectCreation": { # A multiformatMessageString object (§3.12).
            "text": "{0} creates a new instance of {1} which is never used.
                Pass the instance as an argument to another method,
                assign the instance to a variable,
                or remove the object creation if it is unnecessary."
        },
        "stringReturnValue": {
            "text": "{0} calls {1} but does not use the new string
                instance that the method returns.
                Pass the instance as an argument to another method,
                assign the instance to a variable,
                or remove the call if it is unnecessary."
        }
    }
}
```

3.49.12 `helpUri` Property

A `reportingDescriptor` object **MAY** contain a property named `helpUri` whose value is a localizable string (§3.5.1) containing the absolute URI [RFC3986] of the primary documentation for the reporting item.

NOTE 1: The documentation might include examples, contact information for the authors, and links to additional information.

NOTE 2: This property is localizable so that help information in different languages can be viewed at different URIs.

3.49.13 `help` Property

A `reportingDescriptor` object **MAY** contain a property named `help` whose value is a localizable `multiformatMessageString` object (§3.12, §3.12.2) which provides the primary documentation for the reporting item.

NOTE: This property is useful when help information is not available at a URI, for example, in the case of a custom rule written by a developer, as opposed to one supplied by the tool vendor.

3.49.14 `defaultConfiguration` Property

A `reportingDescriptor` object **MAY** contain a property named `defaultConfiguration` whose value is a `reportingConfiguration` object (§3.50).

If this property is absent, it **SHALL** be taken to be present, and its properties **SHALL** be taken to have the default values specified in §3.50.

The rule- or notification-specific configuration parameters for a `reportingDescriptor`, if any, **SHALL NOT** be stored in its property bag (§3.8) Rather, they **SHALL** be stored in `defaultConfiguration.parameters` (§3.50.5).

3.49.15 **relationships** Property

A `reportingDescriptor` object **MAY** contain a property named `relationships` whose value is an array of zero or more unique (§3.7.3) `reportingDescriptorRelationship` objects (§3.53) each of which declares one or more directed relationships from `thisObject` to another `reportingDescriptor` object, which we refer to as the `target`, specified by `reportingDescriptorRelationship.target` (§3.53.2). The natures of the relationships between `thisObject` and the `target` are specified by `reportingDescriptorRelationship.kinds` (§3.53.3).

3.50 **reportingConfiguration** Object

3.50.1 General

A `reportingConfiguration` object contains the information in a `reportingDescriptor` (§3.49) that a SARIF producer can modify at runtime, before executing its scan. We refer to the `reportingDescriptor` object whose configuration is established or modified by a `reportingConfiguration` object as the `descriptor`.

When a `reportingConfiguration` object appears as the value of `theDescriptor.defaultConfiguration` (§3.49.14), it specifies the `reportingDescriptor`'s default configuration. When a `reportingConfiguration` object appears as the value of `configurationOverride.configuration` (§3.51.3), it overrides the default values in the `reportingDescriptor` identified by `configurationOverride.descriptor` (§3.51.2).

For an example, see §3.50.5.

3.50.2 **enabled** Property

A `reportingConfiguration` object **MAY** contain a property named `enabled` whose value is a Boolean that specifies whether the condition described by the `descriptor` was checked for during the scan.

If this property is absent, it **SHALL** default to `true`.

EXAMPLE 1: In this example, a tool allows the user to enable or disable rules or notifications:

```
SecurityScanner --disable "SEC4002,SEC4003" --enable SEC6012
```

3.50.3 **level** Property

A `reportingConfiguration` object **MAY** contain a property named `level` whose value is one of the strings "warning", "error", "note", or "none", with the same meanings as when those strings appear as the value of `result.level` (§3.27.10) or `notification.level` (§3.58.6).

If `level` is absent, it **SHALL** default to "warning".

If the `descriptor` describes a rule, then if `level` is present, it **SHALL** provide the value for the `level` property of any `result` object (§3.27) whose `ruleIndex` (§3.27.6) or `rule` property (§3.27.7), either explicitly supplied or inferred from its default, identifies the `descriptor` and which does not itself specify a `level` property. For details of the configuration property resolution procedure, see §3.27.10 (which illustrates the procedure for the specific case of the `result.level` property).

If the `descriptor` describes a notification, then if `level` is present, it **SHALL** provide the value for the `level` property of any `notification` object (§3.58) whose `descriptor` property (§3.58.2) identifies the `descriptor` and which does not itself specify a `level` property.

EXAMPLE 1: In this example, a tool allows the user to override a rule or notification's default level:

```
WebScanner --level "WEB1002:error,WEB1005:warning"
```

3.50.4 rank Property

A `reportingConfiguration` object **MAY** contain a property named `rank` whose value is a number between `0.0` and `100.0` inclusive, with the same interpretation as the value of the `result.rank` (§3.27.25).

If `rank` is absent, it **SHALL** default to `-1.0`, which indicates that the value is unknown (not set).

If the `Descriptor` describes a rule, then if `rank` is present, it **SHALL** provide the value for the `rank` property of any `result` object (§3.27) whose `ruleIndex` (§3.27.6) or `rule` property (§3.27.7), either explicitly supplied or inferred from its default, identifies the `Descriptor` and which does not itself specify a `rank` property.

`rank` is not applicable to notifications.

3.50.5 parameters Property

A `reportingConfiguration` object **MAY** contain a property named `parameters` whose value is a property bag (§3.8). This allows a `reportingDescriptor` object (§3.49) to define configuration information that is specific to that descriptor.

EXAMPLE 1: In this example, a rule that specifies the maximum permitted source line length is parameterized by the maximum length.

```
{
  # A reportingDescriptor object (§3.49).
  "id": "SA2707",
  "name": {
    "text": "LimitSourceLineLength"
  },
  "shortDescription": {
    "text": "Limit source line length for readability."
  },
  "defaultConfiguration": {
    "enabled": true,
    "level": "warning",
    "parameters": {
      "maxLength": 120
    }
  }
}
```

The rule provides a default value, but the tool allows the user to override it:

```
StyleScanner *.c --rule-config "SA2707:maxLength=80"
```

3.51 configurationOverride Object

3.51.1 General

A `configurationOverride` object modifies the effective runtime configuration of a specified `reportingDescriptor` object (§3.49), which we refer to as the `Descriptor`.

NOTE: Together with `toolComponent.rules` (§3.19.23), the `configurationOverride` object allows the SARIF consumer to determine exactly how the tool's analysis rules were configured during the run. This is useful in compliance scenarios where, for example, an auditor might want to confirm that a particular rule was reconfigured from a warning to an error. It might also be useful for reproducing a run.

The `configurationOverride` object's `descriptor` property (§3.51.2) identifies the `Descriptor`. Its `configuration` property (§3.51.3) overrides the values specified in `Descriptor.defaultConfiguration` (§3.49.14).

EXAMPLE 1: In this example, rule CA2101 is treated as a warning rather than an error.

```
{
  # A run object (§3.14).
  "tool": {
    "driver": {
      # See §3.14.6.
      "name": "CodeScanner",
      # See §3.18.2.
      "rules": [
        # See §3.19.23.
        # A reportingDescriptor object
      ]
    }
  }
```

```

    "id": "CA2101",                # (§3.49).
    "defaultConfiguration": {
      "level": "error"
    }
  ],
},
"invocations": [                 # See §3.14.11.
  {                               # An invocation object (§3.20).
    "ruleConfigurationOverrides": [ # See §3.20.5.
      {                           # A configurationOverride object
        "descriptor": {           # (§3.51).
          "index": 0              # See §3.51.2.
        },
        "configuration": {        # See §3.51.3.
          "level": "warning"
        }
      },
      ...
    ]
  }
]
}

```

3.51.2 descriptor Property

A configurationOverride object **SHALL** contain a property named descriptor whose value is a reportingDescriptorReference object (§3.52) that identifies the reportingDescriptor (§3.49) whose runtime configuration is to be modified, which we refer to as theDescriptor.

3.51.3 configuration Property

A configurationOverride object **SHALL** contain a property named configuration whose value is a reportingConfiguration object (§3.50) each of whose properties overrides the corresponding property in theDescriptor.defaultConfiguration (§3.49.14). If any property of configuration is absent, the corresponding property of theDescriptor.defaultConfiguration is respected.

3.52 reportingDescriptorReference Object

3.52.1 General

A reportingDescriptorReference object identifies a particular reportingDescriptor object (§3.49), which we refer to as theDescriptor, among all reportingDescriptor objects defined by theTool, including those defined by theTool.driver (§3.18.2) and theTool.extensions (§3.18.3).

In some cases, there is no reportingDescriptor object associated with a reportingDescriptorReference object. In that case, the reportingDescriptorReference object **SHALL** contain only the id property (§3.52.4), and theDescriptor does not exist.

EXAMPLE 1: In this example, a tool emits a tool execution notification that refers to a rule. The tool does not provide rule metadata. Therefore, associatedRule (§3.58.3) contains only an id property, whose value is the id of the rule that failed. Similarly, the tool does not provide metadata about its notifications, so "descriptor" (§3.58.2) contains only the id of the notification.

```

{                               # An invocation object (§3.20).
  "toolExecutionNotifications": [ # See §3.20.21.
    {                             # A notification object (§3.58).
      "descriptor": {             # See §3.58.2.
        "id": "CTN9999"
      },
      "associatedRule": {         # See §3.58.3.
        "id": "C2001"
      },
      "level": "error",
      "message": {
        "text": "Exception evaluating rule 'C2001'. Rule disabled;
                  run continues."
      }
    }
  ]
}

```

3.52.2 Constraints

If metadata is present, at least one of `index` (§3.52.5) and `guid` (§3.52.6) **SHALL** be present. If both are present, they **SHALL** identify the same `reportingDescriptor` object (§3.49).

3.52.3 Reportingdescriptor Lookup

`theDescriptor` **SHALL** be located within the `toolComponent` object (§3.19) identified by the `toolComponent` property (§3.52.7), which we refer to as `theComponent`. The procedure for looking up a `toolComponent` from a `toolComponentReference` is described in §3.54.2.

`theDescriptor` **SHALL** be located either within `theComponent.rules` (§3.19.23) or `theComponent.notifications` (§3.19.24), according to this table:

If the <code>reportingDescriptorReference</code> occurs in:	... then <code>theDescriptor</code> is an element of:
<code>invocation.ruleConfigurationOverrides</code> (§3.20.5)	<code>rules</code>
<code>invocation.notificationConfigurationOverrides</code> (§3.20.6)	<code>notifications</code>
<code>result.rule</code> (§3.27.7)	<code>rules</code>
<code>notification.descriptor</code> (§3.58.2)	<code>notifications</code>
<code>notification.associatedRule</code> (§3.58.3)	<code>rules</code>

3.52.4 id Property

A `reportingDescriptorReference` object **MAY** contain a property named `id` whose value is a hierarchical string (§3.5.4) that either equals `theDescriptor.id` (§3.49.3) or equals `theDescriptor.id` plus one additional hierarchical component.

NOTE: This property does not participate in the lookup, but its presence improves the readability of the log file at the expense of increased file size.

If `id` is absent and `theResult.ruleId` (§3.27.5) is present, then `id` **SHALL** default to `theResult.ruleId`. If both are present, they **SHALL** be equal.

For more information about the semantics of `id` when `theDescriptor` is a rule, in particular the usage of the hierarchical components of `id`, see the description of `result.ruleId` (§3.27.5).

EXAMPLE 1: In this example, the first `result` object is valid because `rule.id` (inherited from `ruleId`) equals `theDescriptor.id`. The second `result` object is also valid because `rule.id` (this time specified directly) equals `theDescriptor.id` plus one additional hierarchical component ("ghi"). The third `result` object is invalid because `theDescriptor.id` is not a "component-wise" prefix of `rule.id`. The fourth `result` object is invalid because `ruleId` does not equal `rule.id`.

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "abc/def",
          ...
        },
        ...
      ]
    }
  },
  "results": [
    {
      "ruleId": "abc/def",
      "rule": {
        "index": 0
      }
    },
    {
      "rule": {
        "id": "abc/def/ghi",
        "index": 0
      }
    },
    {
      "ruleId": "abc/def/ghi",
      "rule": {
        "index": 0
      }
    },
    {
      "ruleId": "abc/def",
      "rule": {
        "index": 0
      }
    }
  ]
}
```

A run object (§3.14).
See §3.14.6.
See §3.18.2.
See §3.19.23.
A reportingDescriptor object (§3.49).
See §3.49.3.
See §3.14.23.
A result object (§3.27).
See §3.27.5.


```

    },
    {
      "rule": {
        "id": "abc/defg",      # INVALID: theDescriptor.id is not a
        "index": 0            # "component-wise" prefix of id.
      }
    },
    {
      "ruleId": "abc/def",
      "rule": {
        "id": "abc/defg/hij", # INVALID: Not equal to ruleId.
        "index": 0
      }
    }
  ]
}

```

3.52.5 index Property

A `reportingDescriptorReference` object **MAY** contain a property named `index` whose value is the array index (§3.7.4) into the `theComponent.rules` (§3.19.23) or the `theComponent.notifications` (§3.19.24), according to the table in §3.52.3.

EXAMPLE 1: In this example, there is more than one rule with id CA1711. `index` uniquely specifies the relevant rule, whether or not there are multiple rules with the same id.

```

{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "CA1711",
          ...
        },
        {
          "id": "CA1711",
          ...
        }
      ]
    }
  },
  "results": [
    {
      "ruleId": "CA1711",
      ...
      "rule": {
        "index": 1
      }
    }
  ]
}

```

If `index` is absent and `theResult.ruleIndex` (§3.27.6) is present, `index` **SHALL** default to `theResult.ruleIndex`. If both are present, they **SHALL** be equal.

3.52.6 guid Property

A `reportingDescriptorReference` object **MAY** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) equal to the `theDescriptor.guid` (§3.49.5).

3.52.7 toolComponent Property

A `reportingDescriptorReference` object **MAY** contain a property named `toolComponent` whose value is a `toolComponentReference` object (§3.54) that identifies the `theComponent`.

If `toolComponent` is absent, the `theComponent` shall be taken to be `theTool.driver` (§3.18.2).

3.53 reportingDescriptorRelationship Object

3.53.1 General

A reportingDescriptorRelationship object specifies one or more directed relationships from one reportingDescriptor object (§3.49), which we refer to as theSource, to another one, which we refer to as theTarget.

reportingDescriptorRelationship objects appear as elements of the reportingDescriptor.relationships array (§3.49.15). The reportingDescriptor object containing this property is theSource.

reportingDescriptorRelationship objects are useful in various scenarios:

1. In relating analysis rules to taxonomic categories (“taxa”; see §3.19.3).

EXAMPLE 1: In this example, the definition of rule CA1000 states that every result that violates this rule falls into the taxonomic category (“taxon”) specified by ID 327 of the Common Weakness Enumeration [CWE]:

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "CA1000",
          "relationships": [
            {
              "target": {
                "id": "327",
                "guid": "33333333-0000-1111-8888-111111111111",
                "toolComponent": {
                  "name": "CWE",
                  "guid": "33333333-0000-1111-8888-000000000000"
                }
              },
              "kinds": [
                "superset"
              ]
            }
          ]
        }
      ]
    }
  },
  "taxonomies": [
    {
      "name": "CWE",
      "guid": "33333333-0000-1111-8888-000000000000",
      "taxa": [
        {
          "id": "327",
          "guid": "33333333-0000-1111-8888-111111111111",
          "name": "BrokenOrRiskyCryptographicAlgorithm",
          ...
        }
      ]
    }
  ],
  ...
}
```

2. In relating one analysis rule to another.

EXAMPLE 2: In this example, the definition of rule CA1000 states that every violation of this rule will lead to a violation of rule CA2000.

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "CA1000",
          "guid": "11111111-0000-1111-8888-000000000001",
          "relationships": [
            {
              "target": {
                "id": "CA2000",
                "guid": "11111111-0000-1111-8888-000000000002"
              },
              "kinds": [
                "willFollow"
              ]
            }
          ]
        }
      ]
    }
  },
  ...
}
```

```

    ]
  },
  {
    "id": "CA2000",
    "guid": "11111111-0000-1111-8888-000000000002"
    ...
  }
]
},
...

```

3.53.2 **target** Property

A `reportingDescriptorRelationship` object **SHALL** contain a property named `target` whose value is a `reportingDescriptorReference` object which identifies the `target` (see §3.53.1).

3.53.3 **kinds** Property

A `reportingDescriptorRelationship` object **MAY** contain a property named `kinds` whose value is an array of one or more unique (§3.7.3) strings each of which specifies a relationship between `theSource` and `theTarget` (see §3.53.1). If `kinds` is absent, it **SHALL** default to [`"relevant"`] (see below for the meaning of `"relevant"`).

When possible, SARIF producers **SHOULD** use the following values, with the specified meanings.

- `"equal"`: `theTarget` identifies essentially the same set of items as does `theSource` (for example, a taxonomic category that identifies the same set of results as this rule).
- `"superset"`: `theTarget` identifies a superset of the items identified by `theSource` (for example, a taxonomic category that identifies a superset of the results identified by this rule).
- `"subset"`: `theTarget` identifies a subset of the items identified by `theSource` (for example, a taxonomic category that identifies a subset of the results identified by this rule).
- `"disjoint"`: The sets of items identified by `theTarget` does not intersect with the set of items identified by `theSource`.
- `"incomparable"`: The sets of items identified by `theTarget` intersects with the set of items identified by `theSource` but is neither a superset nor a subset.
- `"canFollow"`: Items identified by `theTarget` can be caused by, or occur downstream of, items identified by `theSource`.
- `"canPrecede"`: Items identified by `theSource` can be caused by, or occur downstream of, items identified by `theTarget`.
- `"willFollow"`: Items identified by `theTarget` will be caused by, or occur downstream of, items identified by `theSource`.
- `"willPrecede"`: Items identified by `theSource` will be caused by, or occur downstream of, items identified by `theTarget`.
- `"relevant"`: `theTarget` is relevant to `theSource` in a way not covered by other relationship kinds.

If none of these values are appropriate, a SARIF producer **MAY** use any value.

NOTE 1: Although `"relevant"` is a catch-all for any relationship not described by the other values, a producer might still wish to define its own more specific values.

NOTE 2: The values `"equal"` and `"superset"` are special in that they allow certain elements of `result.taxa` (§3.27.8) to be elided. See §3.27.8, paragraph 2, for more information on this point.

3.53.4 **description** Property

A `reportingDescriptorRelationship` object **MAY** contain a property named `description` whose value is a message object (§3.11) that describes the relationship.

3.54 **toolComponentReference** Object

3.54.1 General

A `toolComponentReference` object identifies a particular `toolComponent` object (§3.19), either the `tool.driver` (§3.18.2) or an element of the `tool.extensions` (§3.18.3). We refer to the identified `toolComponent` object as the `Component`.

3.54.2 Toolcomponent Lookup

If neither `index` (§3.54.4) nor `guid` (§3.54.5) is present, the `Component` **SHALL** be the `tool.driver` (§3.18.2).

If `index` is present, the `Component` **SHALL** be the object at array index `index` within the `tool.extensions` (§3.18.3).

If `index` is absent and `guid` is present, the `Component` **SHALL** be either the `tool.driver` or an element of the `tool.extensions`, whichever one has a matching `guid` property.

3.54.3 **name** Property

A `toolComponentReference` object **MAY** contain a property named `name` whose value is a string equal to the `Component.name` (§3.19.8).

NOTE: This property does not participate in the lookup, but its presence improves the readability of the log file at the expense of increased file size.

3.54.4 **index** Property

If the `Component` is an element of the `tool.extensions` (§3.18.3), a `toolComponentReference` object **MAY** contain a property named `index` whose value is the array index (§3.7.4) of that element. Otherwise, `index` **SHALL** be absent.

3.54.5 **guid** Property

A `toolComponentReference` object **MAY** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) equal to the `Component.guid` (§3.19.6).

3.55 **fix** Object

3.55.1 General

A `fix` object represents a proposed fix for the problem indicated by the `result`. It specifies a set of artifacts to modify. For each artifact, it specifies regions to remove, and provides new content to insert.

EXAMPLE 1:

```
{
  "fixes": [
    {
      "description": {
        "text": "Private member names begin with '_' "
      },
      "artifactChanges": [
        {
          # A result object (§3.27).
          # See §3.27.30.
          # A fix object.
          # See §3.55.2.
          # See §3.55.3.
          # An artifactChange object (§3.56).
```

```

    ...
  }
],
...
}

```

3.55.2 **description** Property

A **fix** object **SHOULD** contain a property named **description** whose value is a **message** object (§3.11) that describes the proposed fix.

NOTE: The purpose of the **description** property is to enable a SARIF viewer to present the proposed fix to the end user.

EXAMPLE 1:

```

"fix": {
  "description": {
    "text": "Combine declaration and initialization of variable 'x'."
  },
  ...
}

```

3.55.3 **artifactChanges** Property

A **fix** object **SHALL** contain a property named **artifactChanges** whose value is an array of one or more unique (§3.7.3) **artifactChange** objects (§3.56) each of which describes the changes to a single artifact that are necessary to effect the fix.

NOTE: **artifactChanges** is an array because a fix might require changes to multiple artifacts.

The array elements **SHALL** refer to distinct artifacts.

EXAMPLE 1: In this example, two **artifactChange** objects make identical changes (commenting out the first line) in two distinct C-language files, **src/a.c** and **src/b.c**.

```

{
  "artifactChanges": [
    {
      "artifactLocation": {
        "uri": "src/a.c"
      },
      "replacements": [
        {
          "deletedRegion": {
            "startLine": 1,
            "startColumn": 1,
            "endColumn": 1
          },
          "insertedContent": {
            "text": "// "
          }
        }
      ]
    },
    {
      "artifactLocation": {
        "uri": "src/b.c"
      },
      "replacements": [
        {
          "deletedRegion": {
            "startLine": 1,
            "startColumn": 1,
            "endColumn": 1
          },
          "insertedContent": {
            "text": "// "
          }
        }
      ]
    }
  ]
}

```

EXAMPLE 2: This example represents invalid SARIF because the two `artifactChange` objects refer to the same file, `src/a.c`. It is invalid even though the `artifactChange` objects are distinguished by their `replacements` properties.

```
{
  "artifactChanges": [
    {
      "artifactLocation": {
        "uri": "src/a.c"
      },
      "replacements": [
        {
          "deletedRegion": {
            "startLine": 1,
            "startColumn": 1,
            "endColumn": 1
          },
          "insertedContent": {
            "text": "// "
          }
        }
      ]
    },
    {
      "artifactLocation": {
        "uri": "src/a.c"
      },
      "replacements": [
        {
          "deletedRegion": {
            "startLine": 2,
            "startColumn": 1,
            "endColumn": 1
          },
          "insertedContent": {
            "text": "// "
          }
        }
      ]
    }
  ]
}
```

A fix object.
An artifactChange object (§3.56).
See §3.56.2.
See §3.56.3.
A replacement object (§3.57).
See §3.57.3.
See §3.57.4.
Invalid: refers to the same file.
Invalid even though it affects a different line.

3.56 artifactChange Object

3.56.1 General

An `artifactChange` object represents a change to a single artifact.

EXAMPLE 1:

```
{
  "artifactChanges": [
    {
      "artifactLocation": {
        "uri": "a.h"
      },
      "replacements": [
        {
          ...
        },
        {
          ...
        }
      ]
    }
  ]
}
```

A fix object (§3.55).
See §3.55.3.
See §3.56.2.
See §3.56.3.
A replacement object (§3.57).
Another replacement object.

3.56.2 artifactLocation Property

An `artifactChange` object **SHALL** contain a property named `artifactLocation` whose value is an `artifactLocation` object (§3.4) that represents the location of the artifact.

3.56.3 replacements Property

An `artifactChange` object **SHALL** contain a property named `replacements` whose value is an array of one or more replacement objects (§3.57) each of which represents the replacement of a single region of the artifact specified by the `artifactLocation` property (§3.56.2).

3.57 replacement Object

3.57.1 General

A replacement object represents the replacement of a single region of an artifact. If the region's length is zero, it represents an insertion point.

If a replacement object specifies both the removal of a region by means of the `deletedRegion` property (§3.57.3) and the insertion of new content by means of the `insertedContent` property (§3.57.4), then the effect of the replacement **SHALL** be as if the removal were performed before the insertion.

If a single `artifactChange` object (§3.56) specifies more than one replacement, then the effect of the replacements **SHALL** be as if they were performed in the order they appear in the `replacements` array (§3.56.3). The `deletedRegion` property of each replacement object **SHALL** specify the location of the replacement in the unmodified artifact.

EXAMPLE 1: Suppose an `artifactChange` object contains a `replacements` property whose value is the following array of replacement objects:

```
"artifactChanges": [
  {
    "deletedRegion": {
      "byteOffset": 12,
      "byteLength": 5
    },
    "insertedContent": {
      "binary": "ZXhhbXBsZQ=="
    }
  },
  {
    "deletedRegion": {
      "byteOffset": 20,
      "byteLength": 3
    }
  },
  {
    "deletedRegion": {
      "byteOffset": 312,
      "byteLength": 0
    },
    "insertedContent": {
      "binary": "ZXhhbXBsZQ=="
    }
  }
]
```

The first replacement object removes 5 bytes starting at offset 12; that is, it removes bytes 12–16. Then it inserts the 7 bytes specified by the MIME Base64-encoded string in the `insertedContent.binary` property at the same offset.

The second replacement object removes 3 bytes starting at offset 20 *with respect to the unmodified file*. Since 5 bytes were removed and 7 bytes inserted *before* byte 20, the 3 bytes removed actually start at byte 22 of the contents after the first change. Since the `insertedContent` property is absent, no content is inserted in place of the deleted bytes.

In the third replacement object, the length of the region specified by the `deletedRegion` property is zero, so the region represents an insertion point. The 7 bytes specified by the `insertedContent.binary` property are inserted at offset 312 with respect to the unmodified artifact.

A replacement object can represent either a textual replacement or a binary replacement, depending on whether the `deletedRegion` property (§3.57.3) specifies a text region (§3.30.2) or a binary region (§3.30.3).

EXAMPLE 2: In this example, the `replacements` property specifies a replacement in a text file.

```
"replacements": [
  {
    "deletedRegion": { # The region object represents a text region (§3.30.2).
      "startLine": 12,
      "startColumn": 5,
      "endColumn": 9
    },
    "insertedContent": {
      "text": "example" # The insertedContent property contains a text
                       # property instead of a binary property.
    }
  }
]
```


When performing a replacement in a text artifact, the SARIF producer **SHOULD** specify a text replacement rather than a binary replacement. This allows the SARIF producer to specify the region without regard to whether the artifact starts with a byte order mark (BOM).

3.57.2 Constraints

If the `deletedRegion` property (§3.57.3) specifies a text region (§3.30.2) and the `insertedContent` property (§3.57.4) is present, then the `insertedContent` property **SHOULD** contain a `text` property (§3.3.2).

If the `deletedRegion` property specifies a binary region (§3.30.3) and the `insertedContent` property is present, then the `insertedContent` property **SHALL** contain a `binary` property (§3.3.3).

Although it is possible to construct a `replacement` object that neither removes nor adds any content, a `replacement` object **SHOULD** have a material effect on the target artifact, either because `deletedRegion` denotes a non-empty region to delete, or because `insertedContent` specifies non-empty content to insert, or both.

3.57.3 `deletedRegion` Property

A `replacement` object **SHALL** contain a property named `deletedRegion` whose value is a `region` object (§3.30) specifying the region to delete.

If the length of the region specified by `deletedRegion` is zero, then `deletedRegion` specifies an insertion point, and the SARIF consumer performing the replacement **SHALL NOT** remove any content.

3.57.4 `insertedContent` Property

A `replacement` object **MAY** contain a property named `insertedContent` whose value is an `artifactContent` object (§3.3) that specifies the content to insert in place of the region specified by the `deletedRegion` property (or at the point specified by `deletedRegion`, if `deletedRegion` has a length of zero and therefore specifies an insertion point).

If the inserted content is specified as text, the text **SHALL** be transcoded from UTF-8 (the encoding of all text in all SARIF log files) to the encoding of the target artifact before being inserted.

NOTE: This implies that a text fix cannot be safely applied unless the target artifact's encoding is known.

If `insertedContent` is absent or its properties specify content whose length is zero, the SARIF consumer performing the replacement **SHALL NOT** insert any content.

3.58 notification Object

3.58.1 General

A `notification` object describes a condition encountered during the execution of an analysis tool which is relevant to the operation of the tool itself, as opposed to being relevant to an artifact being analyzed by the tool. Conditions relevant to artifacts being analyzed by a tool are represented by `result` objects (§3.27).

3.58.2 `descriptor` Property

A `notification` object **SHOULD** contain a property named `descriptor` whose value is a `reportingDescriptorReference` object (§3.52) that identifies this notification.

If the `reportingDescriptor` object (§3.49) `theDescriptor` to which `descriptor` refers exists (that is, if `theTool` contains a `reportingDescriptor` object that describes this notification), then `descriptor` **SHOULD** refer to `theDescriptor`.

NOTE: If theDescriptor exists but descriptor does not refer to it, a SARIF consumer will not be able to locate the metadata for this notification.

3.58.3 associatedRule Property

If the condition described by the notification object is relevant to a particular analysis rule, the notification object **SHOULD** contain a property named `associatedRule` whose value is a `reportingDescriptorReference` object (§3.52) that identifies the rule.

EXAMPLE 1: In this example, there is more than one rule with id CA1711. `associatedRule.index` uniquely specifies the relevant rule.

```
{
  "tool": {
    "driver": {
      "name": "CodeScanner",
      "rules": [
        {
          "id": "CA1711",
          ...
        },
        {
          "id": "CA1711",
          ...
        }
      ]
    }
  },
  "invocations": [
    {
      "toolConfigurationNotifications": [
        {
          "descriptor": {
            "id": "CFG0001"
          },
          "message": {
            "text": "Rule configuration is missing."
          },
          "associatedRule": {
            "id": "CA1711",
            "index": 1
          }
        }
      ]
    },
    ...
  ]
}
```

A run object (§3.14).
 # See §3.14.6.
 # See §3.18.2.
 # See §3.19.23.
 # A reportingDescriptor object (§3.49).
 # Another reportingDescriptor object
 # with the same id. associatedRule.id
 # identifies this one.
 # See §3.14.11.
 # An invocation object (§3.20).
 # See §3.20.22.
 # A notification object (§3.58).

3.58.4 locations Property

If the condition described by the notification object is relevant to one or more locations, the notification object **MAY** contain a property named `locations` whose value is an array of zero or more unique (§3.7.3) `location` objects (§3.28) that identify those locations to which the condition described by the notification applies.

3.58.5 message Property

A notification object **SHALL** contain a property named `message` whose value is a `message` object (§3.11) that describes the condition that was encountered. See §3.11.7 for the procedure for looking up a message string from a `message` object, in particular, for the case where the `message` object occurs as the value of `notification.message`.

3.58.6 level Property

A notification object **MAY** contain a property named `level` whose value is one of a fixed set of strings that specify the severity level of the notification.

If present, the `level` property **SHALL** have one of the following values, with the specified meanings:

- "error": A serious problem was found. The condition encountered by the tool resulted in the analysis being halted or caused the results to be incorrect or incomplete.

- "warning": A problem that is not considered serious was found. The condition encountered by the tool is such that it is uncertain whether a problem occurred, or is such that the analysis might be incomplete but the results that were generated are probably valid.
- "note": The notification is purely informational. There is no required action.
- "none": This is a trace notification (typically, debug output from the tool).

If `level` is absent, it **SHALL** default to the value determined by the procedure defined for `result.level` (§3.27.10), except throughout the procedure, replace `ruleConfigurationOverrides` with `notificationConfigurationOverrides`.

Analysis tools **SHOULD** treat notifications whose `level` property is "error" as failures and treat the entire run as having failed (for example, by settings the exit code to the value that the tool uses to indicate failure, typically a non-zero value).

Because a notification whose `level` property is "error" describes a failed run, an analysis tool **SHALL NOT** override the severity of such a notification.

3.58.7 `threadId` Property

A `notification` object **MAY** contain a property named `threadId` whose value is an integer which identifies the thread associated with this notification.

3.58.8 `timeUtc` Property

A `notification` object **MAY** contain a property named `timeUtc` whose value is a string in the format specified §3.9, specifying the UTC date and time at which the analysis tool generated the notification.

3.58.9 `exception` Property

If the notification is a result of a runtime exception, the `notification` object **MAY** contain a property named `exception` whose value is an `exception` object (§3.59).

If the notification is not the result of a runtime exception, the `exception` property **SHALL** be absent.

3.58.10 `relatedLocations` Property

A `notification` object **MAY** contain a property named `relatedLocations` whose value is an array of zero or more unique (§3.7.3) `location` objects (§3.28) that identify those locations relevant to understanding the `notification`.

The `relatedLocations` property **SHOULD** allow `notification` objects to distinguish between the following types of locations:

- Locations to which the condition described by the `notification` object **SHALL** apply.
- Other locations to which the condition described by the `notification` object **SHALL NOT** apply but are relevant to understanding the result.

3.59 `exception` Object

3.59.1 General

An `exception` object describes a runtime exception encountered during the execution of an analysis tool. This includes signals in POSIX-conforming operating systems

3.59.2 **kind** Property

An `exception` object **SHOULD** contain a property named `kind` whose value is a string describing the exception.

If the exception represents a thrown object, `kind` **SHALL** be the fully qualified type name of the object that was thrown, if that information is available.

EXAMPLE 1: C#: "System.ArgumentNullException"

If the exception represents a POSIX signal, `kind` **SHALL** be the symbolic name of the signal as specified in `<signal.h>`.

EXAMPLE 2: POSIX: "SIGFPE"

If the tool does not have access to information about the object that was thrown, the `kind` property **SHALL** be absent.

3.59.3 **message** Property

An `exception` object **SHOULD** contain a property named `message` whose value is a string that describes the exception.

If the tool does not have access to an appropriate property of the thrown object, the `message` property **SHALL** be absent.

EXAMPLE 1: C++: The tool might populate `message` with the string returned from the `what()` method of any object derived from `std::exception`.

EXAMPLE 2: C#: The tool might populate `message` with the value returned from the `ToString()` method of the `System.Exception` object, or (less informatively) from that object's `Message` property.

3.59.4 **stack** Property

An `exception` object **MAY** contain a property named `stack` whose value is a `stack` object (§3.44) that describes the sequence of function calls leading to the exception.

3.59.5 **innerExceptions** Property

An `exception` object **MAY** contain a property named `innerExceptions` whose value is an array of zero or more `exception` objects each of which is considered a cause of the containing exception.

NOTE: There is commonly no more than one inner exception. This property is an array to accommodate platforms that provide a mechanism for aggregating exceptions, such as the `System.AggregateException` class from the .NET Framework.

4 External property file format

4.1 General

External property files (see §3.15.2) conform to a schema distinct from that of the root file. External property files contain information that makes it possible for a consumer to determine which properties are contained in the file, to parse their contents, and to associate the external properties with the run to which they belong.

An external property file **SHALL** contain one or more externalized properties. A SARIF consumer **SHALL** treat the value of an externalized property exactly as if it had appeared inline in the root file as the value of the corresponding property.

4.2 External Property File Naming Convention

The file name of an external property file **SHOULD** end with the extension `".sarif-external-properties"`.

EXAMPLE 1: scan-results.sarif-external-properties

The file name **MAY** end with the additional extension ".json".

EXAMPLE 2: scan-results.sarif-external-properties.json

4.3 externalProperties Object

4.3.1 General

The top-level element of an external property file **SHALL** be an object which we refer to as an `externalProperties` object.

EXAMPLE 1: In this example, `run.artifacts` and `run.properties` have been externalized to a file with these contents. Note that `run.properties` has been externalized under the property name `externalizedProperties`, as explained in §3.15.3.

```
{
  "version": "2.1.0",           # An externalProperties object
                                # See §4.3.3.
  "$schema":                   # See §4.3.2.
    "https://docs.oasis-open.org/sarif/sarif/v2.1.0/errata01/csd01/schemas/sarif-external-property-file-schema-2.1.0.json",
                                # See §4.3.4.
  "guid": "00001111-2222-1111-8888-555566667777",
                                # See §4.3.5.
  "runGuid": "88889999-AAAA-1111-8888-DDDDEEEFFFFF",
  "artifacts": [               # See §4.3.6.
    {
      "location": {
        "uri": "apple.png"
      },
      "mimeType": "image/png"
    },
    {
      "location": {
        "uri": "banana.png"
      },
      "mimeType": "image/png"
    }
  ],
  "externalizedProperties": {
    "team": "Security Assurance Team"
  }
}
```

4.3.2 \$schema Property

An `externalProperties` object **MAY** contain a property named `\$schema` whose value is a string containing an absolute URI from which a JSON schema document describing the version of the external property file format to which this external property file conforms can be obtained.

If the `\$schema` property is present, the JSON schema obtained from the specified URI **SHALL** describe the version of the external property file format corresponding to the SARIF version specified by the `version` property (§4.3.3).

NOTE 1: The purpose of the `\$schema` property is to allow JSON schema validation tools to locate an appropriate schema against which to validate the external property file. This is useful, for example, for tool authors who wish to ensure that external property files produced by their tools conform to the external property file format.

NOTE 2: The SARIF external property file schema is available at <https://docs.oasis-open.org/sarif/sarif/v2.1.0/errata01/csd01/schemas/sarif-external-property-file-schema-2.1.0.json>.

4.3.3 version Property

Depending on the circumstances, an `externalProperties` object either **SHALL** or **MAY** contain a property named `version` whose value is a string designating the version of the SARIF specification to which this external property file conforms. If present, this string **SHALL** have the value "2.1.0".

If this `externalProperties` object is the root element of an external property file (see §3.15.2), then `version` **SHALL** be present.

Otherwise (that is, if this `externalProperties` object is an element of the `SarifLog.inlineExternalProperties` (§3.13.5)), then `version` **MAY** be present. If absent, it **SHALL** default to the value of the `SarifLog.version` (§3.13.2).

Although the order in which properties appear in a JSON object value is not semantically significant, the `version` property **SHOULD** appear first.

NOTE: This will make it easier for parsers to handle multiple versions of the external property file format if new versions are defined in the future.

4.3.4 guid Property

An `externalProperties` object **SHOULD** contain a property named `guid` whose value is a GUID-valued string (§3.5.3) that equals the `guid` property (§3.16.4) of the corresponding `externalPropertyFileReference` object (§3.16) in the `run.externalPropertyFiles` property (§3.14.2) in the root file.

4.3.5 runGuid Property

If the externalized properties contained in this `externalProperties` object are associated with a single run object (§3.14) `theRun`, and if `theRun` contains an `automationDetails.guid` property (§3.14.3, §3.17.4), the `externalProperties` object **MAY** contain a property named `runGuid` whose value is a GUID-valued string (§3.5.3) that equals `theRun.automationDetails.guid`. Otherwise (that is, if this `externalProperties` object is associated with more than one run object, or if `theRun` does not define `automationDetails.guid`), then `runGuid` **SHALL** be absent.

4.3.6 The Property Value Properties

An `externalProperties` object **SHALL** contain zero or more externalized properties. The property names in this object, and the names of the corresponding externalized properties, are given in the table in §3.15.3.

The corresponding property values are the values of the externalized properties, exactly as they would have appeared had they occurred inline in the root file.

NOTE 2: See the EXAMPLE in §4.3.1, where the externalized properties are `run.artifacts` and `run.properties`, the externalized value of `run.artifacts` is stored in a property named `artifacts`, and the externalized value of `run.properties` is stored in a property named `externalizedProperties`.

5 Conformance

5.1 Conformance Targets

This document defines requirements for the SARIF file format and for certain software components that interact with it. The entities (“conformance targets”) for which this document defines requirements are:

- **SARIF log file:** A log file in the format defined by this document.
- **SARIF producer:** A program which emits output in the SARIF format.
- **Direct producer:** An analysis tool which acts as a SARIF producer.
- **Converter:** A SARIF producer that transforms the output of an analysis tool from its native output format into the SARIF format.
- **SARIF post-processor:** A SARIF producer that transforms an existing SARIF log file into a new SARIF log file, for example, by removing or redacting security-sensitive elements.
- **SARIF consumer:** A program that reads and interprets a SARIF log file.
- **Viewer:** A SARIF consumer that reads a SARIF log file, displays a list of the results it contains, and allows an end user to view each result in the context of the artifact in which it occurs.
- **Result management system:** a software system that consumes the log files produced by analysis tools, produces reports that enable engineering teams to assess the quality of their software artifacts at a point in time and to observe trends in the quality over time, and performs functions such as filing bugs and displaying information about individual results.
- **Engineering system:** a software development environment within which analysis tools execute. It might include a build system, a source control system, a result management system, a bug tracking system, a test execution system, and so on.

The normative content in this document defines requirements for SARIF log files, except for those normative requirements that are explicitly designated as defining the behavior of another conformance target.

5.2 Conformance Clause 1: SARIF Log File

A text file satisfies the “SARIF log file” conformance profile if:

- It conforms to the syntax and semantics defined in §3

5.3 Conformance Clause 2: SARIF Producer

A program satisfies the “SARIF producer” conformance profile if:

- It produces output in the SARIF format, according to the semantics defined in §3
- It satisfies those normative requirements in §3 that are designated as applying to SARIF producers.

5.4 Conformance Clause 3: Direct Producer

An analysis tool satisfies the “Direct producer” conformance profile if:

- It satisfies the “SARIF producer” conformance profile.
- It additionally satisfies those normative requirements in §3 that are designated as applying to “direct producers” or to “analysis tools”.
- It does not emit any objects, properties, or values which, according to §3, are intended to be produced only by converters.

5.5 Conformance Clause 4: Converter

A converter satisfies the “Converter” conformance profile if:

- It satisfies the “SARIF producer” conformance profile.
- It additionally satisfies those normative requirements in §3 that are designated as applying to converters.
- It does not emit any objects, properties, or values which, according to §3, are intended to be produced only by direct producers.

5.6 Conformance Clause 5: SARIF Post-Processor

A SARIF post-processor satisfies the “SARIF post-processor” conformance profile if:

- It satisfies the “SARIF consumer” conformance profile.
- It satisfies the “SARIF producer” conformance profile.
- It additionally satisfies those normative requirements in §3 that are designated as applying to post-processors.

5.7 Conformance Clause 6: SARIF Consumer

A consumer satisfies the “SARIF consumer” conformance profile if:

- It reads SARIF log files and interprets them according to the semantics defined in §3
- It satisfies those normative requirements in §3 that are designated as applying to SARIF consumers.

5.8 Conformance Clause 7: Viewer

A viewer satisfies the “viewer” conformance profile if:

- It satisfies the “SARIF consumer” conformance profile.
- It additionally satisfies the normative requirements in §3 that are designated as applying to viewers.

5.9 Conformance Clause 8: Result Management System

A result management system satisfies the “result management system” conformance profile if:

- It satisfies the “SARIF consumer” conformance profile.
- It additionally satisfies the normative requirements in §3 and §Appendix B. (“Use of fingerprints by result management systems”) that are designated as applying to result management systems.

5.10 Conformance Clause 9: Engineering System

An engineering system satisfies the “engineering system” conformance profile if:

- It satisfies the normative requirements in §3 that are designated as applying to engineering systems.

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Appendix B. (Normative) Use of Fingerprints by Result Management Systems

On large software projects, a single run of a set of analysis tools can produce hundreds of thousands of results or more. To deal with so many results, some engineering teams adopt a strategy whereby they first prevent the introduction of new problems into their code, and then work to address the existing problems.

To prevent the introduction of new problems, it is necessary first to record the results from a designated run. We refer to this as a baseline. It is then necessary to compare the results from a subsequent run with the baseline.

To determine whether a result from a subsequent run is logically the same as a result from the baseline, there must be a way to use information contained in the result to construct a stable identifier for the result. We refer to this identifier as a fingerprint.

A result management system **SHOULD** construct a fingerprint by using information contained in the SARIF file such as

- the name of the tool that produced the result.
- the rule id.
- the file system path to the analysis target.

There are situations where information that would be helpful in uniquely identifying a result is not easily detectable by the result management system. For example, consider a tool which checks documentation for words that are culturally or politically sensitive. The word would most likely occur only in `result.message`, for example: "The word xxx should not be used in documentation."

The SARIF format provides the `partialFingerprints` property to allow analysis tools and other components in the SARIF ecosystem to provide additional information which a result management system can incorporate into the fingerprint that it constructs for each result. In this example, the tool might set the value of a property in the `partialFingerprints` object to the prohibited word. A result management system **SHOULD** include the information in `partialFingerprints` in its fingerprint computation. See §3.27.17 for more requirements on how a result management system decides which partial fingerprints to use.

An analysis tool **SHOULD NOT** include in `partialFingerprints` information that a result management system could deduce from other information in the SARIF file, for example, file hashes. Rather, the result management would use such information, along with `partialFingerprints`, in its computation of `fingerprints`.

Some information contained in the result is not useful in constructing a fingerprint. For example, suppose the fingerprint were to include the line number where the result was located, and suppose that after the baseline was constructed, a developer inserted additional lines of code above that location. Then in the next run, the result would occur on a different line, the computed fingerprint would change, and the result management system would erroneously report it as a new result.

A result management system **SHOULD NOT** include an absolute line number (or an absolute byte location in a binary artifact) in its fingerprint computation.

NOTE: The inclusion of non-deterministic file format elements (§Appendix F., §F.2) or non-deterministic absolute URIs (§Appendix F., §F.4) in the fingerprint computation will compromise the usefulness of fingerprints for distinguishing logically identical from logically distinct results.

It is difficult to devise an algorithm that constructs a truly stable fingerprint for a result. Fortunately, for practical purposes, the fingerprint does not need to be absolutely stable; it only needs to be stable enough to reduce the number of results that are erroneously reported as “new” to a low enough level that the development team can manage the erroneously reported results without too much effort.

Appendix C. (Informative) Use of SARIF by Log File Viewers

It is frequently useful for an end user to view the results produced by an analysis tool in the context of the artifacts in which they occur. A log file viewer is a program that allows an end user to do this.

Typically, the user opens a log file in the viewer, which presents a list of the results in the log file. When the user selects a result from the list, the viewer displays the source code from the file specified in the result, and displays information about the result in the vicinity of the region where the result occurred. For example, the viewer might interleave result information between lines of source code.

There are various reasons why a viewer might need to know the type of information contained in a source file that it displays:

- If the viewer knows the programming language, it can provide services such as syntax highlighting.
- If the result occurs in a source file that is nested within (for example) a compressed container file, then the viewer needs to know the file type of the container so that it can extract the source file.

There are various ways that a viewer might obtain file type information. In the SARIF format, the `mimeType` (§3.24.7) and `sourceLanguage` (§3.24.10) properties of the `artifact` object (§3.24) provides this information. In the absence of these properties, a viewer can fall back to examining the filename extension, for example “.c”.

Appendix D. (Normative) Production of SARIF by Converters

There are two broad categories of tools that can produce output in the SARIF format. Analysis tools produce SARIF as a result of performing a scan on a set of analysis targets. Converters translate existing data from a non-SARIF format into the SARIF format. That data might come from an analysis tool that produces output in a non-SARIF format, from a bug database, or from any other source.

A converter **SHOULD** populate those elements of the SARIF format for which a direct equivalent exists in the input data.

If the input data includes information for which there is no SARIF equivalent, a converter **MAY** use it to populate the various property bags (§3.8) and tag lists (§3.8.2) defined by the SARIF format, or they **MAY** simply omit it from the output. When populating a property bag with such information, a converter **SHOULD** use a property name that matches the name of that piece of information in the native tool format, even if that name does not conform to the camelCase convention used in the rest of this document.

NOTE: This makes it easier to match these properties with the source data in the native tool format.

When serializing SARIF as JSON, a converter **SHALL** replace any characters in string-valued properties that cannot occur in a JSON string with the appropriate escape sequence as defined by JSON [RFC8259].

If the input data does not include an equivalent for any SARIF element, a converter **MAY** attempt to synthesize that element. (For example, a converter might heuristically extract a rule id from the text of an unstructured error message.)

Since each converter might synthesize SARIF elements differently (notably the rule id; see §3.27.5), a SARIF consumer **SHOULD NOT** attempt to combine results produced by different converters for the same tool.

A converter **SHOULD** populate its own semantic version [SEMVER] property `theRun.conversion.tool.driver.semanticVersion` (§3.19.12). If it does, and if a subsequent version of the converter synthesizes SARIF elements in a semantically incompatible way, it **SHALL** increment the major version component of its semantic version.

Notwithstanding this general guidance recommending that a converter synthesize SARIF elements where possible:

- A converter that knows which artifact a result was detected in, but not which artifact the analysis tool was originally instructed to scan, **SHOULD** populate `result.locations` (§3.27.12), but **SHOULD NOT** attempt to populate `result.analysisTarget` (§3.27.13).
- A converter **SHOULD NOT** populate the analysis tool's `toolComponent.semanticVersion` (§3.19.12) unless it knows that the tool component's version string is intended to be interpreted as a semantic version [SEMVER] version string.

Appendix E. (Informative) Locating Rule and Notification Metadata

The SARIF format allows rule and notification metadata to be included in a SARIF log file (see §3.19.23 and §3.19.24). A SARIF log file does not need to include any metadata. This raises the questions of when metadata should be included in a log file, and how to locate the metadata if it is not included in the log file.

Metadata should be included in a log file in the following circumstances:

- The log file is intended to be viewed in a tool such as a log file viewer that needs to display metadata related to each result or notification even when the tool is not connected to a network.
- The log file is intended to be uploaded to a result management system which requires information about every rule specified by every result, and which might not have prior knowledge of the rules specified by the results in this log file.
- Neither of the above applies, but the increased log file size due to the metadata is not considered significant.

If metadata is not included in the log file, and if external property files (see §3.15.2) are not used, this document does not specify a mechanism for locating the metadata. If the SARIF log file is produced in the context of an engineering system that provides a service from which metadata can be obtained (for example, a result management system, or a web service dedicated to metadata), then tooling can be created to merge a log file with the relevant metadata when required (for example, when presenting the results in a log file viewer).

Appendix F. (Informative) Producing Deterministic SARIF Log Files

F.1. General

In certain circumstances, it is desirable for an analysis tool to produce deterministic output; that is, for it to produce identical output when run repeatedly with identical inputs.

For example, this is useful in a build system that caches the output from each build step. If the build is rerun and the inputs to a given step are identical (which the build system might determine, for example, by comparing timestamps, or by computing a hash of the inputs to the step and storing it along with the output from the step), then the build system can save time by not re-running the step, and simply using the existing outputs.

Consider this sequence of build steps:

1. A binary analysis tool analyzes A.dll and produces A.sarif.
2. A bug database ingestion tool reads A.sarif and files bugs for any new results.

If A.sarif has not changed between this build and the previous one, the build system does not have to execute Step 2.

Authors of analysis tools are encouraged to provide a mechanism (for example, a command line option such as `--deterministic`) which instructs the tool to produce deterministic output.

There are several issues to consider when producing deterministic output:

- Avoiding elements of the SARIF file format whose values are non-deterministic.
- Emitting array and dictionary elements in a deterministic order.
- Avoiding absolute paths.
- Handling baseline information

F.2. Non-Deterministic File Format Elements

Certain optional elements of the SARIF format are non-deterministic in most situations. A log file that includes these elements will not be deterministic except under special circumstances. For example:

- If a build system always runs on the same machine under the same account, `invocation.machine` and `invocation.account` is deterministic.
- If a binary analysis tool runs in an environment that guarantees the same memory layout from run to run (for example, an environment that allows a binary to be loaded at a fixed address and that does not use address space layout randomization (ASLR)), then `physicalLocation.address` and `run.addresses` are deterministic.

Authors of analysis tools are encouraged to provide a mechanism (for example, a command line option such as `--known-deterministic-properties:<property name>...`) which allows the tool to emit specified properties even when producing deterministic output.

Avoiding these elements, in conjunction with the techniques described in subsequent sections of this Appendix, makes it more likely that the analysis tool will produce deterministic output:

- Non-deterministic elements in property bag properties.

- Non-deterministic elements in user-facing messages, for example, a timestamp in a result message.
- The trailing component of `run.automationDetails.id`
- `run.automationDetails.guid`
- `run.baselineGuid`
- `run.originalUriBaseIds`
- `run.addresses`, because security measures such as address space layout randomization (ASLR) might place the same code at different addresses from run to run.
- `invocation.commandLine`, because it might specify non-deterministic absolute file paths or other non-deterministic elements.
- `invocation.arguments`, for the same reason.
- `invocation.processId`
- `invocation.startTimeUtc`
- `invocation.endTimeUtc`
- `invocation.machine`
- `invocation.account`
- `invocation.workingDirectory`, because the tool might be launched from different directories on different machines.
- `invocation.environmentVariables`
- `invocation.stdin`, `stdout`, `stderr`, or `stdoutStderr`, because the tool's console output might include non-deterministic elements such as timestamps.
- `versionControlDetails.revisionId`
- `versionControlDetails.asOfTimeUtc`
- `versionControlDetails.mappedTo`, because a repository might be downloaded to different directories on different machines.
- `threadFlow.threadId`
- `threadFlowLocation.executionTimeUtc`
- `notification.threadId`
- `notification.timeUtc`
- `result.guid`
- `stackFrame.threadId`
- `physicalLocation.address`, for the same reason as `run.addresses`.

F.3. Array and Dictionary Element Ordering

One obstacle to determinism in SARIF log files is the ordering of array elements and object properties.

For some arrays, SARIF requires a specific ordering. For example, within `stack.frames`, SARIF requires the `location` object representing the most deeply nested function call to appear first.

For other arrays, for example `properties.tags`, SARIF does not require a specific ordering. For such arrays, a tool can ensure the order by sorting the array elements before writing them to the log file. For example, it might sort the tags in locale-insensitive alphabetical order.

The array of `result` objects in the `run.results` array presents more of a problem. A multi-threaded analysis tool analyzing multiple artifacts in parallel might produce results in any order, and there is no natural order for the results. A tool might choose to order them, for example, first alphabetically by analysis target URI, then numerically by line number, then by column number, then alphabetically by rule id.

For dictionaries such as the `artifact.hash`s object, a tool might order the property names alphabetically, using a locale-insensitive ordering.

F.4. Absolute Paths

Another obstacle to determinism is the use of absolute paths which might differ from machine to machine. For example:

- Different build machines might be configured to use different source directories.
- A single build machine might use a different directory for each build.

Tools can avoid the use of absolute file paths by emitting URIs that are relative to one or more root directories (for example, a source root directory and an output root directory), and accompanying each `artifactLocation.uri` property with the corresponding `artifactLocation.uriBaseId` property.

F.5. Inherently Non-Deterministic Tools

The algorithms used by some tools are inherently non-deterministic because, for example, they perform random sampling or random traversals of the graphs that represent the code. Generally, these tools produce mostly the same result set, but there might be small differences between runs.

Such tools can avoid this source of non-determinism by, for example, providing a command-line argument to specify the random number generator seed.

F.6. Compensating for Non-Deterministic Output

If an analysis tool does not produce deterministic output, a build system can add additional processing steps to compensate.

There are two scenarios to consider:

- Log equality is determined by a simple comparison of file contents, or by comparing file hashes.
- Log equality is determined by an “intelligent” comparison.

In the first scenario, a post-processing step could produce deterministic output by creating a new file that omits non-deterministic elements, reorders array elements and object properties, removes file path prefixes, and introduces `artifactLocation.uriBaseId` properties.

In the second scenario, a post-processing step could intelligently compare the newly produced log to the log from a previous build by ignoring non-deterministic elements, ensuring that arrays have the same elements regardless of order, and ignoring file path prefixes.

F.7. Interaction Between Determinism and Baselining

SARIF's baselining feature poses a particular challenge for determinism. We illustrate the problem with the following scenario:

On a particular date, a project's nightly build runs an analysis tool ToolX, which produces a log file, say, `log_20170914.sarif`. The next day, a developer modifies one of the files scanned by the tool in a way that introduces a new problem. That night, the nightly build tool runs again, this time producing a log file which compares the current set of results to those that appeared in the previous run:

```
ToolX --input a.c b.c --baseline log_20170914.sarif --output log_20170915.sarif
```

Because a new problem has been introduced, `log_20170915.sarif` will contain a result object whose `baselineState` is "new". The next night, without any further changes to the source files, the tool is run yet again:

```
ToolX --input a.c b.c --baseline log_20170915.sarif --output log_20170916.sarif
```

The result object that first appeared in `log_20170915.sarif` still appears in `log_20170916.sarif`, but since it existed in the baseline, its `baselineState` will now be "unchanged" or "updated" as appropriate (see §3.27.24).

The result is that even though none of the analysis target files have changed, the log file has changed, or at least, a simple file comparison (such as comparing the hash of the new log with the hash of the baseline) will report that it has changed.

Strictly speaking, this does not violate determinism. After all, the baseline file has changed, and the baseline file is one of the inputs to the analysis. But from a practical standpoint, this is still a problem, albeit a small one.

If the build uses a simple mechanism such as hash value comparison to determine if a file has changed, then on those occasions when the only difference between the newest log and the baseline is that some results that were previously "new" are now "unchanged", subsequent build steps which consume the SARIF log file will run, even if they might not actually be necessary. For example, a build step which automatically files bugs for new results will run, even though the log contains no new results. Or a build step which tracks the number of open issues will run, even though the number of open issues has not actually changed.

If the build engineers for a project wish to absolutely minimize the execution of unnecessary build steps, they have various options. They might perform an "intelligent" comparison between the baseline and the new log, treating "new" results in the baseline as equivalent to "unchanged" results. Or they might rewrite the baseline (marking all "new" results as "unchanged") before performing the comparison. Of course, there is no guarantee that such an "intelligent" comparison or baseline rewriting process will actually take less time than the unnecessary build steps it is intended to avoid.

Appendix G. (Informative) Guidance on Fixes

Tools that produce SARIF files which include `fix` objects should take care to structure those fixes in such a way as to affect a minimal range of content. This maximizes the likelihood that an automated tool can safely apply multiple fixes to the same artifact.

The following example will clarify what this means and why it is important. Consider an XML file containing the following element:

```
<lineItem partNumber=A3101 />
```

Suppose that a (domain-specific) XML scanning tool reported two results:

- The value of the `partNumber` attribute is not enclosed in quotes.
- The part numbering scheme has changed, and part numbers beginning with “A” now begin with “AA”.

Fixing only result #1 would produce the element

```
<lineItem partNumber="A3101" />
```

Fixing only result #2 would produce the element

```
<lineItem partNumber=AA3101 />
```

Fixing both results should produce the element

```
<lineItem partNumber="AA3101" />
```

The fix for result #1 might be specified in various ways, for example:

1. As a single replacement:
 - Replace the characters `A3101` with the characters `"A3101"`.
2. As a sequence of two replacements:
 - a. Insert a quotation mark before `A3101`.
 - b. Insert a quotation mark after `A3101`.

The fix for result #2 is most simply specified as a single replacement:

- Replace the characters `A3101` with the characters `AA3101`.

Suppose there exists an automated tool which reads a SARIF file containing `fix` objects and applies as many of the specified fixes as possible to the source files.

If the fix for result #1 were structured as a single replacement, then after applying the fix, the tool would not be able to fix result #2, because the range of characters specified by the fix for result #2 would have been replaced. On the other hand, if the fix for result #1 were structured as two replacements (with a separate insertion for each quotation mark), the tool would still be able to apply the fix for result #2, because the targeted range of characters would still exist.

Therefore, structuring fixes as sequences of minimal, disjoint replacements maximizes the amount of work that can be done by automated fixup tools.

Appendix H. (Informative) Diagnosing Results in Generated Files

Sometimes it is desirable to analyze files generated by the build. These files are usually not under source control, and the build might even overwrite them multiple times. This Appendix offers guidance on how to persist enough information in a SARIF log file to facilitate the diagnosis of results in these files.

In what follows, we will refer to files that are generated only once as “singly generated,” and files that are generated multiple times as “multiply generated.”

It can be difficult to diagnose results in generated files for the following reasons:

- The file might not be available to the engineer who diagnoses the result (for example, the engineer might not have a build environment).
- If the file is multiply generated, then at best only the last version is available, but results might have been found in previous versions.
- It might be difficult to tell which instance of a multiply generated file contained the result.

For both singly and multiply generated files, there are two options (which can be used together):

1. Use the `physicalLocation` object's (§3.29) `region` (§3.29.4) and `contextRegion` (§3.29.5) properties to store enough of the generated file's contents to facilitate diagnosis. The `region` object's (§3.30) `snippet` property (§3.30.13) holds the relevant portion of the file contents.
2. Use the `artifact` object's (§3.24) `contents` (§3.24.8) property to persist the entire contents of the file in the `run.artifacts` (§3.14.15).

The first option is more compact; the second allows a SARIF viewer to present results with greater context.

EXAMPLE 1: In this example, the analysis tool populates `region.snippet` and `contextRegion.snippet`, allowing a SARIF viewer to display just enough context (one hopes) to diagnose the result.

```
{
  "originalUriBaseIds": {
    "GENERATED": {
      "uri": "file:///C:/code/browser/obj/"
    }
  },
  "results": [
    {
      "ruleId": "CS6789",
      "message": {
        "text": "Division by 0"
      },
      "locations": [
        {
          "physicalLocation": {
            "artifactLocation": {
              "uri": "ui/window.g.cs",
              "uriBaseId": "GENERATED"
            },
            "region": {
              "startLine": 42,
              "snippet": {
                "text": "    int z = x / y;\r\n"
              }
            },
            "contextRegion": {
              "startLine": 40,
              "endLine": 42,
              "snippet": {
                "text": "    int x = 54;\r\n    int y = 0;\r\n    int z = x / y;\r\n"
              }
            }
          }
        }
      ]
    }
  ],
  ...
}
```

EXAMPLE 2: In this example, the analysis tool populates `artifact.contents`, allowing a SARIF viewer to present the result in a larger context at the expense of a larger log file.

```
{
  "originalUriBaseIds": {
    "GENERATED": {
      "uri": "file:///dev-1.example.com/code/browser/obj/"
    }
  },
  "results": [
    {
      "ruleId": "CS6789",
      "message": {
        "text": "Division by 0"
      },
      "locations": [
        {
          "physicalLocation": {
            "artifactLocation": {
              "uri": "ui/window.g.cs",
              "uriBaseId": "GENERATED",
              "index": 0
            },
            "region": {
              "startLine": 42
            },
            "contextRegion": {
              "startLine": 40,
              "endLine": 42
            }
          }
        }
      ]
    }
  ],
  "artifacts": [
    {
      "location": {
        "uri": "ui/window.g.cs",
        "uriBaseId": "GENERATED"
      },
      "contents": {
        "text": "..."
      }
    }
  ]
}
```

See §3.14.15.
An artifact object (§3.24).
See §3.24.2.
See §3.24.8.
See §3.3.2.

Multiply generated files are treated similarly, but they present an additional problem: if more than one version of a given multiply generated file appears in `theRun.artifacts` – either because the analysis tool wishes to persist the file contents, or for any other reason – then there must be a way to distinguish them.

The recommended solution is for the analysis tool to create a new entry in `theRun.artifacts` for each version of the generated files. The result might look like the following example.

EXAMPLE 3: In this example, `ui/window.g.cs` is multiply generated. The analysis tool creates distinct entries in `theRun.artifacts` to distinguish the two versions.

```
{
  "originalUriBaseIds": {
    "GENERATED": {
      "uri": "file:///dev-1.example.com/code/browser/obj/"
    }
  },
  "results": [
    {
      "ruleId": "CS6789",
      "message": {
        "text": "Division by 0"
      },
      "locations": [
        {
          "physicalLocation": {
            "artifactLocation": {
              "uri": "ui/window.g.cs",
              "uriBaseId": "GENERATED",
              "index": 0
            },
            "region": {
              "startLine": 42
            },
            "contextRegion": {
              "startLine": 40,
              "endLine": 42
            }
          }
        }
      ]
    }
  ],
  "artifacts": [
    {
      "location": {
        "uri": "ui/window.g.cs",
        "uriBaseId": "GENERATED"
      },
      "contents": {
        "text": "..."
      }
    }
  ]
}
```

Points to the appropriate instance
of the generated file.

```

    },
    ],
    "artifacts": [
      {
        "location": {
          "uri": "ui/window.g.cs",
          "uriBaseId": "GENERATED"
        },
        "lastModifiedTimeUtc": "2019-04-13T11:45:23.477",
        "contents": {
          "text": "... "
        }
      },
      {
        "location": {
          "uri": "ui/window.g.cs",
          "uriBaseId": "GENERATED"
        },
        "lastModifiedTimeUtc": "2019-04-13T11:46:27.013",
        "contents": {
          "text": "... "
        }
      }
    ]
  }
}

```

Appendix I. (Informative) Detecting Incomplete Result Sets

This document describes three conditions that inform the SARIF consumer that the tool has failed to produce a comprehensive set of results. For convenience, this Appendix gathers those conditions together in one place:

- If any invocation object (§3.20) in `theRun.invocations` (§3.14.11) has a value of `false` for its `executionSuccessful` property (§3.20.14), the tool either failed to start, terminated with an exit code that denotes failure, or terminated with an unhandled exception or signal.
- If any notification object (§3.58) in `invocation.toolExecutionNotifications` (§3.20.21) or `toolConfigurationNotifications` (§3.20.22) has a value of `"error"` for its `level` property (§3.58.6), it is possible that the tool was unable to execute every analysis rule on every analysis target. Therefore, the results cannot be assumed to be complete.
- If `theRun.results` (§3.14.23) is `null`, the tool either failed to start or failed to begin its analysis.

These conditions apply separately to each run in the log file.

Appendix J. (Informative) Sample `sourceLanguage` Values

This Appendix contains a list of sample values for the `artifact.sourceLanguage` property (§3.24.10) for some common programming languages. The purpose of this Appendix is to promote interoperability by encouraging SARIF producers to use the same identifiers for these languages.

The names of some of the languages in this list are the trademarks of their respective owners.

- `abap`
- `actionscript`
- `ada`
- `algol68`
- `apex`
- `assembler`
- `c`
- `clojure`
- `cobol`

- coldfusion
- cplusplus
- csharp
- css
- d
- erlang
- fsharp
- fortran
- go
- groovy
- haskell
- java
- javascript
- json
- jsp
- julia
- lisp
- lua
- markdown (variants: markdown/gfm, markdown/cmark)
- objectivec
- objectpascal
- ocaml
- perl
- php
- prolog
- python
- r
- razor
- ruby
- rust
- sarif
- scala
- scheme
- sql (variants: sql/tsql, sql/psql).
- swift

- systemverilog
- typescript
- visualbasic
- visualbasicdotnet
- yaml
- zig
- Markup languages:
 - html
 - sgml
 - xml
- Typesetting languages:
 - latex
 - nroff
 - roff
 - tex
 - troff
- UNIX® shell languages:
 - bash
 - csh
 - ksh
 - sh
 - tcsh
- Windows® shell languages:
 - cmd
 - powershell

Appendix K. (Informative) Examples

This Appendix contains examples of complete, valid SARIF files, to complement the fragments shown in examples throughout this document.

K.1. Minimal Valid SARIF Log File

This is a minimal valid SARIF log file. It contains only those elements required by this document (elements which the document states **SHALL** be present).

The file contains a single run object (§3.14) with an empty `results` array (§3.14.23), as would happen if the tool detected no issues in any of the artifacts it scanned.

```
{
  "version": "2.2",
  "runs": [
    {
      "tool": {
        "driver": {
          "name": "CodeScanner"
        }
      },
      "results": [
      ]
    }
  ]
}
```

K.2. Minimal Recommended SARIF Log File with Source Information

This is a minimal recommended SARIF log file for the case where an analysis tool produced results and source location information is available.

The file contains those elements recommended by this document (elements which the document states “**SHOULD**” be present), in addition to the required elements.

The file contains a single run object (§3.14) with a `results` array (§3.14.23). The results array contains a single `result` object (§3.27) so the recommended elements of the `result` object can be shown.

Its `run.artifacts` property (§3.14.15) specifies only those artifacts in which the tool detected a result.

It does not contain a `run.logicalLocations` property (§3.14.17), because when physical location information is available, that property is optional (it “**MAY**” be present).

This example also includes a `toolComponent.rules` property (§3.19.23) containing rule metadata, even though rule metadata is optional, to show how a SARIF log file can be self-contained, in the sense of containing all the information necessary to interpret the results.

```
{
  "version": "2.2",
  "runs": [
    {
      "tool": {
        "driver": {
          "name": "CodeScanner",
          "rules": [
            {
              "id": "C2001",
              "fullDescription": {
                "text": "A variable was used without being initialized. This can result
in runtime errors such as null reference exceptions."
              },
              "messageStrings": {
                "default": {
                  "text": "Variable \"{0}\" was used without being initialized."
                }
              }
            }
          ]
        }
      },
      "artifacts": [
        {
          "location": {
            "uri": "src/collections/list.cpp",
            "uriBaseId": "SRCROOT"
          },
          "sourceLanguage": "c"
        }
      ]
    }
  ]
}
```

```

    }
  ],
  "results": [
    {
      "ruleId": "C2001",
      "ruleIndex": 0,
      "message": {
        "id": "default",
        "arguments": [
          "count"
        ]
      }
    },
    {
      "locations": [
        {
          "physicalLocation": {
            "artifactLocation": {
              "uri": "src/collections/list.cpp",
              "uriBaseId": "SRCROOT",
              "index": 0
            },
            "region": {
              "startLine": 15
            }
          },
          "logicalLocations": [
            {
              "fullyQualifiedName": "collections::list::add"
            }
          ]
        }
      ]
    }
  ]
}

```

K.3. Minimal Recommended SARIF Log File without Source Information

This is a minimal recommended SARIF file for the case where an analysis tool produced results and source location information is not available.

The file contains those elements recommended by this document (elements which the document states “SHOULD” be present), in addition to the required elements.

The file contains a single `run` object (§3.14) with a `results` array (§3.14.23). The results array contains a single `result` object (§3.27) so the recommended elements of the `result` object can be shown.

Its `run.artifacts` property (§3.14.15) specifies only those artifacts in which the tool detected a result.

It contains a `run.logicalLocations` property (§3.14.17), because when physical location information is not available, that property is recommended.

```

{
  "version": "2.2",
  "runs": [
    {
      "tool": {
        "driver": {
          "name": "BinaryScanner"
        }
      },
      "artifacts": [
        {
          "location": {
            "uri": "bin/example",
            "uriBaseId": "BINROOT"
          }
        }
      ],
      "logicalLocations": [
        {
          "name": "Example",
          "kind": "namespace"
        },
        {
          "name": "Worker",
          "fullyQualifiedName": "Example.Worker",
          "kind": "type",
          "parentIndex": 0
        },
        {
          "name": "DoWork",
          "fullyQualifiedName": "Example.Worker.DoWork",
          "kind": "function",
          "parentIndex": 1
        }
      ],
      "results": [
        {

```

```
"ruleId": "B6412",
"message": {
  "text": "The insecure method \"Crypto.Sha1.Encrypt\" should not be used.",
},
"level": "warning",
"locations": [
  {
    "logicalLocations": [
      {
        "fullyQualifiedName": "Example.Worker.DoWork",
        "index": 2
      }
    ]
  }
]
}
```

K.4. Comprehensive SARIF File

The purpose of this example is to demonstrate the usage of as many SARIF elements as possible. Not all elements are shown, because some are mutually exclusive.

Because the purpose is to present as many elements as possible, the file as a whole does not represent best practices for SARIF usage, nor does it represent the output of a single, coherent analysis. For example, the result presented in the file involves a runtime exception, but at the same time it is marked as suppressed (to demonstrate the `result.suppressions` property), which is unrealistic.

```
{
  "version": "2.2",
  "$schema": "https://docs.oasis-open.org/sarif/sarif/v2.2/schema/sarif.json",
  "runs": [
    {
      "addresses": [
        {
          "properties": {
            "baseAddress": "4194304",
            "section": ".text"
          },
          "fullyQualifiedName": "collections.dll",
          "kind": "module"
        },
        {
          "properties": {
            "offset": 100
          },
          "fullyQualifiedName": "collections.dll!collections::list::add",
          "kind": "function",
          "parentIndex": 0
        },
        {
          "properties": {
            "offset": 22
          },
          "fullyQualifiedName": "collections.dll!collections::list::add+0x16",
          "parentIndex": 1
        }
      ],
      "automationDetails": {
        "guid": "BC650830-A9FE-44CB-8818-AD6C387279A0",
        "id": "Nightly code scan/2018-10-08"
      },
      "baselineGuid": "0A106451-C9B1-4309-A7EE-06988B95F723",
      "runAggregates": [
        {
          "id": "Build/14.0.1.2/Release/20160716-13:22:18",
          "correlationGuid": "26F138B6-6014-4D3D-B174-6E1ACE9439F3"
        }
      ],
      "tool": {
        "driver": {
          "name": "CodeScanner",
          "fullName": "CodeScanner 1.1 for Microsoft Windows (R) (en-US)",
          "version": "2.1",
          "semanticVersion": "2.1.0",
          "dottedQuadFileVersion": "2.1.0.0",
          "releaseDateUtc": "2019-03-17",
          "organization": "Example Corporation",
          "product": "Code Scanner",
          "productSuite": "Code Quality Tools",
          "shortDescription": {
            "text": "A scanner for code."
          },
          "fullDescription": {
            "text": "A really great scanner for all your code."
          },
          "informationUri": "https://www.examplecorp.com/products/codescanner",
          "properties": {
            "copyright": "Copyright (c) 2017 by Example Corporation."
          }
        }
      }
    }
  ]
}
```

```

"globalMessageStrings": {
  "variableDeclared": {
    "text": "Variable \"{0}\" was declared here.",
    "markdown": " Variable `{0}` was declared here."
  }
},
"rules": [
  {
    "id": "C2001",
    "deprecatedIds": [
      "CA2000"
    ],
    "defaultConfiguration": {
      "level": "error",
      "rank": 95
    },
    "shortDescription": {
      "text": "A variable was used without being initialized."
    },
    "fullDescription": {
      "text": "A variable was used without being initialized. This can result in runtime errors such as null reference exceptions."
    },
    "messageStrings": {
      "default": {
        "text": "Variable \"{0}\" was used without being initialized. It was declared [here]({1}).",
        "markdown": "Variable `{0}` was used without being initialized. It was declared [here]({1})."
      }
    }
  }
],
"notifications": [
  {
    "id": "start",
    "shortDescription": {
      "text": "The run started."
    },
    "messageStrings": {
      "default": {
        "text": "Run started."
      }
    }
  },
  {
    "id": "end",
    "shortDescription": {
      "text": "The run ended."
    },
    "messageStrings": {
      "default": {
        "text": "Run ended."
      }
    }
  }
],
"language": "en-US"
},
"extensions": [
  {
    "name": "CodeScanner Security Rules",
    "version": "3.1",
    "rules": [
      {
        "id": "S0001",
        "defaultConfiguration": {
          "level": "error"
        },
        "shortDescription": {
          "text": "Do not use weak cryptographic algorithms."
        },
        "messageStrings": {
          "default": {
            "text": "The cryptographic algorithm '{0}' should not be used."
          }
        }
      }
    ]
  }
]
},
"language": "en-US",
"versionControlProvenance": [
  {
    "repositoryUri": "https://github.com/example-corp/browser",
    "revisionId": "5da53fbb2a0aaa12d648b73984acc9aac2e11c2a",
    "mappedTo": {
      "uriBaseId": "PROJECTROOT"
    }
  }
],
"originalUriBaseIds": {
  "PROJECTROOT": {
    "uri": "file://build.example.com/work/"
  },
  "SRCROOT": {
    "uri": "src/",
    "uriBaseId": "PROJECTROOT"
  },
  "BINROOT": {
    "uri": "bin/",
    "uriBaseId": "PROJECTROOT"
  }
},

```

```

"invocations": [
  {
    "commandLine": "CodeScanner @build/collections.rsp",
    "responseFiles": [
      {
        "uri": "build/collections.rsp",
        "uriBaseId": "SRCROOT",
        "index": 0
      }
    ],
    "startTimeUtc": "2016-07-16T14:18:25Z",
    "endTimeUtc": "2016-07-16T14:19:01Z",
    "machine": "BLD01",
    "account": "buildAgent",
    "processId": 1218,
    "workingDirectory": {
      "uri": "file:///home/buildAgent/src"
    },
    "environmentVariables": {
      "PATH": "/usr/local/bin:/bin:/bin:/tools:/home/buildAgent/bin",
      "HOME": "/home/buildAgent",
      "TZ": "EST"
    },
    "toolConfigurationNotifications": [
      {
        "descriptor": {
          "id": "UnknownRule"
        },
        "level": "warning",
        "message": {
          "text": "Could not disable rule \"ABC0001\" because there is no rule with that id."
        }
      }
    ],
    "toolExecutionNotifications": [
      {
        "descriptor": {
          "id": "CTN0001"
        },
        "level": "note",
        "message": {
          "text": "Run started."
        }
      },
      {
        "descriptor": {
          "id": "CTN9999"
        },
        "associatedRule": {
          "id": "C2001",
          "index": 0
        },
        "level": "error",
        "message": {
          "text": "Exception evaluating rule \"C2001\". Rule disabled; run continues."
        },
        "locations": [
          {
            "physicalLocation": {
              "artifactLocation": {
                "uri": "crypto/hash.cpp",
                "uriBaseId": "SRCROOT",
                "index": 4
              }
            }
          }
        ]
      },
      {
        "threadId": 52,
        "timeUtc": "2016-07-16T14:18:43.119Z",
        "exception": {
          "kind": "ExecutionEngine.RuleFailureException",
          "message": "Unhandled exception during rule evaluation.",
          "stack": {
            "frames": [
              {
                "location": {
                  "message": {
                    "text": "Exception thrown"
                  },
                  "logicalLocations": [
                    {
                      "fullyQualifiedName": "Rules.SecureHashAlgorithmRule.Evaluate"
                    }
                  ],
                  "physicalLocation": {
                    "address": {
                      "offsetFromParent": 4244988
                    }
                  }
                },
                "module": "RuleLibrary",
                "threadId": 52
              },
              {
                "location": {
                  "logicalLocations": [
                    {
                      "fullyQualifiedName": "ExecutionEngine.Engine.EvaluateRule"
                    }
                  ],
                  "physicalLocation": {
                    "address": {

```

```

        "offsetFromParent": 4245514
    }
}
},
"module": "ExecutionEngine",
"threadId": 52
}
],
},
"innerExceptions": [
{
"kind": "System.ArgumentException",
"message": "length is < 0"
}
]
}
},
{
"descriptor": {
"id": "CTN0002"
},
"level": "note",
"message": {
"text": "Run ended."
}
}
],
"exitCode": 0,
"executionSuccessful": true
}
],
"artifacts": [
{
"location": {
"uri": "build/collections.rsp",
"uriBaseId": "SRCROOT"
},
"mimeType": "text/plain",
"length": 81,
"contents": {
"text": "-input src/collections/*.cpp -log out/collections.sarif -rules all -disable C9999"
}
},
{
"location": {
"uri": "application/main.cpp",
"uriBaseId": "SRCROOT"
},
"sourceLanguage": "cplusplus",
"length": 1742,
"hashes": {
"sha-256": "cc8e6a99f3eff00adc649fee132ba80fe333ea5a"
}
},
{
"location": {
"uri": "collections/list.cpp",
"uriBaseId": "SRCROOT"
},
"sourceLanguage": "cplusplus",
"length": 980,
"hashes": {
"sha-256": "b13ce2678a8807ba0765ab94a0ecd394f869bc81"
}
},
{
"location": {
"uri": "collections/list.h",
"uriBaseId": "SRCROOT"
},
"sourceLanguage": "cplusplus",
"length": 24656,
"hashes": {
"sha-256": "849be119aaba4e9f88921a99e3036fb6c2a8144a"
}
},
{
"location": {
"uri": "crypto/hash.cpp",
"uriBaseId": "SRCROOT"
},
"sourceLanguage": "cplusplus",
"length": 1424,
"hashes": {
"sha-256": "3ffe2b77dz255cdf95f97d986d7a6ad8f287eae"
}
},
{
"location": {
"uri": "app.zip",
"uriBaseId": "BINROOT"
},
"mimeType": "application/zip",
"length": 310450,
"hashes": {
"sha-256": "df18a5e74b6b46ddaa23ad7271ee2b7c5731cbe1"
}
},
{
"location": {
"uri": "docs/intro.docx"
},

```

```

    "mimeType": "application/vnd.openxmlformats-officedocument.wordprocessingml.document",
    "parentIndex": 5,
    "offset": 17522,
    "length": 4050
  }
],
"logicalLocations": [
  {
    "name": "add",
    "fullyQualifiedName": "collections::list::add",
    "decoratedName": "?add@list@collections@@QAEXH@Z",
    "kind": "function",
    "parentIndex": 1
  },
  {
    "name": "list",
    "fullyQualifiedName": "collections::list",
    "kind": "type",
    "parentIndex": 2
  },
  {
    "name": "collections",
    "kind": "namespace"
  },
  {
    "name": "add_core",
    "fullyQualifiedName": "collections::list::add_core",
    "decoratedName": "?add_core@list@collections@@QAEXH@Z",
    "kind": "function",
    "parentIndex": 1
  },
  {
    "fullyQualifiedName": "main",
    "kind": "function"
  }
],
"results": [
  {
    "ruleId": "C2001",
    "ruleIndex": 0,
    "kind": "fail",
    "level": "error",
    "message": {
      "id": "default",
      "arguments": [
        "ptr",
        "0"
      ]
    },
    "suppressions": [
      {
        "kind": "external",
        "status": "accepted"
      }
    ],
    "baselineState": "unchanged",
    "rank": 95,
    "analysisTarget": {
      "uri": "collections/list.cpp",
      "uriBaseId": "SRCROOT",
      "index": 2
    },
    "locations": [
      {
        "physicalLocation": {
          "artifactLocation": {
            "uri": "collections/list.h",
            "uriBaseId": "SRCROOT",
            "index": 3
          },
          "region": {
            "startLine": 15,
            "startColumn": 9,
            "endLine": 15,
            "endColumn": 10,
            "charLength": 1,
            "charOffset": 254,
            "snippet": {
              "text": "add_core(ptr, offset, val);\n    return;"
            }
          }
        },
        "logicalLocations": [
          {
            "fullyQualifiedName": "collections::list::add",
            "index": 0
          }
        ]
      }
    ],
    "relatedLocations": [
      {
        "id": 0,
        "message": {
          "id": "variableDeclared",
          "arguments": [
            "ptr"
          ]
        },
        "physicalLocation": {
          "artifactLocation": {
            "uri": "collections/list.h",

```

```

        "uriBaseId": "SRCROOT",
        "index": 3
    },
    "region": {
        "startLine": 8,
        "startColumn": 5
    }
},
"logicalLocations": [
    {
        "fullyQualifiedName": "collections::list::add",
        "index": 0
    }
]
},
"codeFlows": [
    {
        "message": {
            "text": "Path from declaration to usage"
        },
        "threadFlows": [
            {
                "id": "thread-52",
                "locations": [
                    {
                        "importance": "essential",
                        "location": {
                            "message": {
                                "text": "Variable `ptr` declared.",
                                "markdown": "Variable `ptr` declared."
                            },
                            "physicalLocation": {
                                "artifactLocation": {
                                    "uri": "collections/list.h",
                                    "uriBaseId": "SRCROOT",
                                    "index": 3
                                },
                                "region": {
                                    "startLine": 15,
                                    "snippet": {
                                        "text": "int *ptr;"
                                    }
                                }
                            },
                            "logicalLocations": [
                                {
                                    "fullyQualifiedName": "collections::list::add",
                                    "index": 0
                                }
                            ]
                        },
                        "module": "platform"
                    },
                    {
                        "state": {
                            "y": {
                                "text": "2"
                            },
                            "z": {
                                "text": "4"
                            },
                            "y + z": {
                                "text": "6"
                            },
                            "q": {
                                "text": "7"
                            }
                        },
                        "importance": "unimportant",
                        "location": {
                            "physicalLocation": {
                                "artifactLocation": {
                                    "uri": "collections/list.h",
                                    "uriBaseId": "SRCROOT",
                                    "index": 3
                                },
                                "region": {
                                    "startLine": 15,
                                    "snippet": {
                                        "text": "offset = (y + z) * q + 1;"
                                    }
                                }
                            },
                            "logicalLocations": [
                                {
                                    "fullyQualifiedName": "collections::list::add",
                                    "index": 0
                                }
                            ]
                        },
                        "annotations": [
                            {
                                "startLine": 15,
                                "startColumn": 13,
                                "endColumn": 19,
                                "message": {
                                    "text": "(y + z) = 42",
                                    "markdown": "(y + z) = 42"
                                }
                            }
                        ]
                    }
                ]
            }
        ]
    }
],

```



```

    "module": "platform"
  },
  {
    "importance": "essential",
    "location": {
      "message": {
        "text": "Uninitialized variable `ptr` passed to method `add_core`.",
        "markdown": "Uninitialized variable `ptr` passed to method `add_core`."
      },
      "physicalLocation": {
        "artifactLocation": {
          "uri": "collections/list.h",
          "uriBaseId": "SRCROOT",
          "index": 3
        },
        "region": {
          "startLine": 25,
          "snippet": {
            "text": "add_core(ptr, offset, val)"
          }
        }
      },
      "logicalLocations": [
        {
          "fullyQualifiedName": "collections::list::add",
          "index": 0
        }
      ]
    },
    "module": "platform"
  }
]
},
"stacks": [
  {
    "message": {
      "text": "Call stack resulting from usage of uninitialized variable."
    },
    "frames": [
      {
        "location": {
          "message": {
            "text": "Exception thrown."
          },
          "physicalLocation": {
            "artifactLocation": {
              "uri": "collections/list.h",
              "uriBaseId": "SRCROOT",
              "index": 3
            },
            "region": {
              "startLine": 110,
              "startColumn": 15
            }
          },
          "address": {
            "offsetFromParent": 4229178
          }
        },
        "logicalLocations": [
          {
            "fullyQualifiedName": "collections::list::add_core",
            "index": 0
          }
        ]
      },
      {
        "module": "platform",
        "threadId": 52,
        "parameters": [
          "null",
          "0",
          "14"
        ]
      }
    ]
  },
  {
    "location": {
      "physicalLocation": {
        "artifactLocation": {
          "uri": "collections/list.h",
          "uriBaseId": "SRCROOT",
          "index": 3
        },
        "region": {
          "startLine": 43,
          "startColumn": 15
        }
      },
      "address": {
        "offsetFromParent": 4229268
      }
    },
    "logicalLocations": [
      {
        "fullyQualifiedName": "collections::list::add",
        "index": 0
      }
    ]
  },
  {
    "module": "platform",
    "threadId": 52,
    "parameters": [

```

```
    "14"  
  ],  
  {  
    "location": {  
      "physicalLocation": {  
        "artifactLocation": {  
          "uri": "application/main.cpp",  
          "uriBaseId": "SRCROOT",  
          "index": 1  
        },  
        "region": {  
          "startLine": 28,  
          "startColumn": 9  
        },  
        "address": {  
          "offsetFromParent": 4229836  
        }  
      },  
      "logicalLocations": [  
        {  
          "fullyQualifiedName": "main",  
          "index": 4  
        }  
      ]  
    },  
    "module": "application",  
    "threadId": 52  
  }  
]  
},  
"fixes": [  
  {  
    "description": {  
      "text": "Initialize the variable to null"  
    },  
    "artifactChanges": [  
      {  
        "artifactLocation": {  
          "uri": "collections/list.h",  
          "uriBaseId": "SRCROOT",  
          "index": 3  
        },  
        "replacements": [  
          {  
            "deletedRegion": {  
              "startLine": 42  
            },  
            "insertedContent": {  
              "text": "A different line\n"  
            }  
          }  
        ],  
      }  
    ],  
  }  
],  
"hostedViewerUri": "https://www.example.com/viewer/3918d370-c636-40d8-bf23-8c176043a2df",  
"workItemUris": [  
  "https://github.com/example/project/issues/42",  
  "https://github.com/example/project/issues/54"  
],  
"provenance": {  
  "firstDetectionTimeUtc": "2016-07-15T14:20:42Z",  
  "firstDetectionRunGuid": "8F62D8A0-C14F-4516-9959-1A663BA6FB99",  
  "lastDetectionTimeUtc": "2016-07-16T14:20:42Z",  
  "lastDetectionRunGuid": "BC650830-A9FE-44CB-8818-AD6C387279A0",  
  "invocationIndex": 0  
}  
}  
}  
}
```

Appendix L. (Informative) Revision History

Revision	Date	Editors	Description
sarif-v2.2-wd20240605-dev	2024-06-05	Stacy Wray and Stefan Hagen	Editor revision implementing proposals #471 and #637.
sarif-v2.2-wd20240808-dev	2024-08-08	Stacy Wray and Stefan Hagen	Editor revision implementing proposals #459, #483, #491, #492, and #634.
sarif-v2.2-wd20250612-dev	2025-06-12	Stefan Hagen	Editor revision for TC meeting.
sarif-v2.2-wd20250710-dev	2025-07-10	Stefan Hagen	Editor revision for TC meeting.
sarif-v2.2-wd20250807-dev	2025-08-07	Stefan Hagen	Editor revision for TC meeting.
sarif-v2.2-wd20251204-dev	2025-12-04	Stefan Hagen	Editor revision for TC meeting.
sarif-v2.2-wd20260205-dev	2026-02-05	Stefan Hagen	Editor revision for TC meeting.
sarif-v2.2-wd20260305-dev	2026-03-05	Stefan Hagen	Editor revision for TC meeting.

Appendix M. (Informative) MIME Types and File Name Extensions

The following is a list of MIME types and file extensions for files that conform to this specification, registered according to [RFC2048].

MIME type	Extension	Description
application/sarif+json	.sarif,.sarif.json	SARIF log files (§3)
application/sarif-external-properties+json	.sarif-external-properties,.sarif-external-properties.json	SARIF external property files (§4)

Appendix N. Safety, Security and Data Protection

All safety, security, and data protection requirements relevant to the context in which SARIF documents are used **MUST** be translated into, and consistently enforced through, SARIF implementations and processes.

Maintainers of SARIF-producing tools determine which data in the SARIF output should be treated as sensitive.

Example: Data protection requirements (together with best practices) rule and guide the handling of sensitive data in SARIF reports.

SARIF documents are based on JSON, thus the security considerations of [RFC8259] apply and are repeated here as service for the reader:

Generally, there are security issues with scripting languages. JSON is a subset of JavaScript but excludes assignment and invocation.

Since JSON's syntax is borrowed from JavaScript, it is possible to use that language's `eval()` function to parse most JSON texts (but not all; certain characters such as U+2028 LINE SEPARATOR and U+2029 PARAGRAPH SEPARATOR are legal in JSON but not JavaScript). This generally constitutes an unacceptable security risk, since the text could contain executable code along with data declarations. The same consideration applies to the use of `eval()`-like functions in any other programming language in which JSON texts conform to that language's syntax.

In addition, SARIF documents may be rendered by consumers in various human-readable formats like HTML or PDF. Thus, for security reasons, SARIF producers and consumers **SHALL** adhere to the following:

- SARIF producers **SHOULD NOT** emit messages that contain HTML, even though GitHub-flavoured Markdown is permitted. To include HTML, source code, or any other content that may be interpreted or executed by a SARIF consumer, e.g. to provide a proof-of-concept, the issuing party **SHALL** use Markdown's fenced code blocks or inline code option.
- Deeply nested markup can cause a stack overflow in the Markdown processor [GFMENG]. To reduce this risk, SARIF consumers **SHALL** use a Markdown processor that is hardened against such attacks. **Note:** One example is the GitHub fork of the `cmark` Markdown processor [GFMCMARK].
- To reduce the risk posed by possibly malicious SARIF files that do contain arbitrary HTML (including, for example, `data:image/svg+xml`), SARIF consumers **SHALL** either disable HTML processing (for example, by using the `--safe` option in the `cmark` Markdown processor) or run the resulting HTML through an HTML sanitizer.
- To reduce the risk posed by possibly malicious links within a SARIF document (including, for example, `javascript:` links), SARIF consumers **SHALL** either remove all actions from links (for example, by displaying them as standard text) or render only those actionable that are known to be safe (for example, determining that via the media type).
- `externalPropertyFileReferences` **MAY** allow confidential data to be read from JSON files to which the SARIF producer does not have access.
- In an `artifactLocation` object, the `uri` property **MAY** point to a file outside the expected directory tree, e.g. `file:///etc/passwd` when the actually analysed files are in a disjoint directory tree. In this case, `fix` objects **MAY** propose edits to such an out-of-tree artifact.

SARIF consumers that are not prepared to deal with the security implications of formatted messages SHALL NOT attempt to render them and SHALL instead fall back to the corresponding plain text messages. As also any other programming code can be contained within a SARIF document, SARIF consumers SHALL ensure that none of the values of a SARIF document is run as code. Moreover, it SHALL be treated as unsafe (user) input.

Additional, supporting mitigation measures like retrieving only SARIF documents from trusted sources and check their integrity before parsing the document SHOULD be in place to reduce the risk further.

The distribution requirements of SARIF data allow to specify domains as the value of the HTTP header `Access-Control-Allow-Origin`. While a wildcard (*) as header value usually prevents implementing browsers from sending credentials during the CORS request, the restriction to specified domains often enables sending credentials. Allowing several specified domains results in using dynamics on the server, which can widen the attack surface by using more code and configuration. Furthermore, this might reveal information about internal structures, e.g. which domains are allowed to send credentials, or which tools are used. Given that credentials from a browser are a potent tool in the event of an attack, restricting the origins seems to imply a higher risk and therefore less secure than allowing all domains without credentials.

As setting the `Access-Control-Allow-Origin` header potentially allows for cross site request forgery, it SHOULD only be served on files and directories containing SARIF data. For any restricted feeds, standard authentication methods SHOULD be used that are not send by web browsers if the wildcard is used as header value.